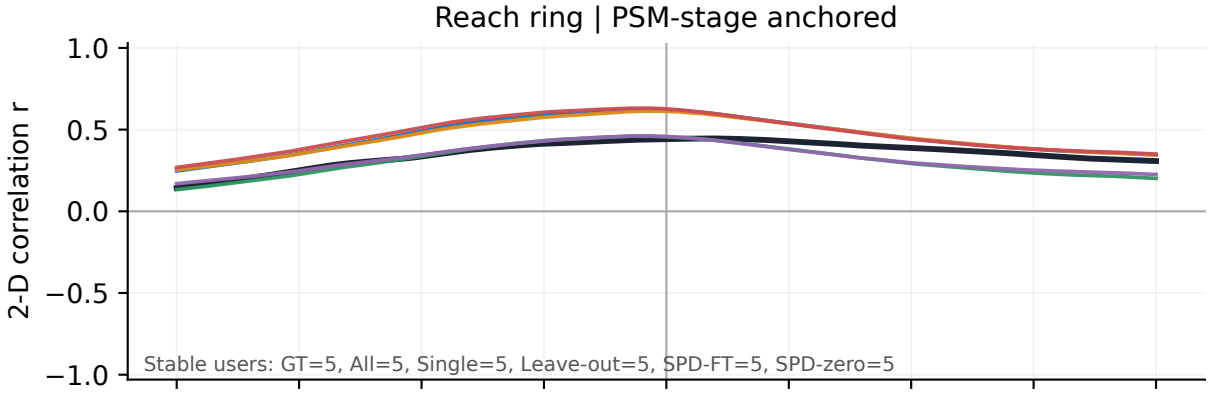
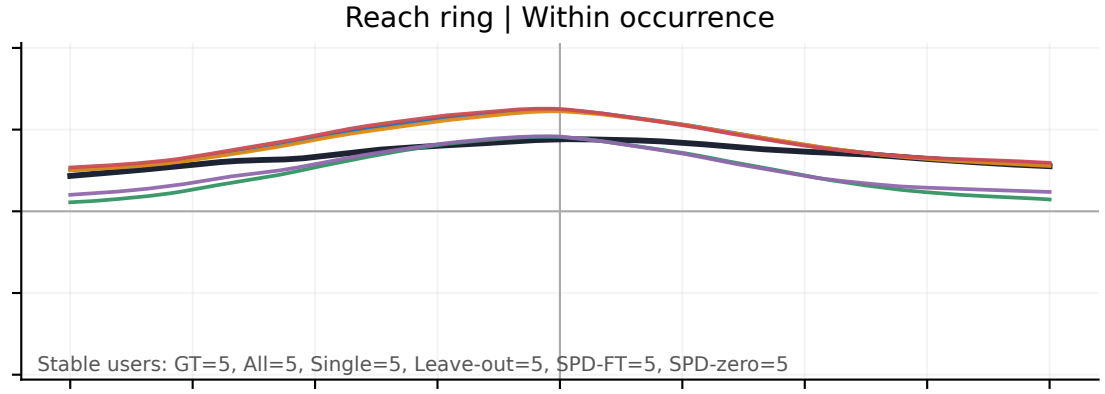
Task A: position xcorr with independently selected DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

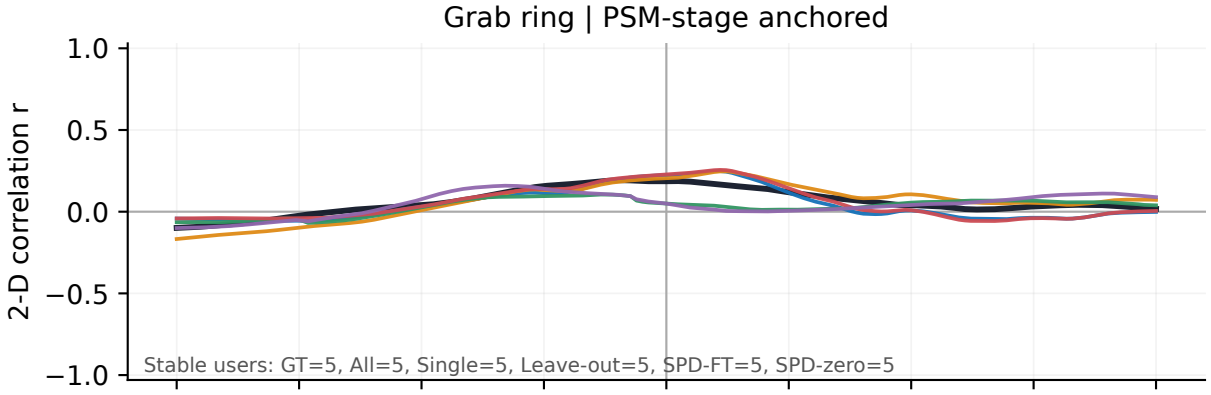
Reach ring | PSM-stage anchored



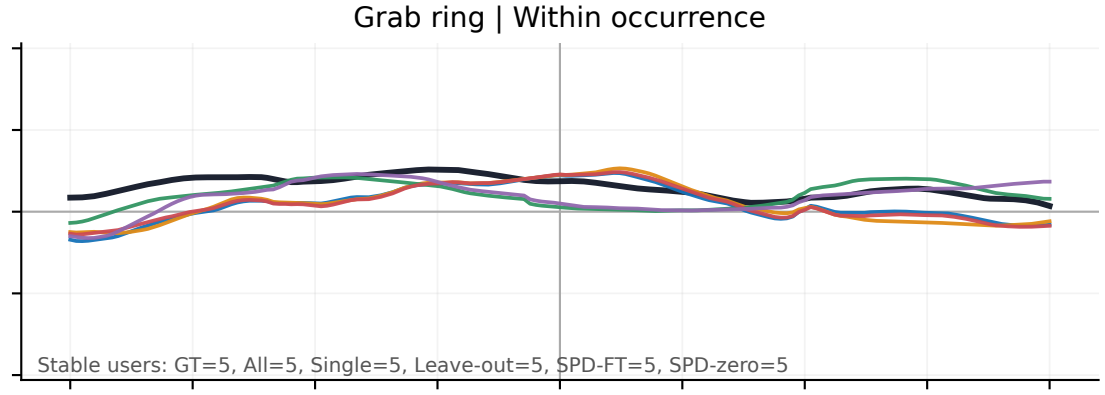
Reach ring | Within occurrence



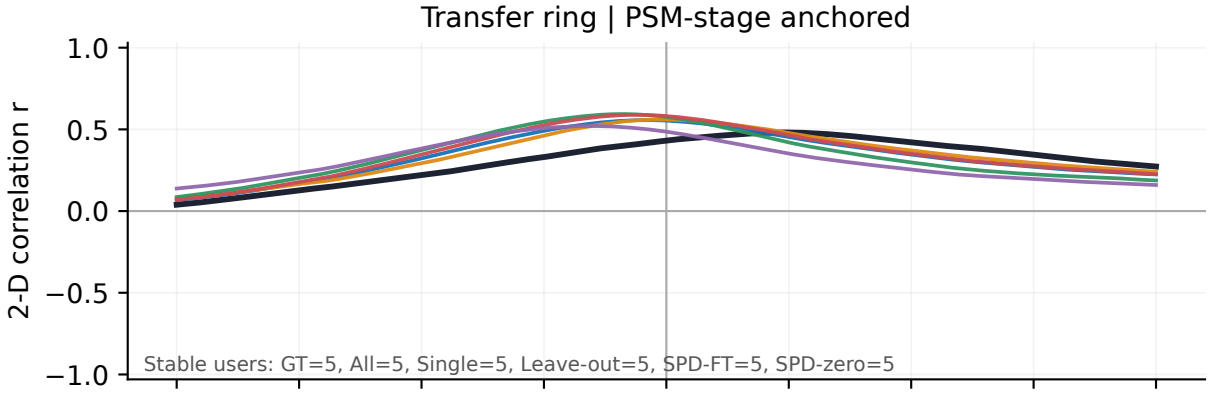
Grab ring | PSM-stage anchored



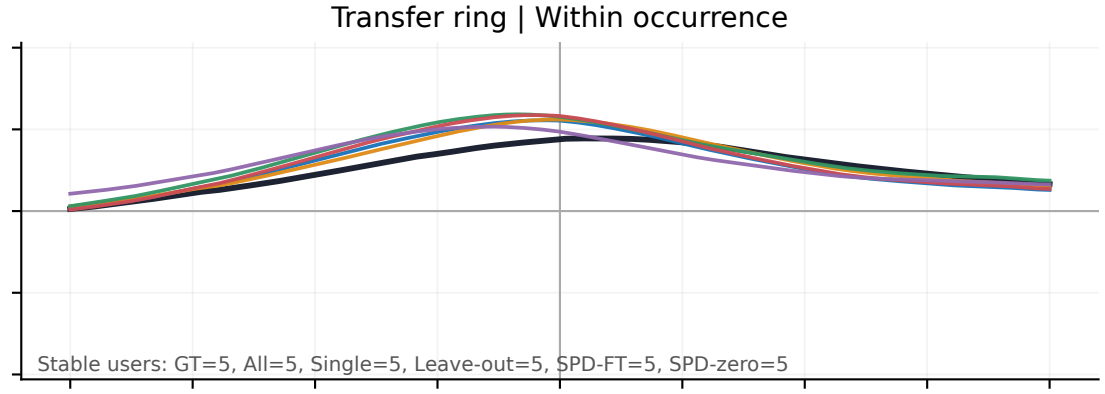
Grab ring | Within occurrence



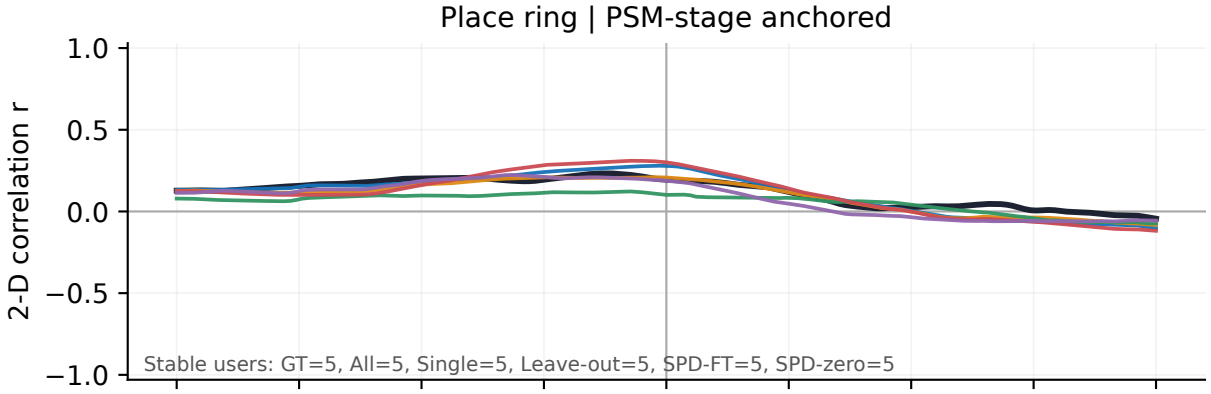
Transfer ring | PSM-stage anchored



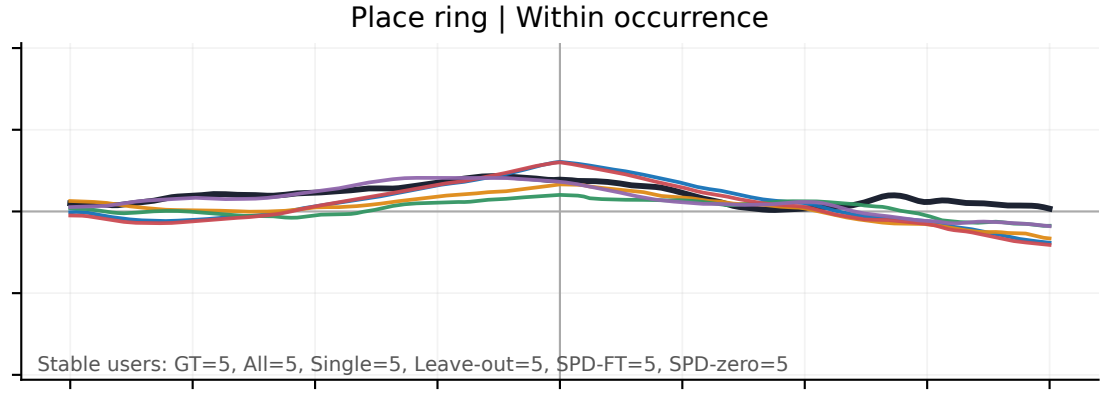
Transfer ring | Within occurrence



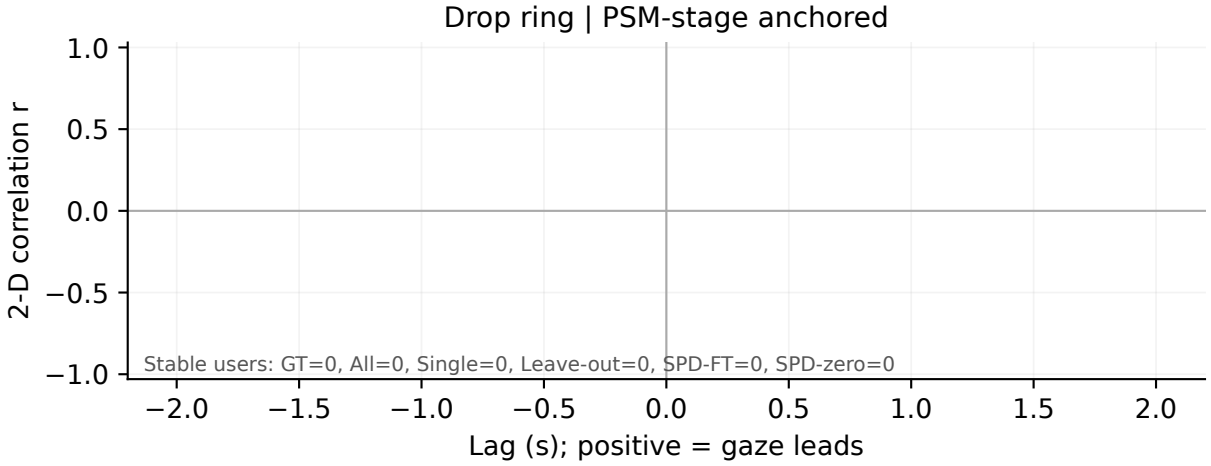
Place ring | PSM-stage anchored



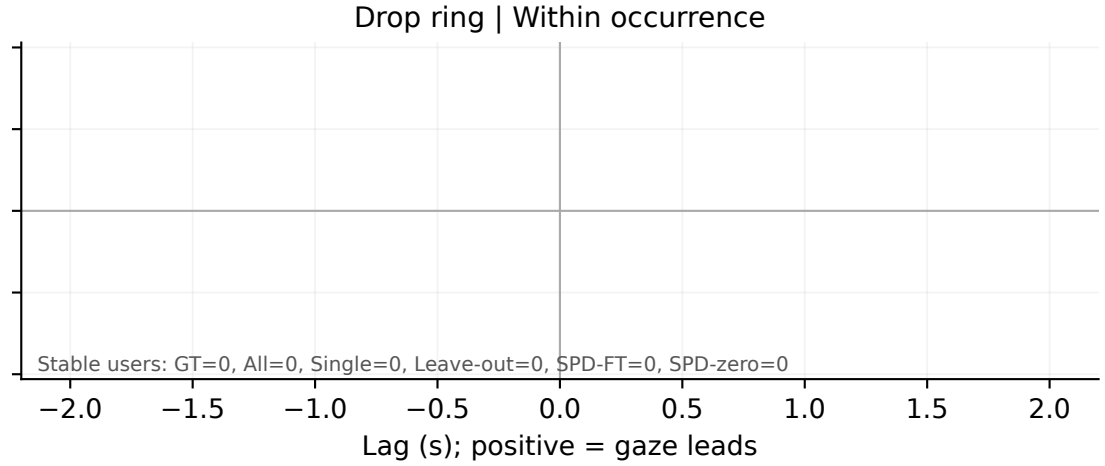
Place ring | Within occurrence



Drop ring | PSM-stage anchored



Drop ring | Within occurrence

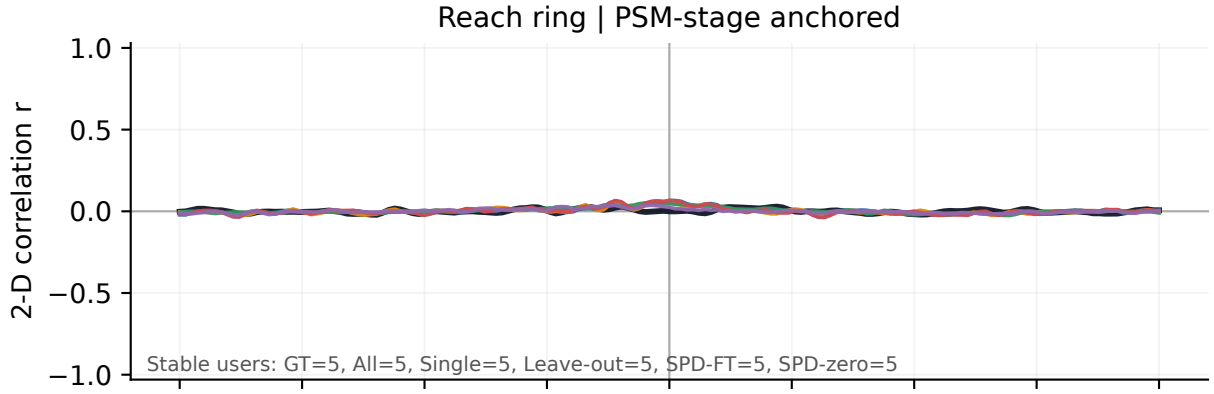


Each source retains its selected side's whole curve. Sides and stable-user support may differ between sources.

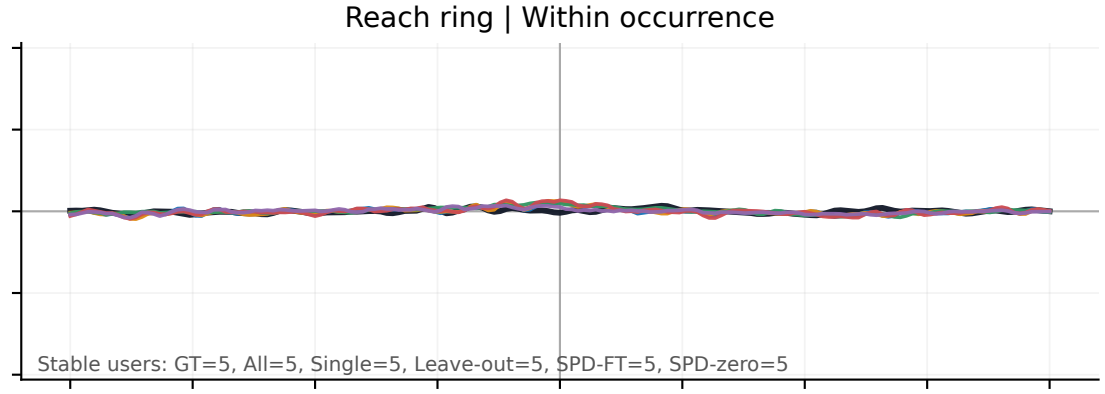
Task A: velocity xcorr with independently selected DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

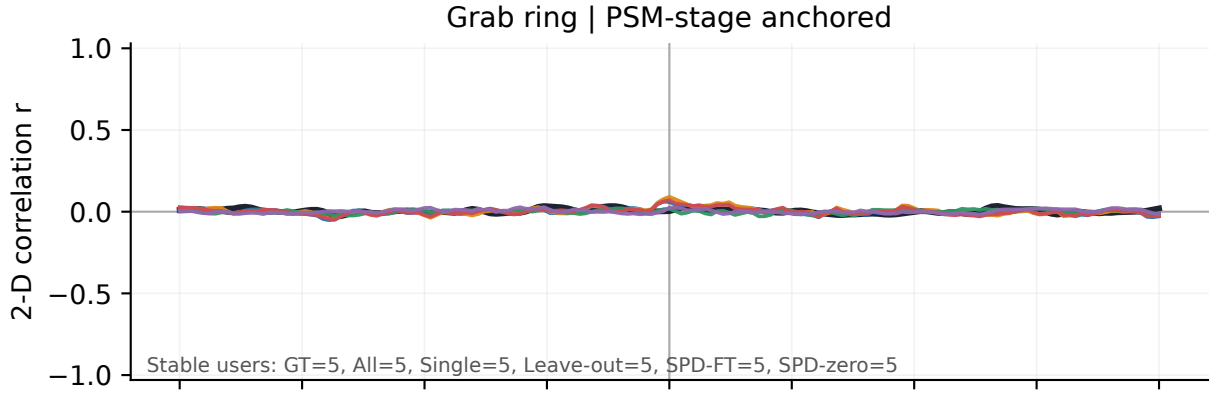
Reach ring | PSM-stage anchored



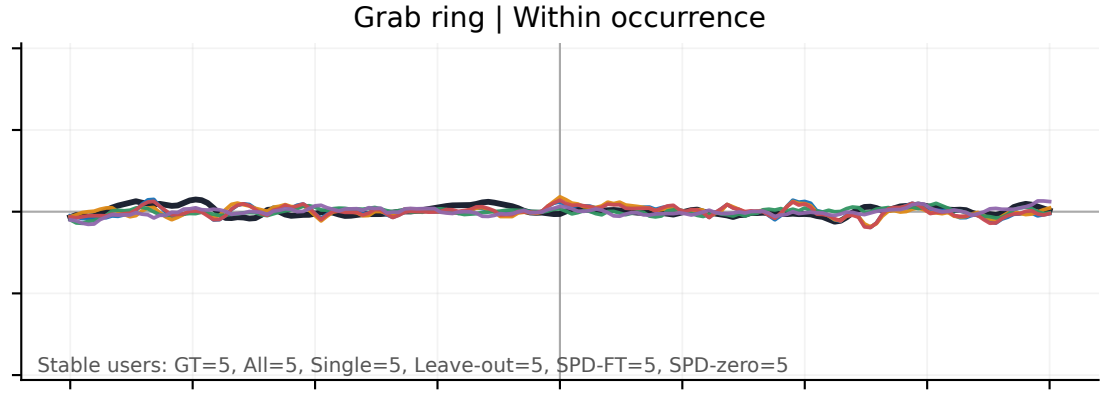
Reach ring | Within occurrence



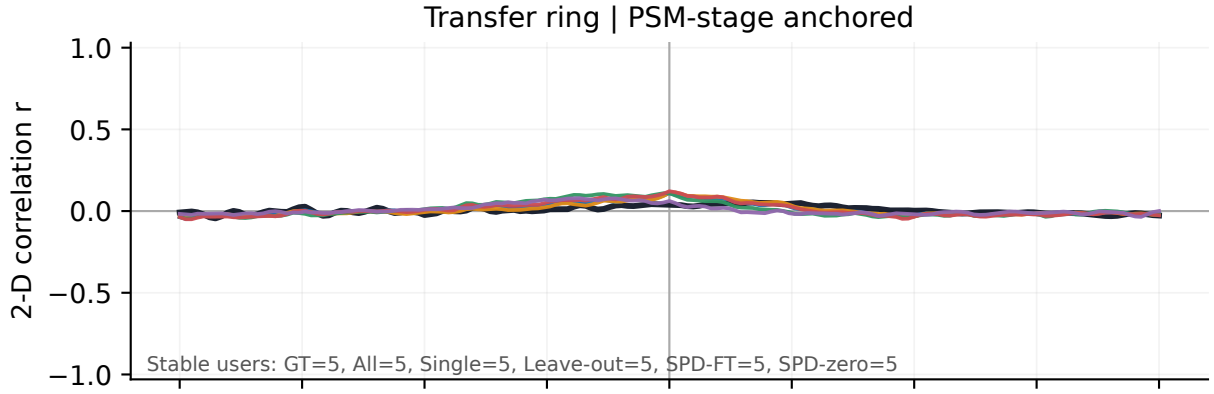
Grab ring | PSM-stage anchored



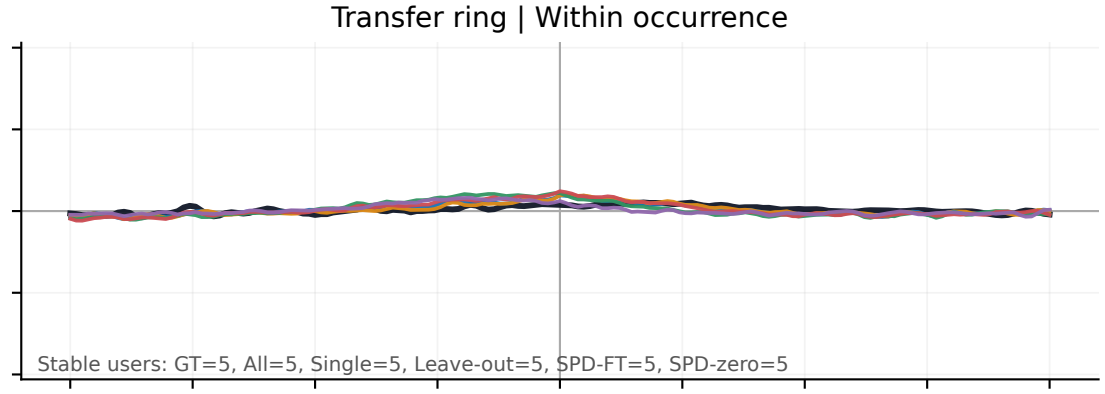
Grab ring | Within occurrence



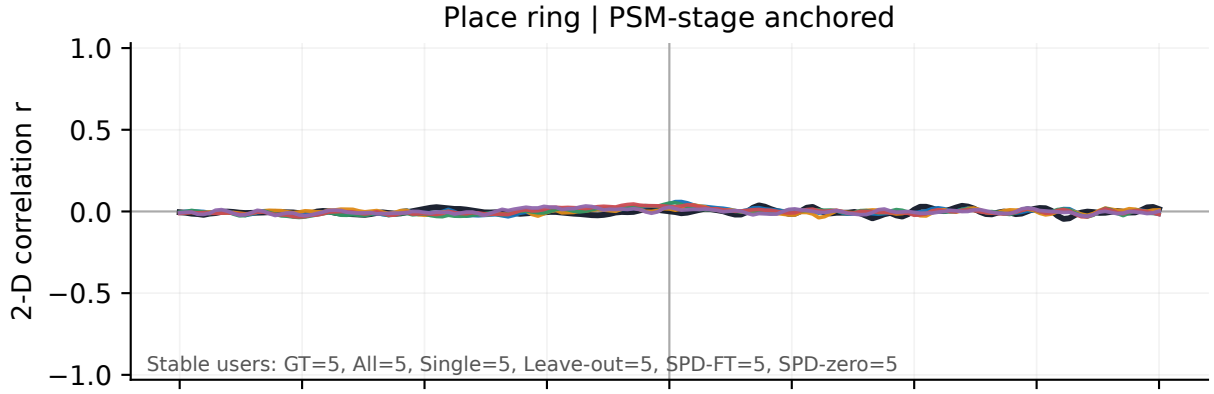
Transfer ring | PSM-stage anchored



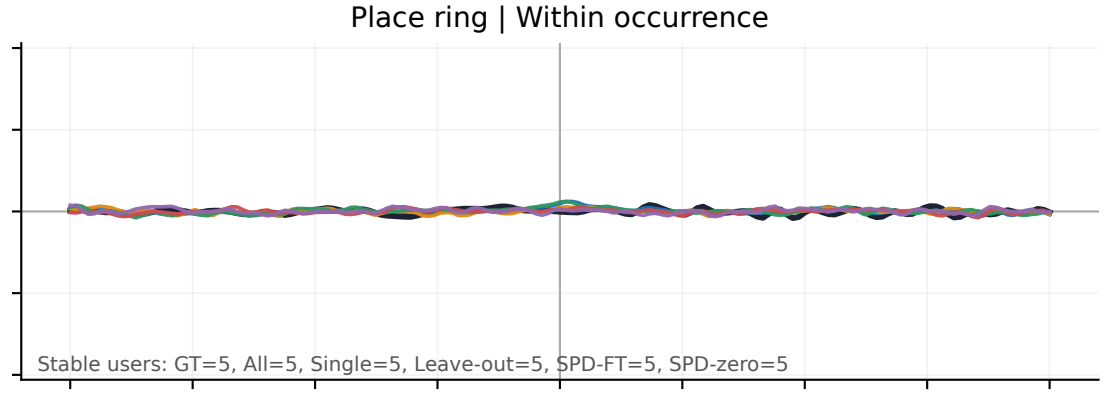
Transfer ring | Within occurrence



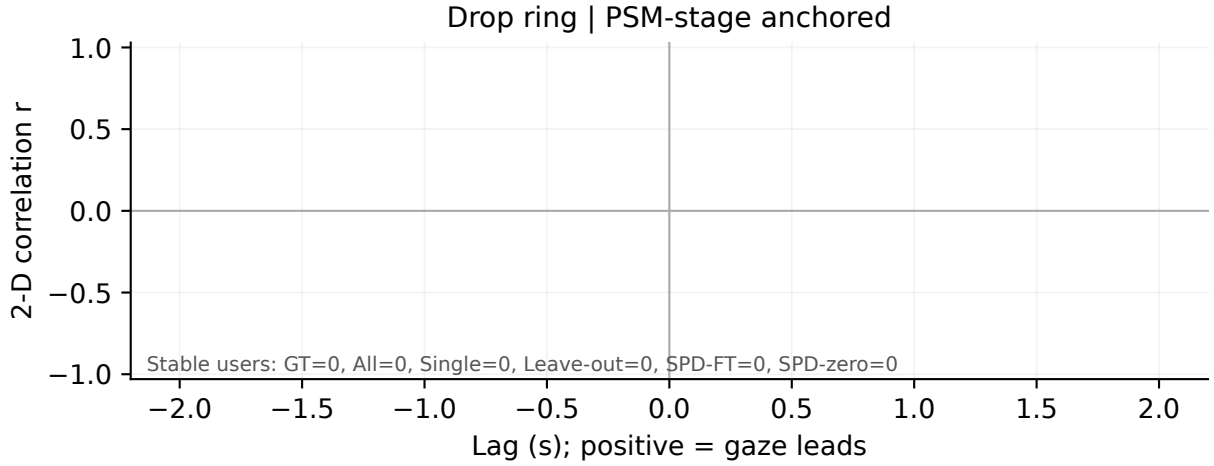
Place ring | PSM-stage anchored



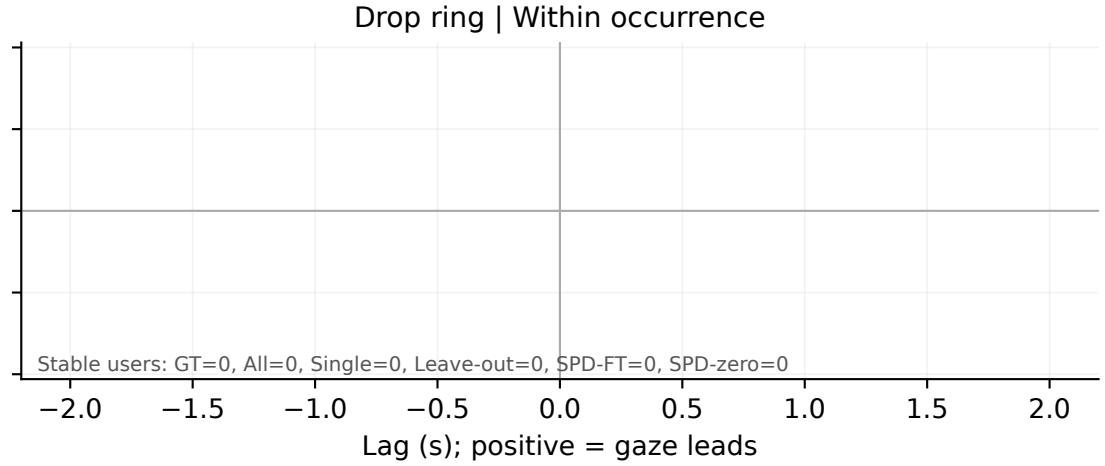
Place ring | Within occurrence



Drop ring | PSM-stage anchored



Drop ring | Within occurrence

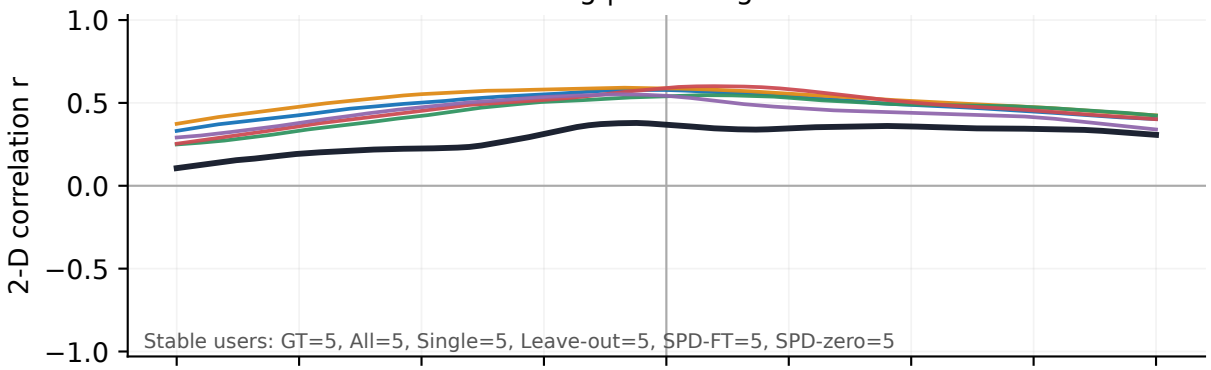


Each source retains its selected side's whole curve. Sides and stable-user support may differ between sources.

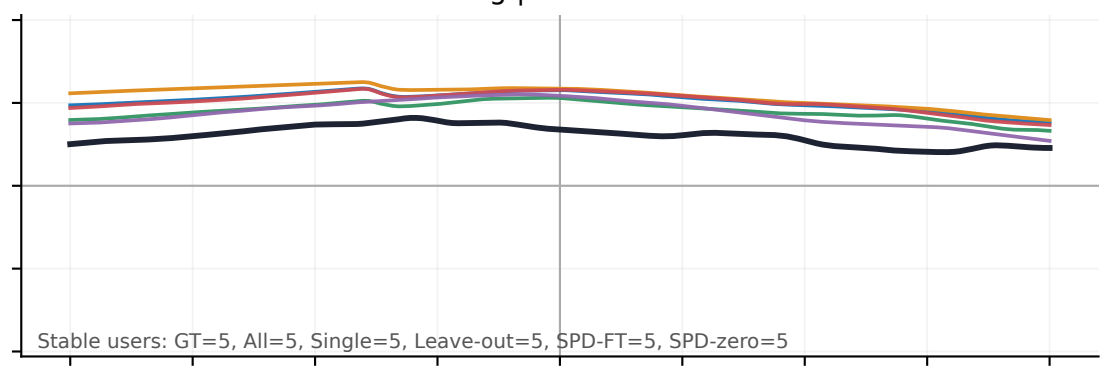
Task C: position xcorr with independently selected DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

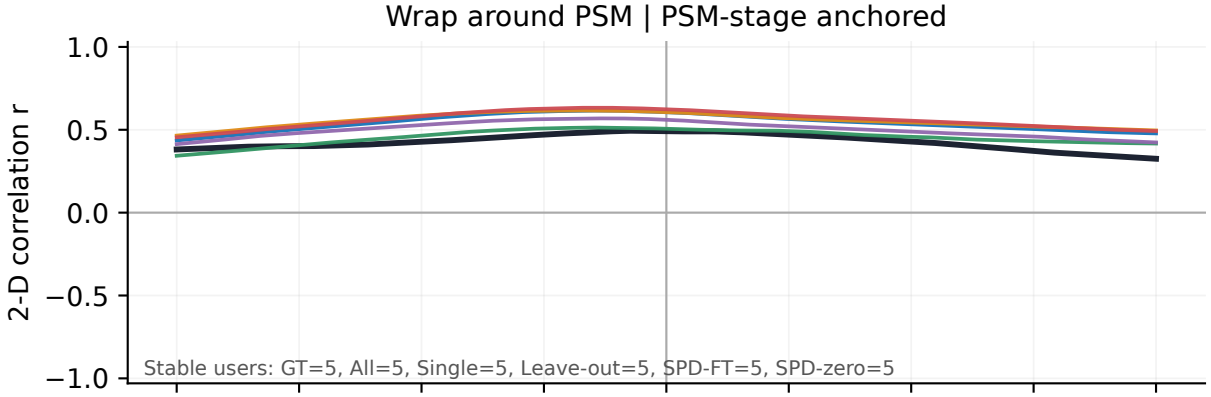
Reach string | PSM-stage anchored



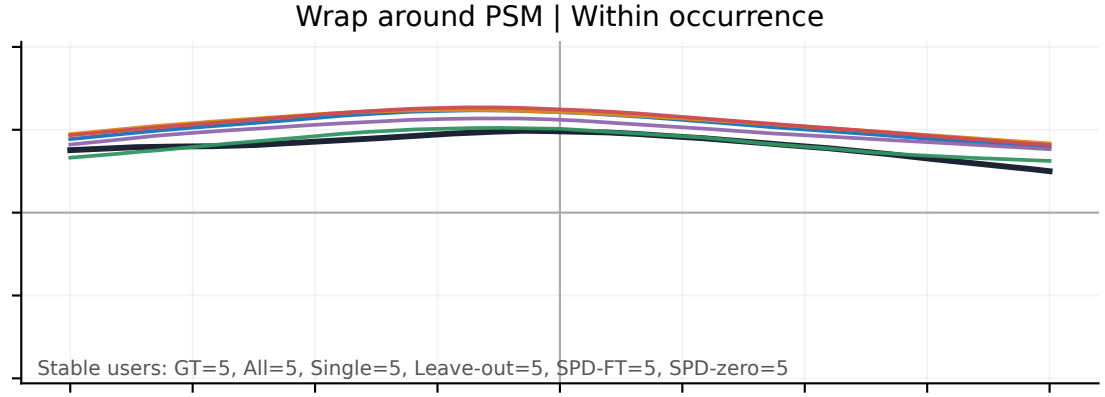
Reach string | Within occurrence



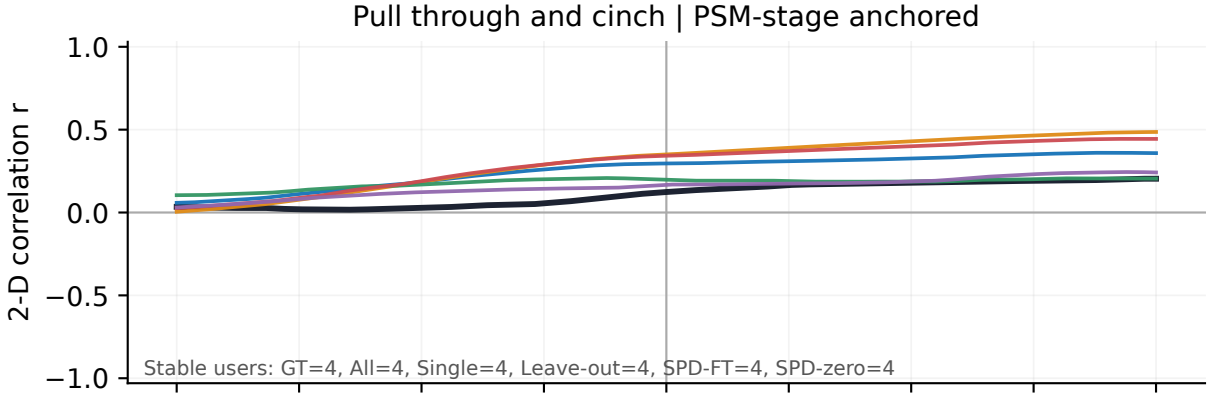
Wrap around PSM | PSM-stage anchored



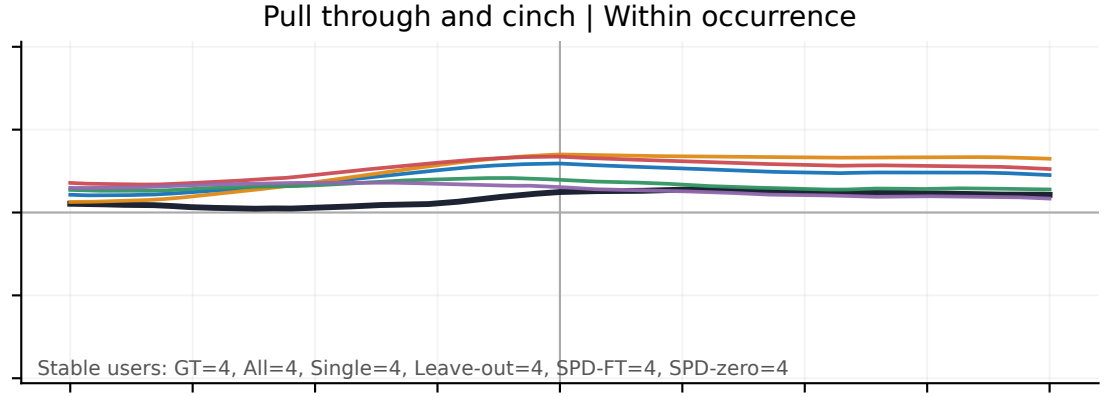
Wrap around PSM | Within occurrence



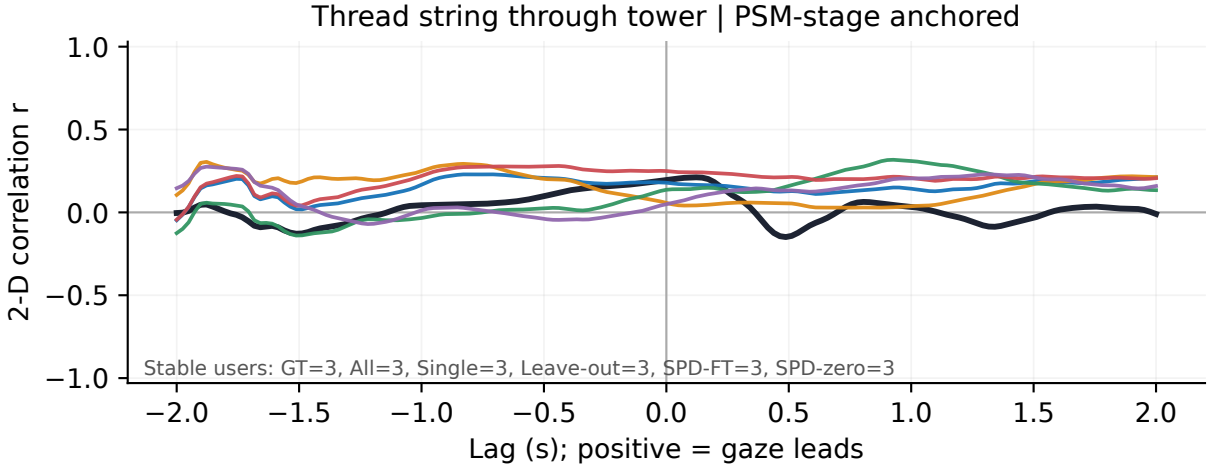
Pull through and cinch | PSM-stage anchored



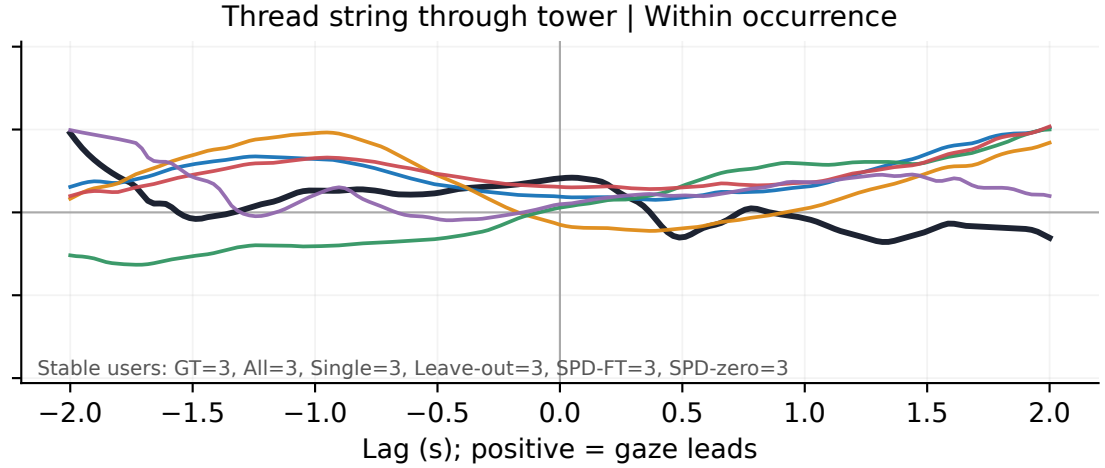
Pull through and cinch | Within occurrence



Thread string through tower | PSM-stage anchored



Thread string through tower | Within occurrence

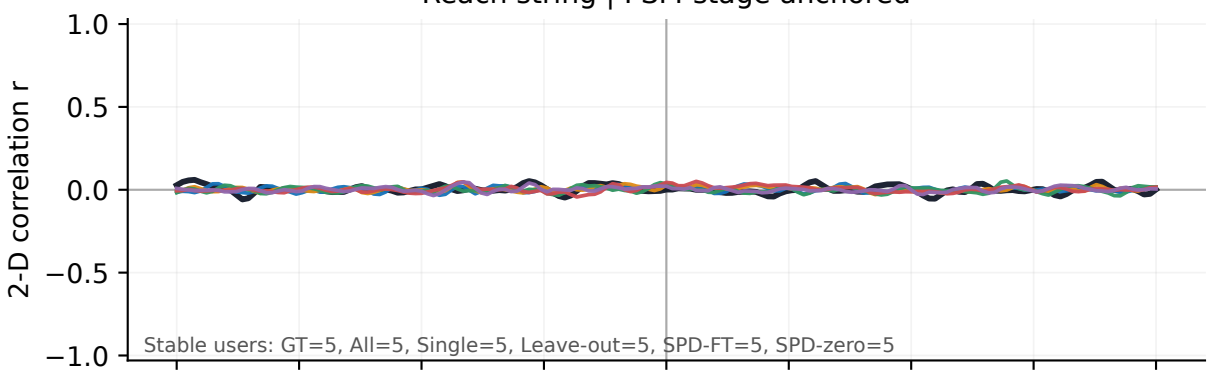


Each source retains its selected side's whole curve. Sides and stable-user support may differ between sources.

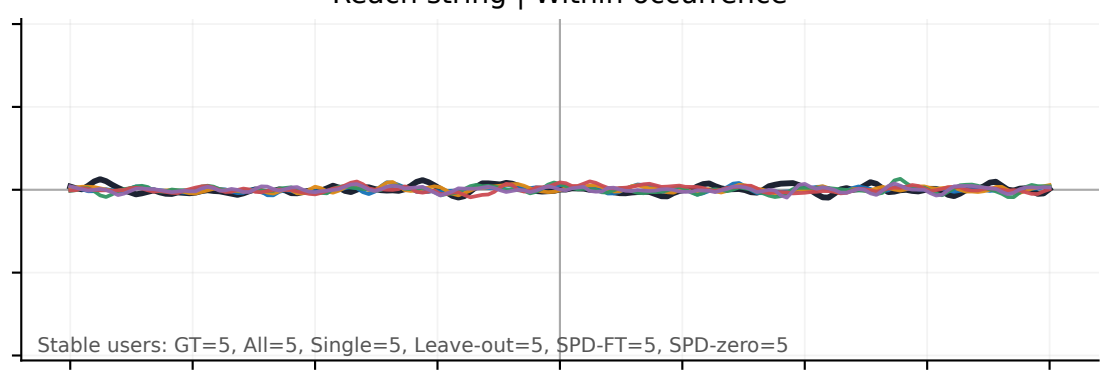
Task C: velocity xcorr with independently selected DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

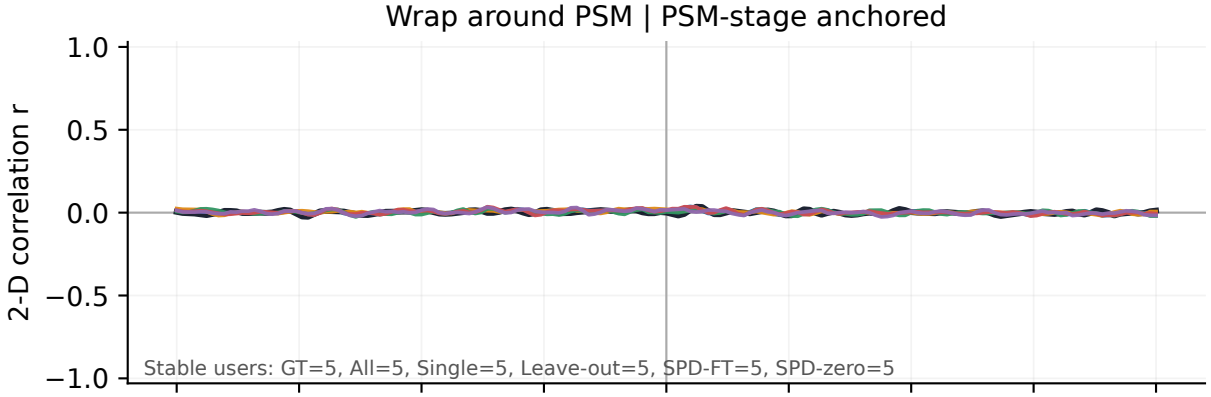
Reach string | PSM-stage anchored



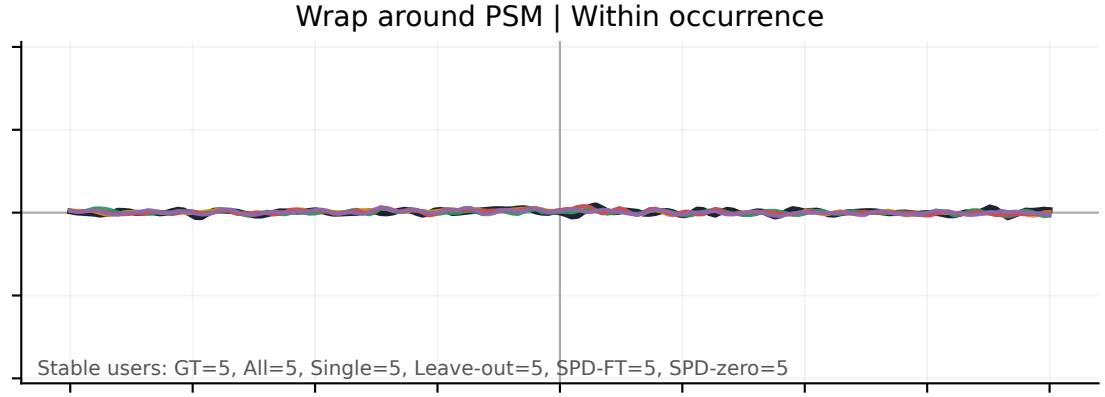
Reach string | Within occurrence



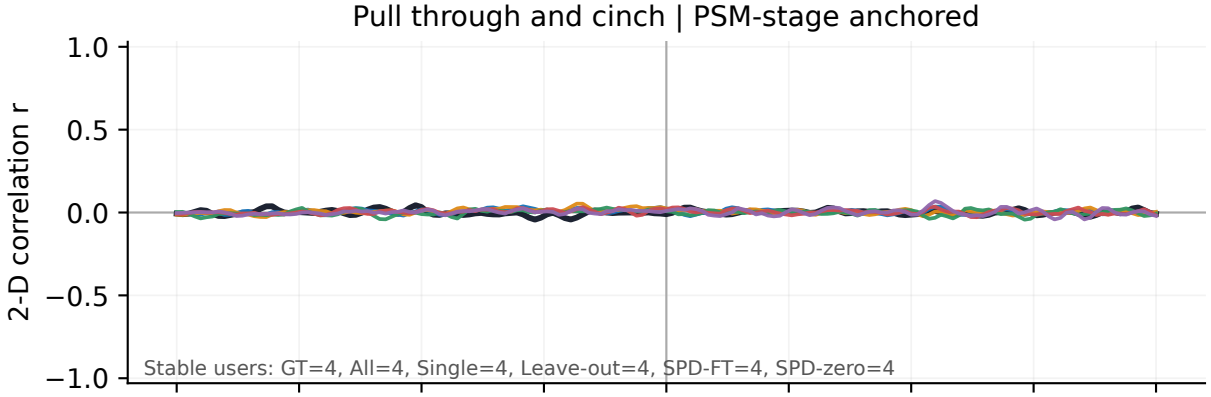
Wrap around PSM | PSM-stage anchored



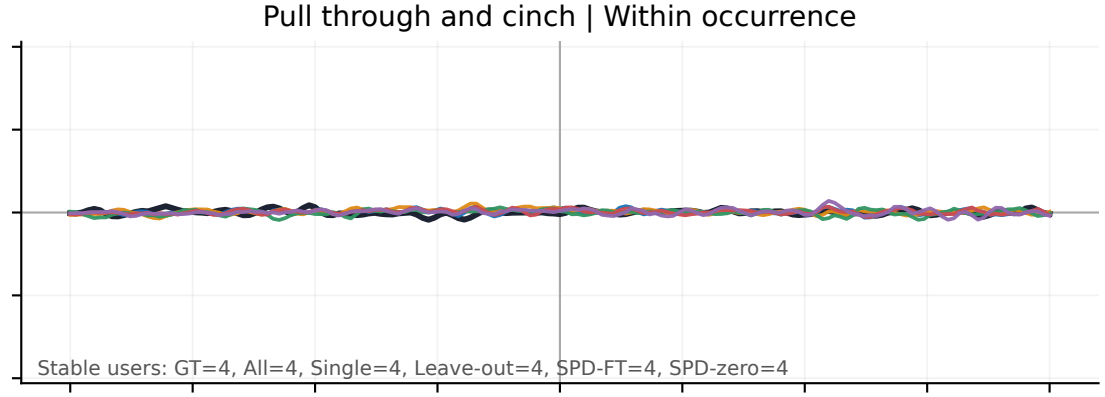
Wrap around PSM | Within occurrence



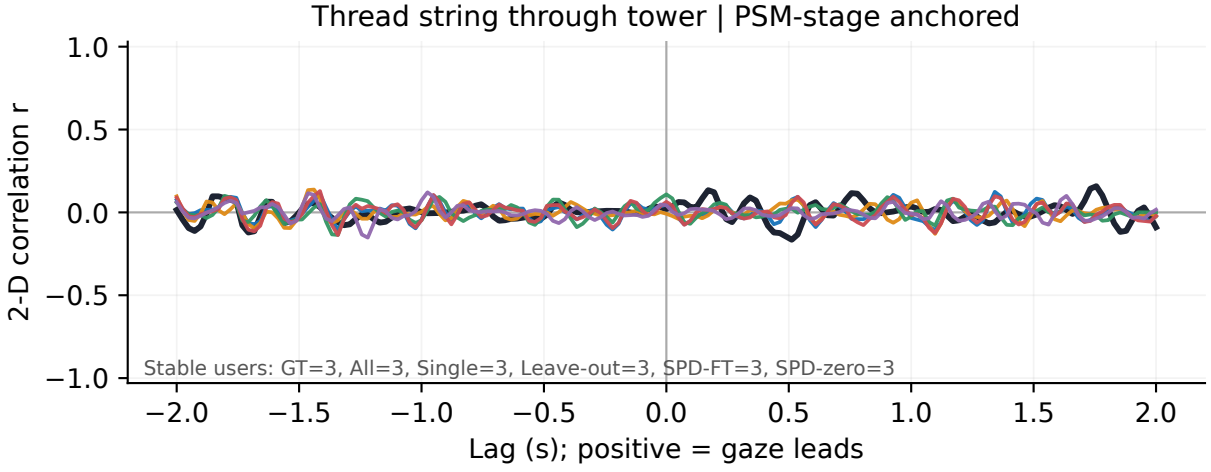
Pull through and cinch | PSM-stage anchored



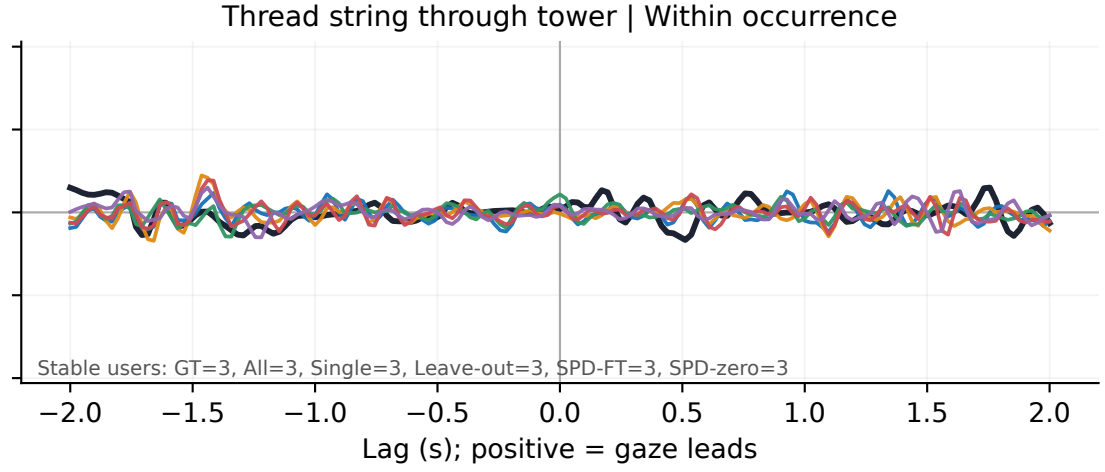
Pull through and cinch | Within occurrence



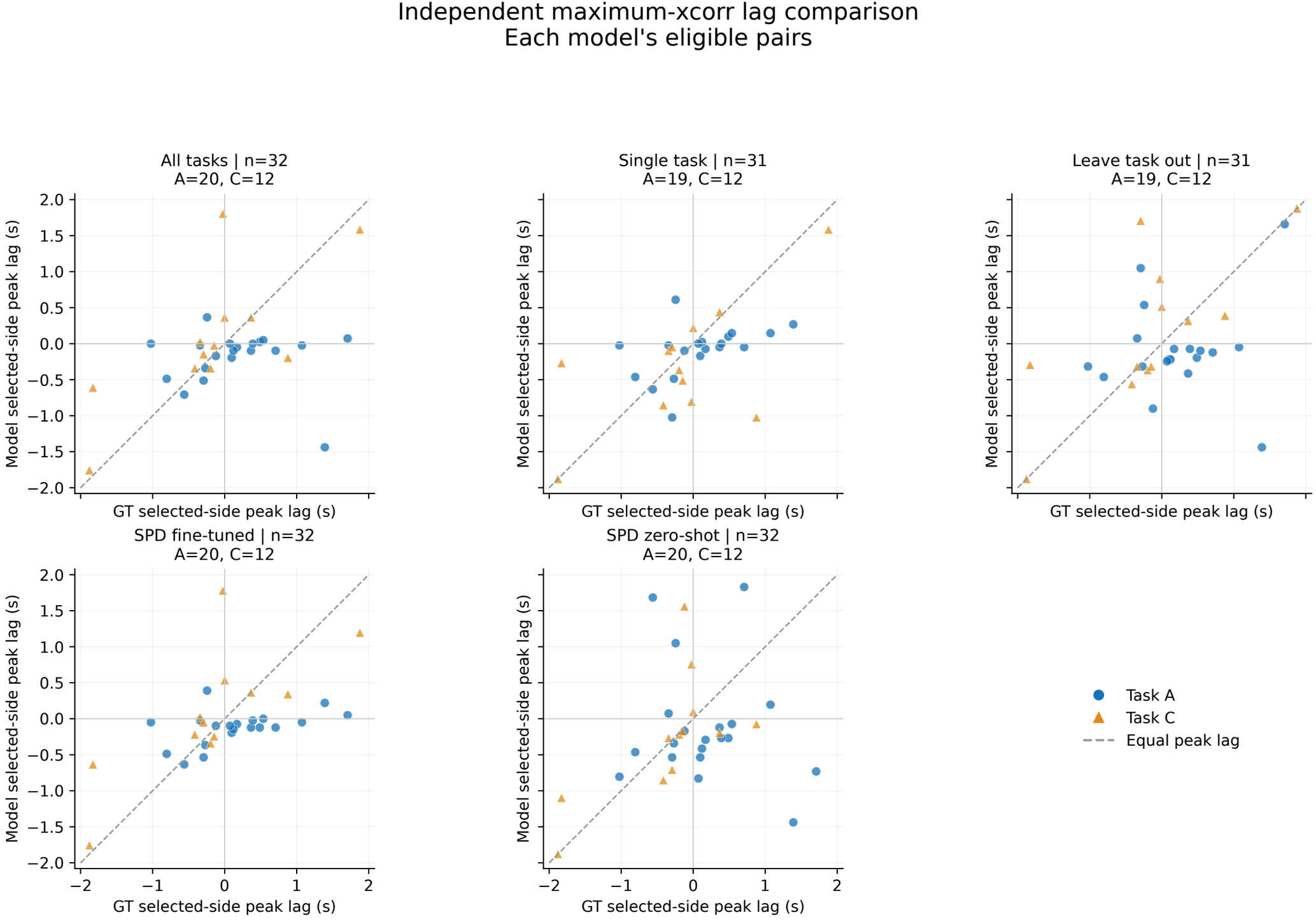
Thread string through tower | PSM-stage anchored

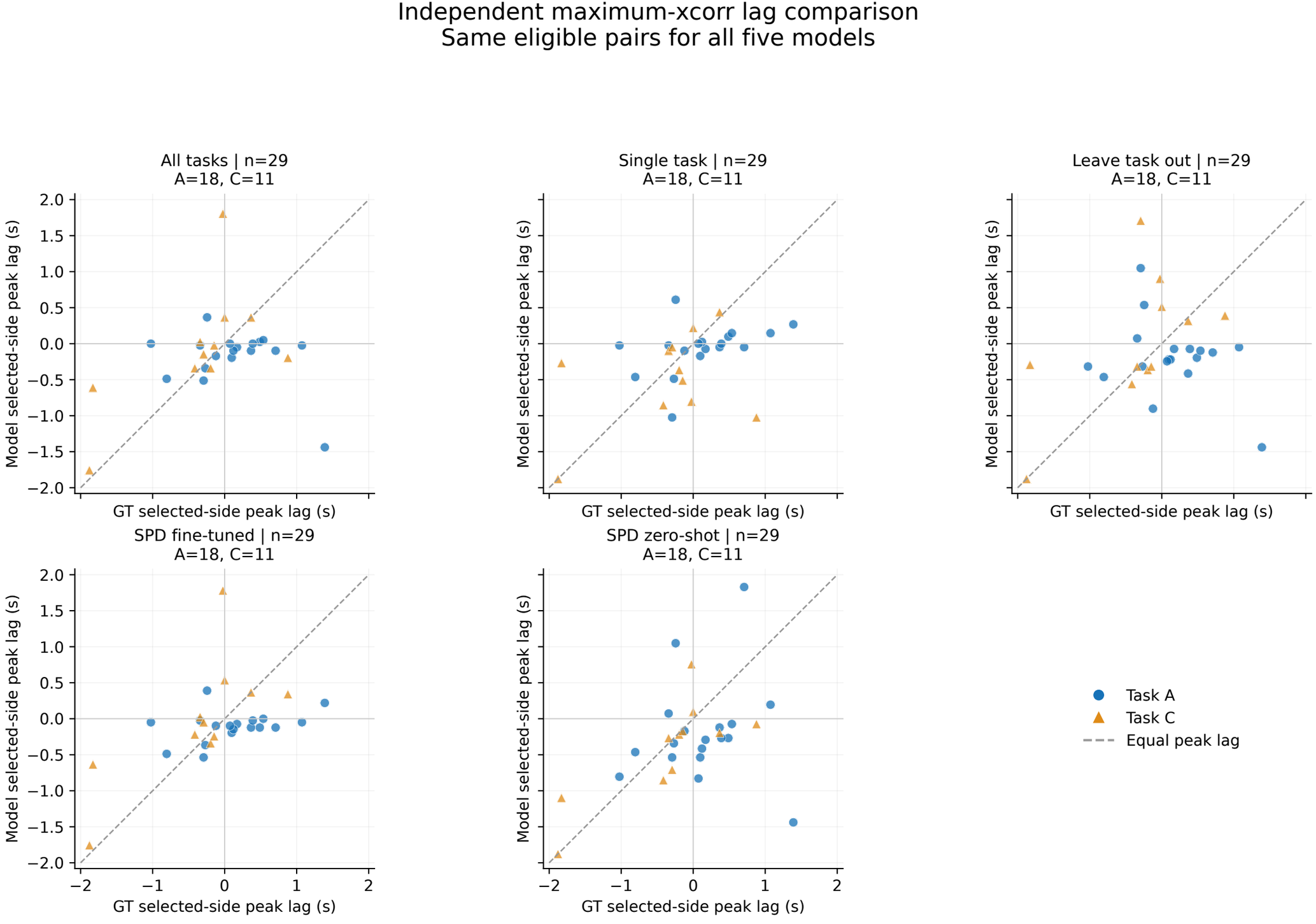


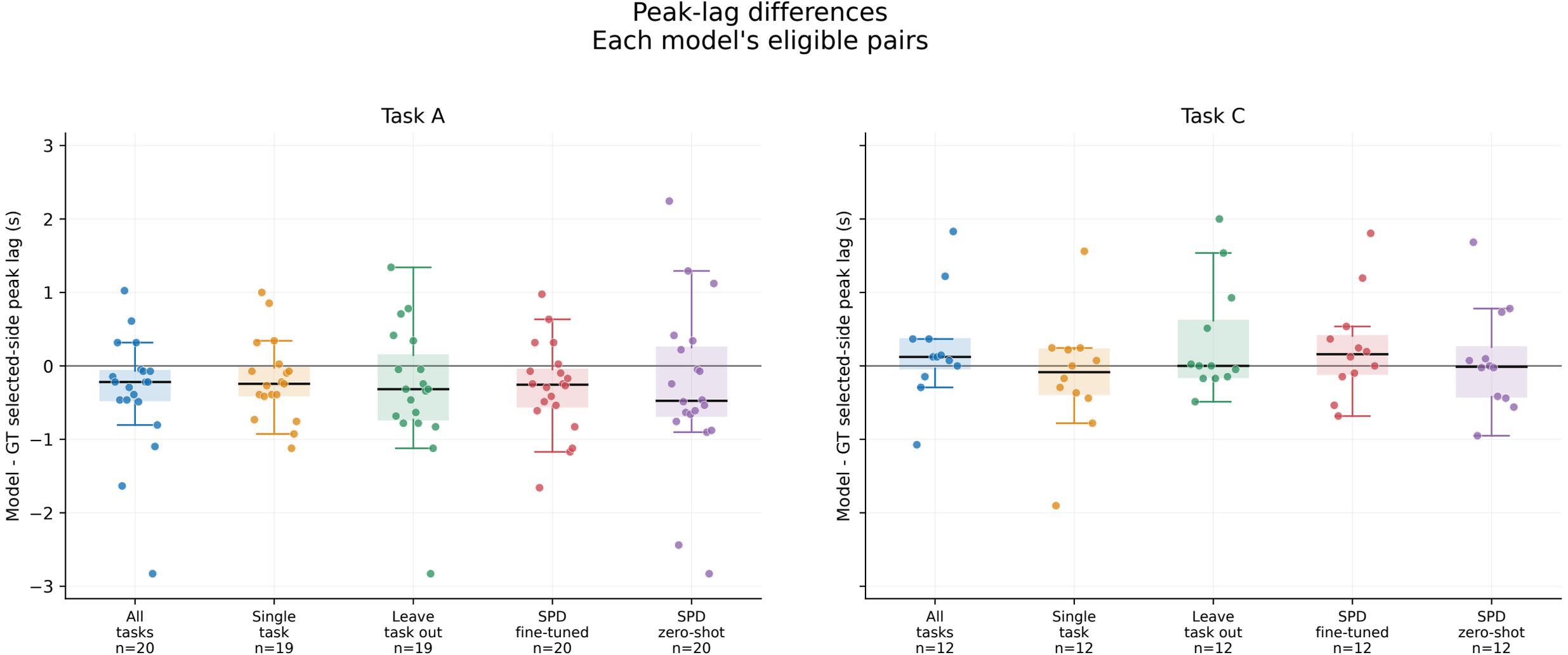
Thread string through tower | Within occurrence

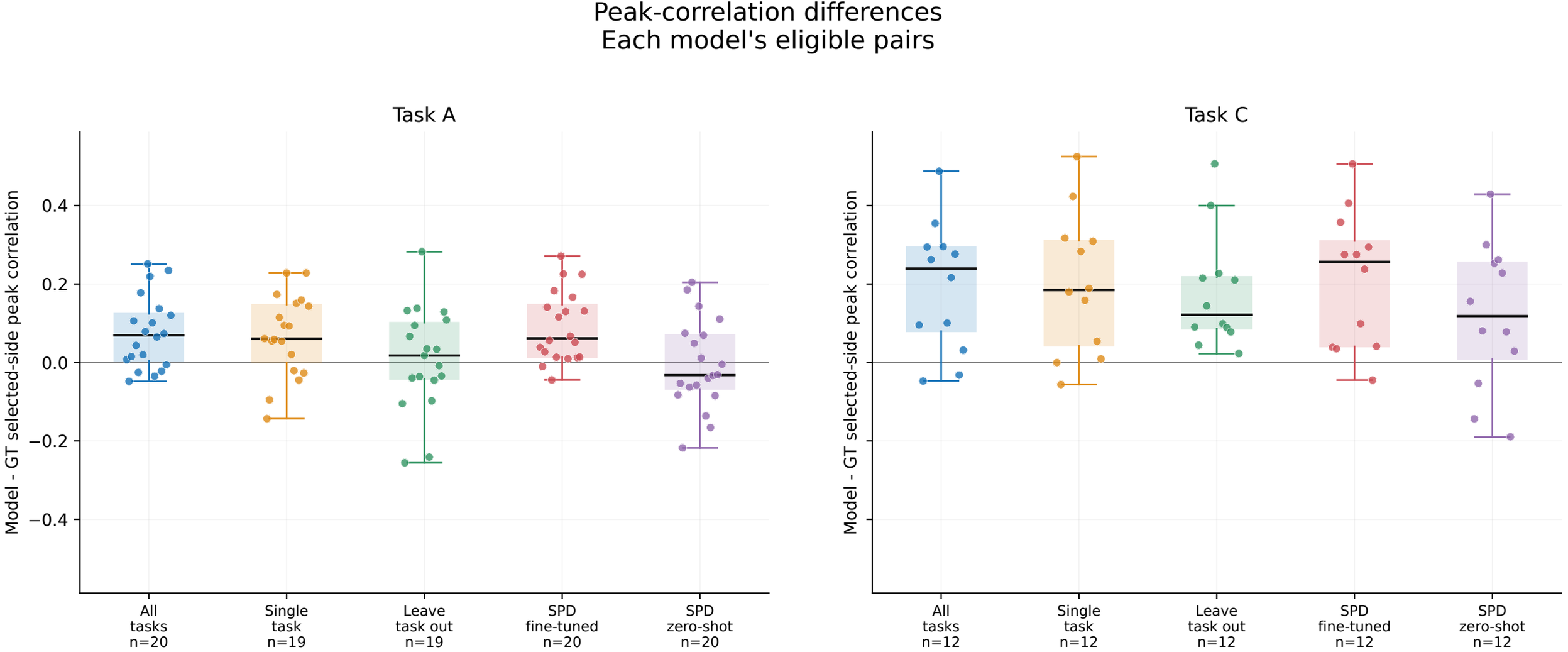


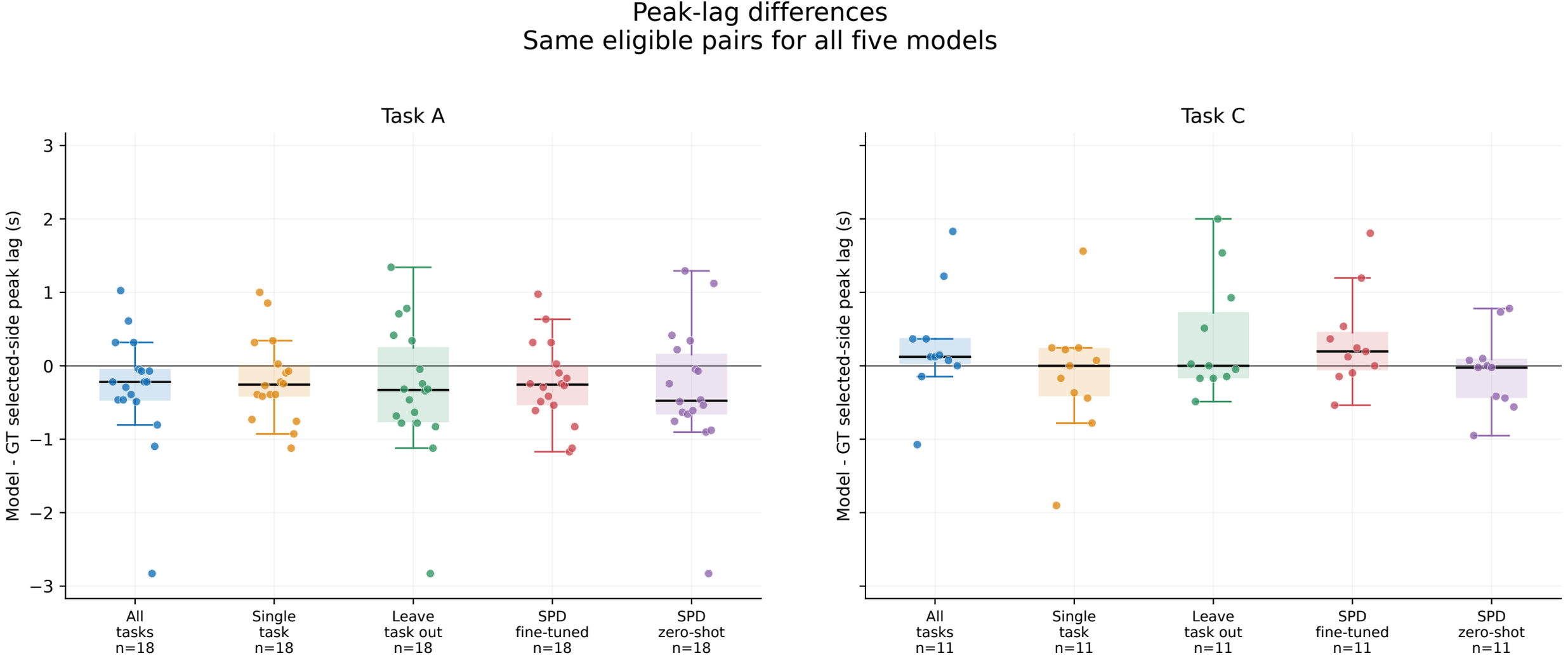
Each source retains its selected side's whole curve. Sides and stable-user support may differ between sources.

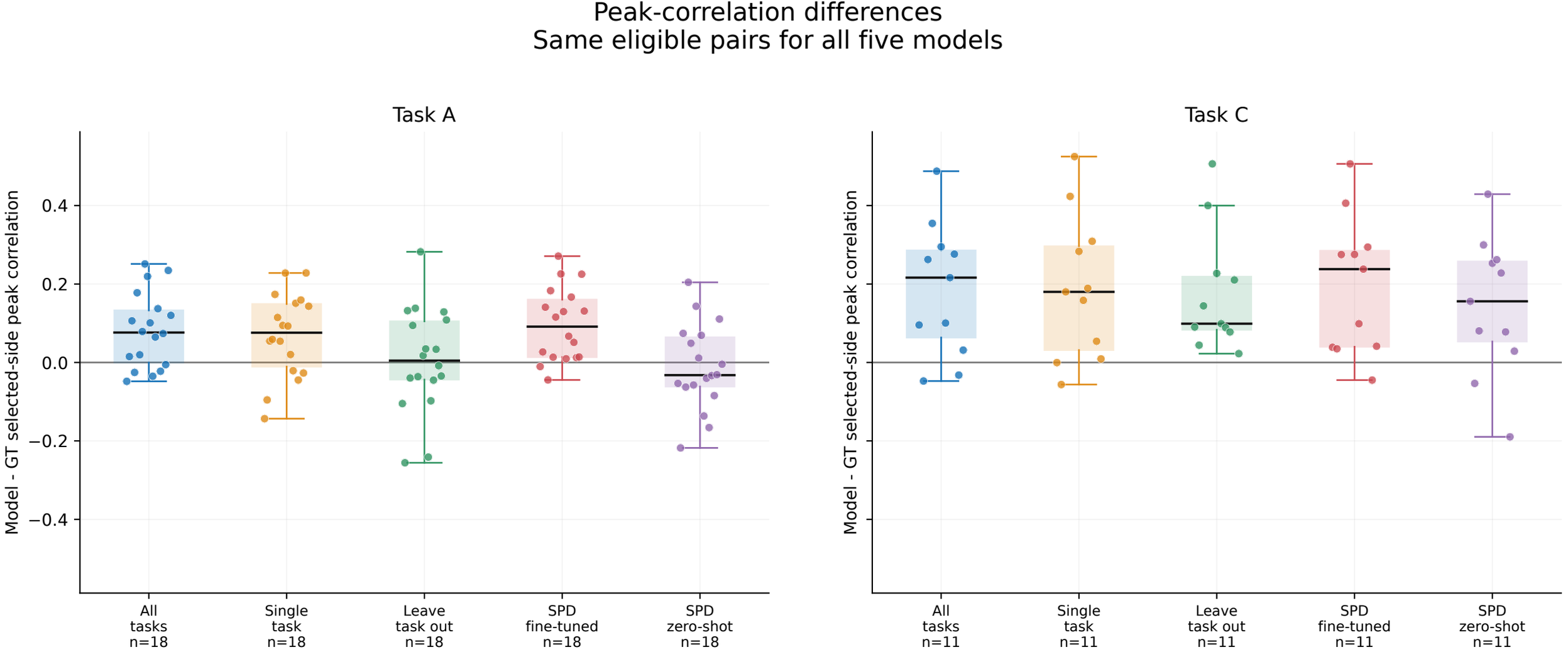

Independent maximum-xcorr lag comparison
Each model's eligible pairs
All tasks | n=32
A=20, C=12
Single task | n=31
A=19, C=12
Leave task out | n=31
A=19, C=12
SPD fine-tuned | n=32
A=20, C=12
SPD zero-shot | n=32
A=20, C=12
Model selected-side peak lag (s)
GT selected-side peak lag (s)
Task A
Task C
Equal peak lag
Position, PSM-stage anchored. A point compares selected-side statistics; GT and model need not select the same instrument.


Independent maximum-xcorr lag comparison
Same eligible pairs for all five models
All tasks | n=29
A=18, C=11
Single task | n=29
A=18, C=11
Leave task out | n=29
A=18, C=11
SPD fine-tuned | n=29
A=18, C=11
SPD zero-shot | n=29
A=18, C=11
Model selected-side peak lag (s)
GT selected-side peak lag (s)
Task A
Task C
Equal peak lag
Position, PSM-stage anchored. A point compares selected-side statistics; GT and model need not select the same instrument.

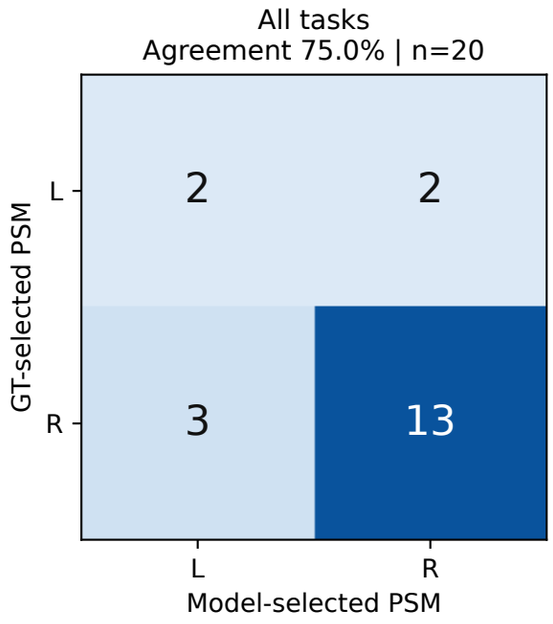

Peak-lag differences
Each model's eligible pairs
Task A
Task C
Model - GT selected-side peak lag (s)
3
2
1
0
−1
−2
−3
All tasks n=20
Single task n=19
Leave task out n=19
SPD fine-tuned n=20
SPD zero-shot n=20
All tasks n=12
Single task n=12
Leave task out n=12
SPD fine-tuned n=12
SPD zero-shot n=12
Each dot is one recording-stage pair. Independently selected sides may differ. Boxes show descriptive spread, not confidence intervals.

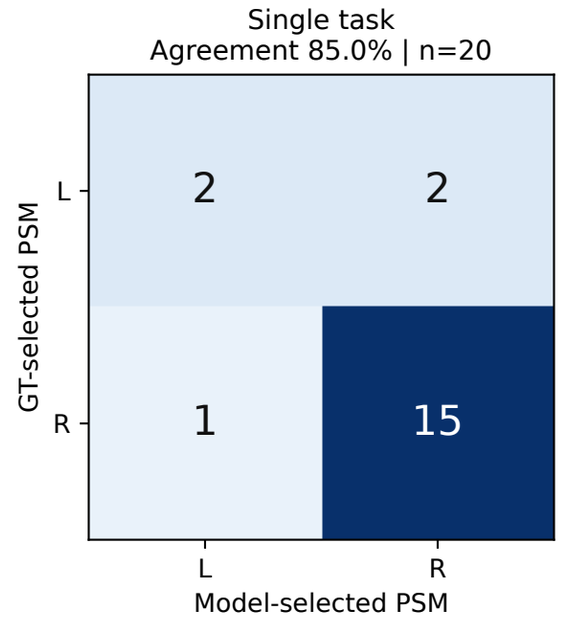

Peak-correlation differences
Each model's eligible pairs
Task A
Task C
Model - GT selected-side peak correlation
0.4
0.2
0.0
−0.2
−0.4
All tasks n=20
Single task n=19
Leave task out n=19
SPD fine-tuned n=20
SPD zero-shot n=20
All tasks n=12
Single task n=12
Leave task out n=12
SPD fine-tuned n=12
SPD zero-shot n=12
Each dot is one recording-stage pair. Independently selected sides may differ. Boxes show descriptive spread, not confidence intervals.

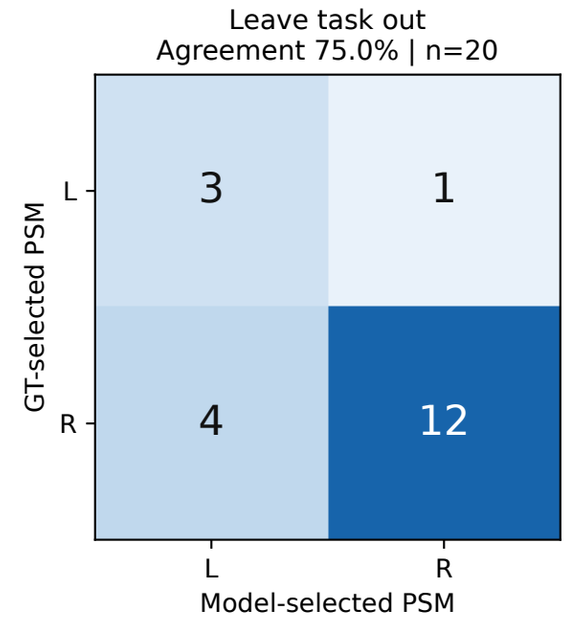

Peak-lag differences
Same eligible pairs for all five models
Task A
Task C
Model - GT selected-side peak lag (s)
3
2
1
0
−1
−2
−3
All tasks n=18
Single task n=18
Leave task out n=18
SPD fine-tuned n=18
SPD zero-shot n=18
All tasks n=11
Single task n=11
Leave task out n=11
SPD fine-tuned n=11
SPD zero-shot n=11
Each dot is one recording-stage pair. Independently selected sides may differ. Boxes show descriptive spread, not confidence intervals.

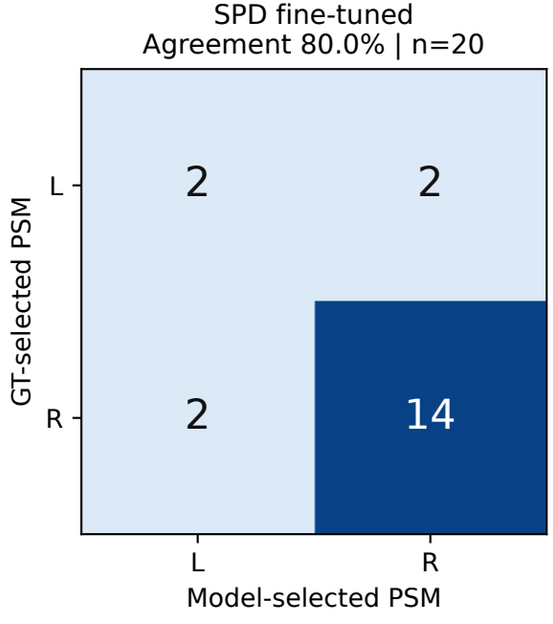

Peak-correlation differences
Same eligible pairs for all five models
Task A
Task C
Model - GT selected-side peak correlation
0.4
0.2
0.0
−0.2
−0.4
All tasks n=18
Single task n=18
Leave task out n=18
SPD fine-tuned n=18
SPD zero-shot n=18
All tasks n=11
Single task n=11
Leave task out n=11
SPD fine-tuned n=11
SPD zero-shot n=11
Each dot is one recording-stage pair. Independently selected sides may differ. Boxes show descriptive spread, not confidence intervals.

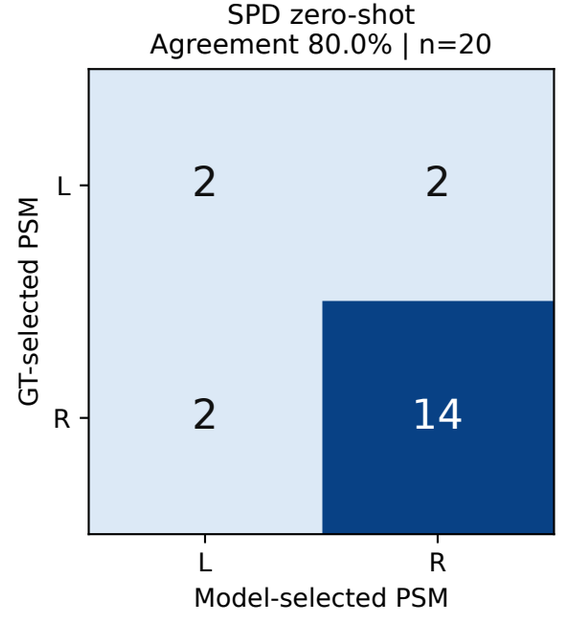
Task A: selected DLC instrument agreement

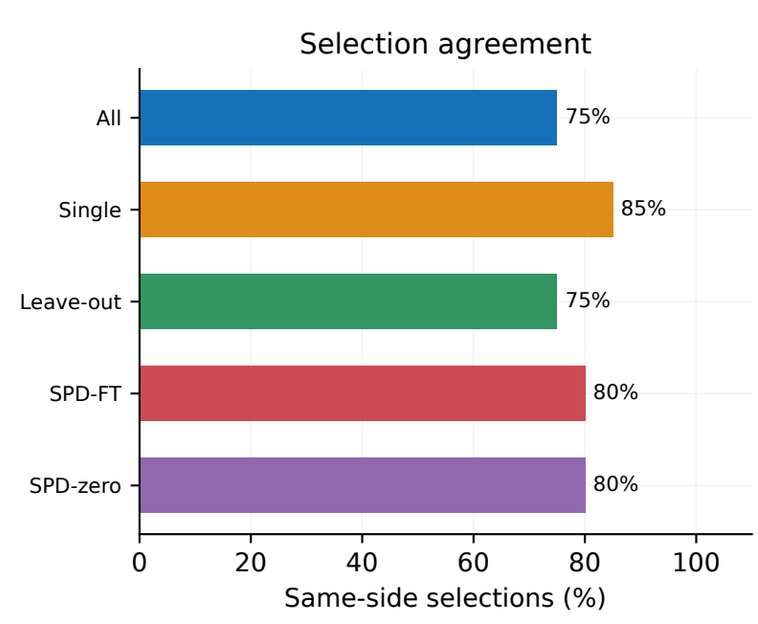






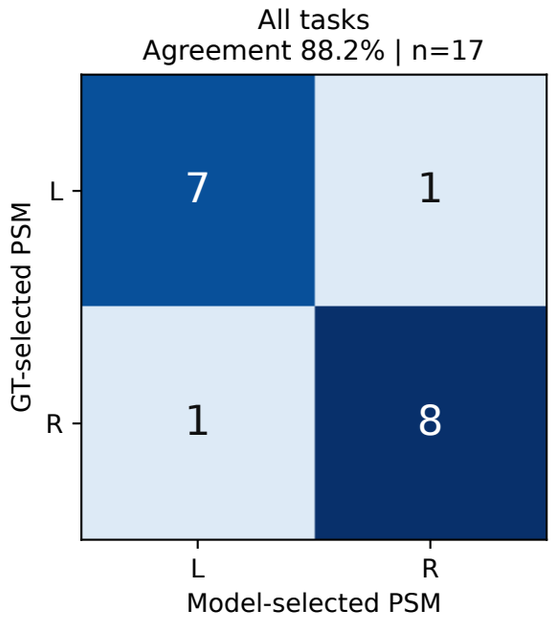


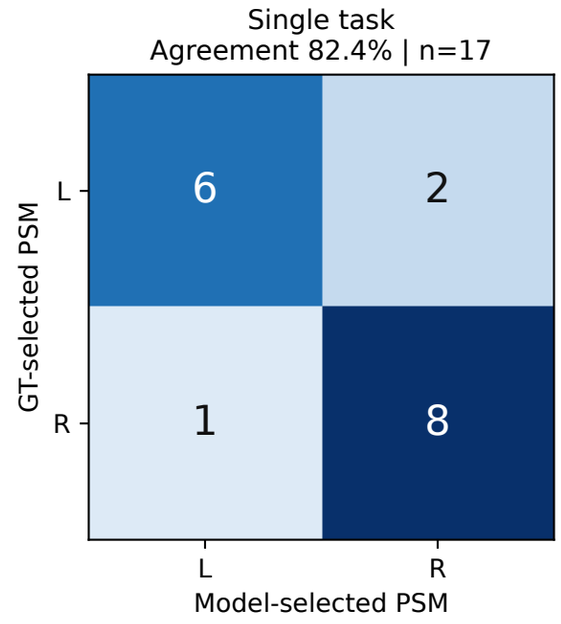


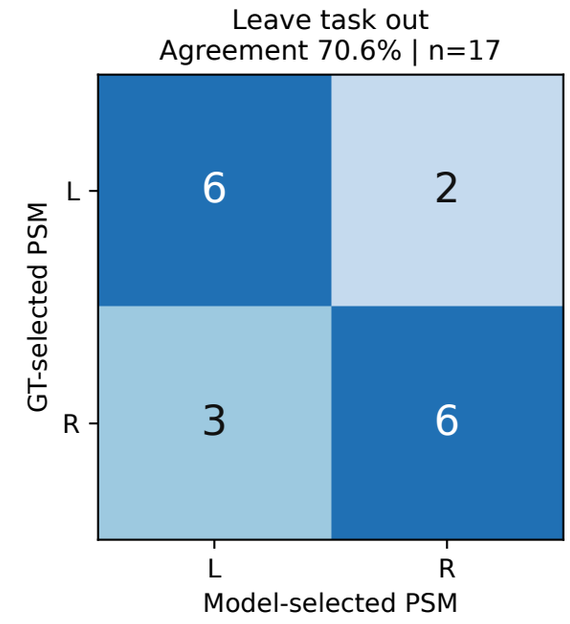


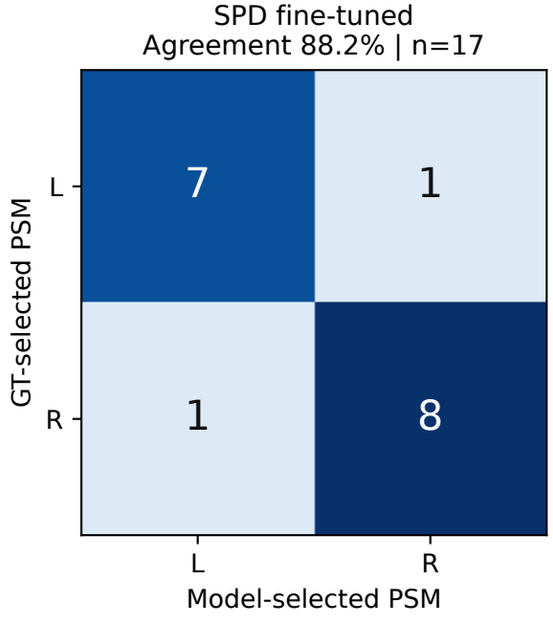
Position, PSM-stage anchored. Counts use all defined L/R selections before peak-eligibility filtering. Selection does not identify the operated instrument.

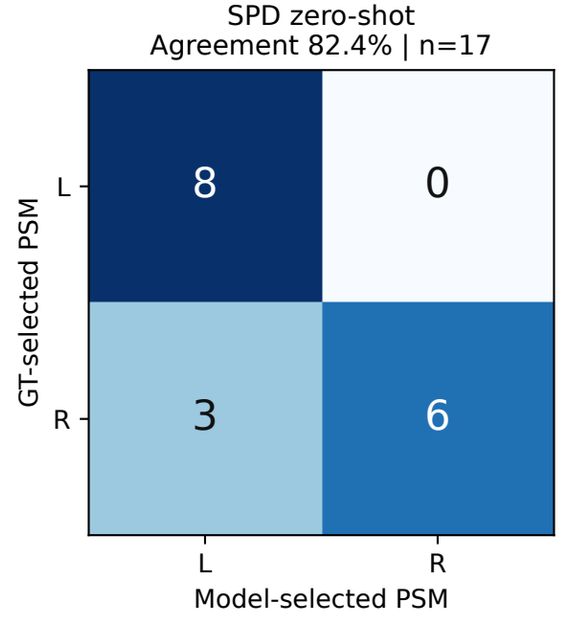
Task C: selected DLC instrument agreement

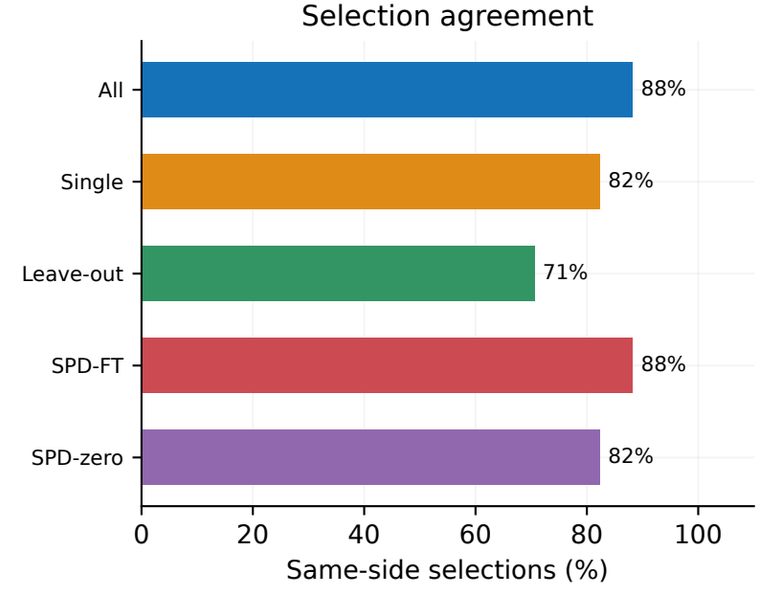






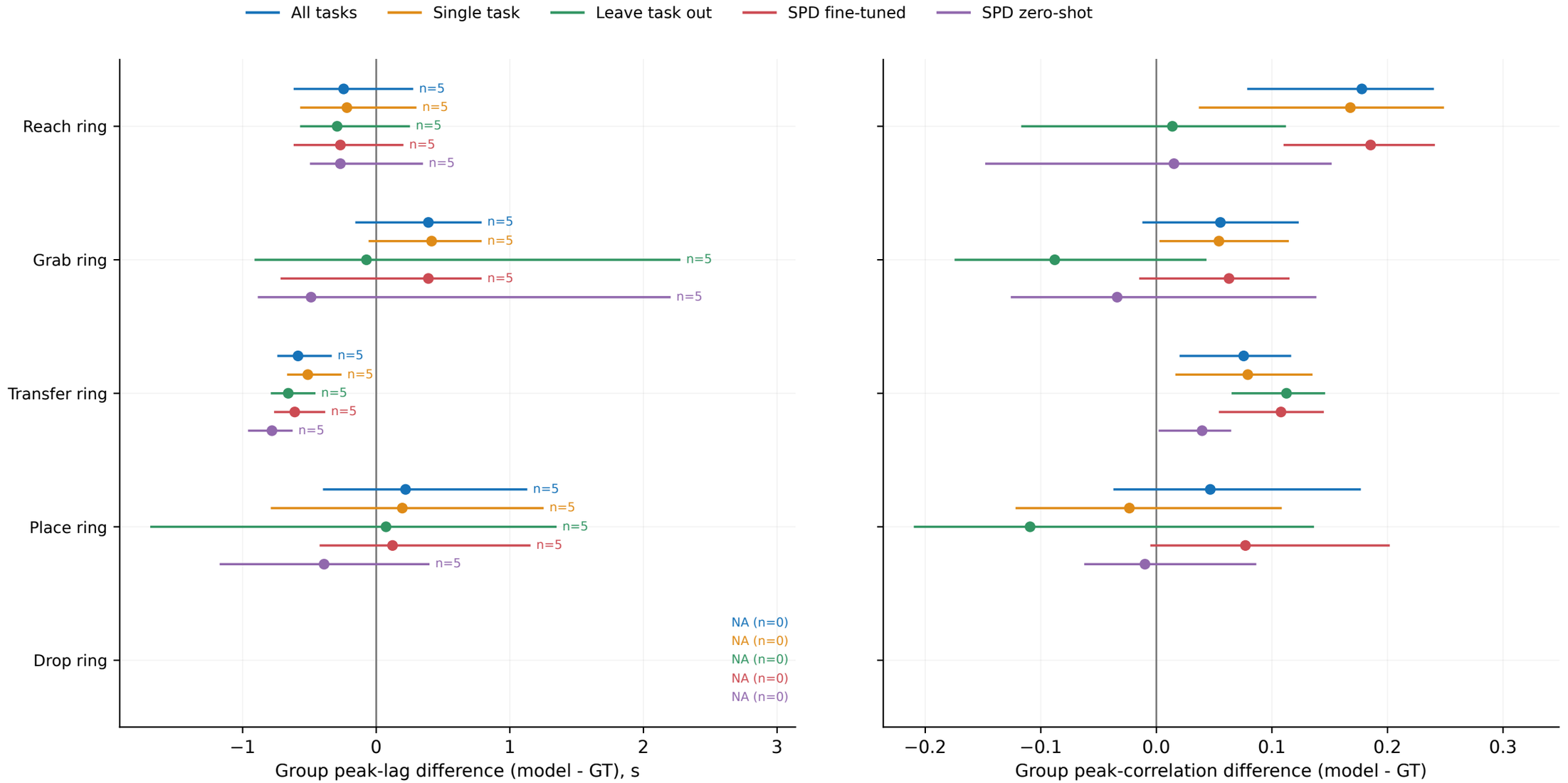




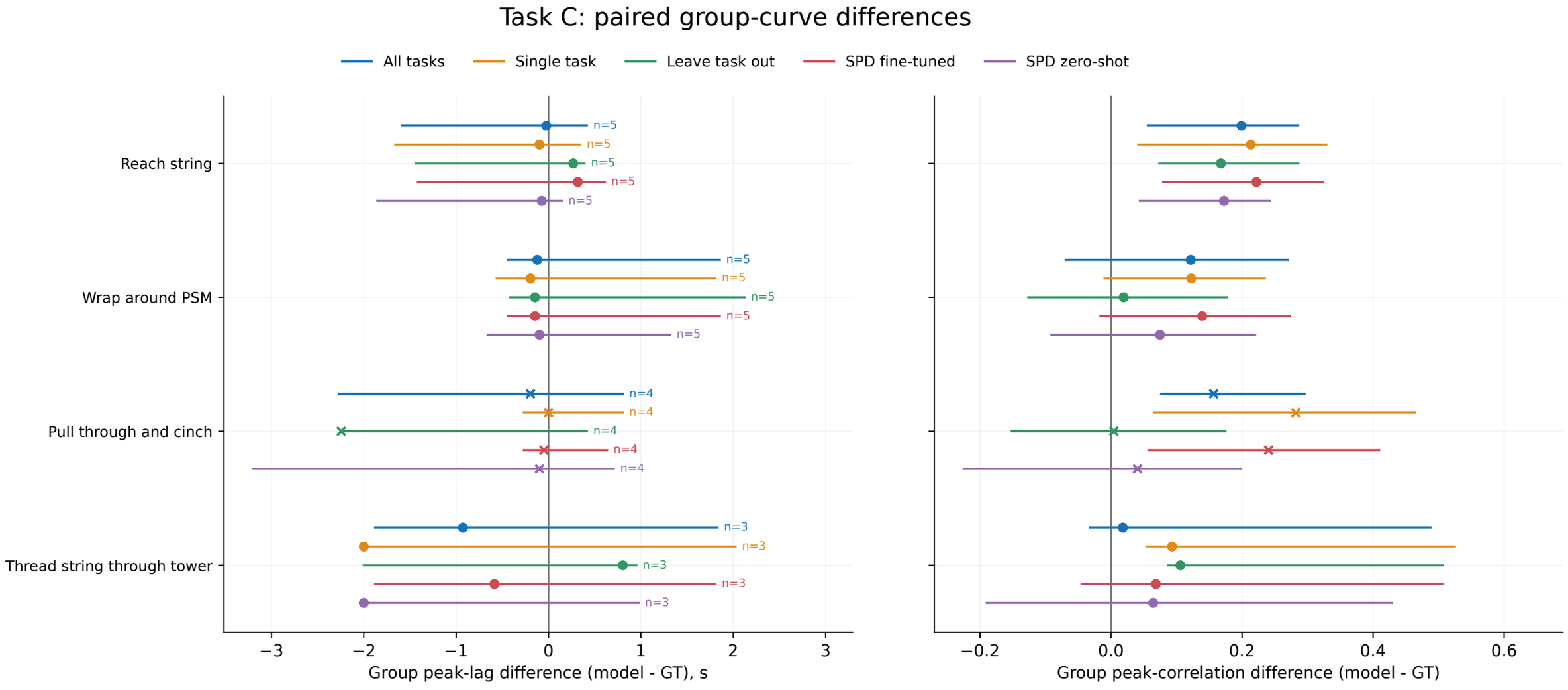


Position, PSM-stage anchored. Counts use all defined L/R selections before peak-eligibility filtering. Selection does not identify the operated instrument.

Task A: paired group-curve differences



Position, PSM-stage anchored. Same complete-lag participants within each model/GT pair; 1,000 paired participant bootstrap draws.
Lines: unadjusted exploratory 95% intervals. Numbers: shared participants. Cross: nonpositive or boundary group peak.
Independent maximum-xcorr selection may use different PSMs. Group-peak difference is not recording-level lag error; pairs may use different participants.

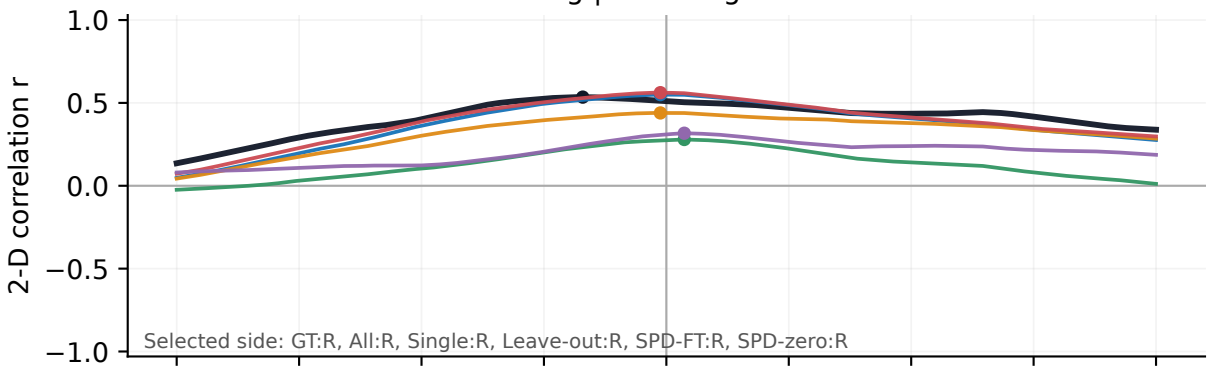


Position, PSM-stage anchored. Same complete-lag participants within each model/GT pair; 1,000 paired participant bootstrap draws.
Lines: unadjusted exploratory 95% intervals. Numbers: shared participants. Cross: nonpositive or boundary group peak.
Independent maximum-xcorr selection may use different PSMs. Group-peak difference is not recording-level lag error; pairs may use different participants.

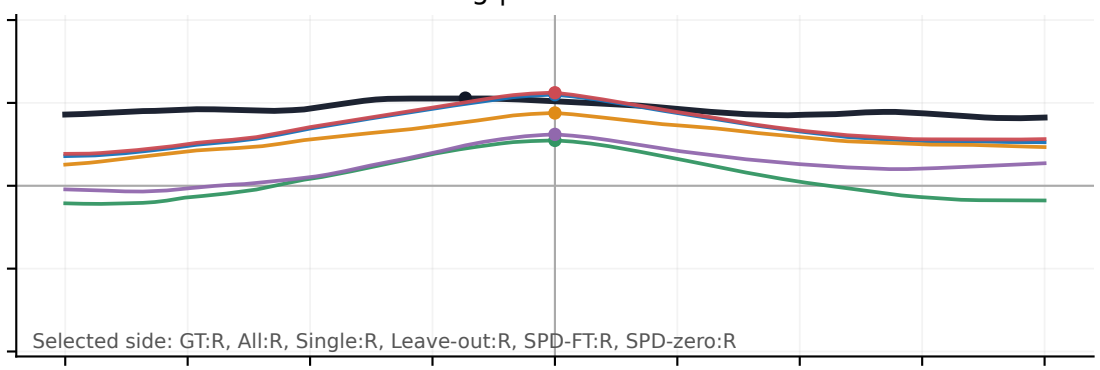
User11_6_1 | Task A: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

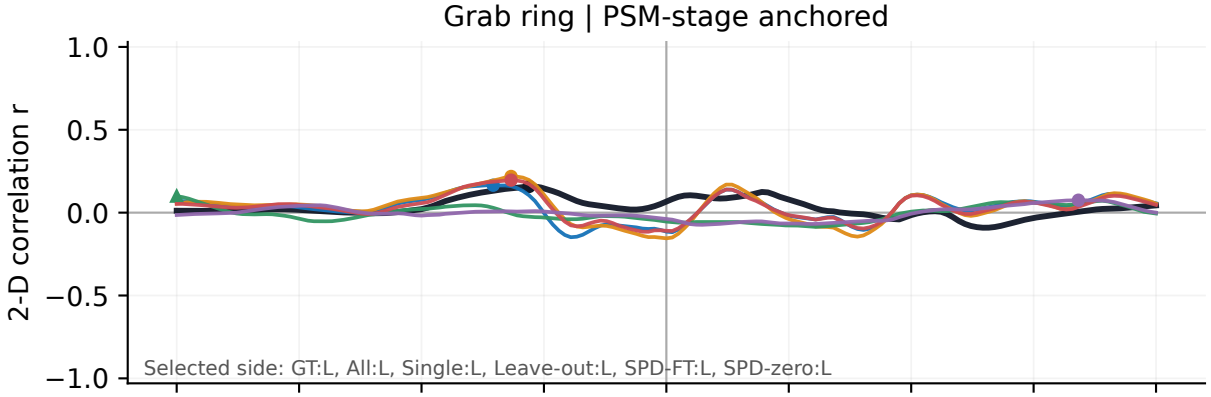
Reach ring | PSM-stage anchored



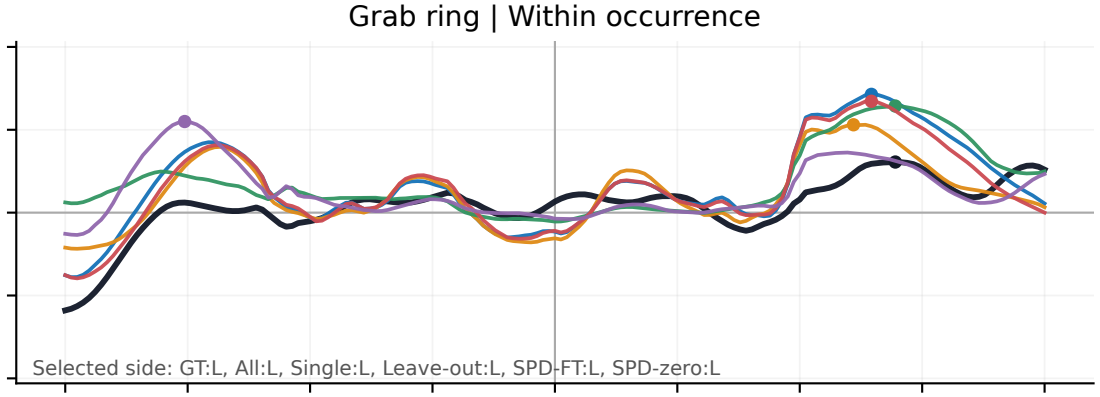
Reach ring | Within occurrence



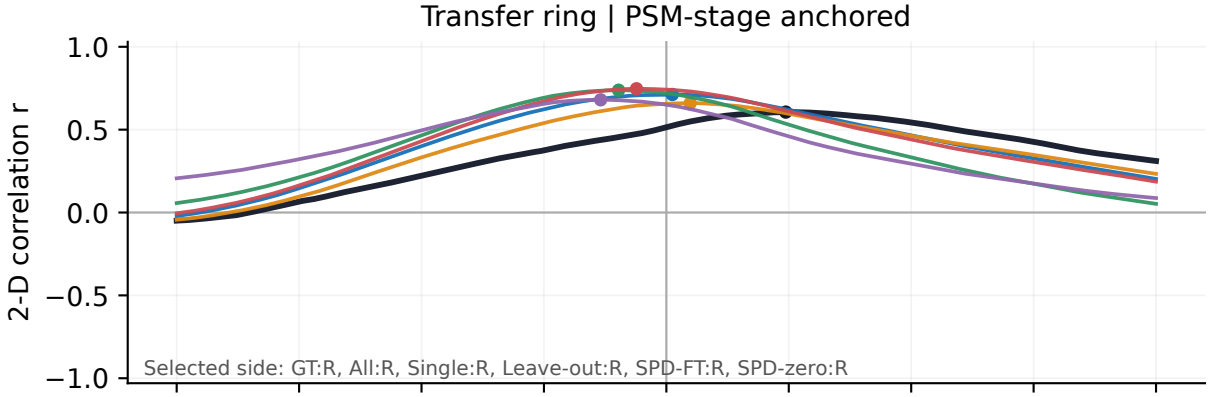
Grab ring | PSM-stage anchored



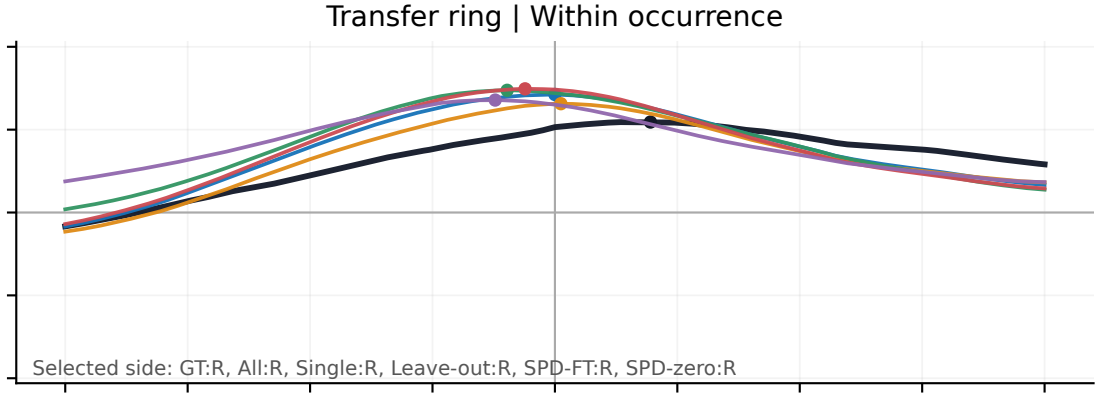
Grab ring | Within occurrence



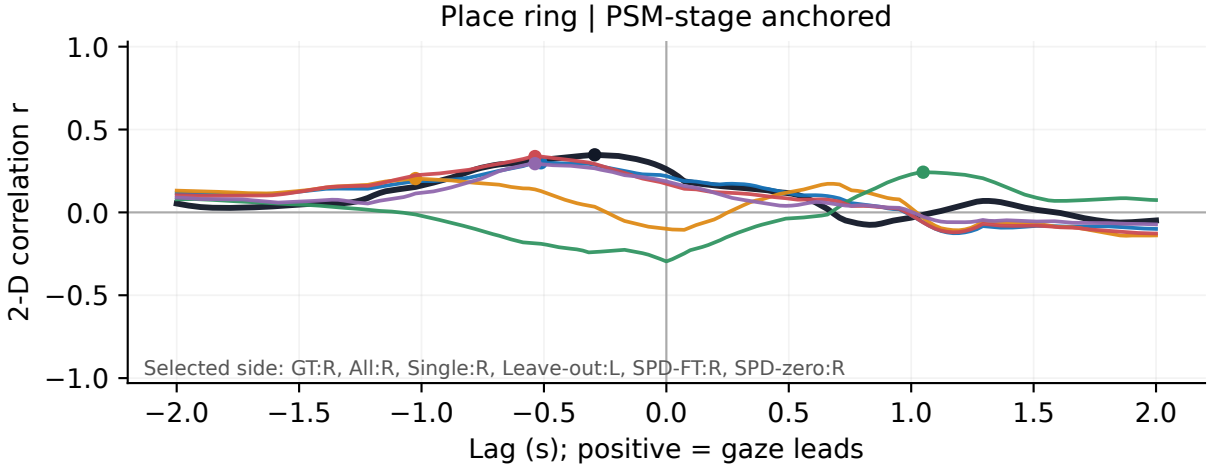
Transfer ring | PSM-stage anchored



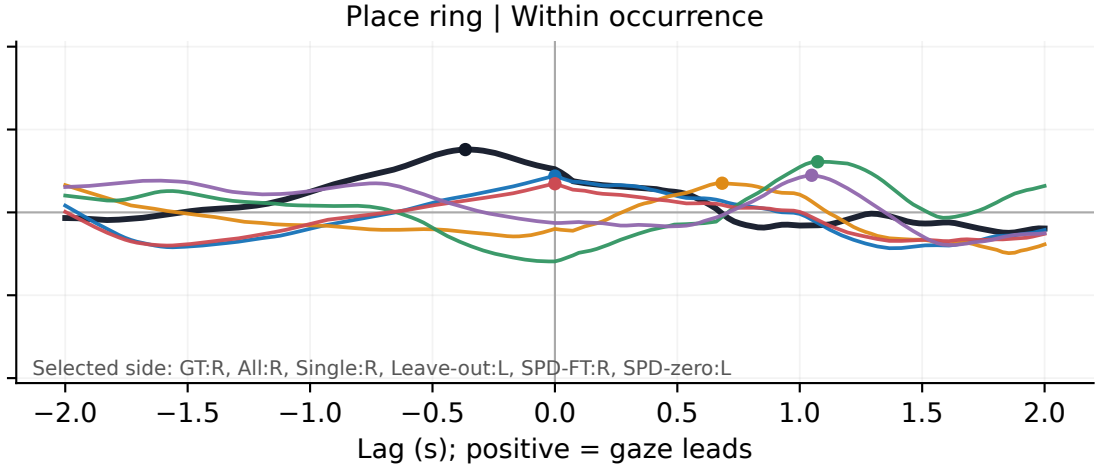
Transfer ring | Within occurrence



Place ring | PSM-stage anchored



Place ring | Within occurrence

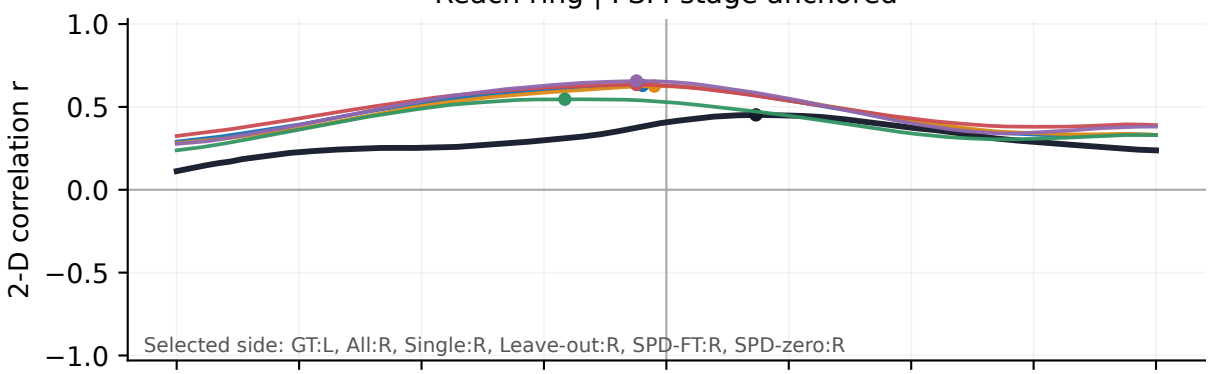


Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

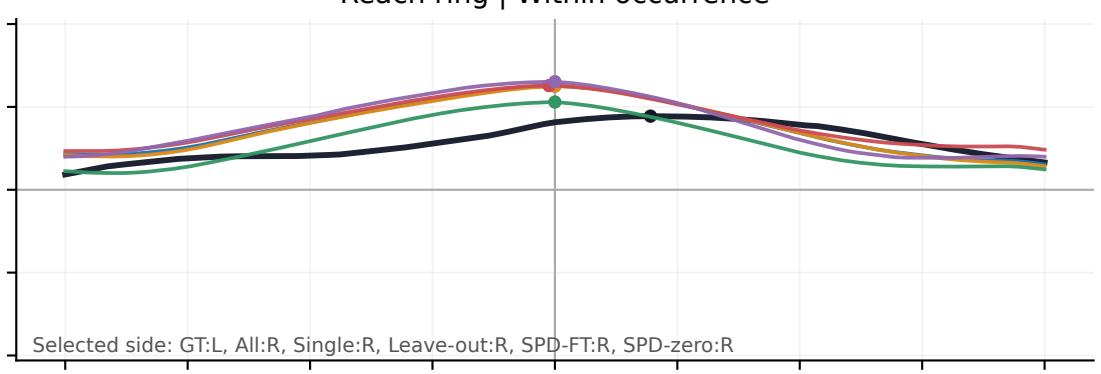
User12_3_1 | Task A: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

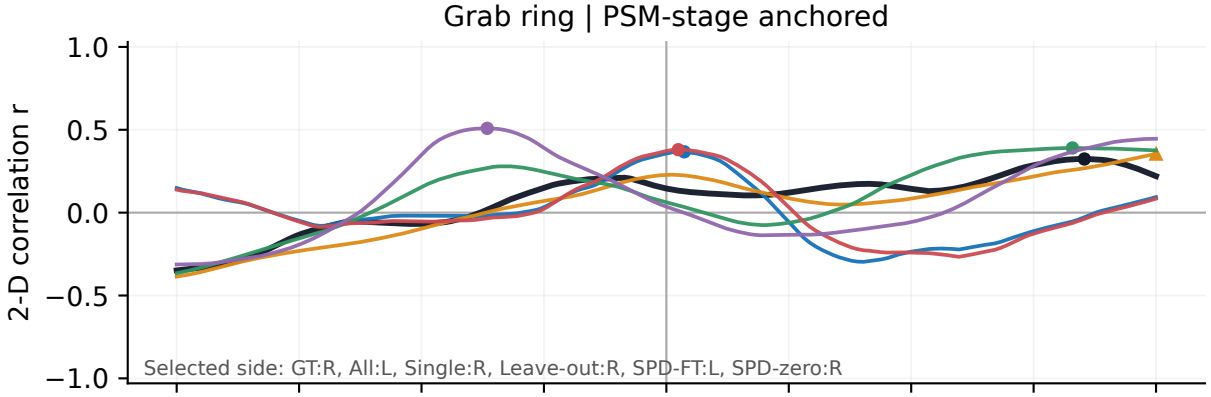
Reach ring | PSM-stage anchored



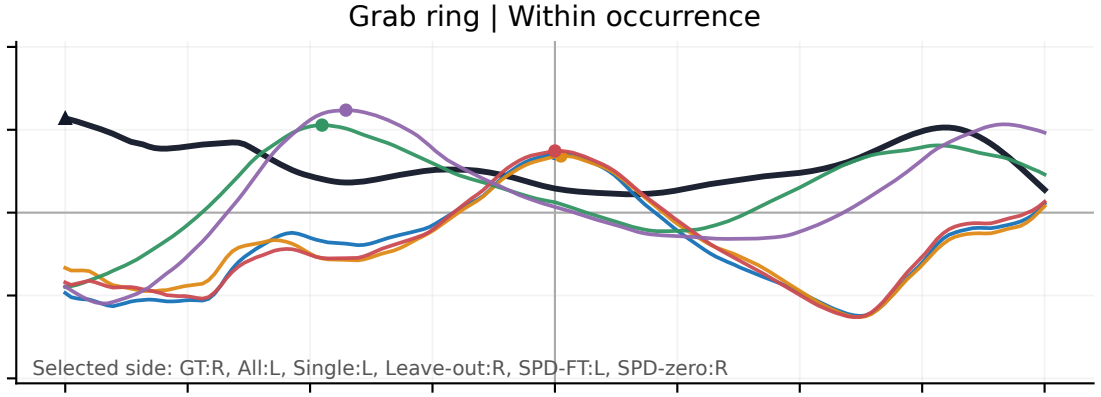
Reach ring | Within occurrence



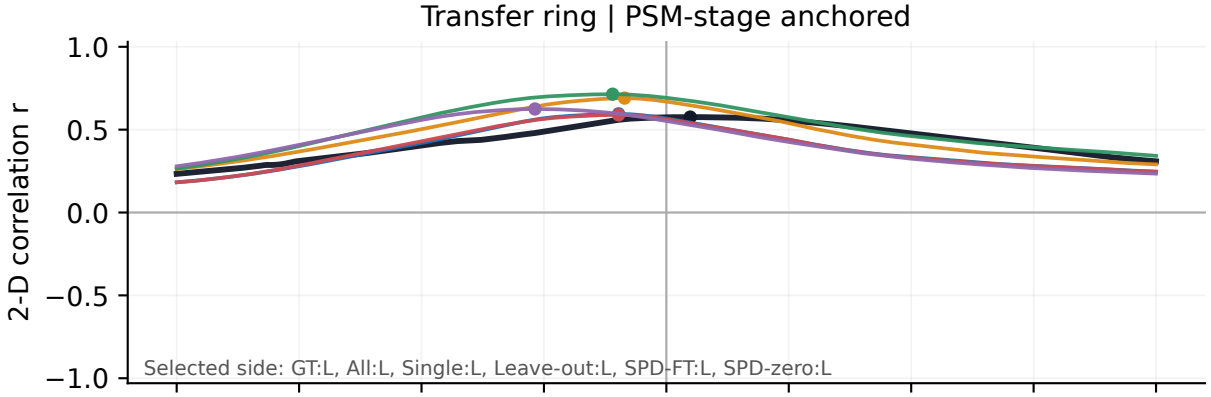
Grab ring | PSM-stage anchored



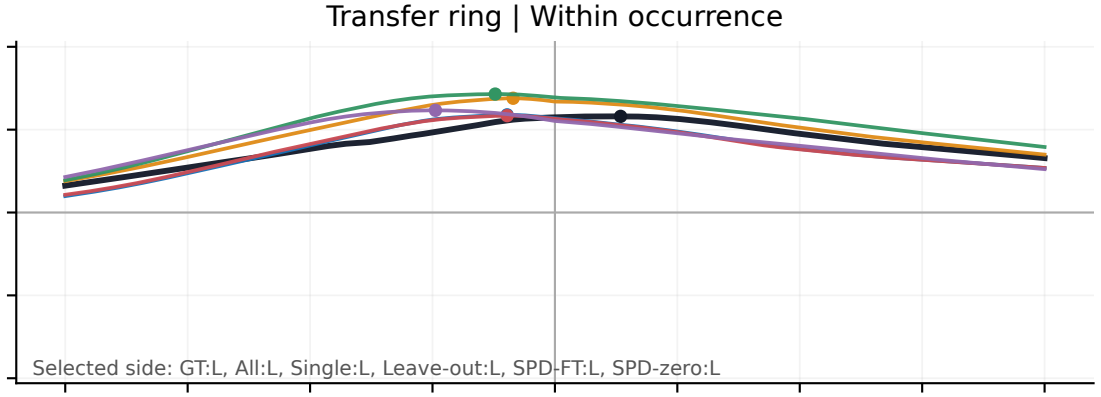
Grab ring | Within occurrence



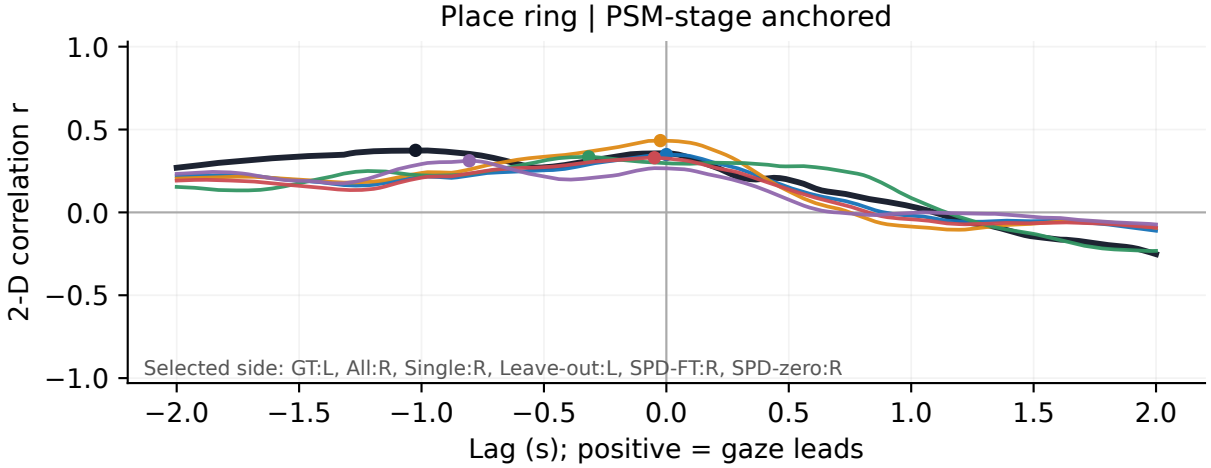
Transfer ring | PSM-stage anchored



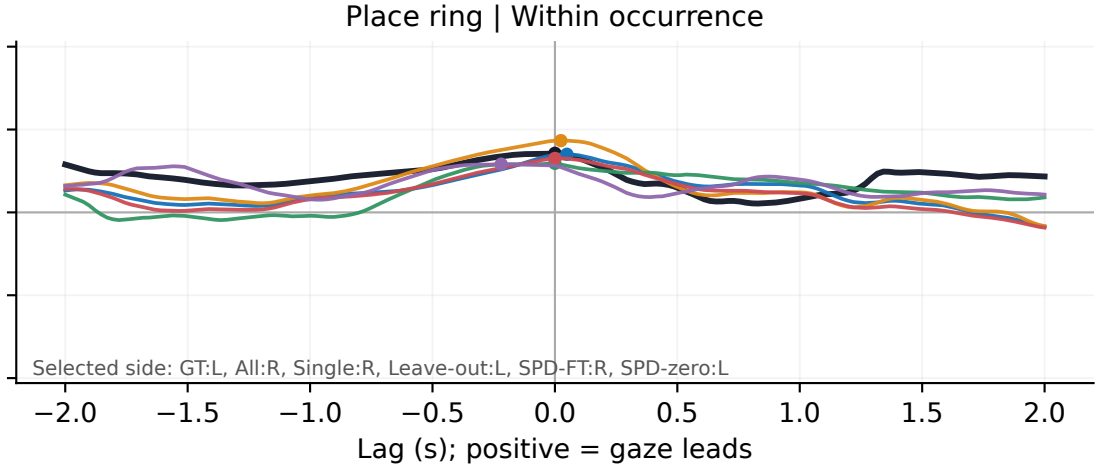
Transfer ring | Within occurrence



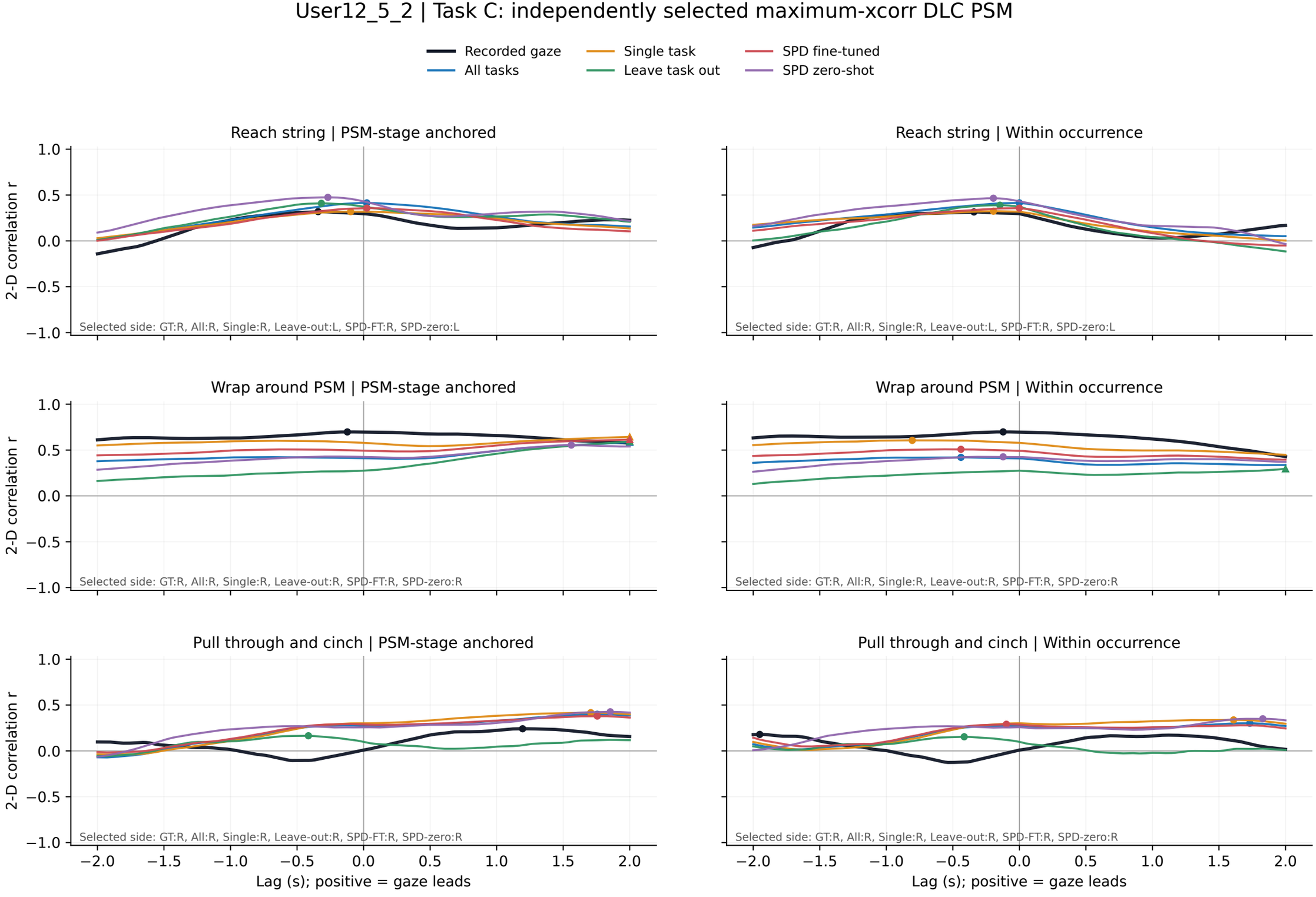
Place ring | PSM-stage anchored



Place ring | Within occurrence



Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

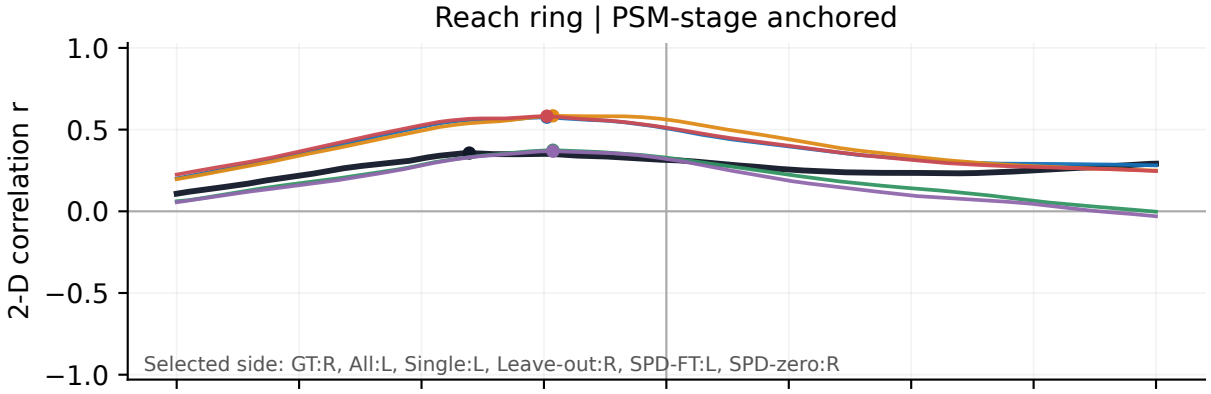


Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

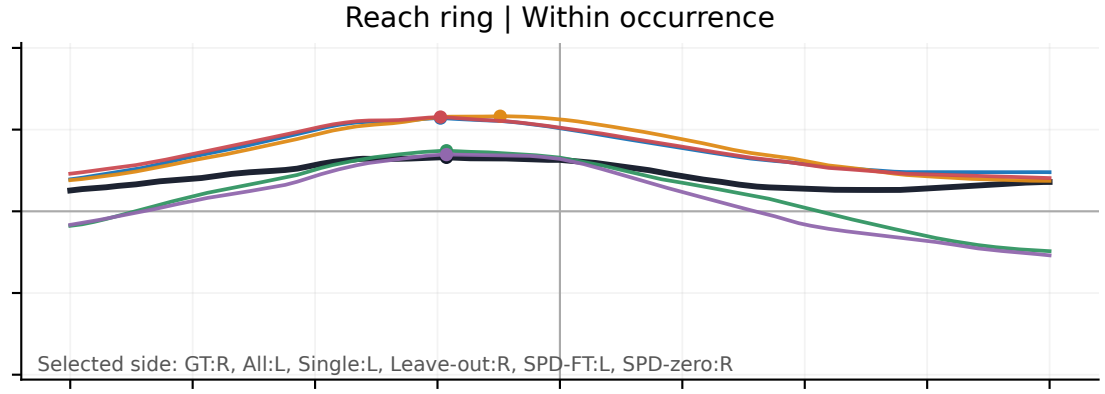
User1_1_1 | Task A: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

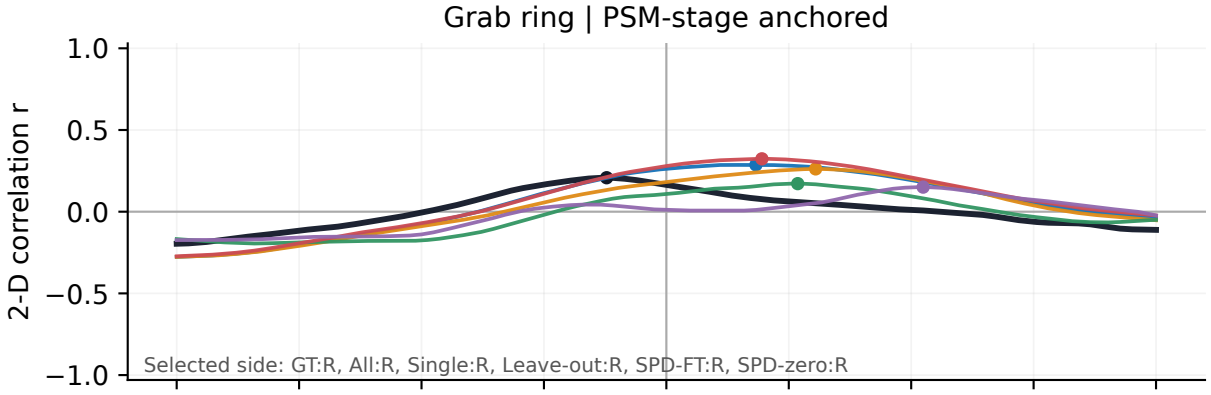
Reach ring | PSM-stage anchored



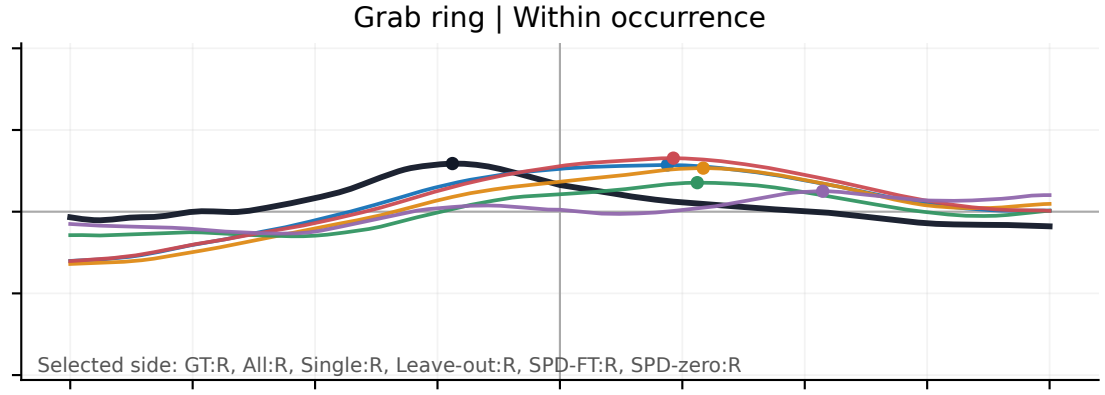
Reach ring | Within occurrence



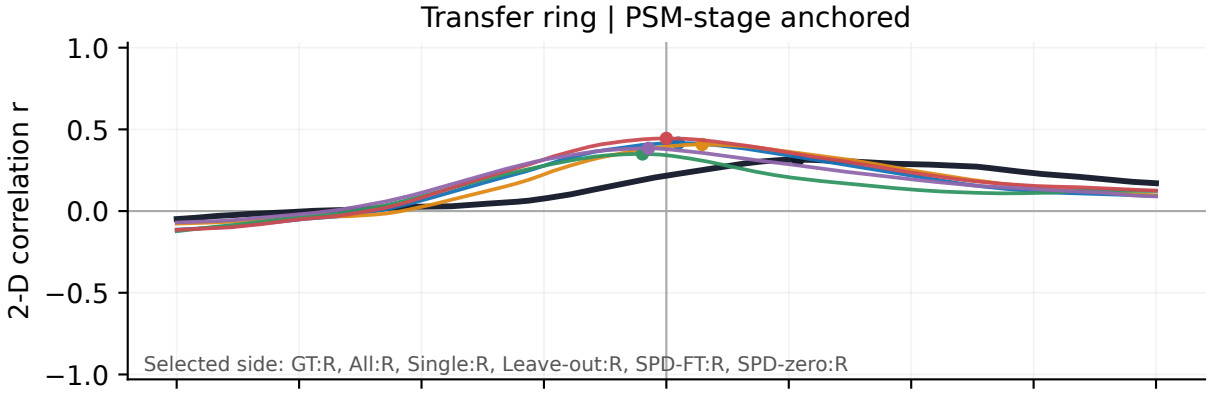
Grab ring | PSM-stage anchored



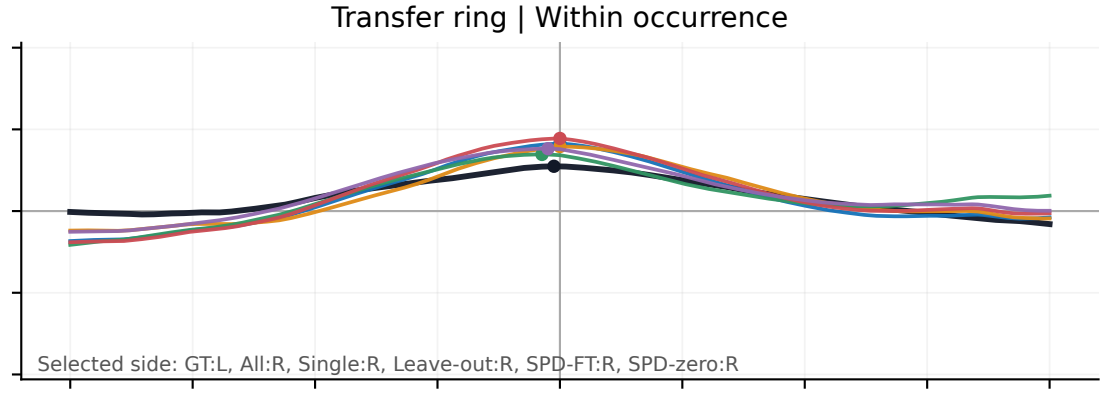
Grab ring | Within occurrence



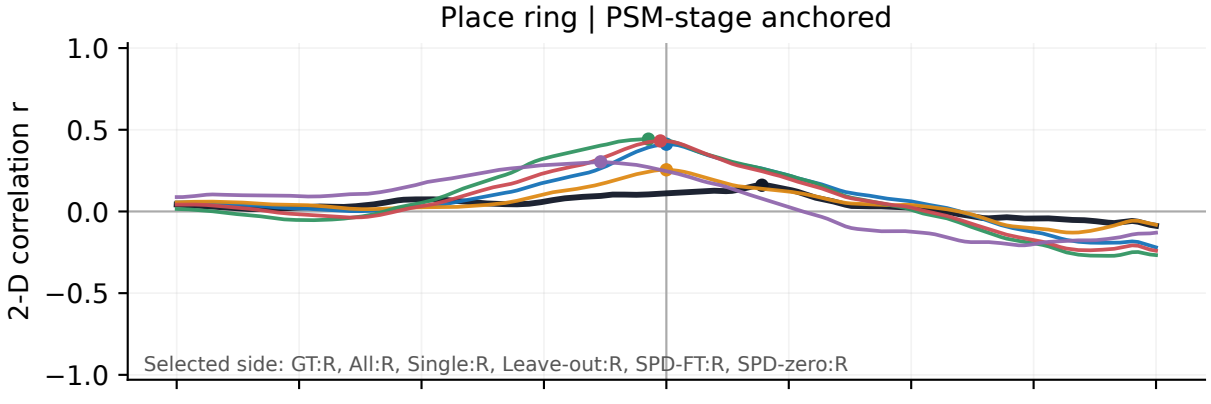
Transfer ring | PSM-stage anchored



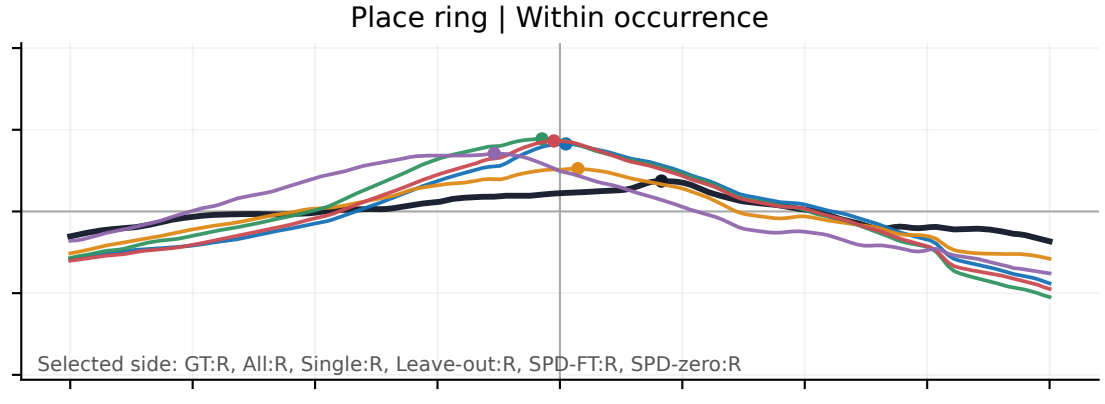
Transfer ring | Within occurrence



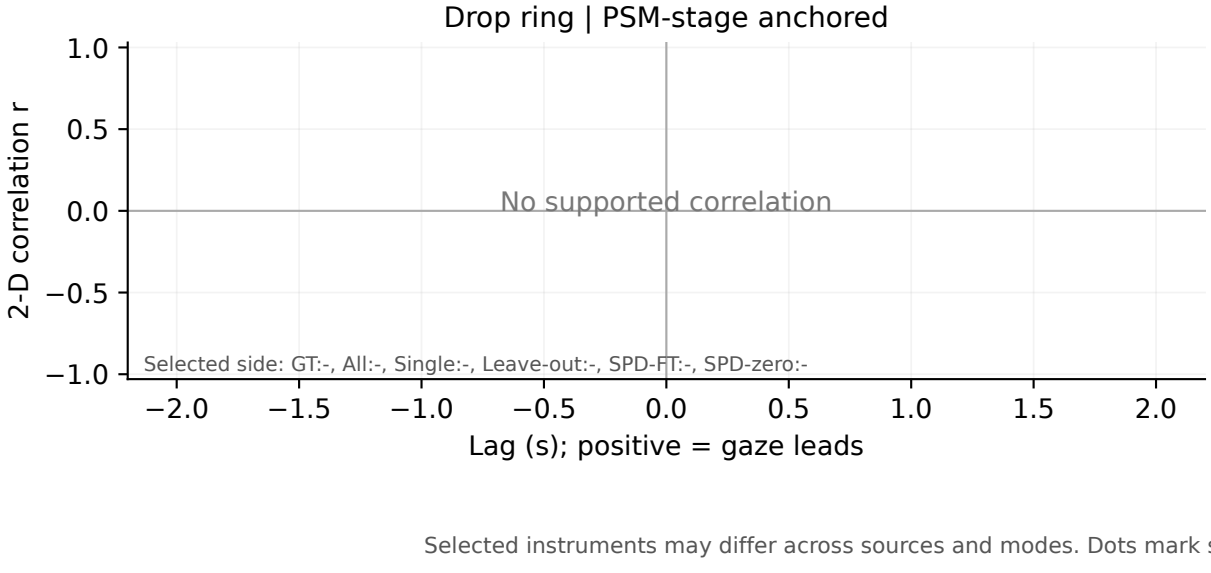
Place ring | PSM-stage anchored



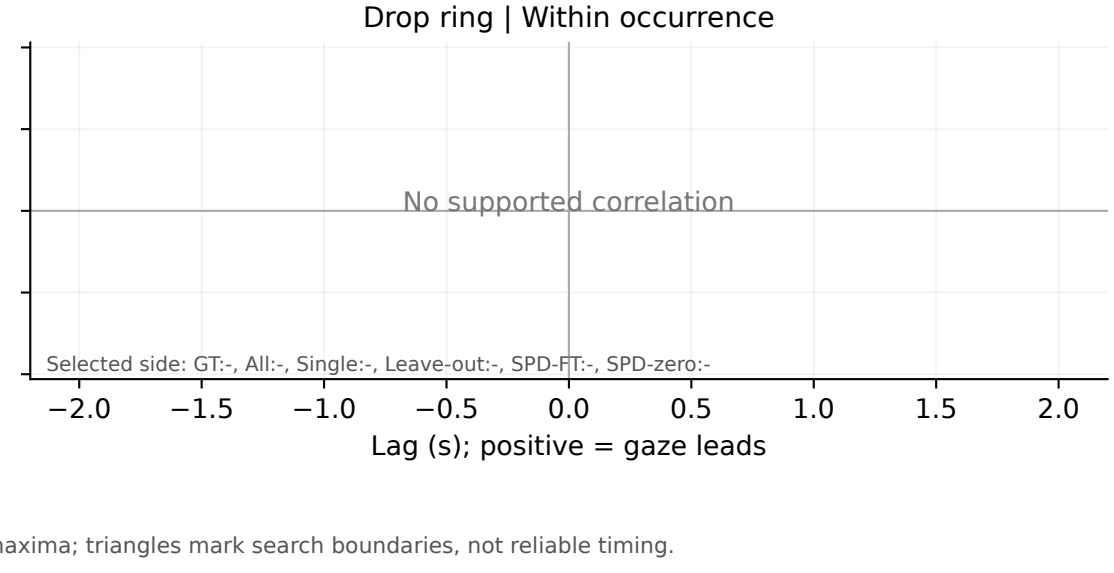
Place ring | Within occurrence



Drop ring | PSM-stage anchored



Drop ring | Within occurrence

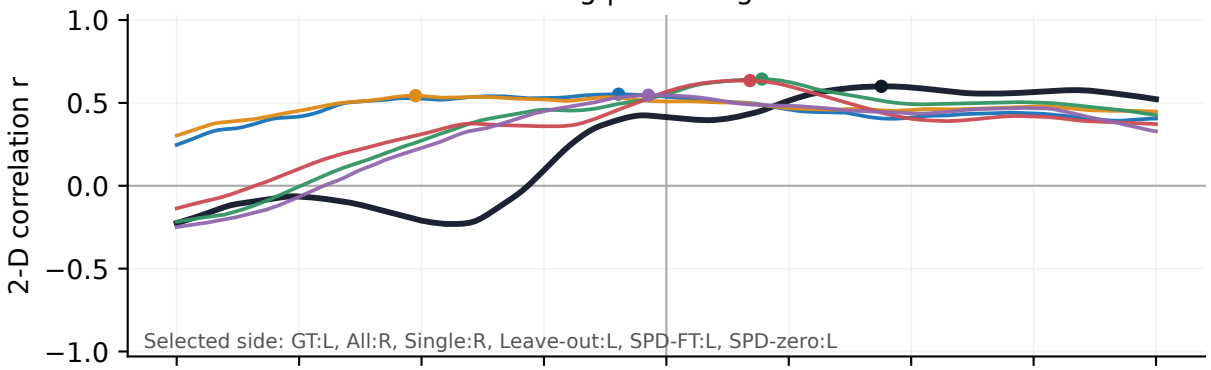


Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

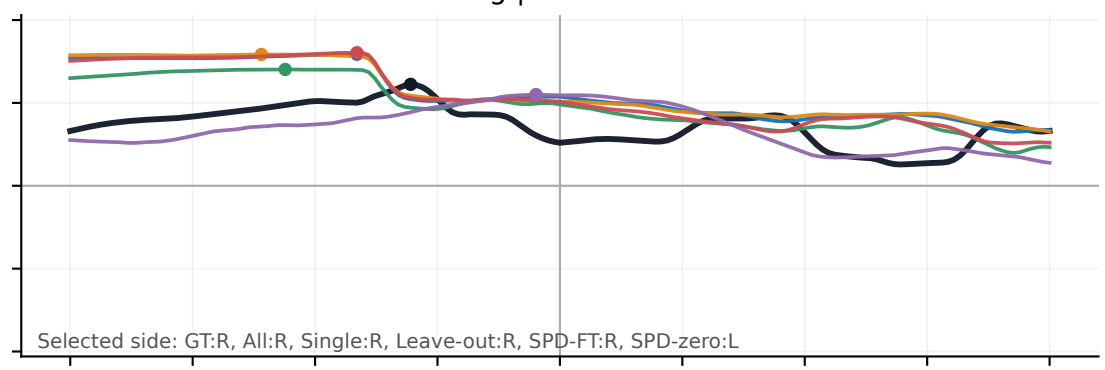
User1_5_2 | Task C: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

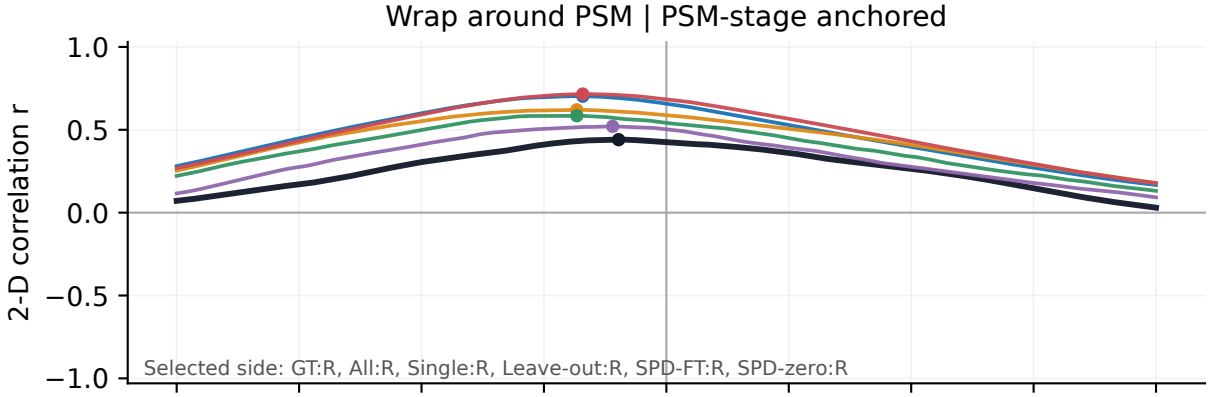
Reach string | PSM-stage anchored



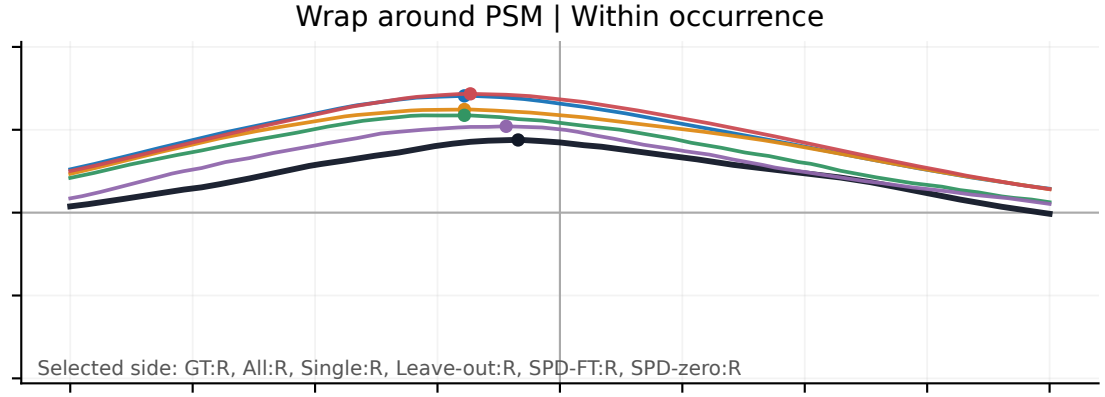
Reach string | Within occurrence



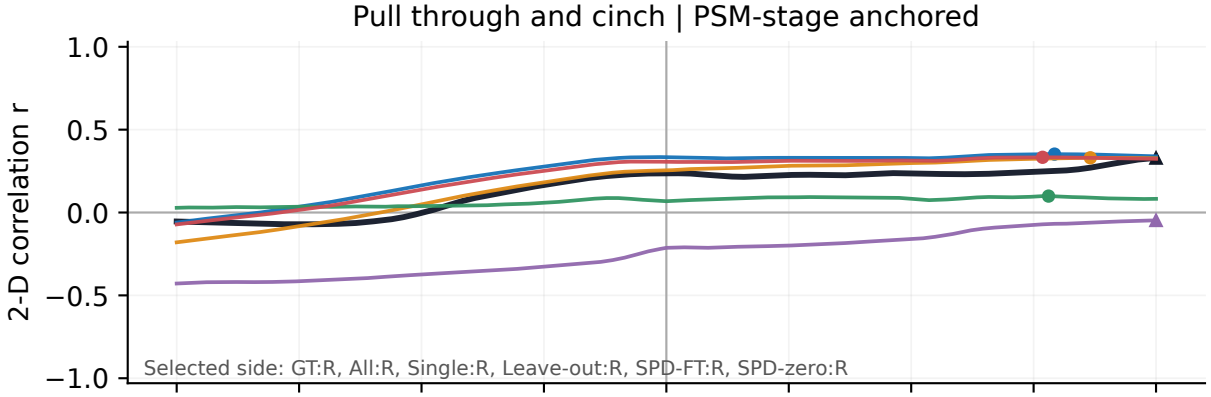
Wrap around PSM | PSM-stage anchored



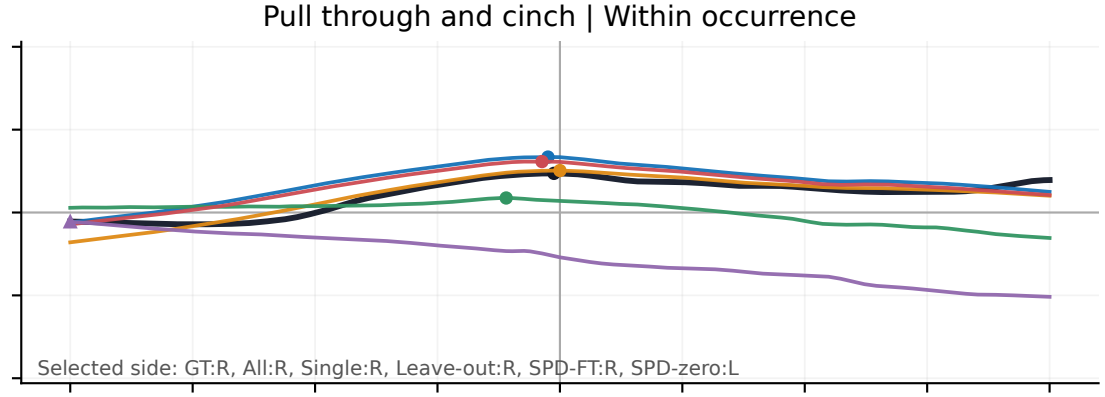
Wrap around PSM | Within occurrence



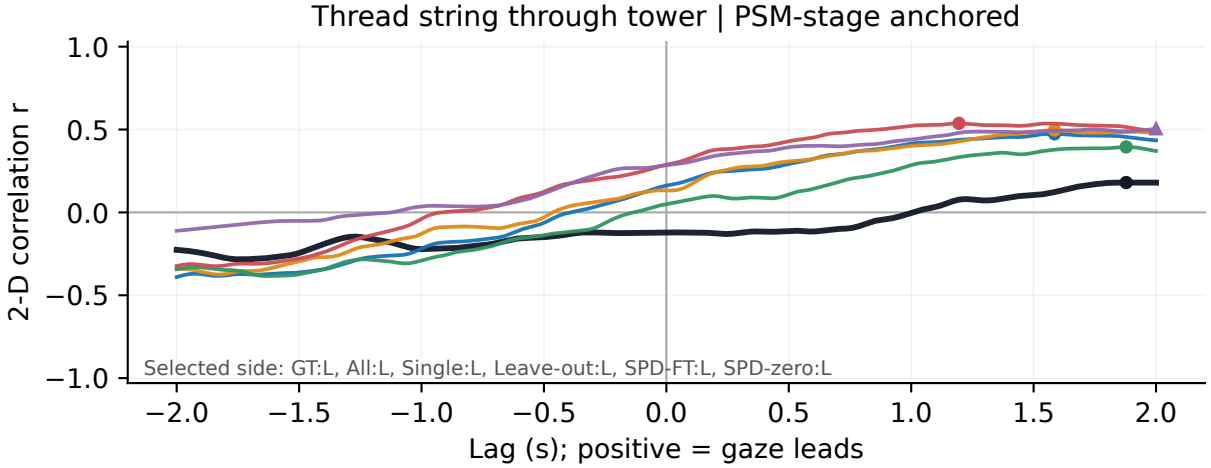
Pull through and cinch | PSM-stage anchored



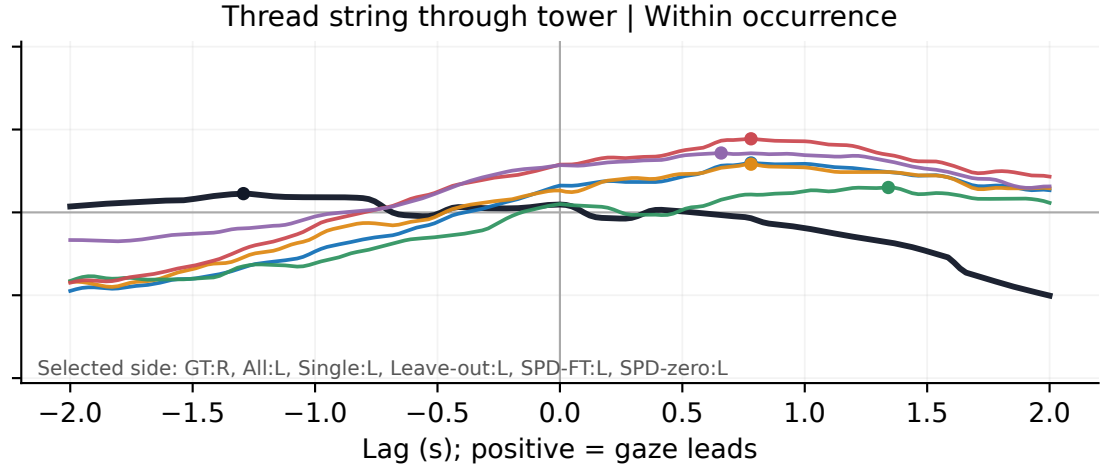
Pull through and cinch | Within occurrence



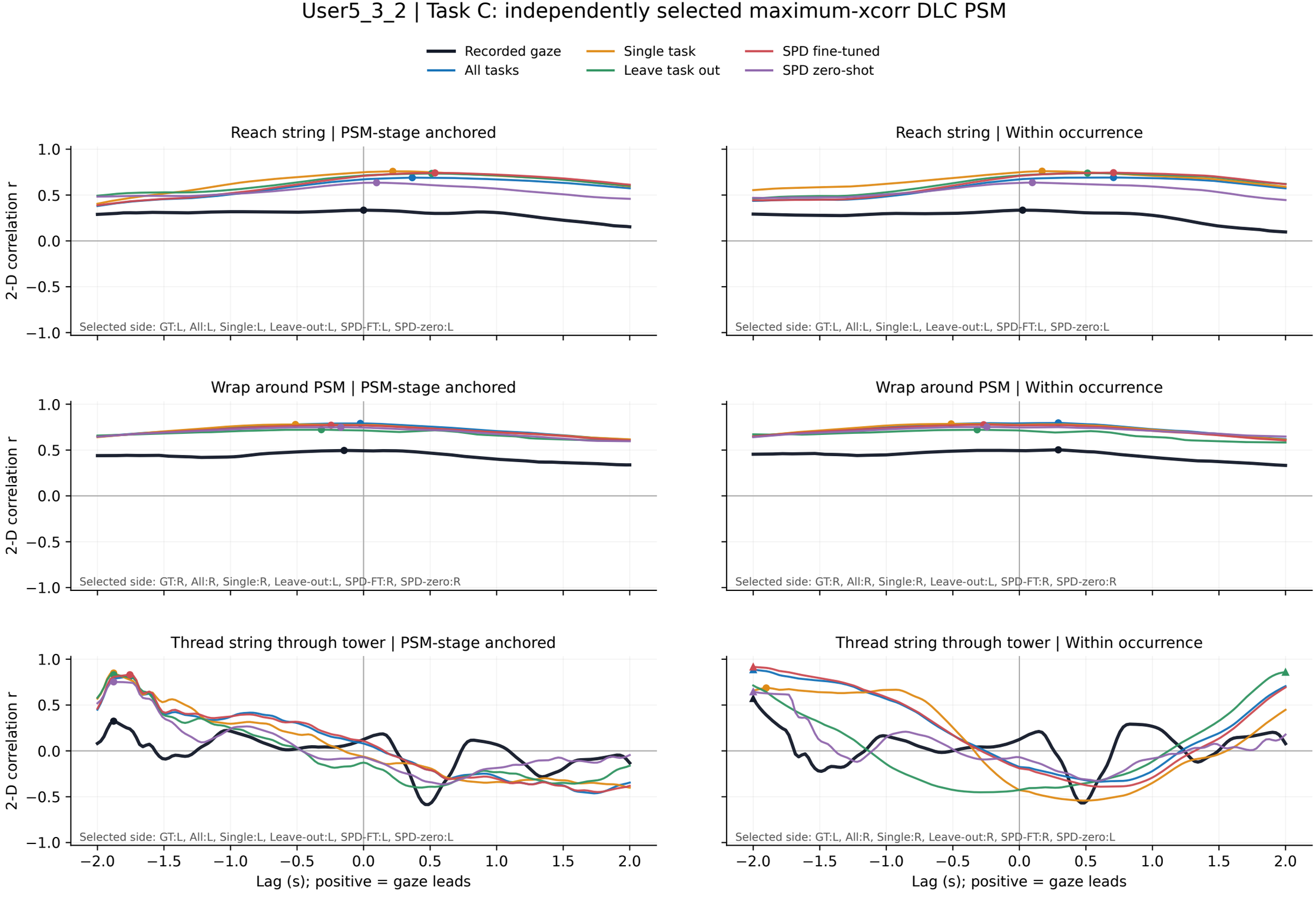
Thread string through tower | PSM-stage anchored

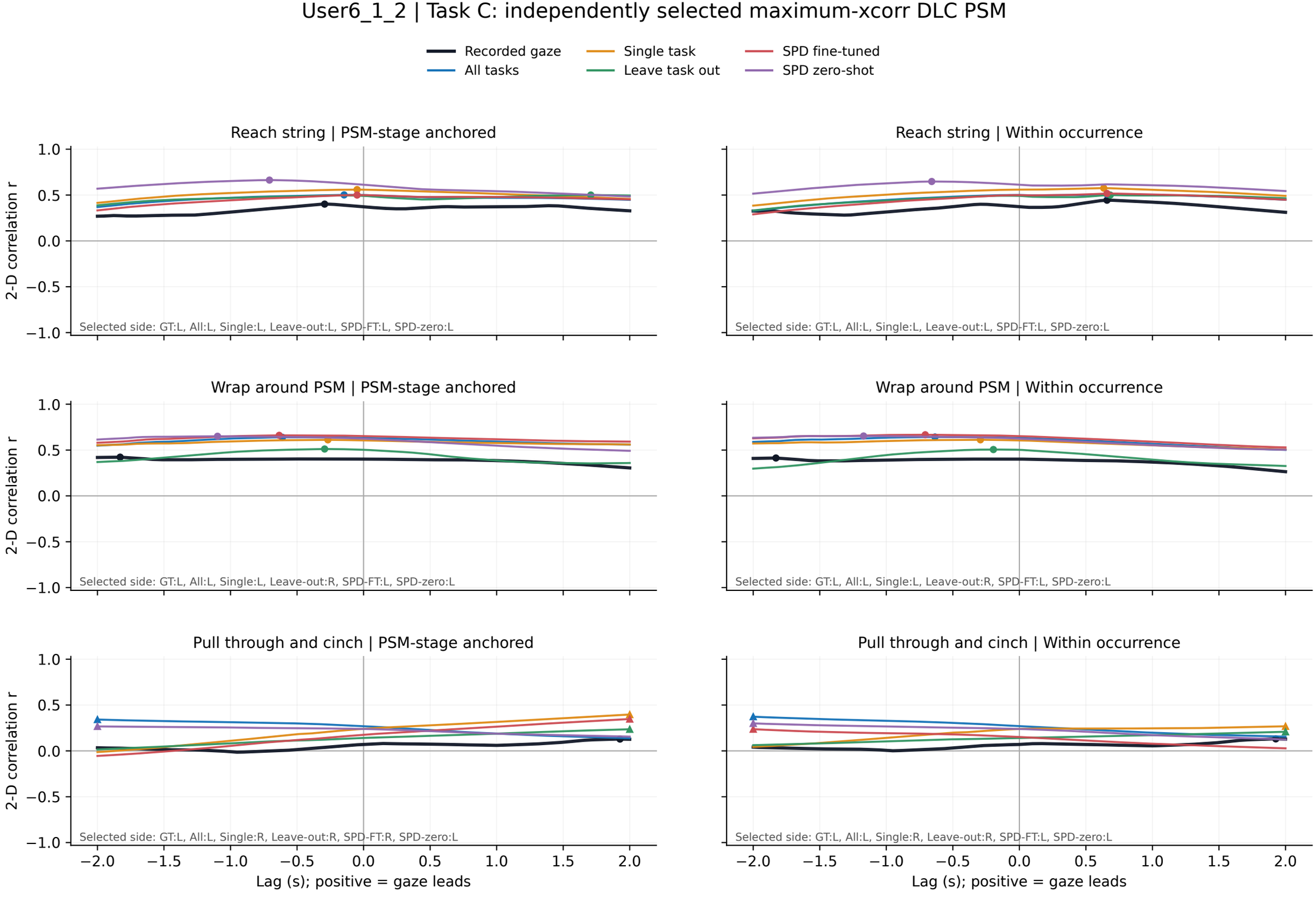


Thread string through tower | Within occurrence



Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

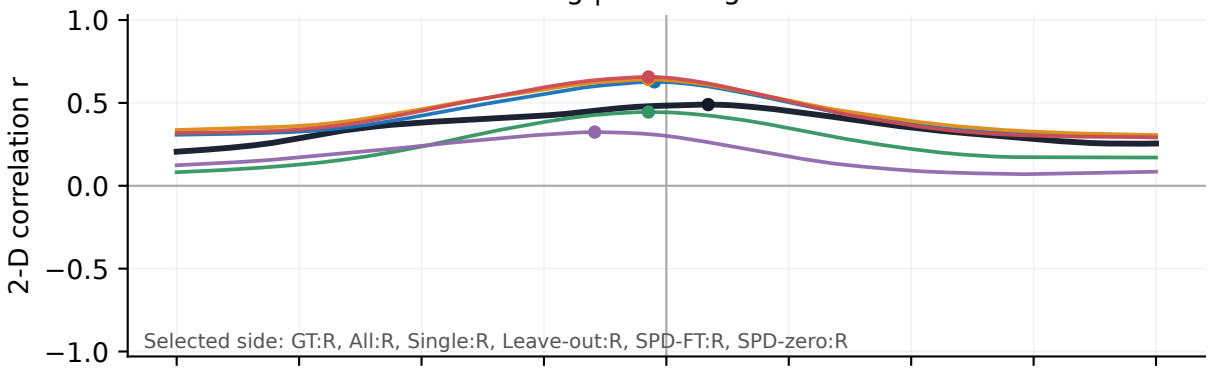

User5_3_2 | Task C: independently selected maximum-xcorr DLC PSM
Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot
Reach string | PSM-stage anchored
Reach string | Within occurrence
Wrap around PSM | PSM-stage anchored
Wrap around PSM | Within occurrence
Thread string through tower | PSM-stage anchored
Thread string through tower | Within occurrence
2-D correlation r
1.0
0.5
0.0
−0.5
−1.0
Selected side: GT:L, All:L, Single:L, Leave-out:L, SPD-FT:L, SPD-zero:L
Selected side: GT:R, All:R, Single:R, Leave-out:L, SPD-FT:R, SPD-zero:R
Selected side: GT:L, All:R, Single:R, Leave-out:R, SPD-FT:R, SPD-zero:L
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Lag (s); positive = gaze leads
Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.


User6_1_2 | Task C: independently selected maximum-xcorr DLC PSM
Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot
Reach string | PSM-stage anchored
Reach string | Within occurrence
Wrap around PSM | PSM-stage anchored
Wrap around PSM | Within occurrence
Pull through and cinch | PSM-stage anchored
Pull through and cinch | Within occurrence
2-D correlation r
1.0
0.5
0.0
−0.5
−1.0
Selected side: GT:L, All:L, Single:L, Leave-out:L, SPD-FT:L, SPD-zero:L
Selected side: GT:L, All:L, Single:L, Leave-out:R, SPD-FT:L, SPD-zero:L
Selected side: GT:L, All:L, Single:R, Leave-out:R, SPD-FT:R, SPD-zero:L
Selected side: GT:L, All:L, Single:R, Leave-out:R, SPD-FT:L, SPD-zero:L
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Lag (s); positive = gaze leads
Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

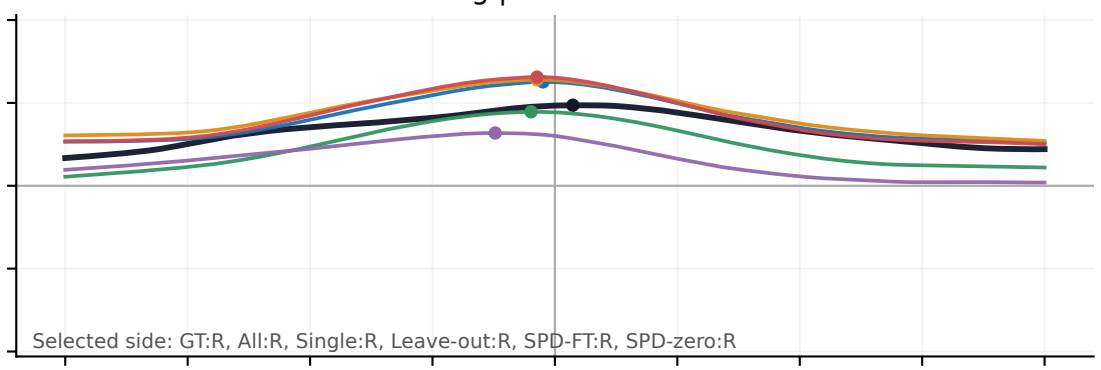
User7_1_1 | Task A: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

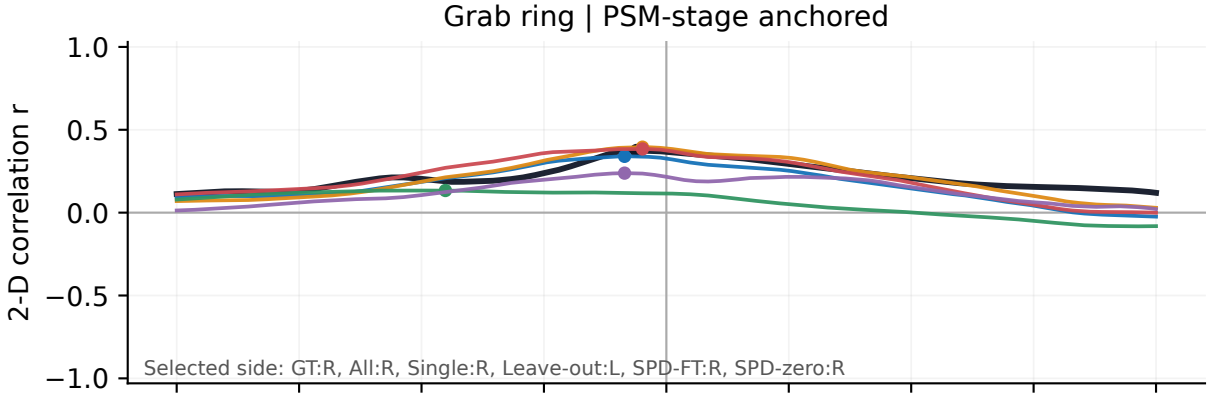
Reach ring | PSM-stage anchored



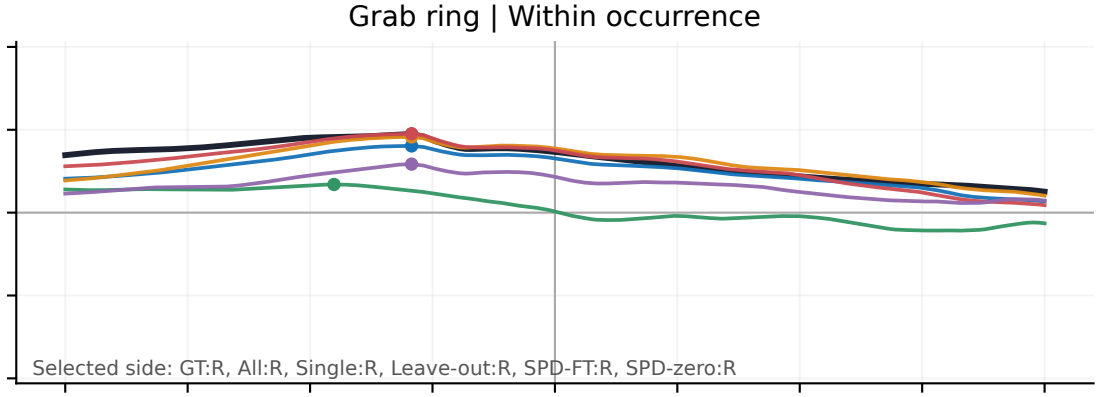
Reach ring | Within occurrence



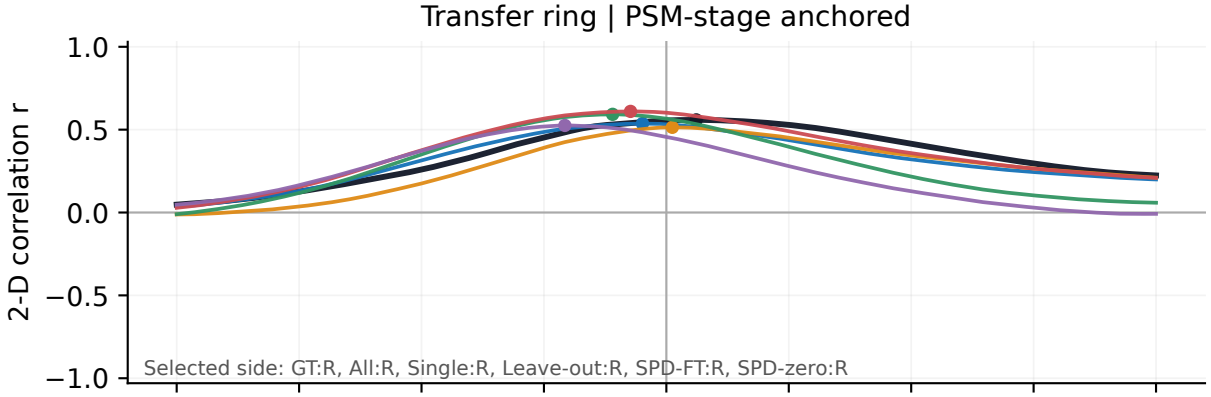
Grab ring | PSM-stage anchored



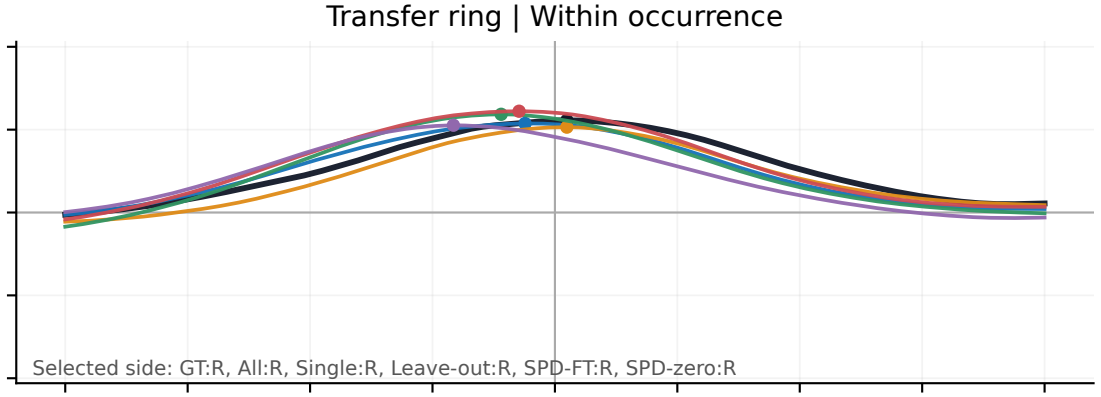
Grab ring | Within occurrence



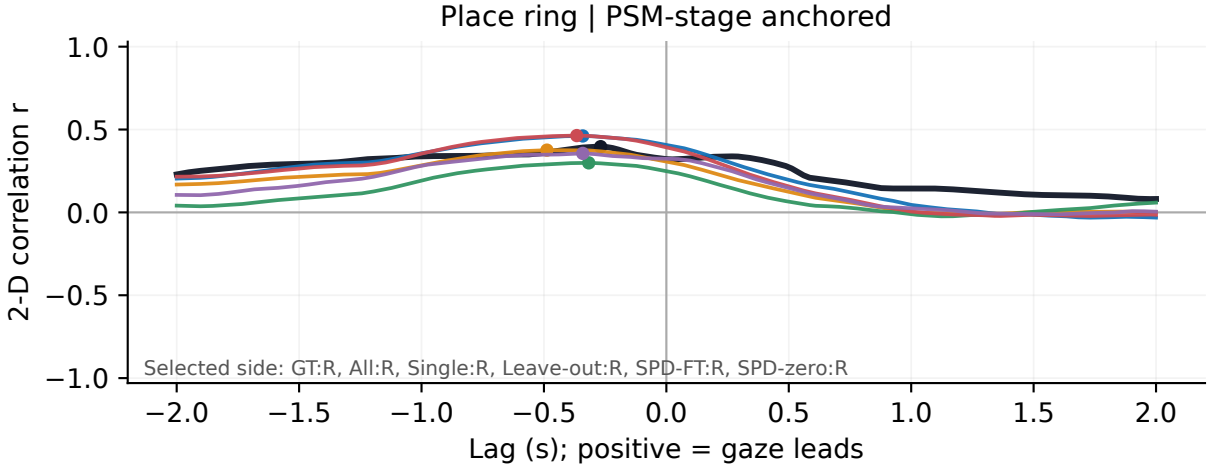
Transfer ring | PSM-stage anchored



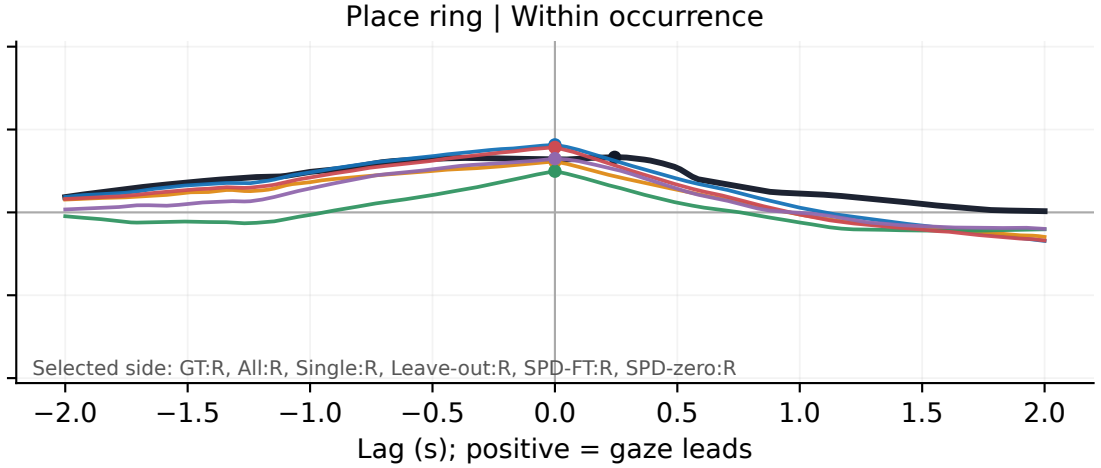
Transfer ring | Within occurrence



Place ring | PSM-stage anchored



Place ring | Within occurrence

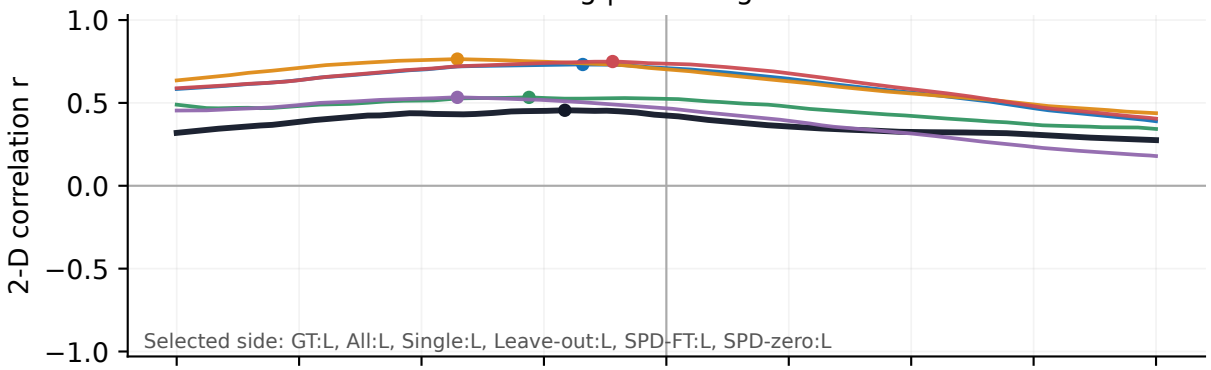


Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

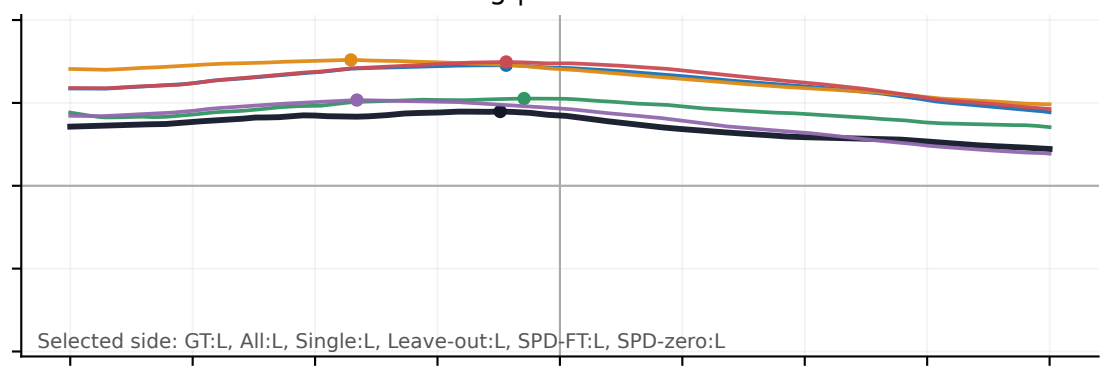
User7_4_1 | Task C: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

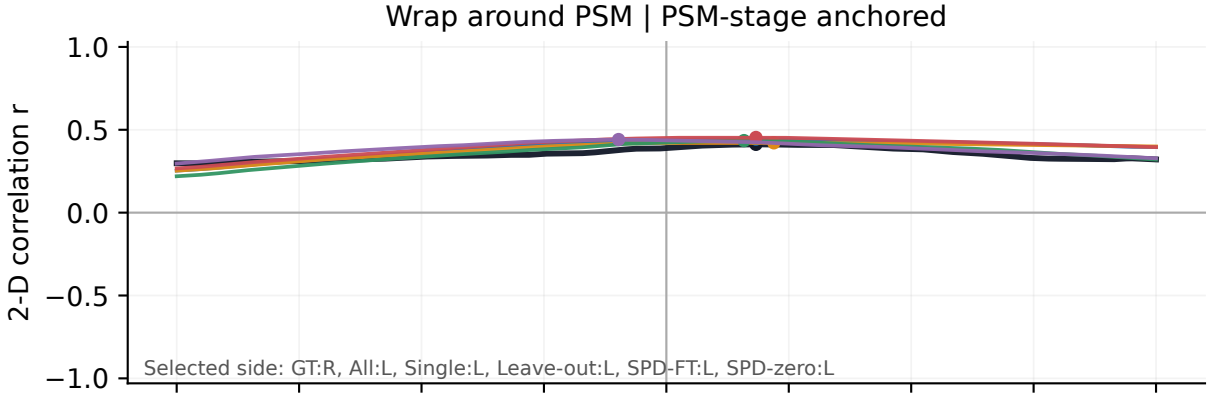
Reach string | PSM-stage anchored



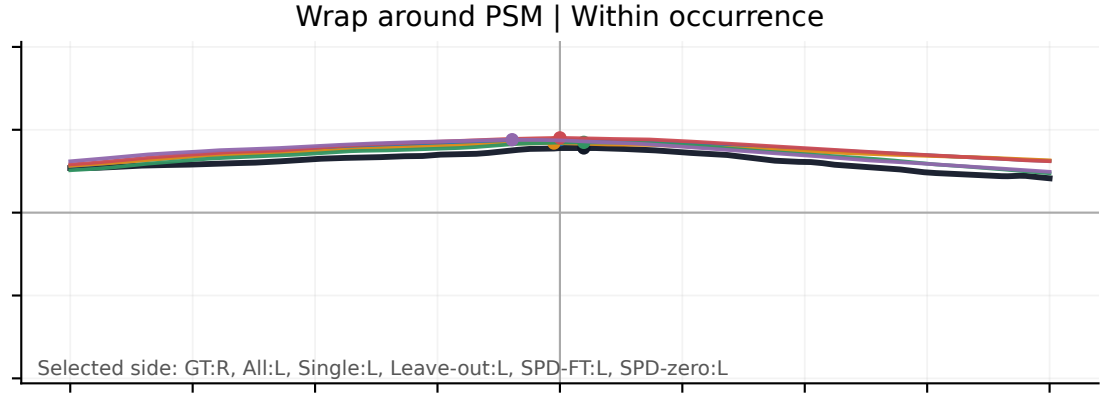
Reach string | Within occurrence



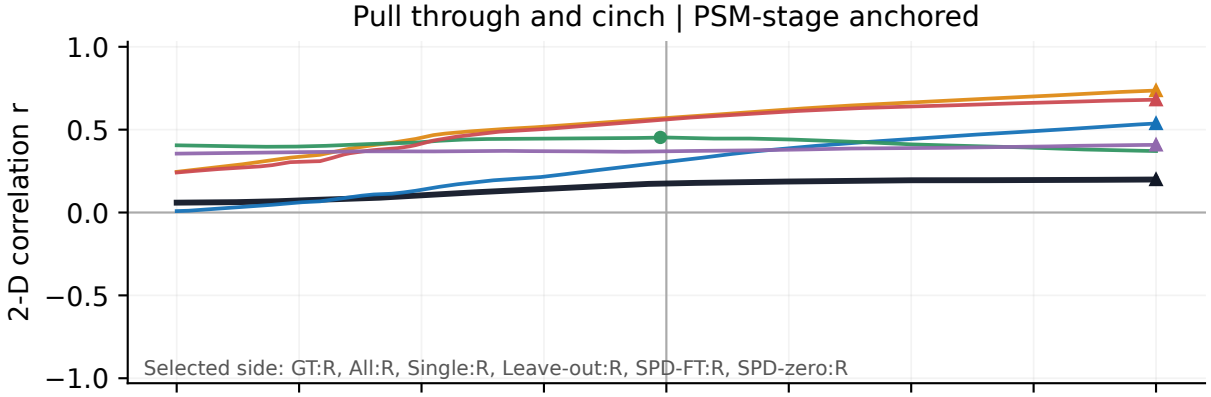
Wrap around PSM | PSM-stage anchored



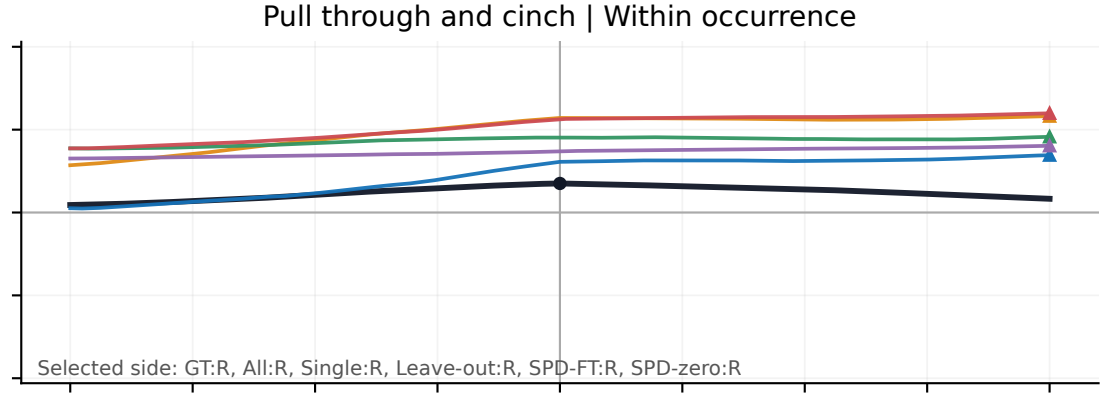
Wrap around PSM | Within occurrence



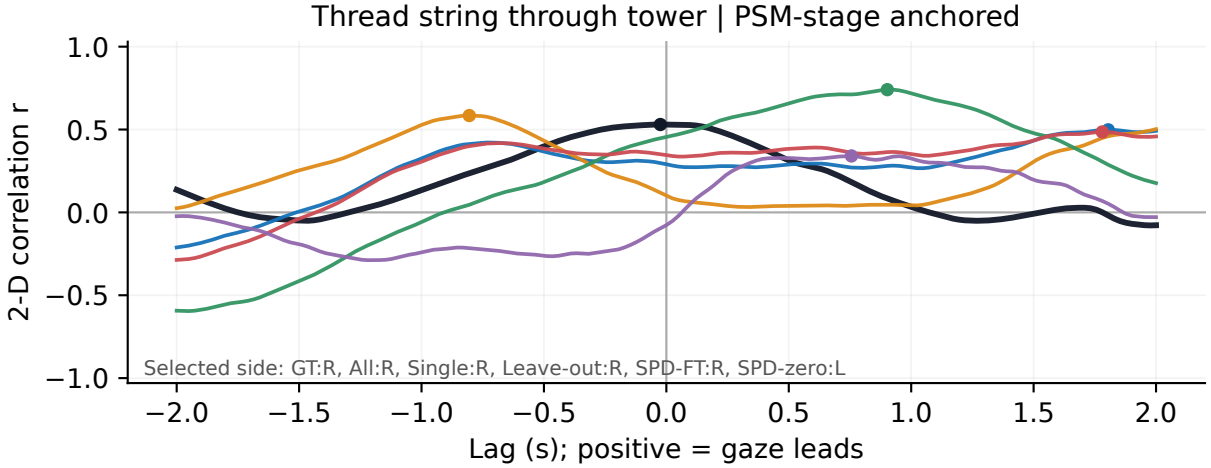
Pull through and cinch | PSM-stage anchored



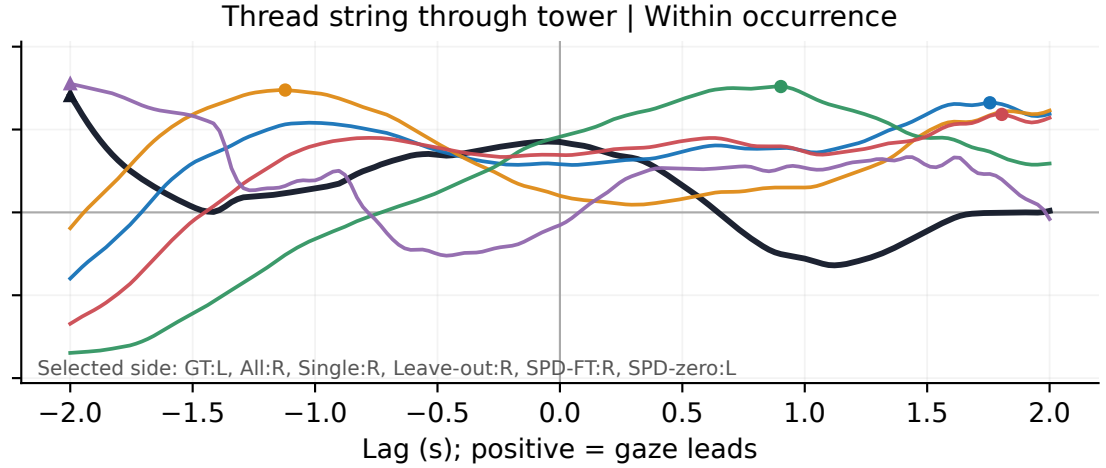
Pull through and cinch | Within occurrence



Thread string through tower | PSM-stage anchored



Thread string through tower | Within occurrence

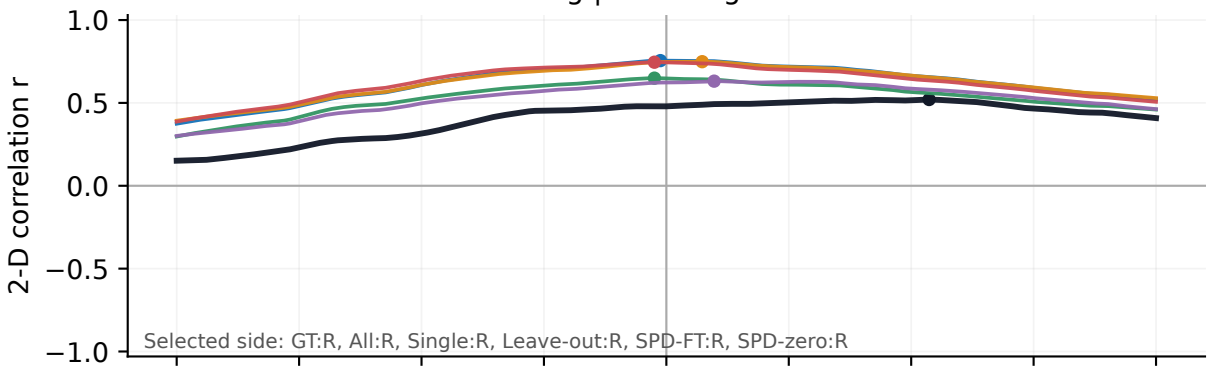


Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.

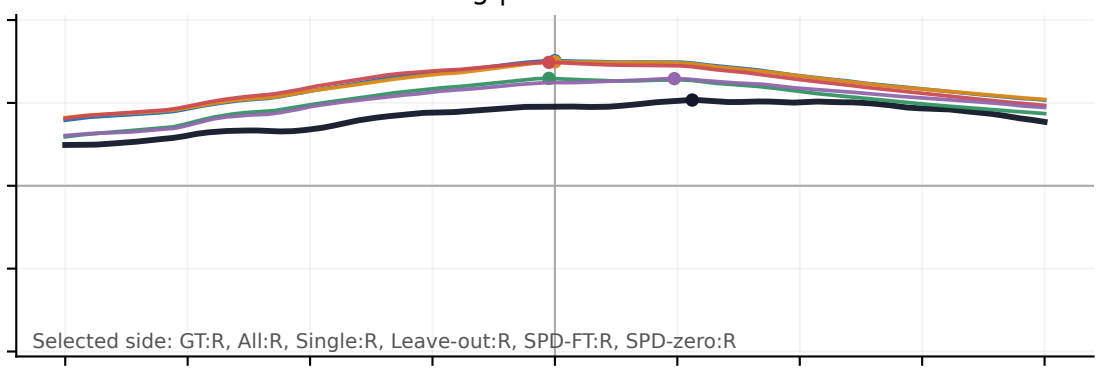
User9_7_1 | Task A: independently selected maximum-xcorr DLC PSM

Recorded gaze
All tasks
Single task
Leave task out
SPD fine-tuned
SPD zero-shot

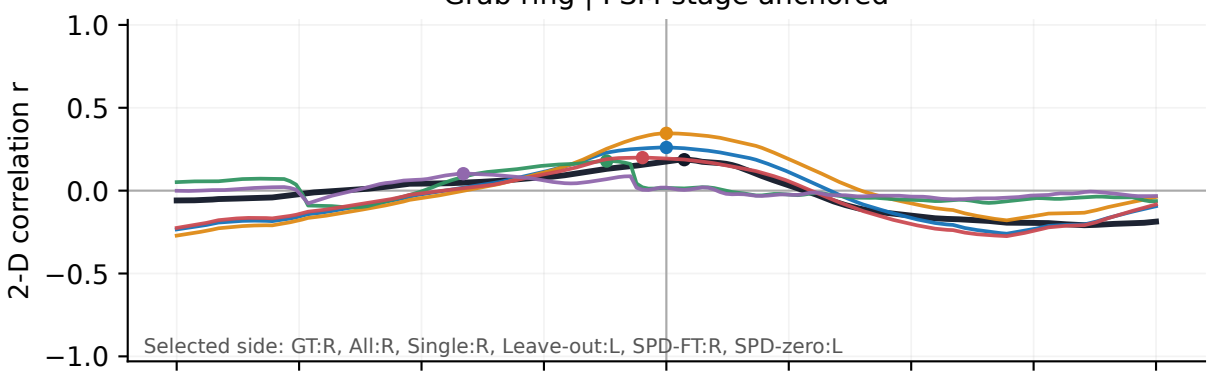
Reach ring | PSM-stage anchored



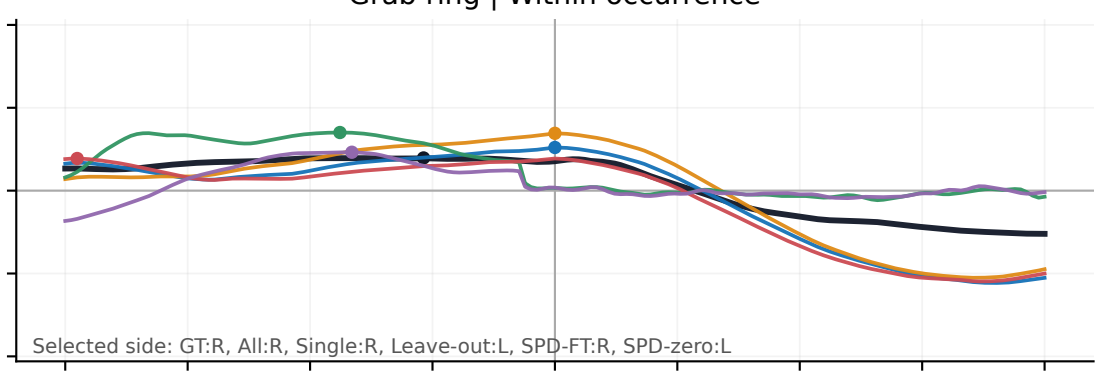
Reach ring | Within occurrence



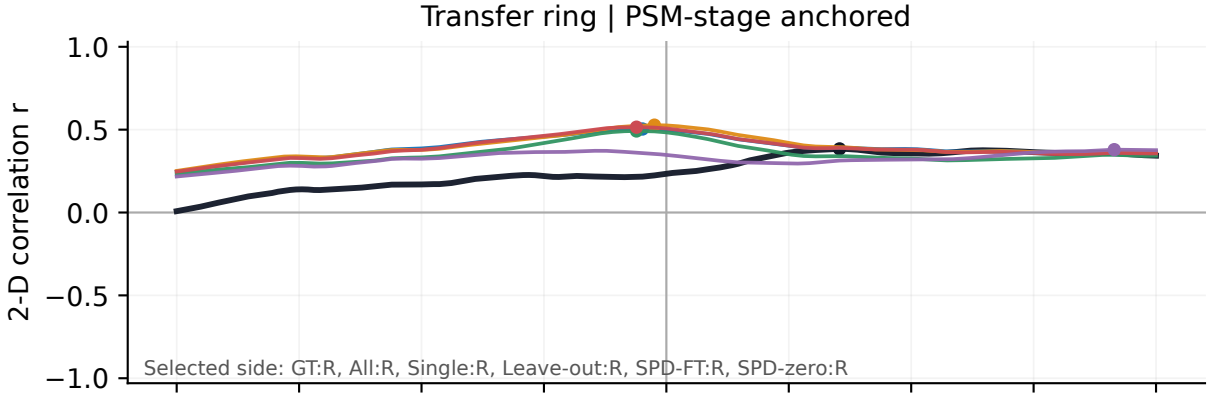
Grab ring | PSM-stage anchored



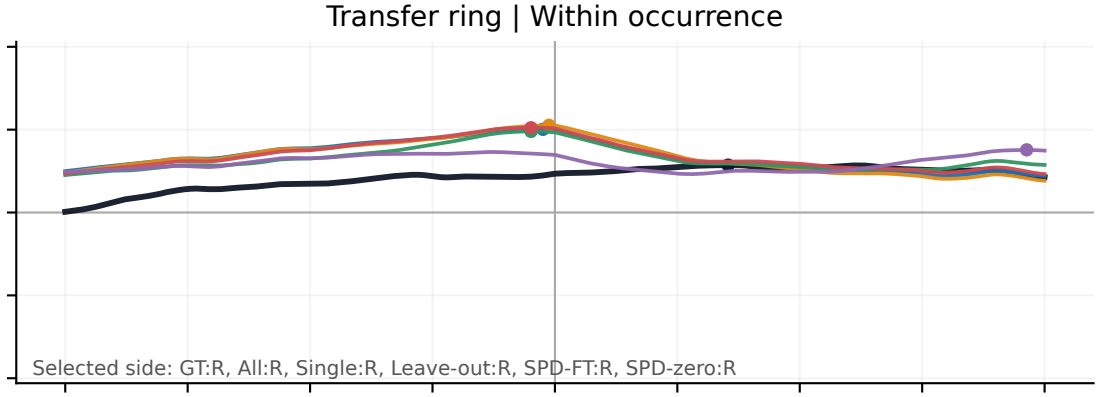
Grab ring | Within occurrence



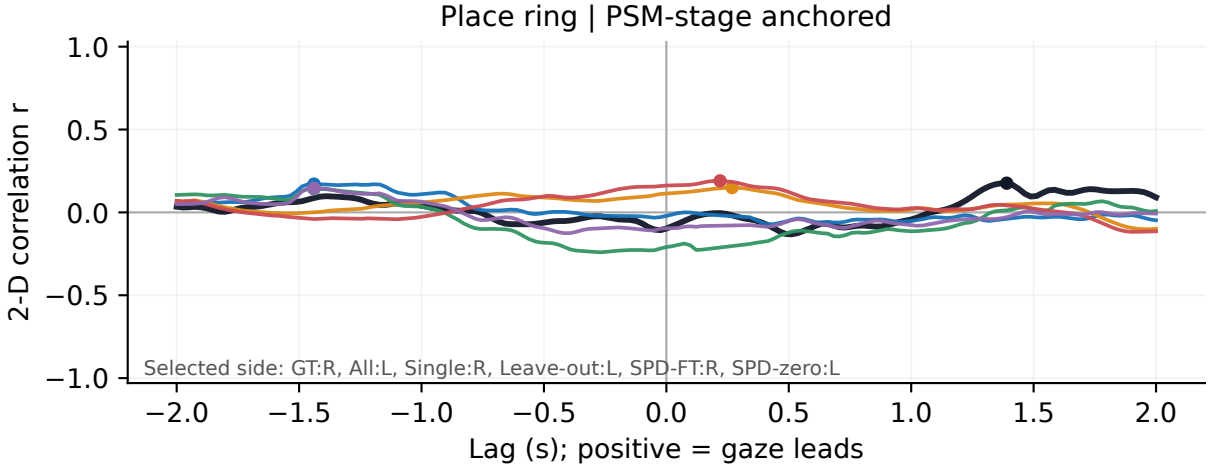
Transfer ring | PSM-stage anchored



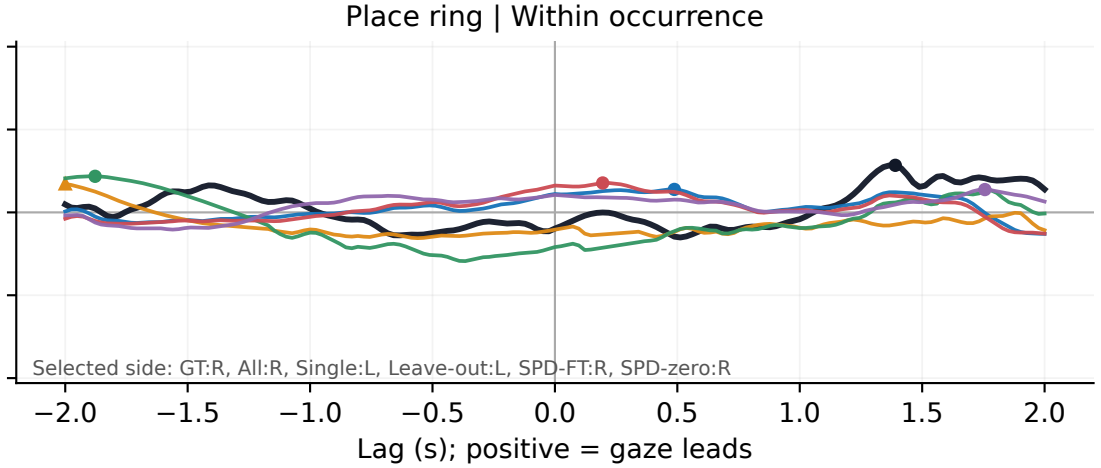
Transfer ring | Within occurrence



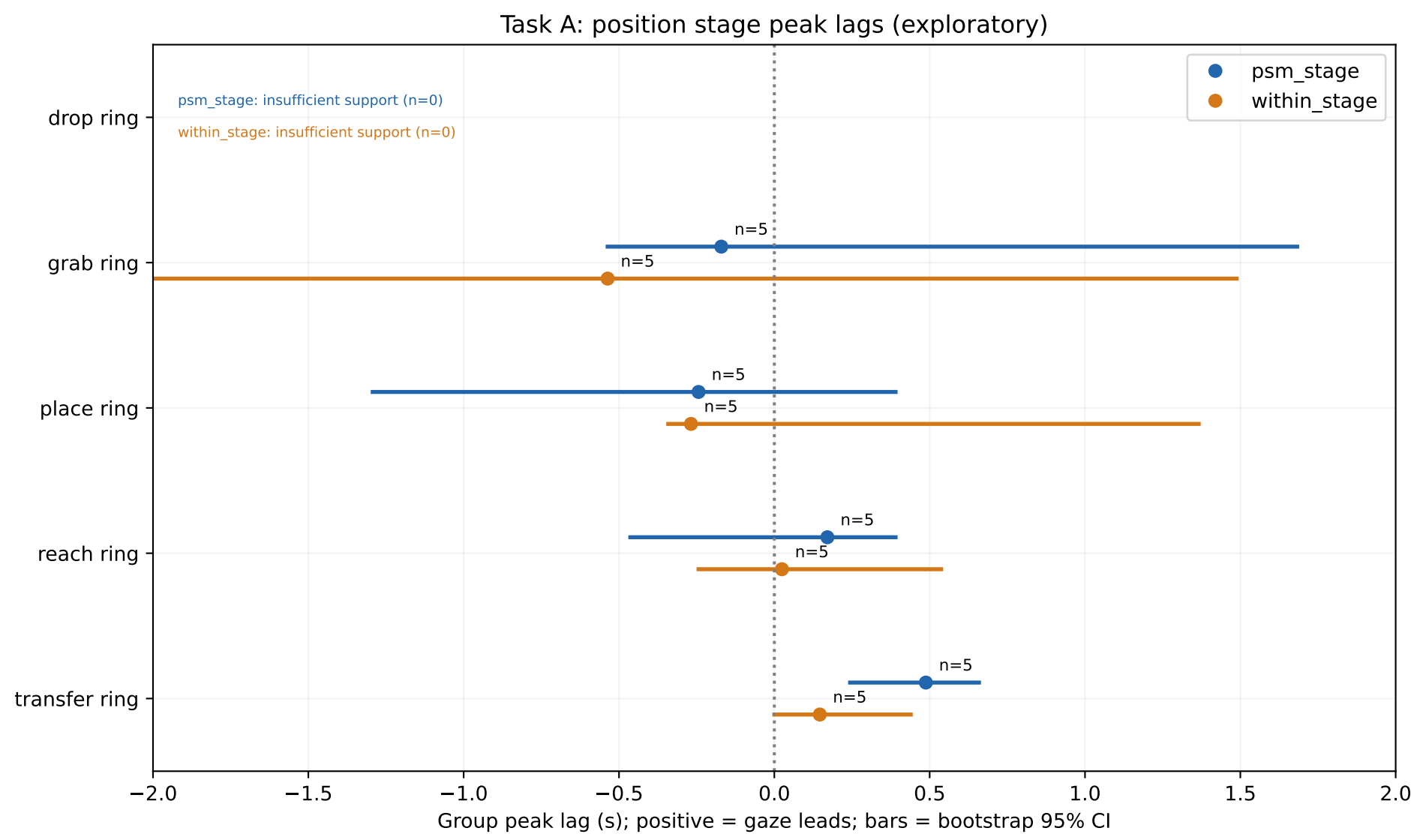
Place ring | PSM-stage anchored



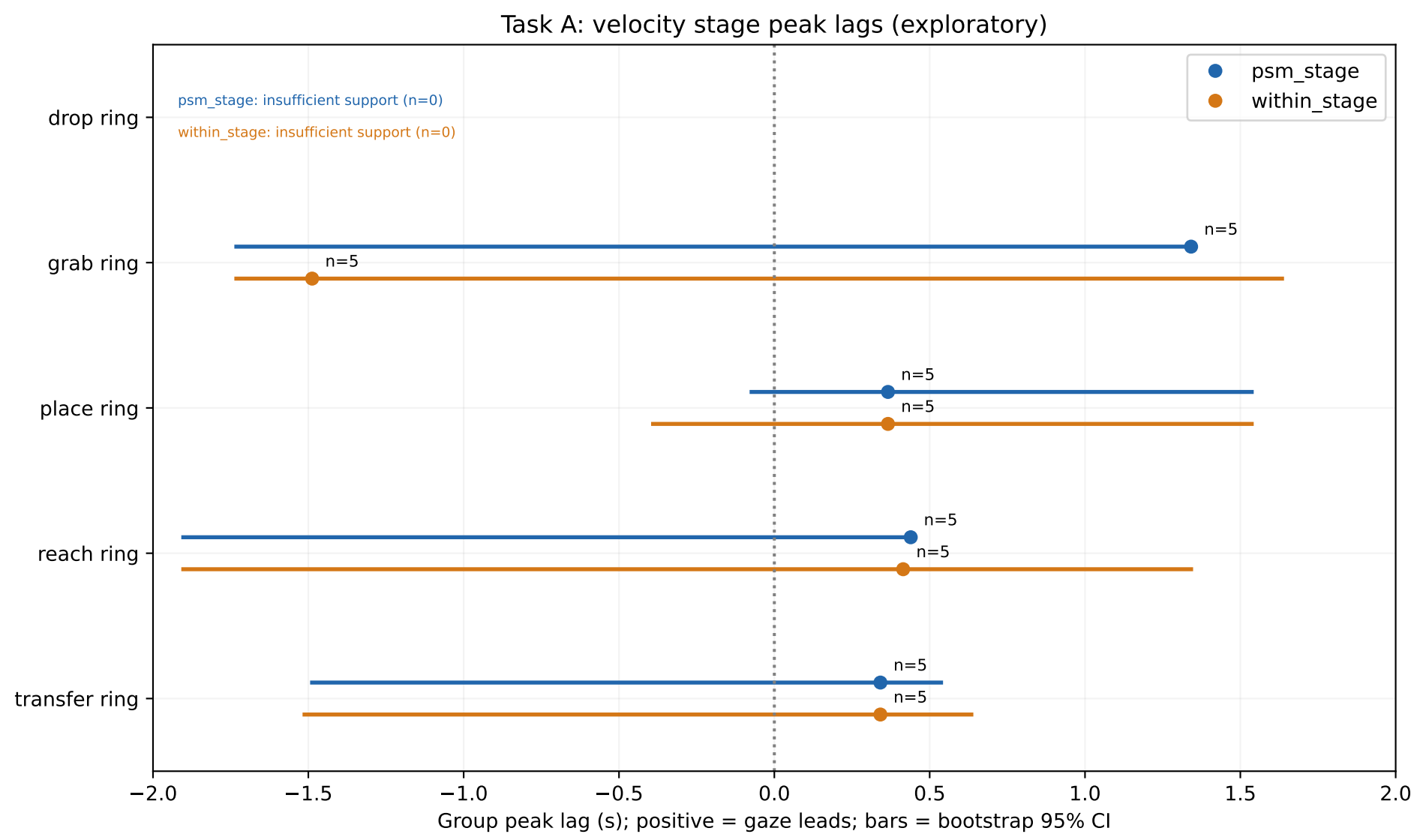
Place ring | Within occurrence



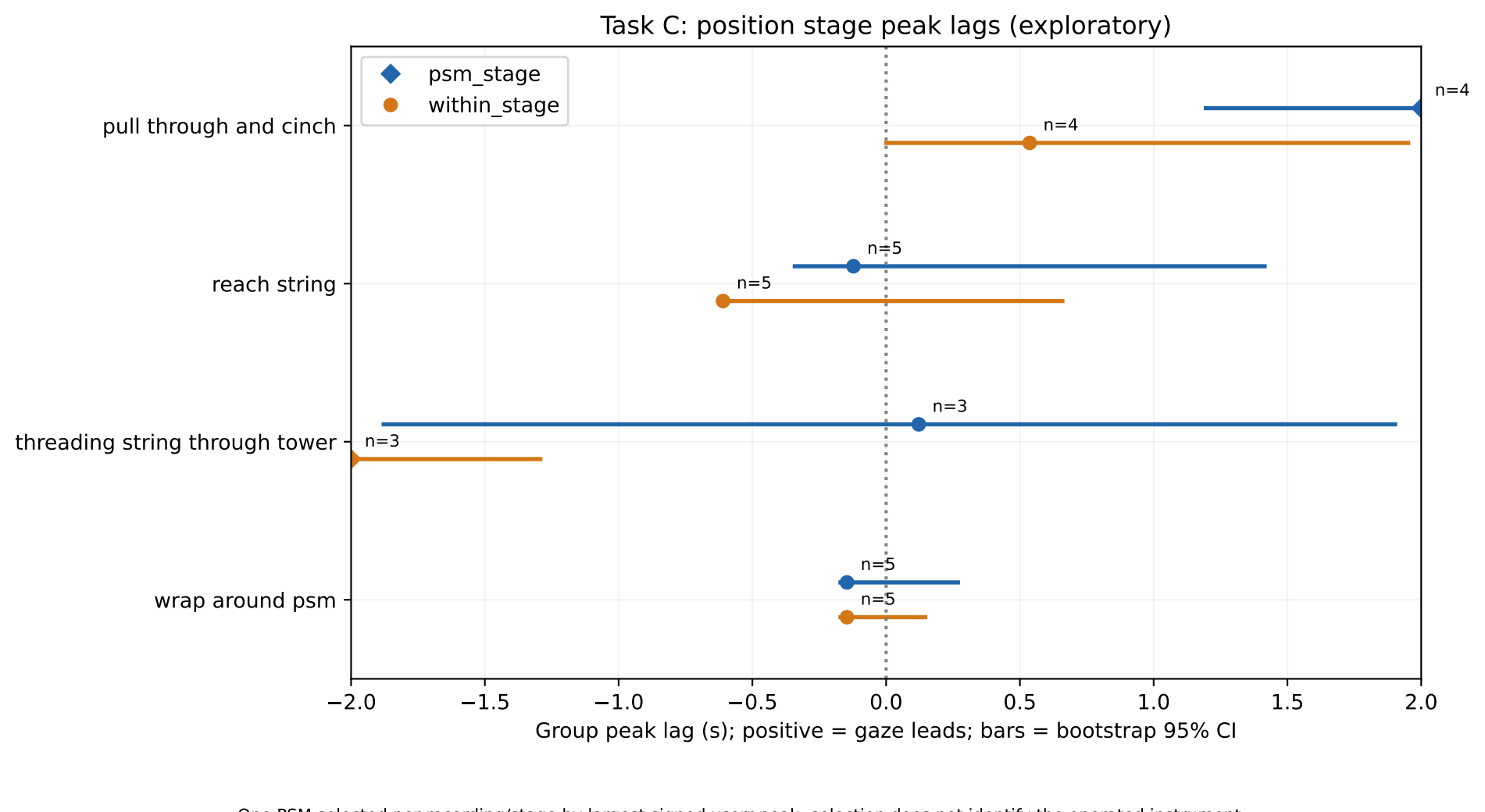
Selected instruments may differ across sources and modes. Dots mark signed maxima; triangles mark search boundaries, not reliable timing.



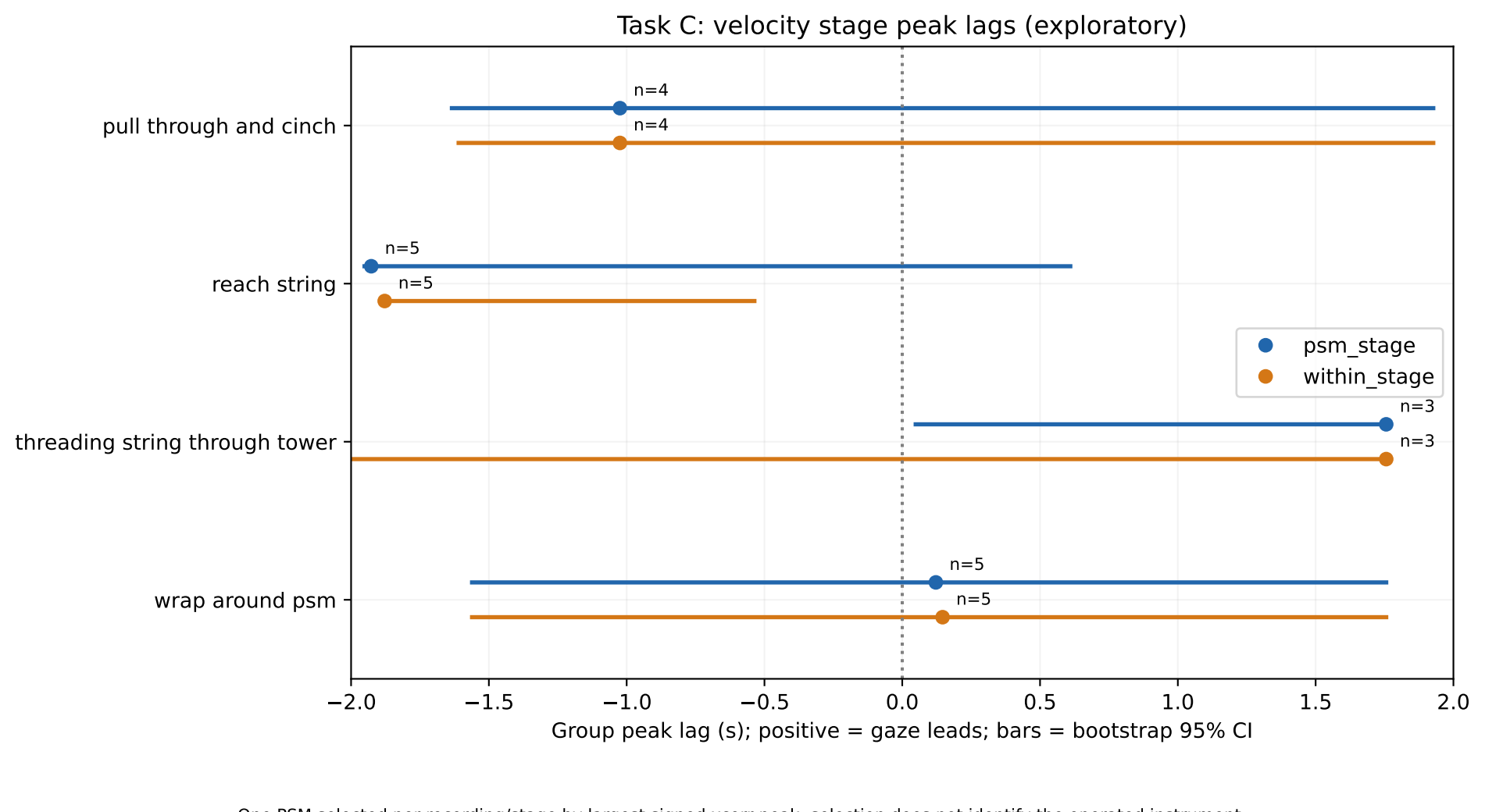
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



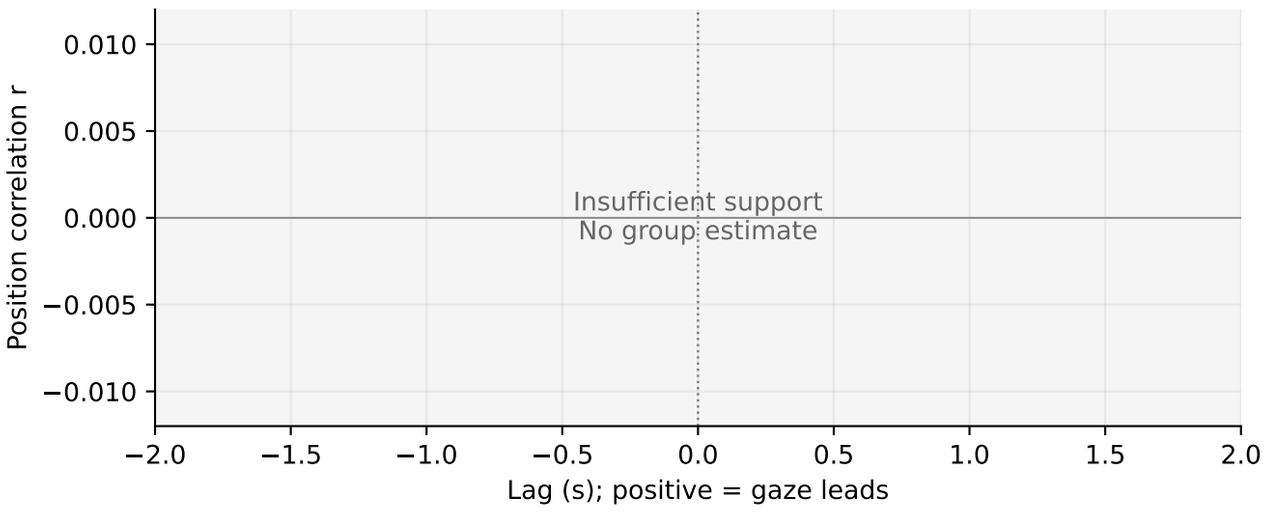
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) · Participant bootstrap (95%) · Group mean

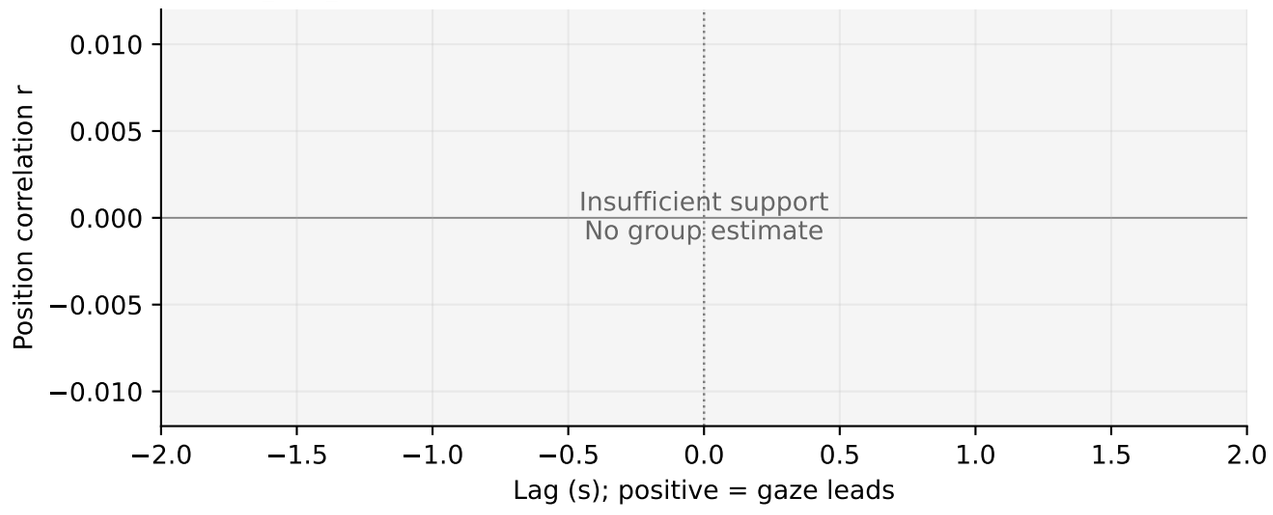
drop ring / psm_stage

n = 0; no_valid_curve



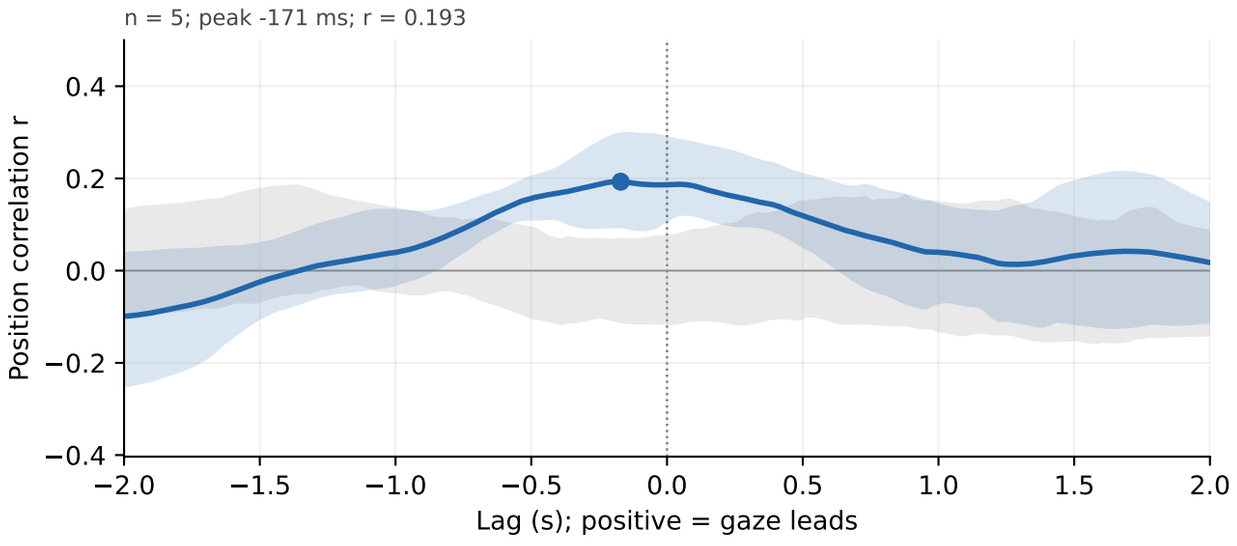
drop ring / within_stage

n = 0; no_valid_curve



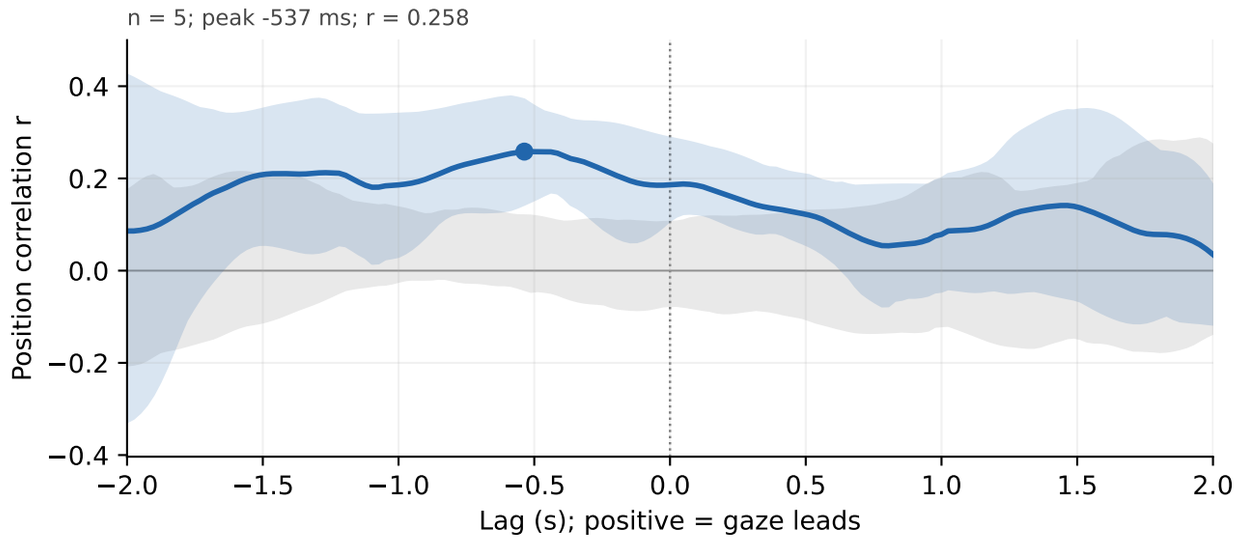
grab ring / psm_stage

n = 5; peak -171 ms; r = 0.193



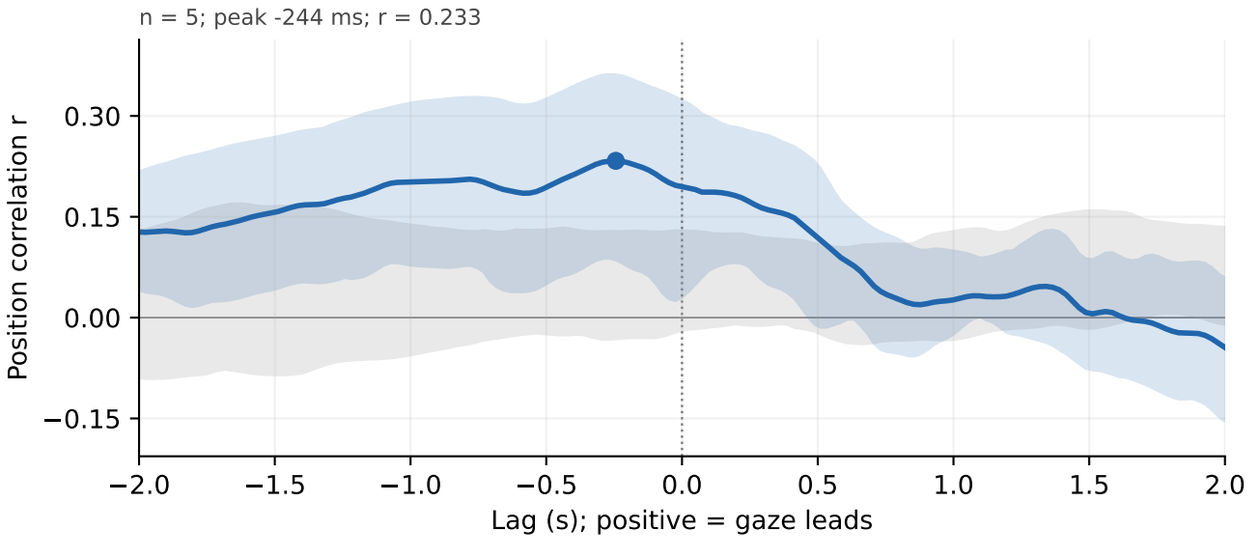
grab ring / within_stage

n = 5; peak -537 ms; r = 0.258



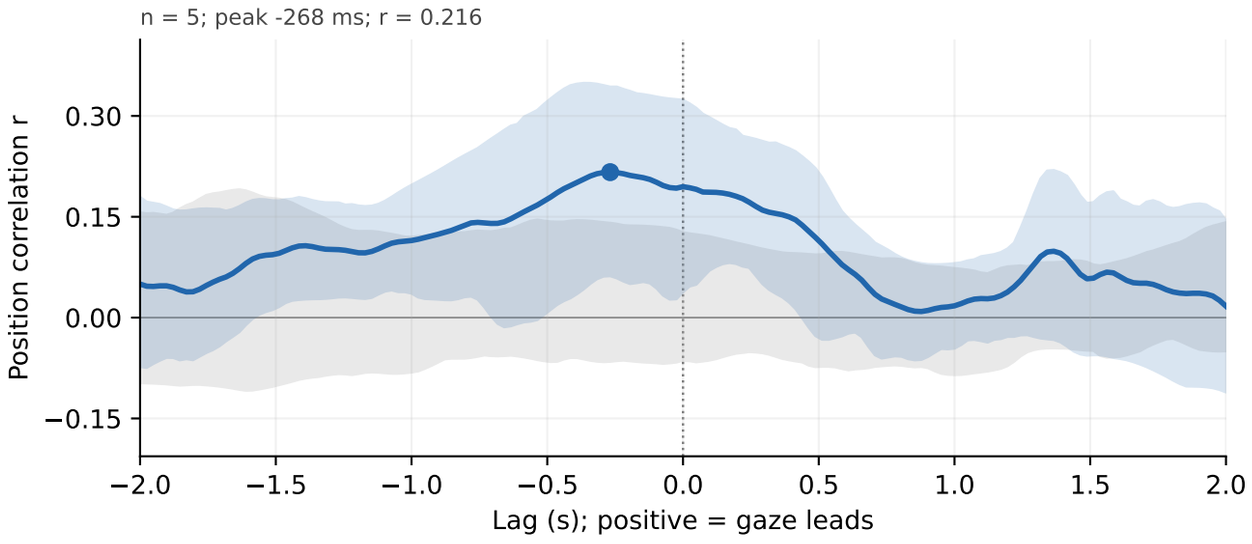
place ring / psm_stage

n = 5; peak -244 ms; r = 0.233



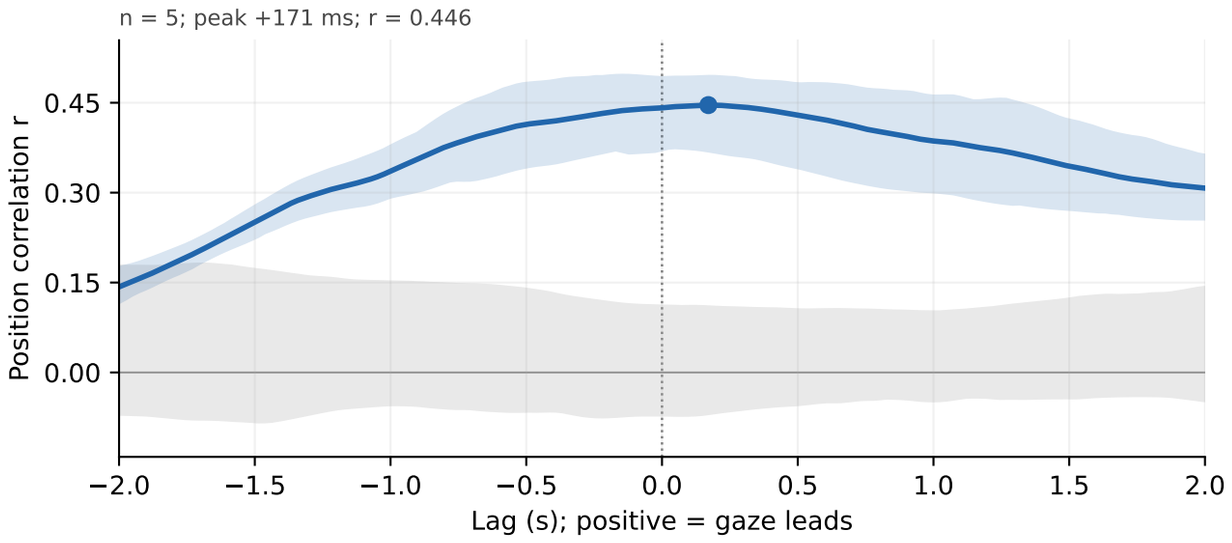
place ring / within_stage

n = 5; peak -268 ms; r = 0.216



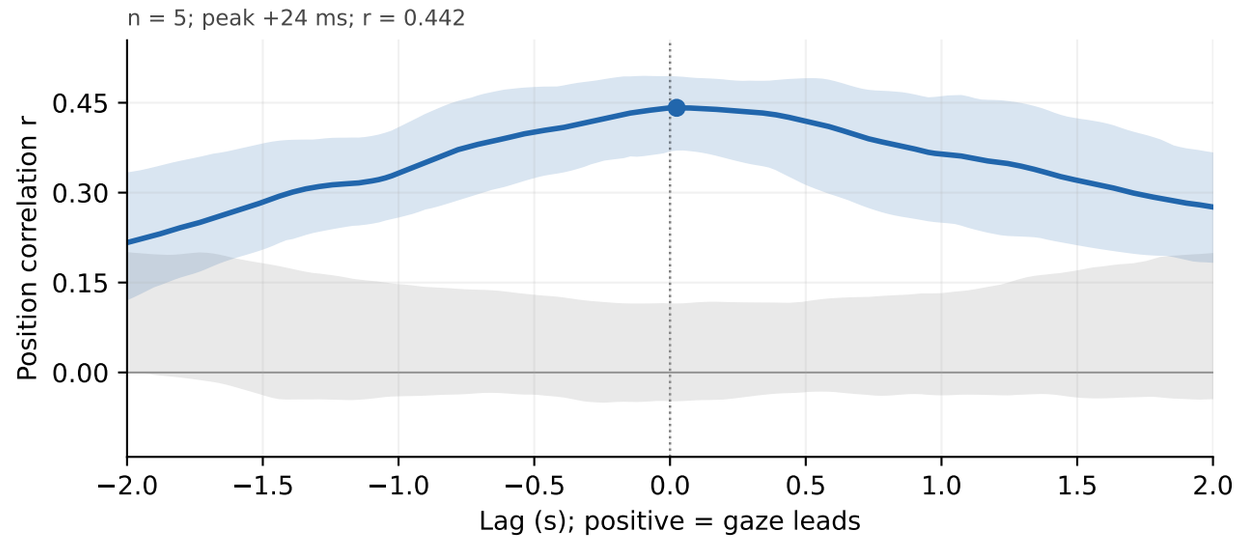
reach ring / psm_stage

n = 5; peak +171 ms; r = 0.446



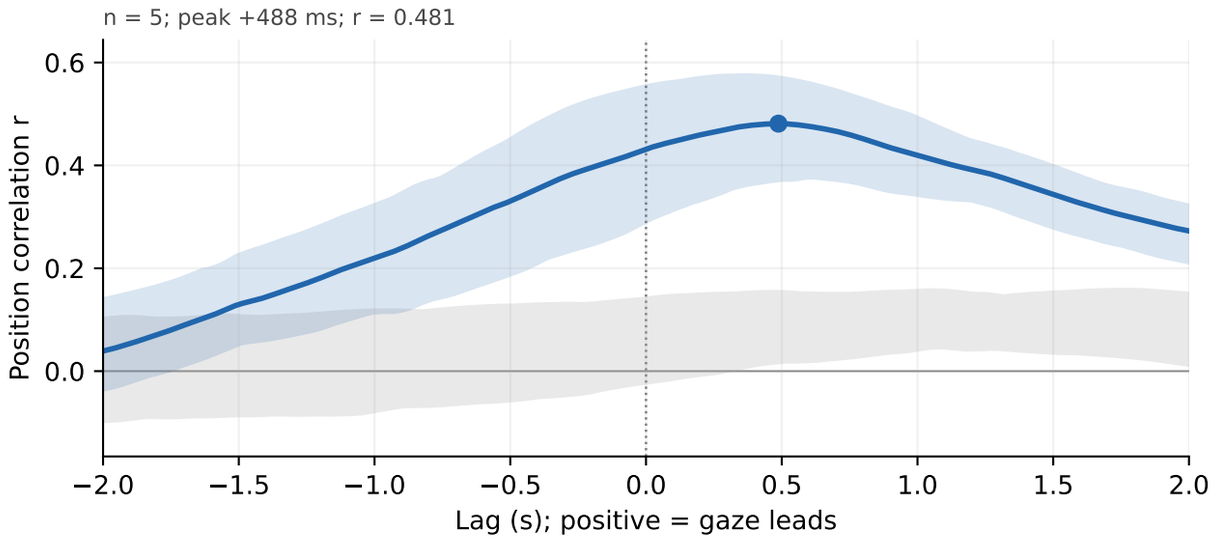
reach ring / within_stage

n = 5; peak +24 ms; r = 0.442



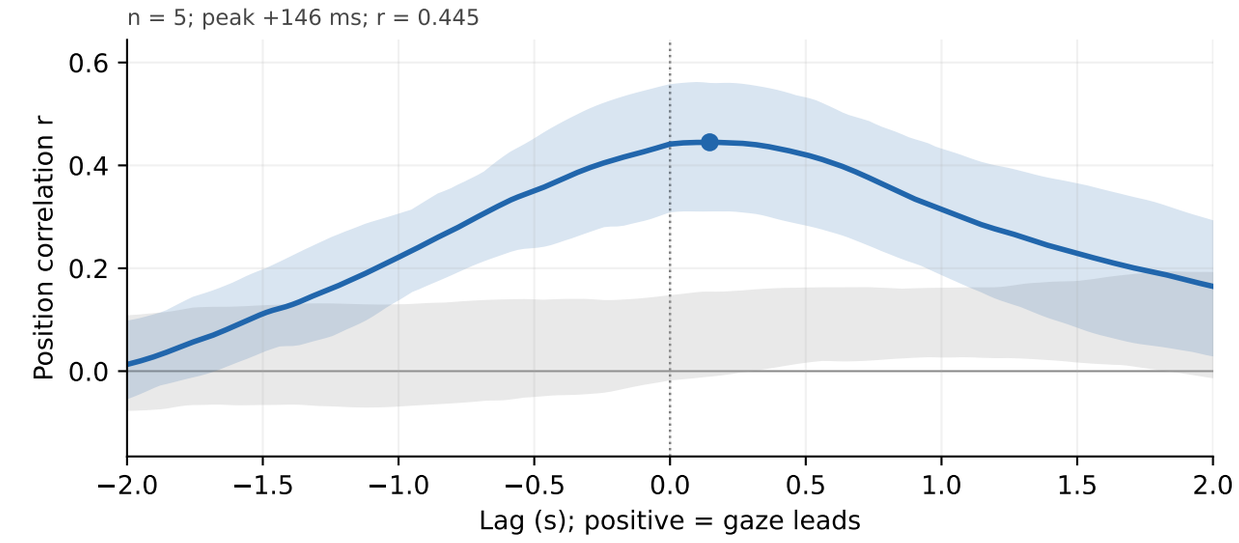
transfer ring / psm_stage

n = 5; peak +488 ms; r = 0.481



transfer ring / within_stage

n = 5; peak +146 ms; r = 0.445



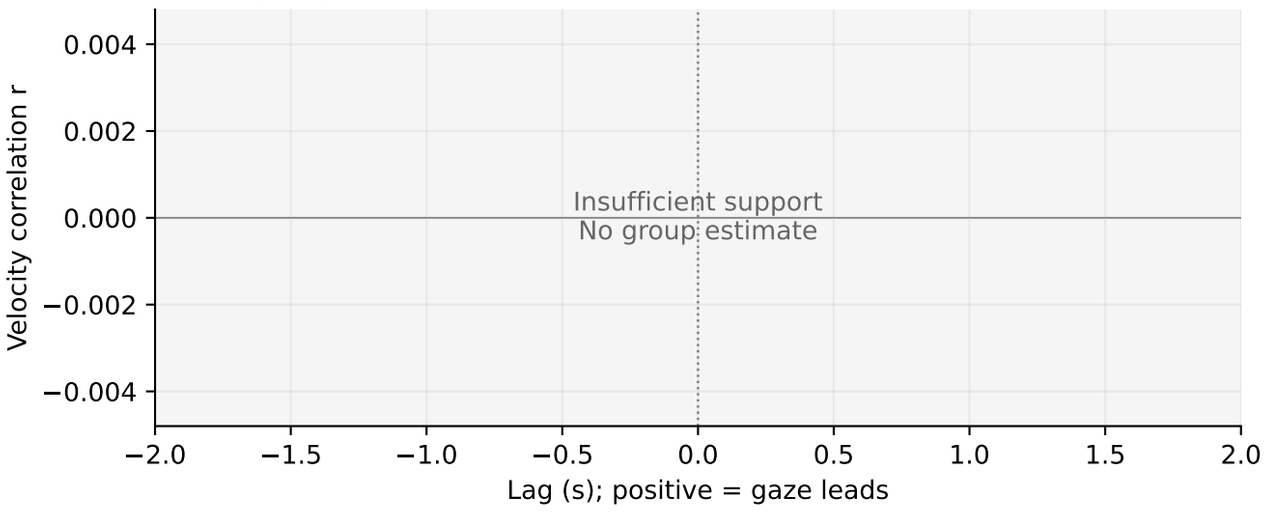
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

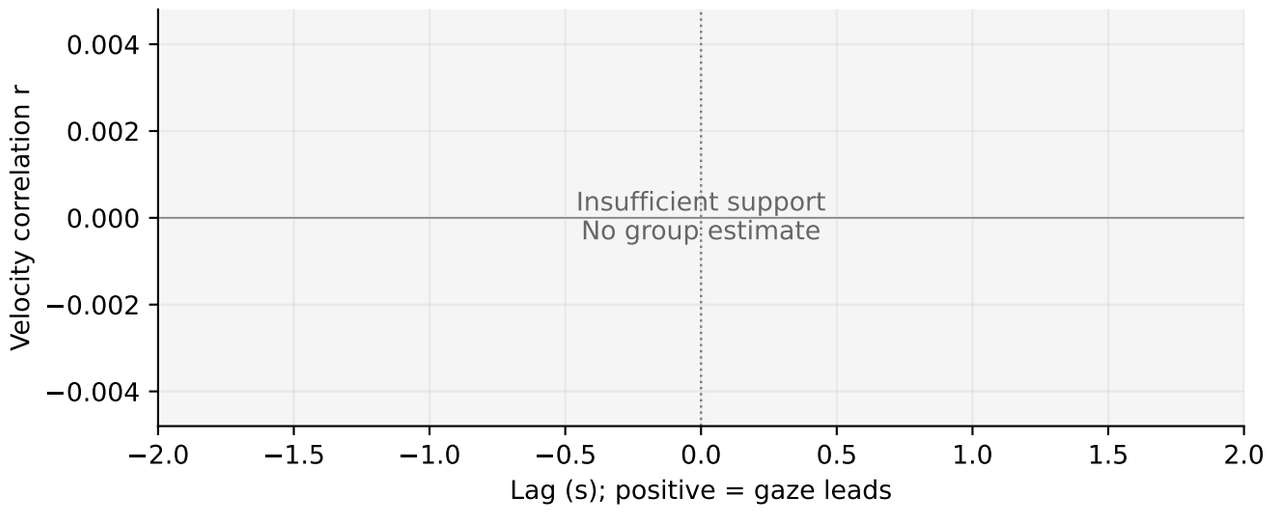
drop ring / psm_stage

n = 0; no_valid_curve



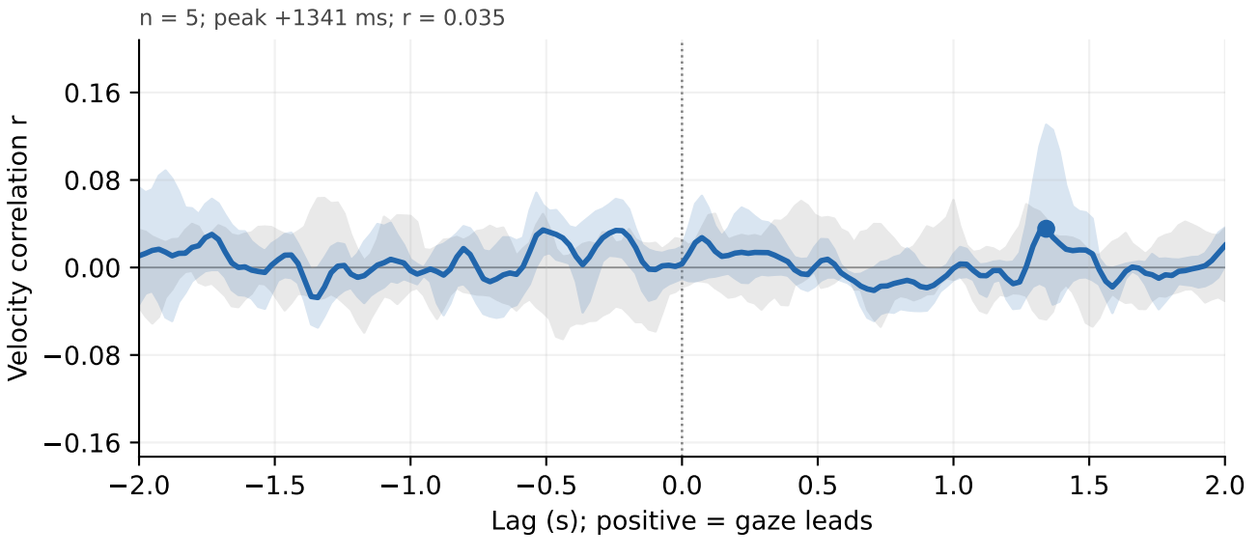
drop ring / within_stage

n = 0; no_valid_curve



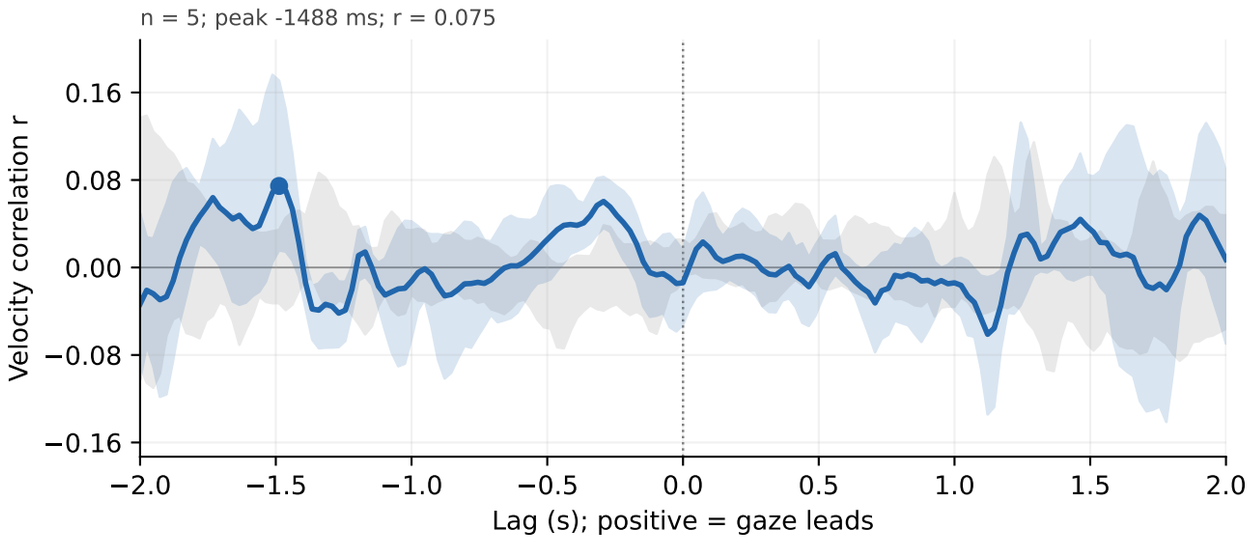
grab ring / psm_stage

n = 5; peak +1341 ms; r = 0.035



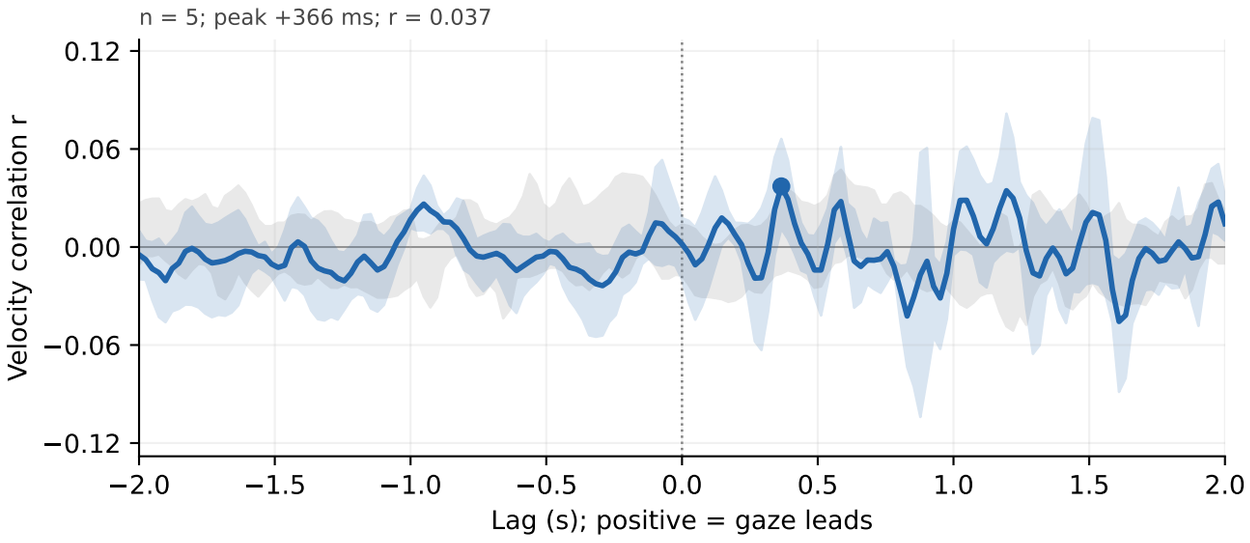
grab ring / within_stage

n = 5; peak -1488 ms; r = 0.075



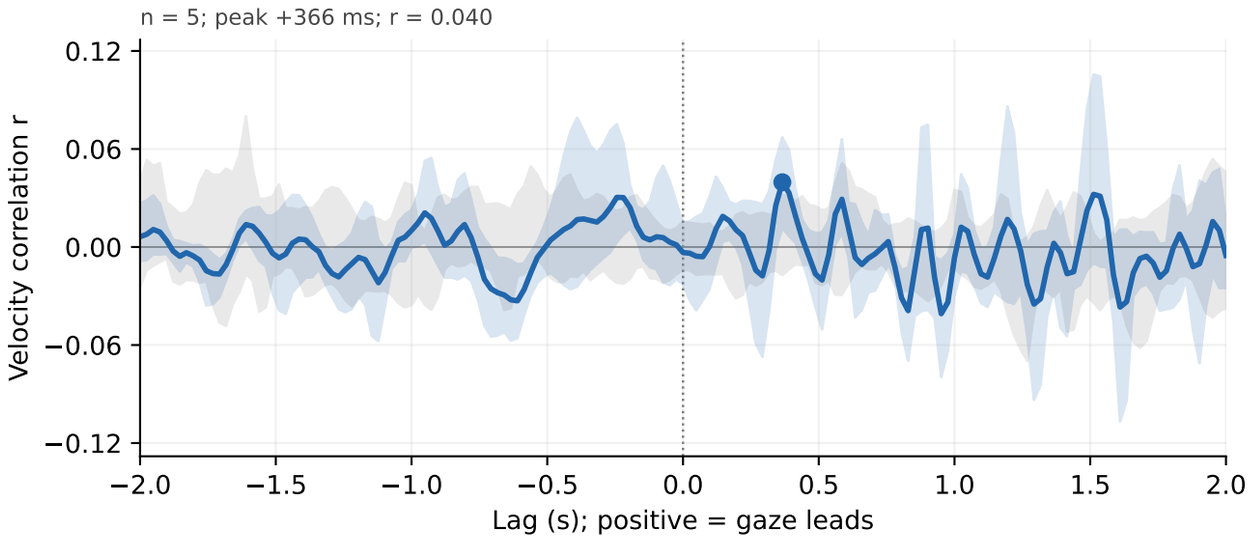
place ring / psm_stage

n = 5; peak +366 ms; r = 0.037



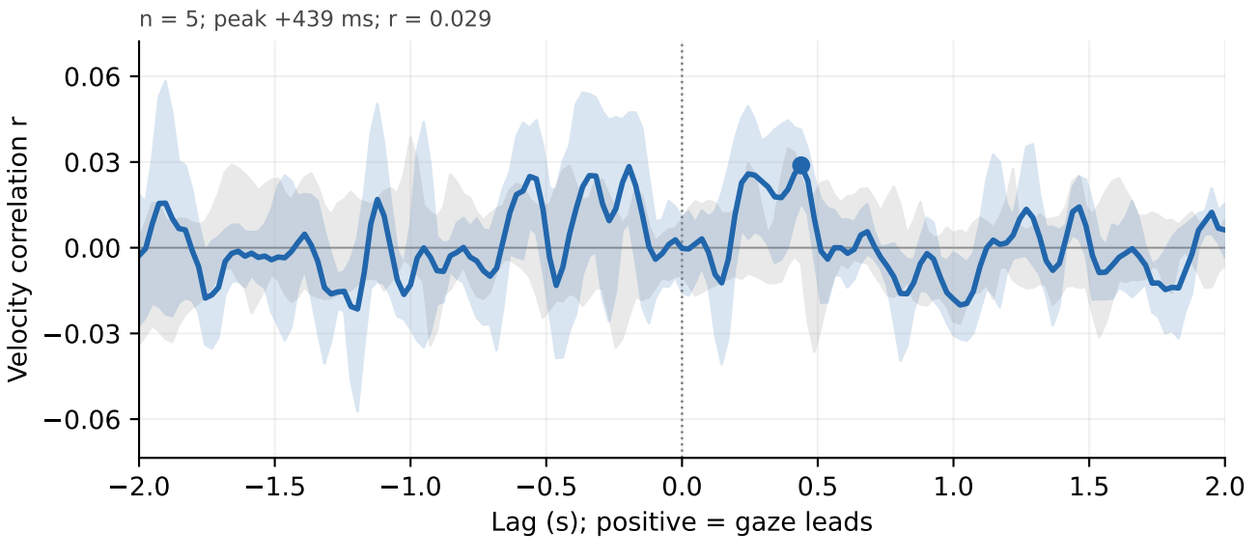
place ring / within_stage

n = 5; peak +366 ms; r = 0.040



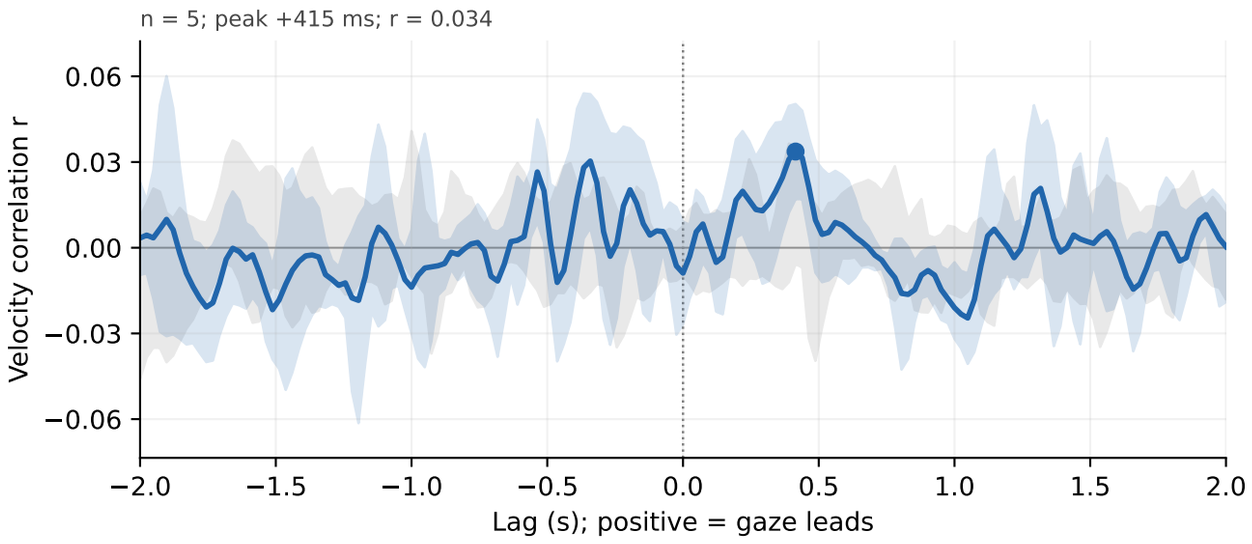
reach ring / psm_stage

n = 5; peak +439 ms; r = 0.029



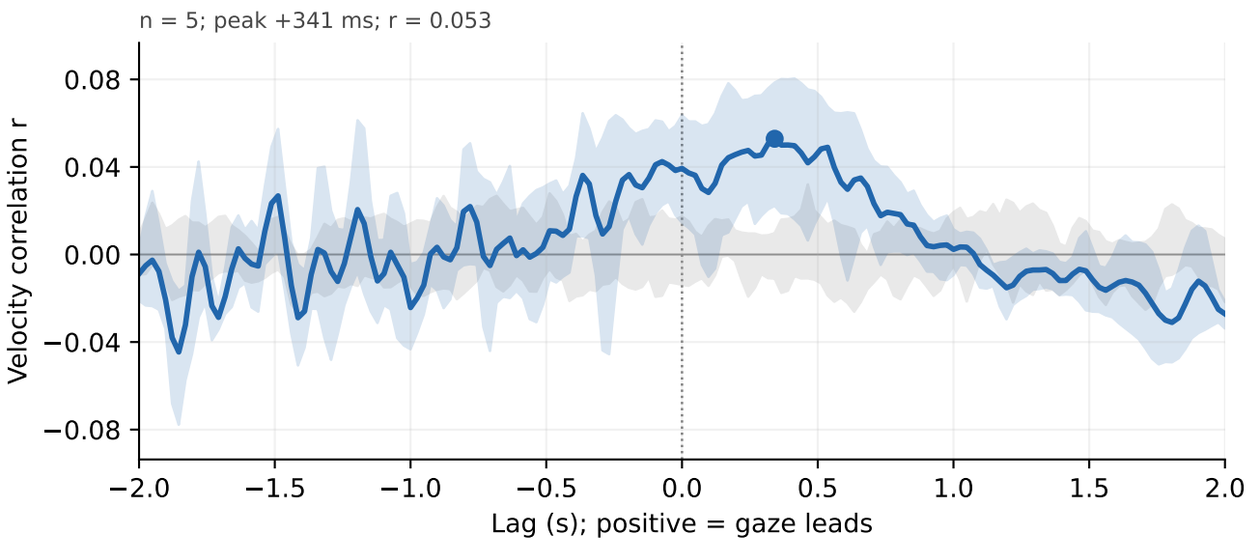
reach ring / within_stage

n = 5; peak +415 ms; r = 0.034



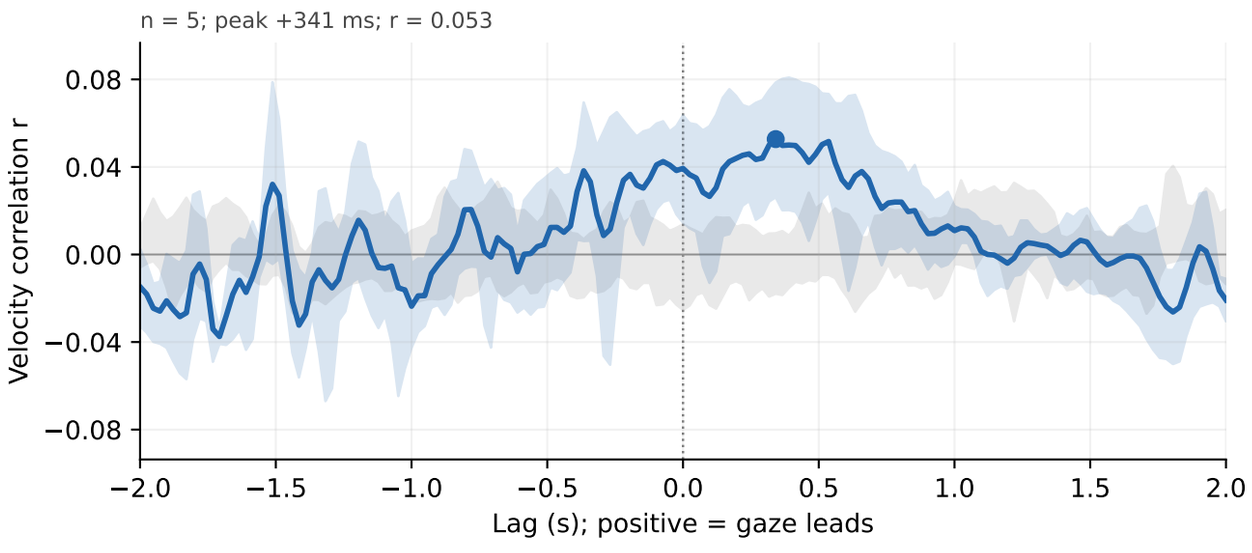
transfer ring / psm_stage

n = 5; peak +341 ms; r = 0.053



transfer ring / within_stage

n = 5; peak +341 ms; r = 0.053

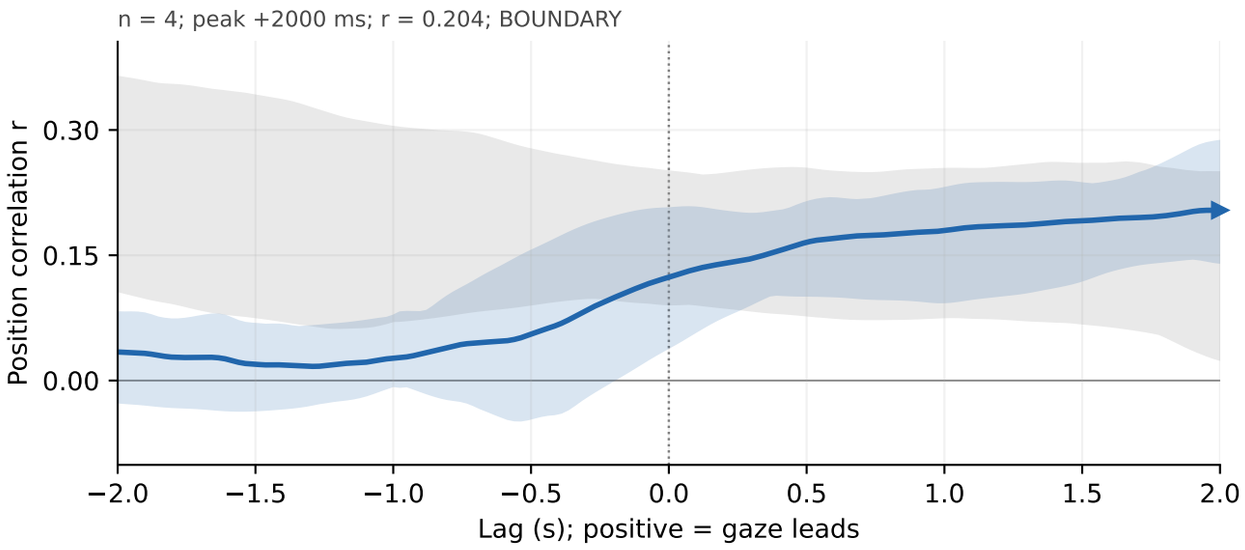


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

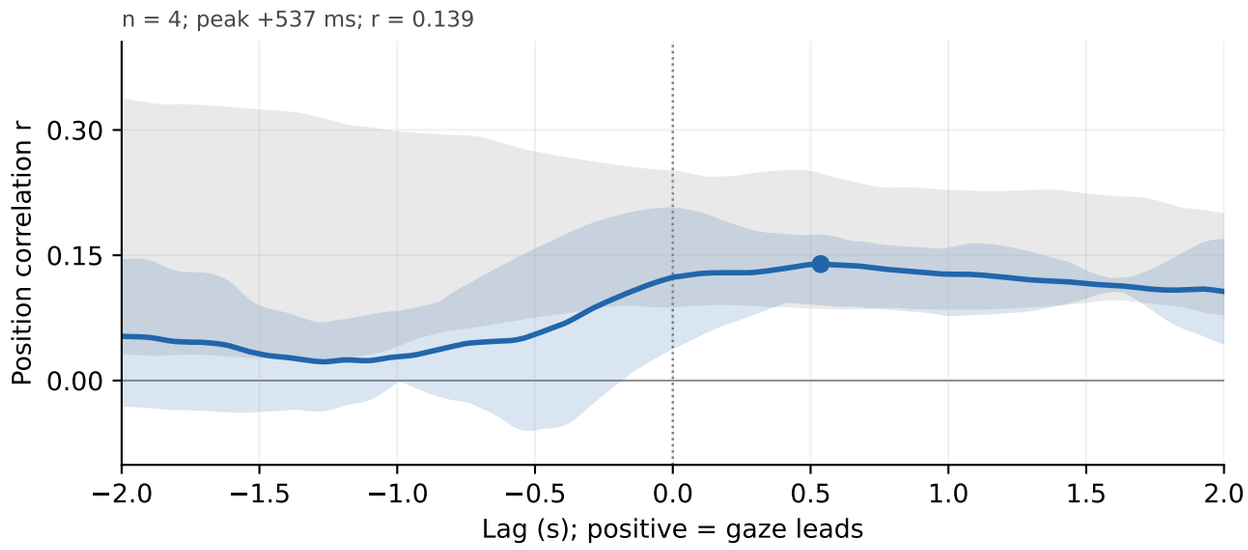
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

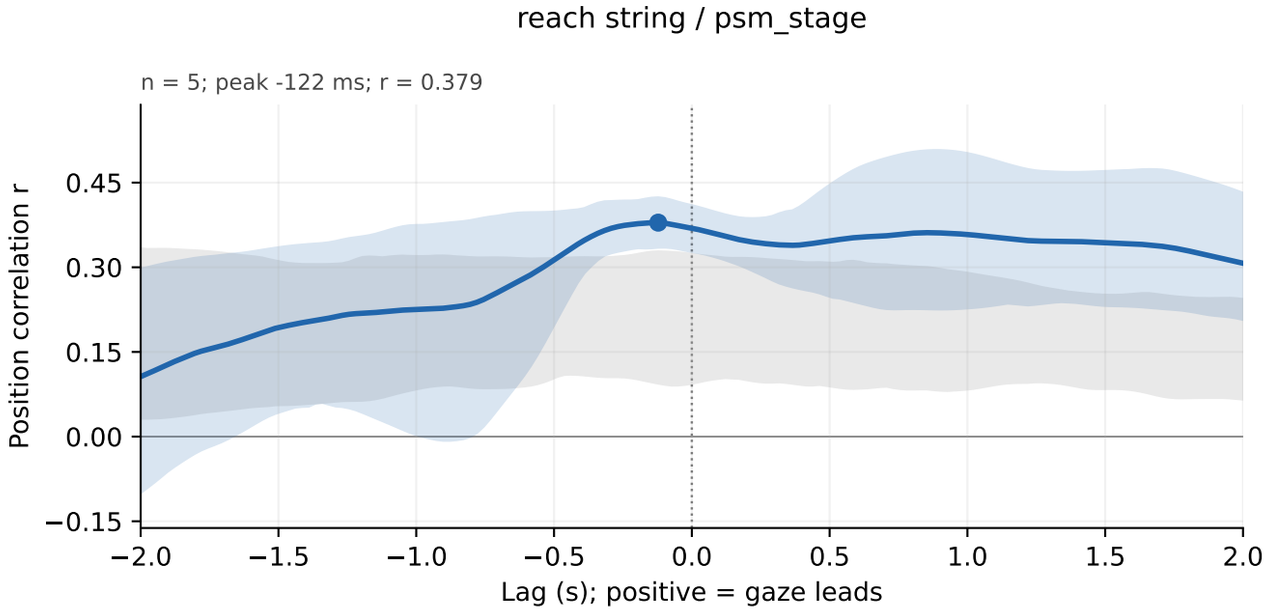
pull through and cinch / psm_stage



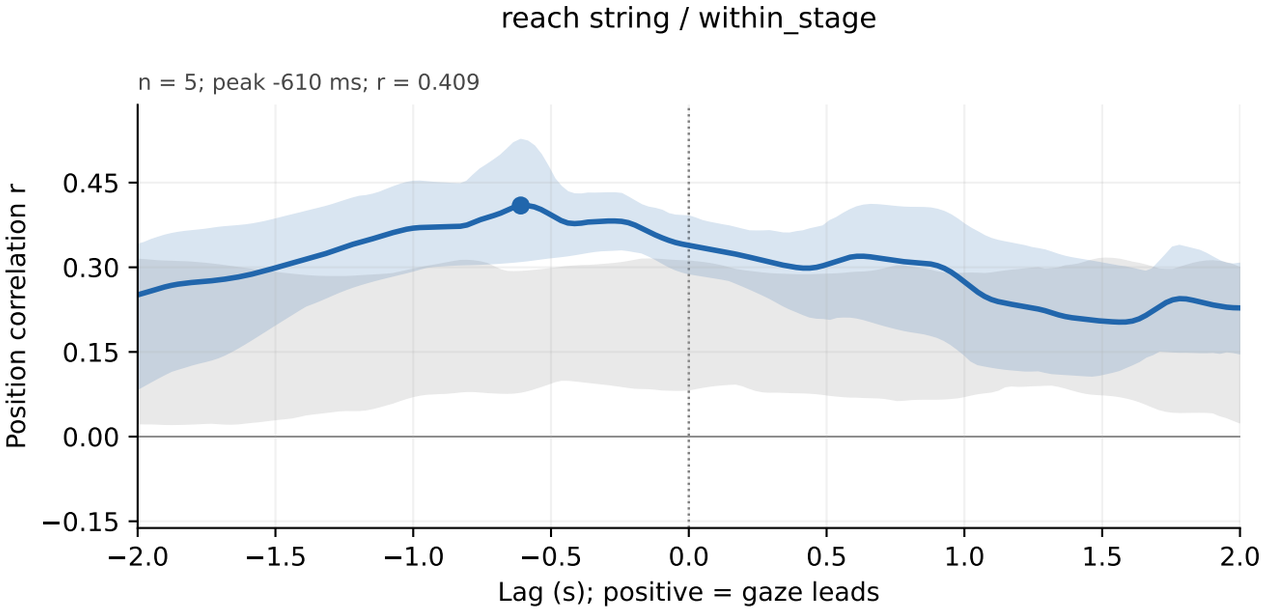
pull through and cinch / within_stage



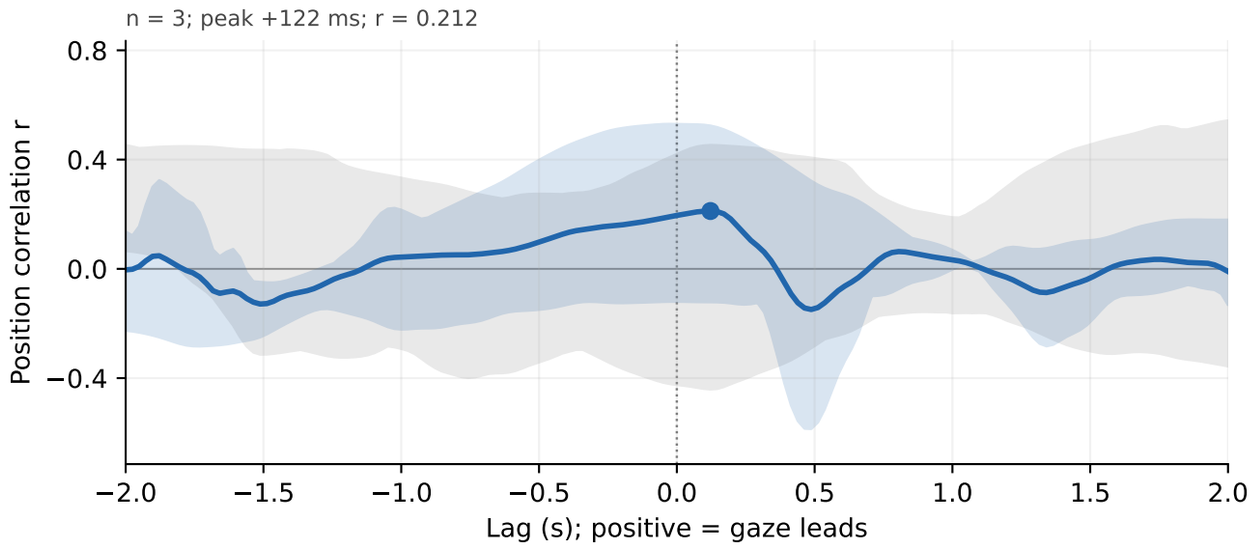
reach string / psm_stage



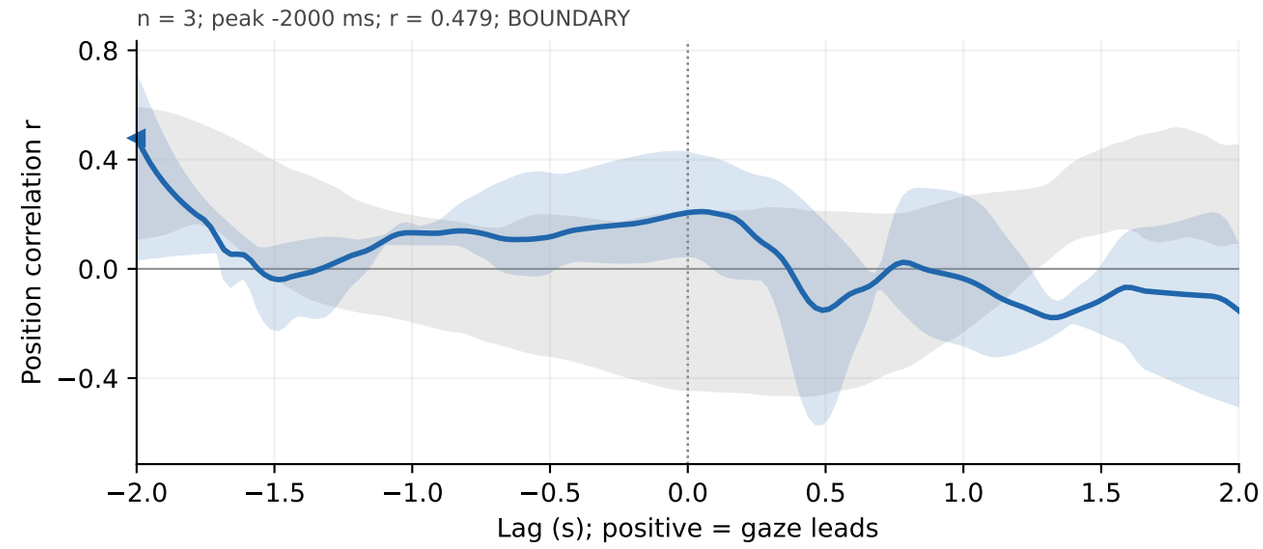
reach string / within_stage



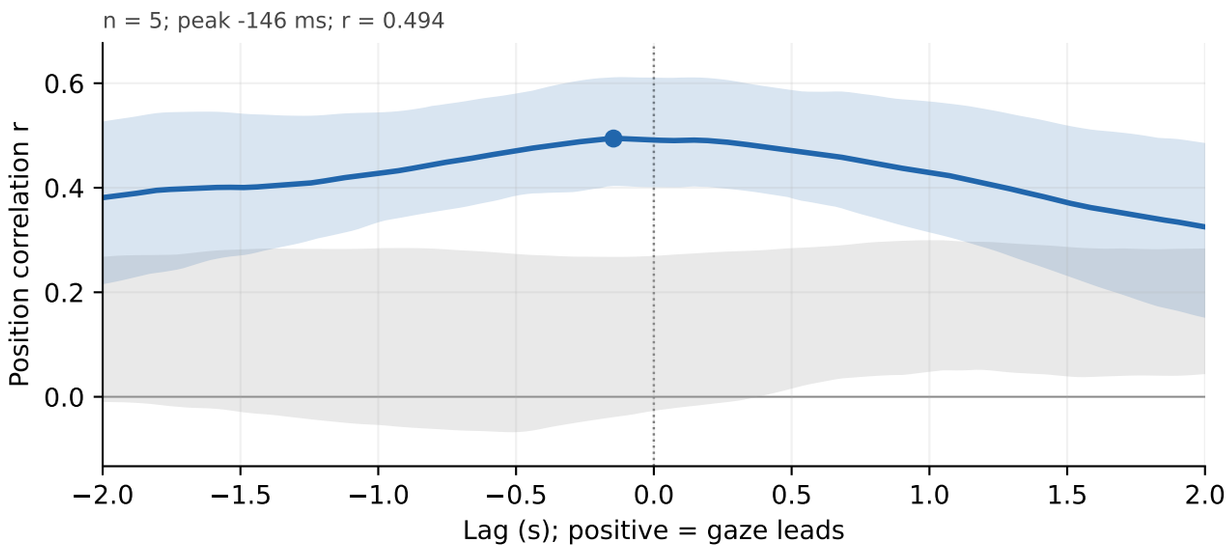
threading string through tower / psm_stage



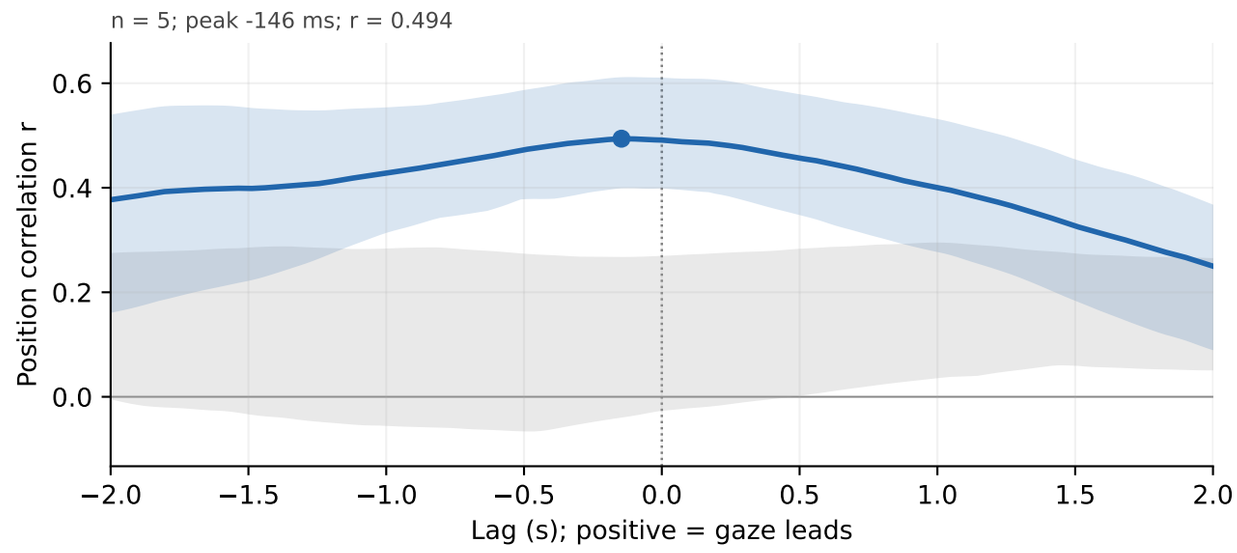
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

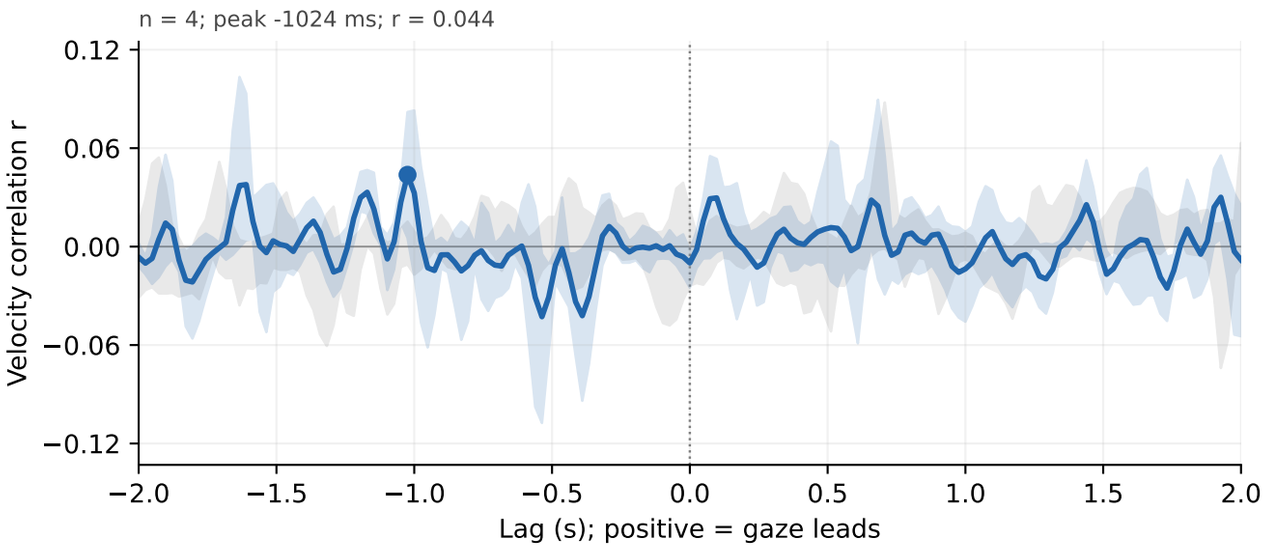


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

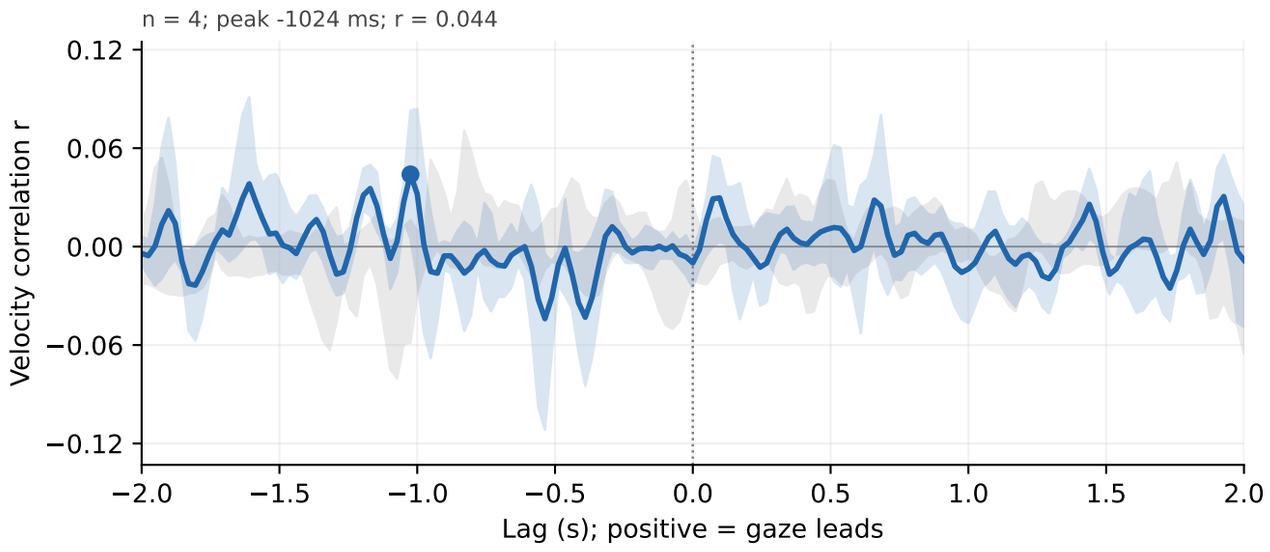
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

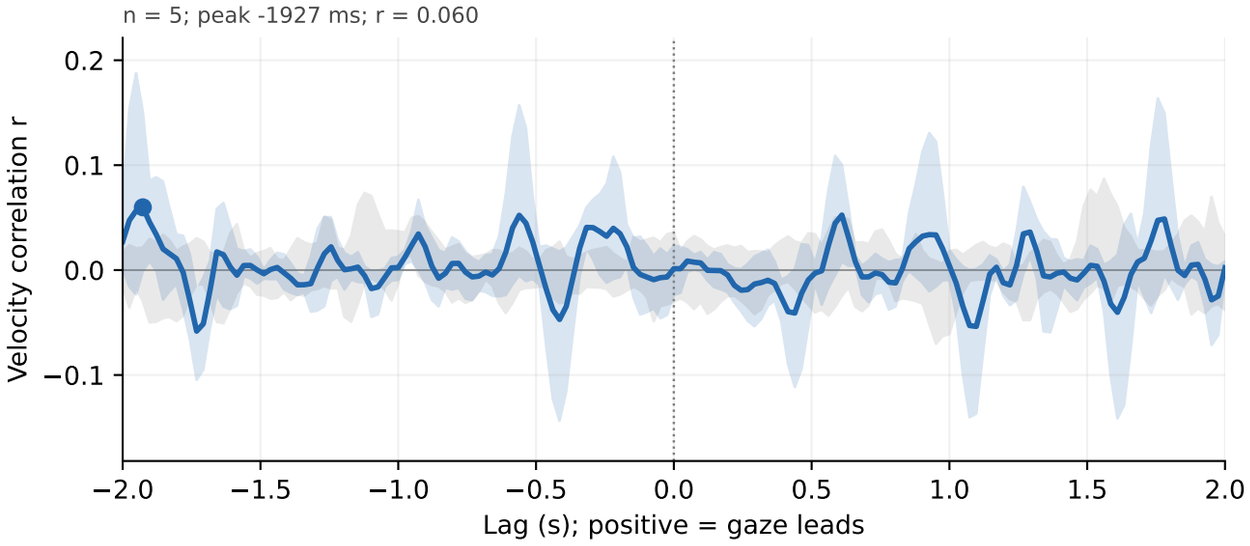
pull through and cinch / psm_stage



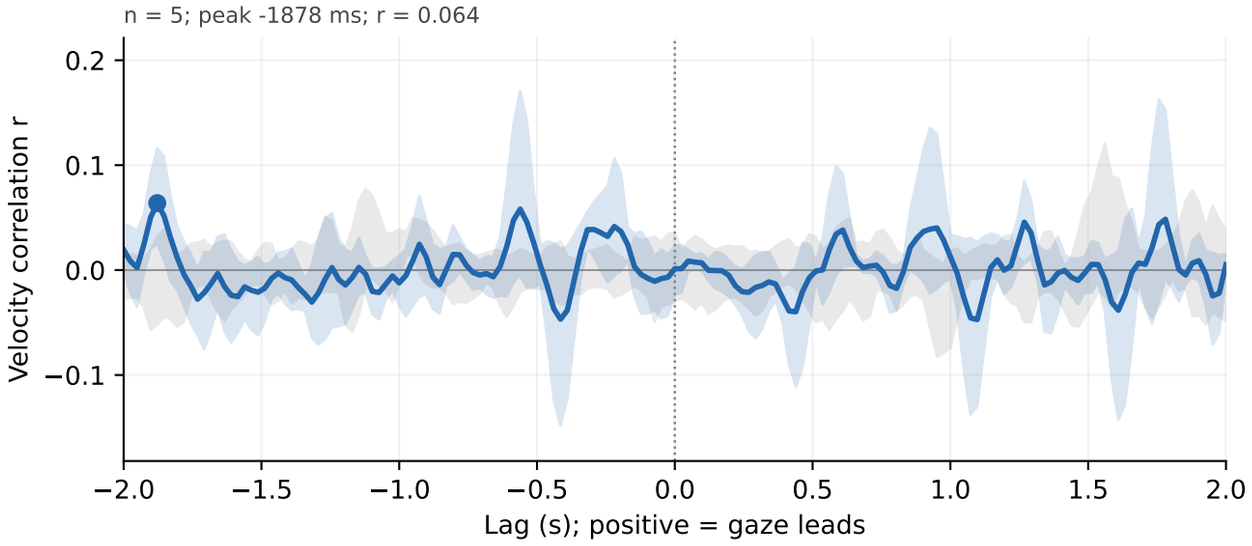
pull through and cinch / within_stage



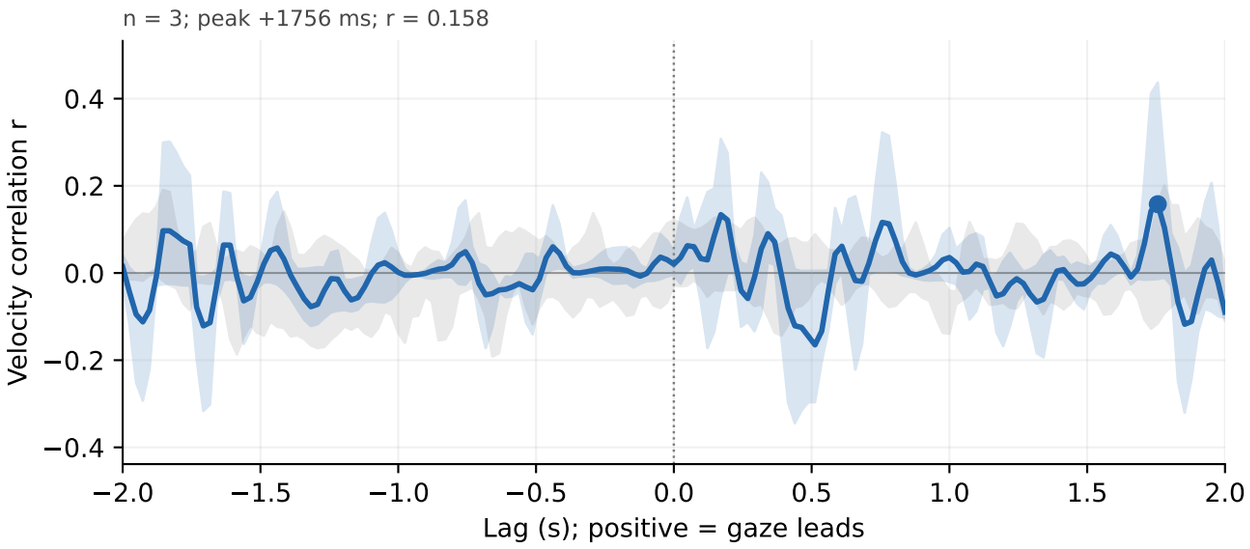
reach string / psm_stage



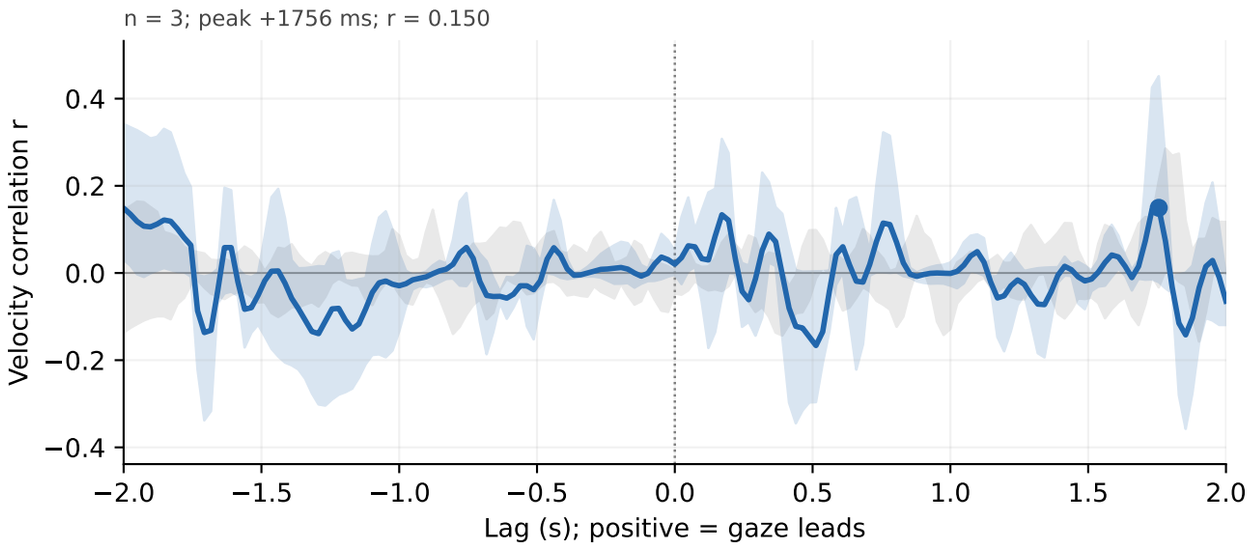
reach string / within_stage



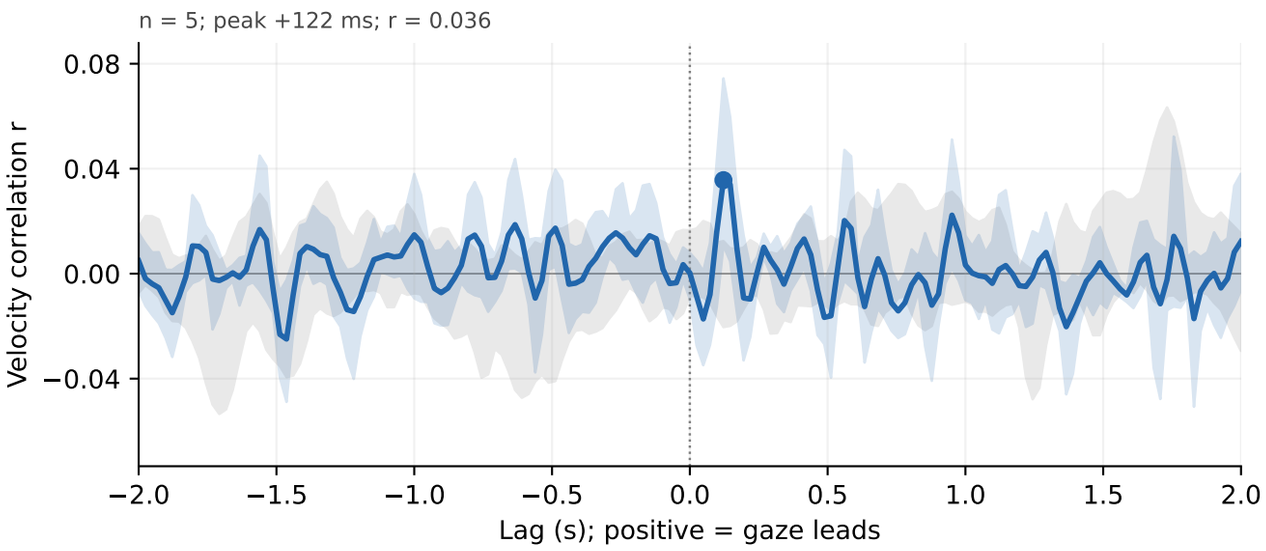
threading string through tower / psm_stage



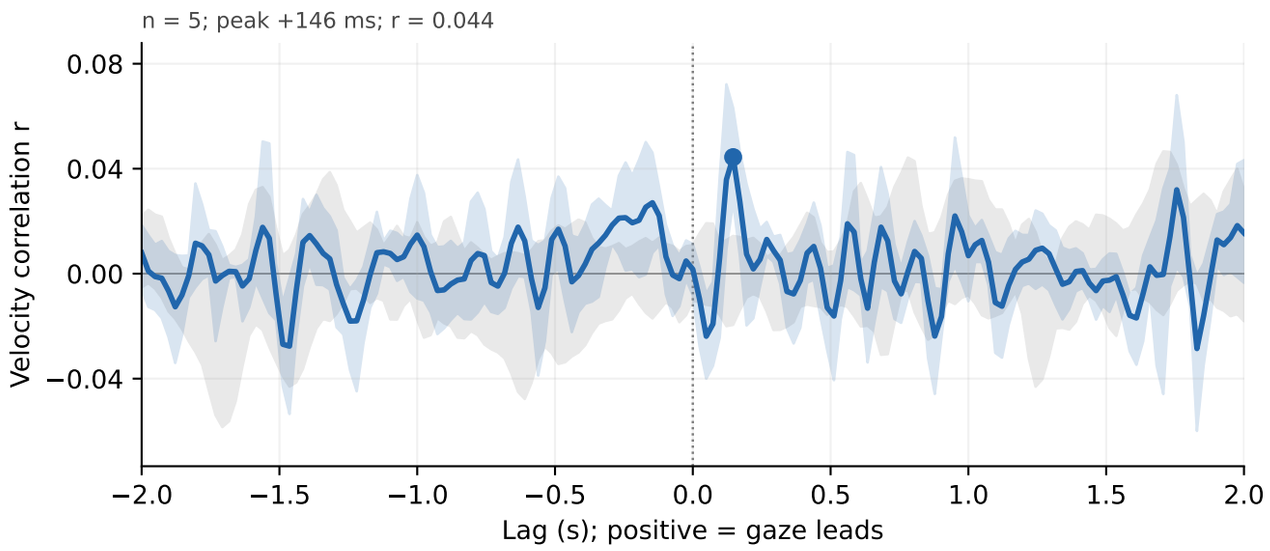
threading string through tower / within_stage



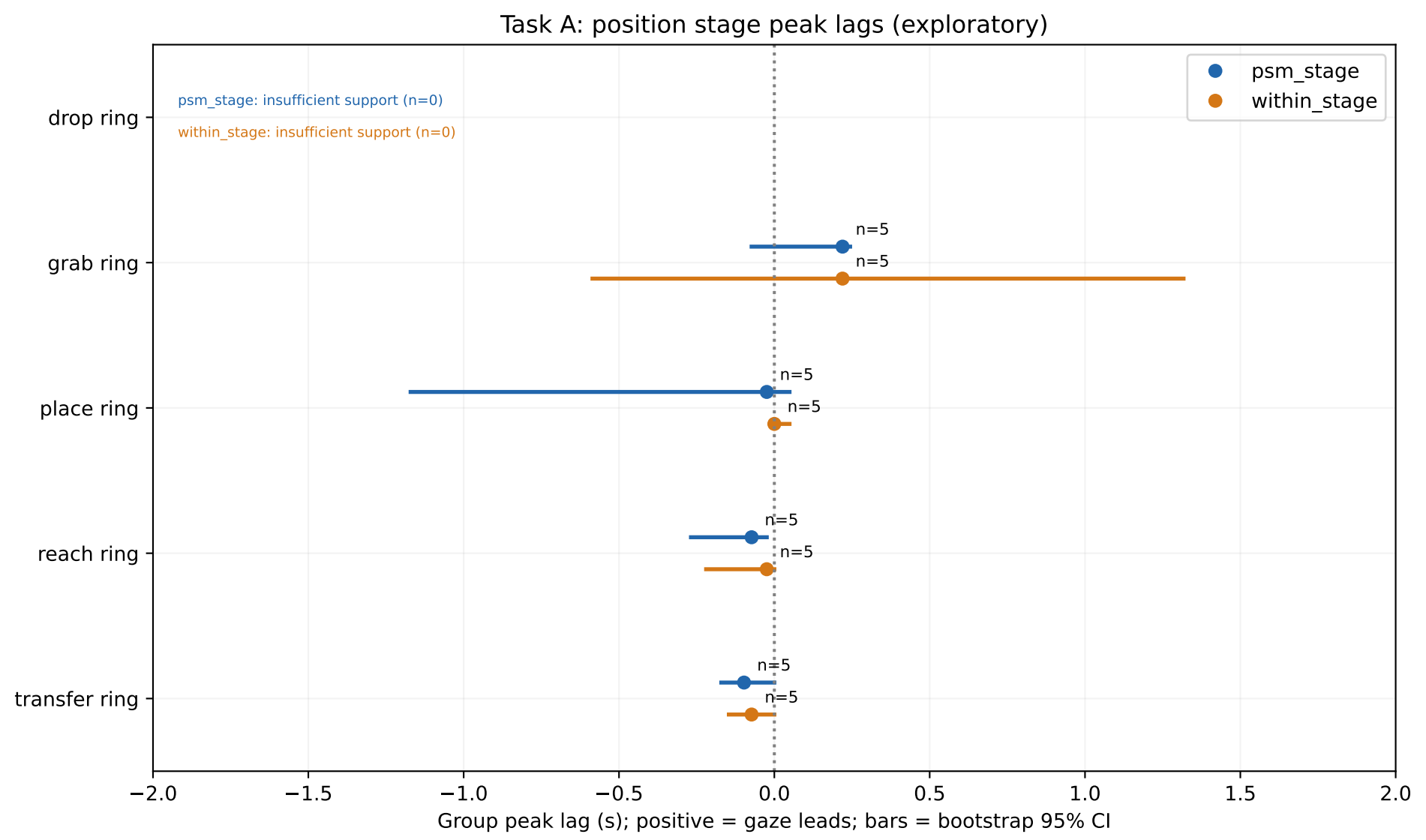
wrap around psm / psm_stage



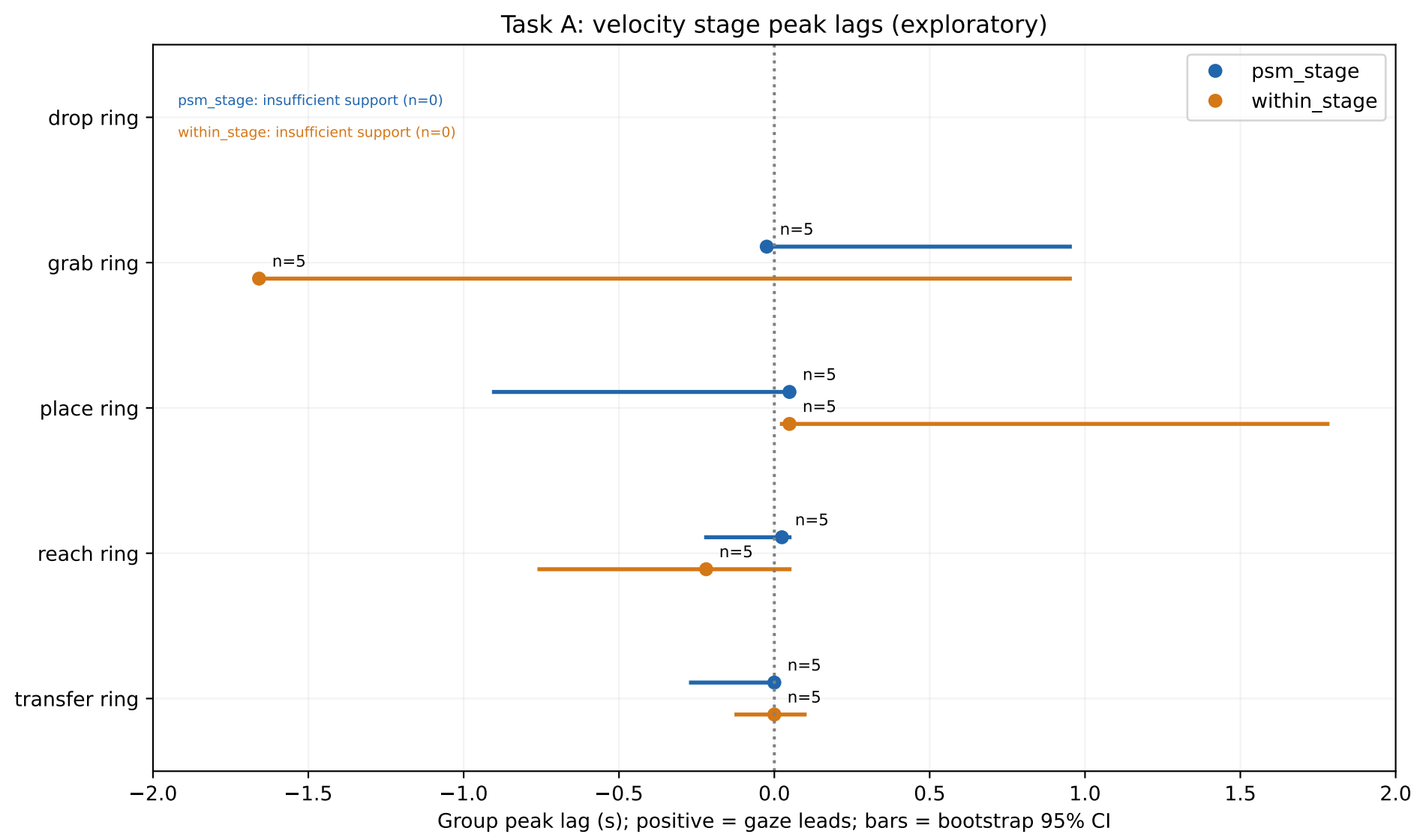
wrap around psm / within_stage



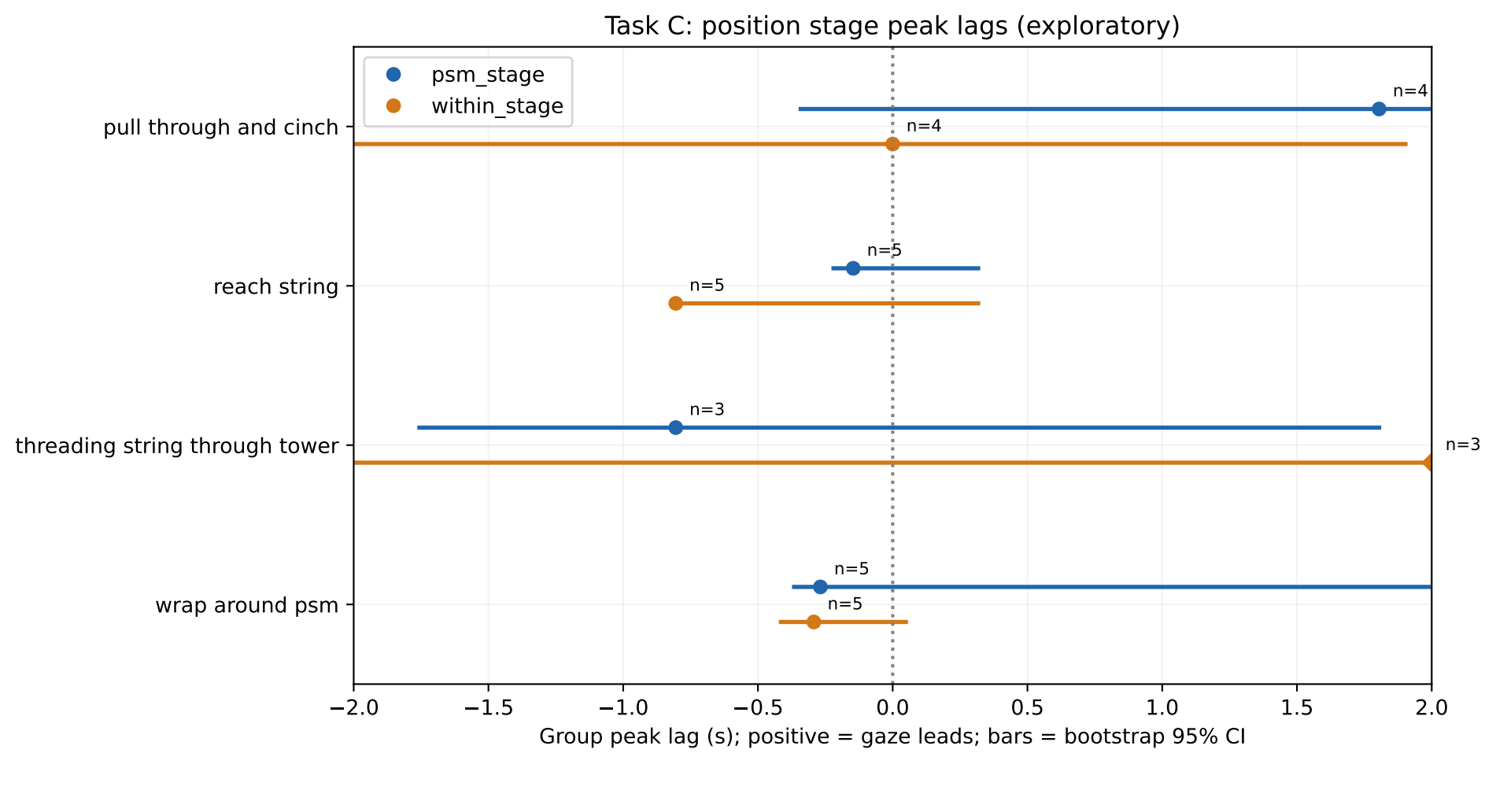
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



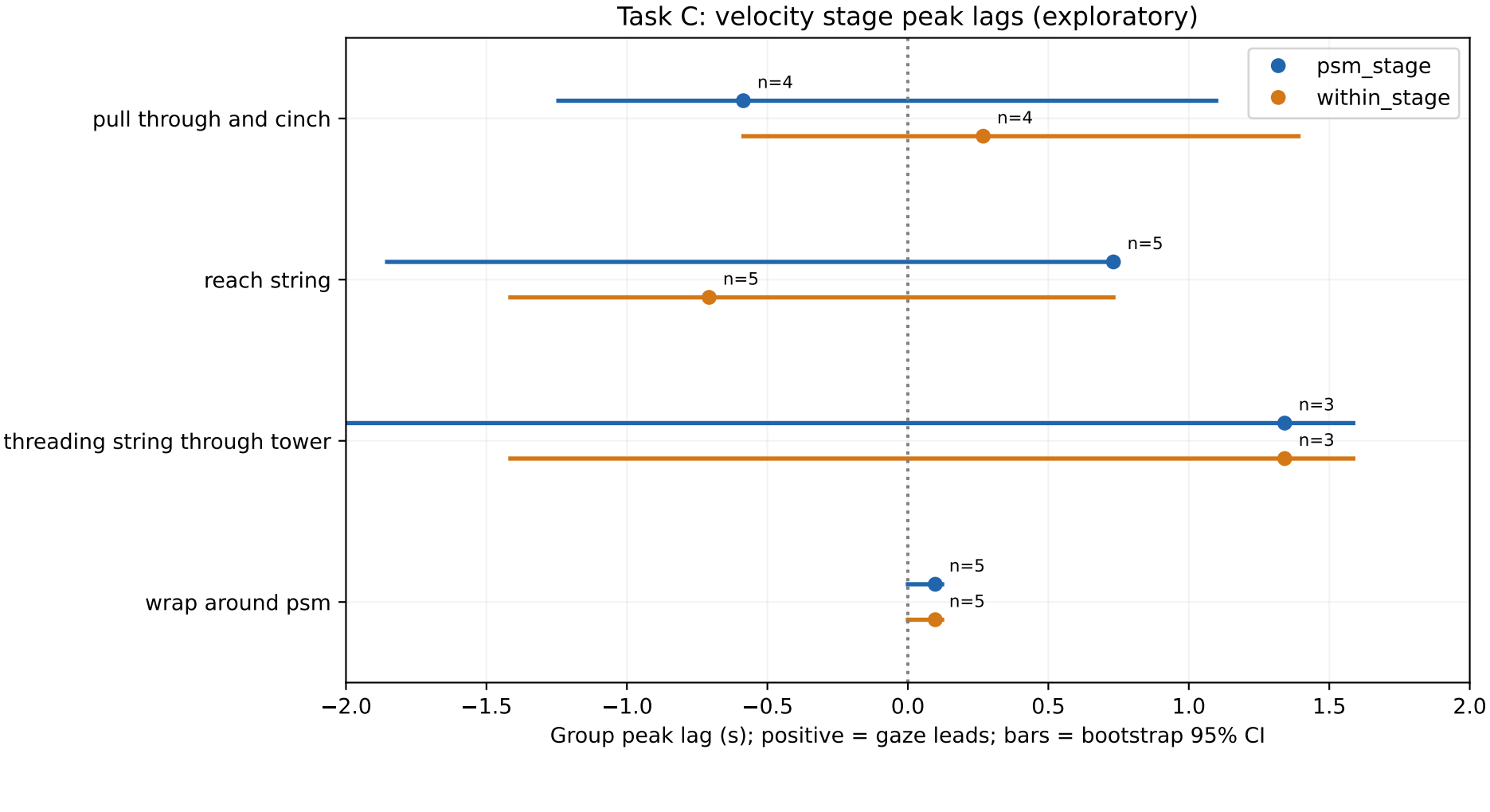
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



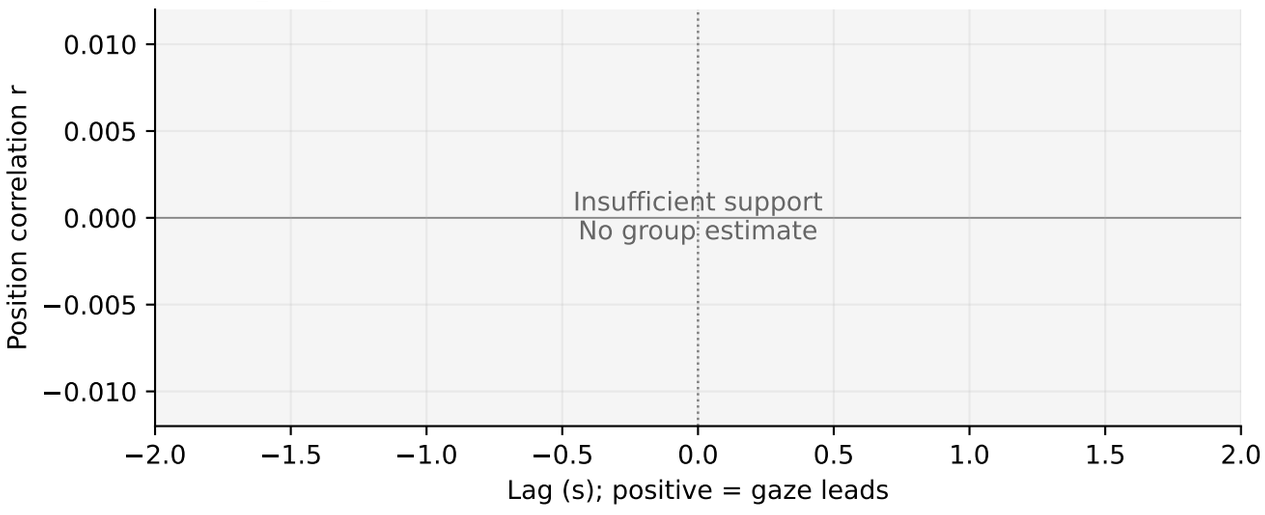
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

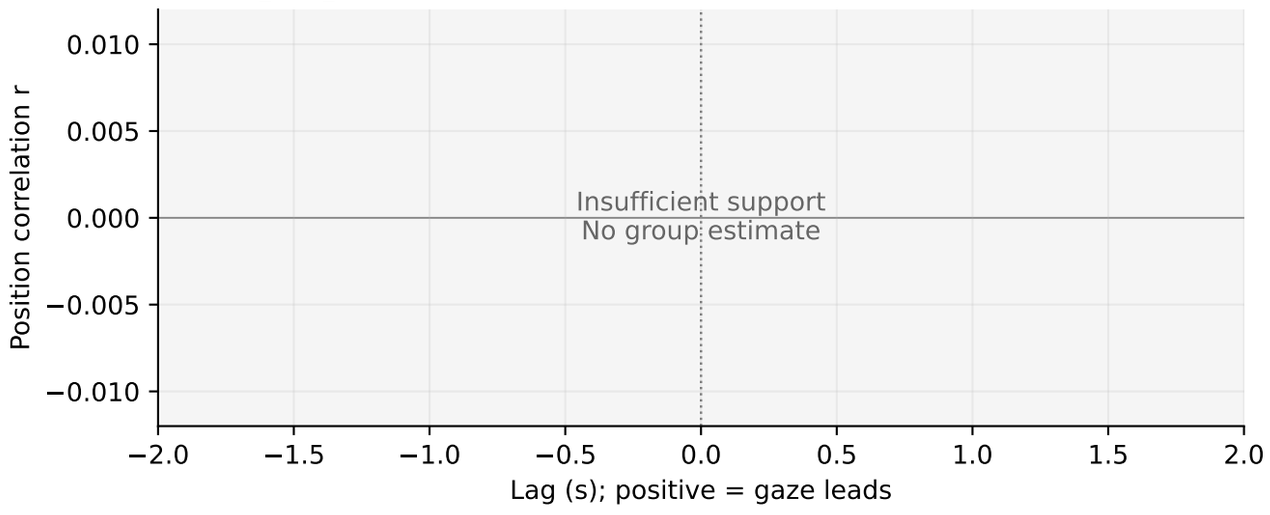
drop ring / psm_stage

n = 0; no_valid_curve



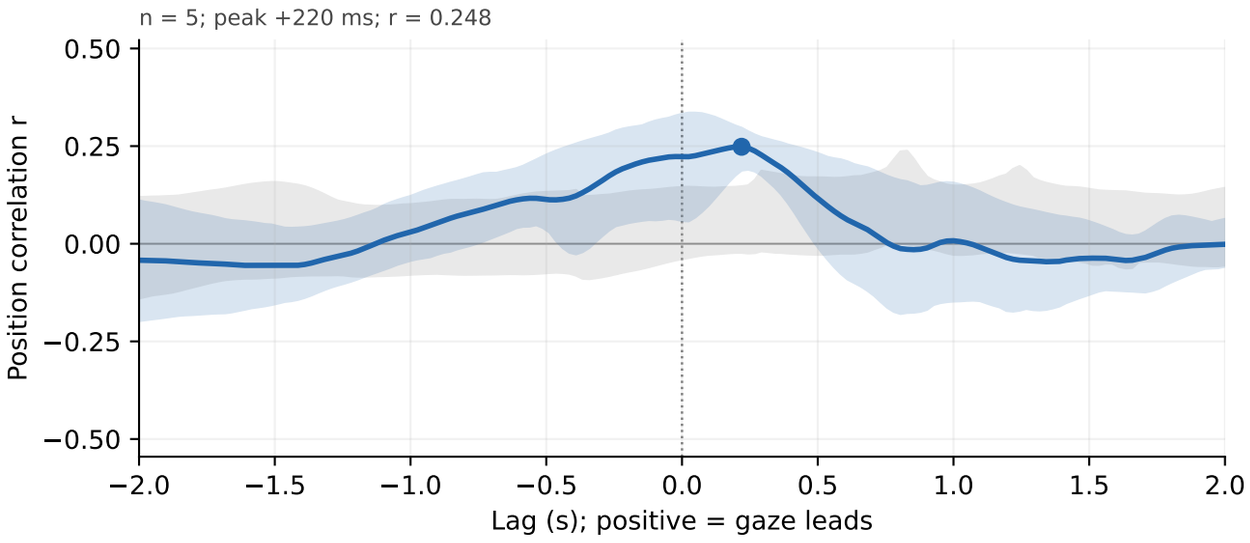
drop ring / within_stage

n = 0; no_valid_curve



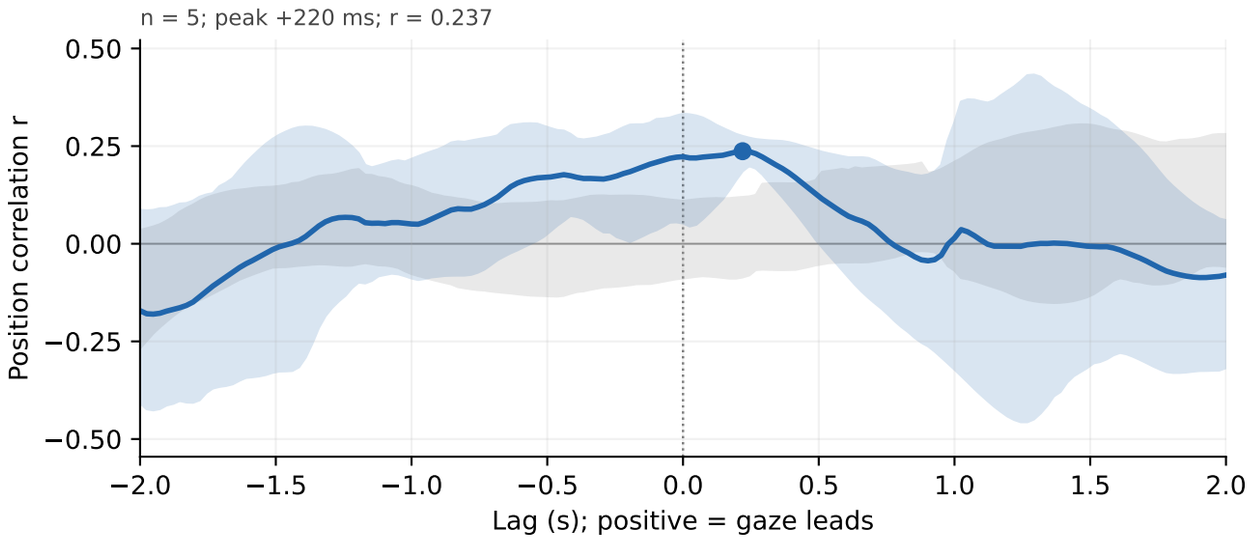
grab ring / psm_stage

n = 5; peak +220 ms; r = 0.248



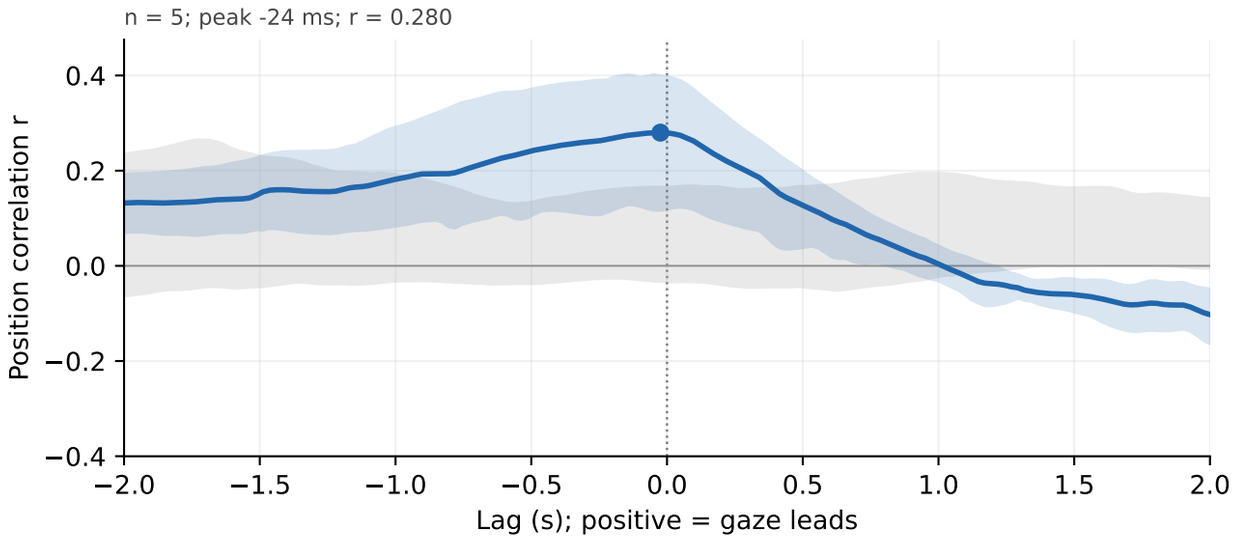
grab ring / within_stage

n = 5; peak +220 ms; r = 0.237



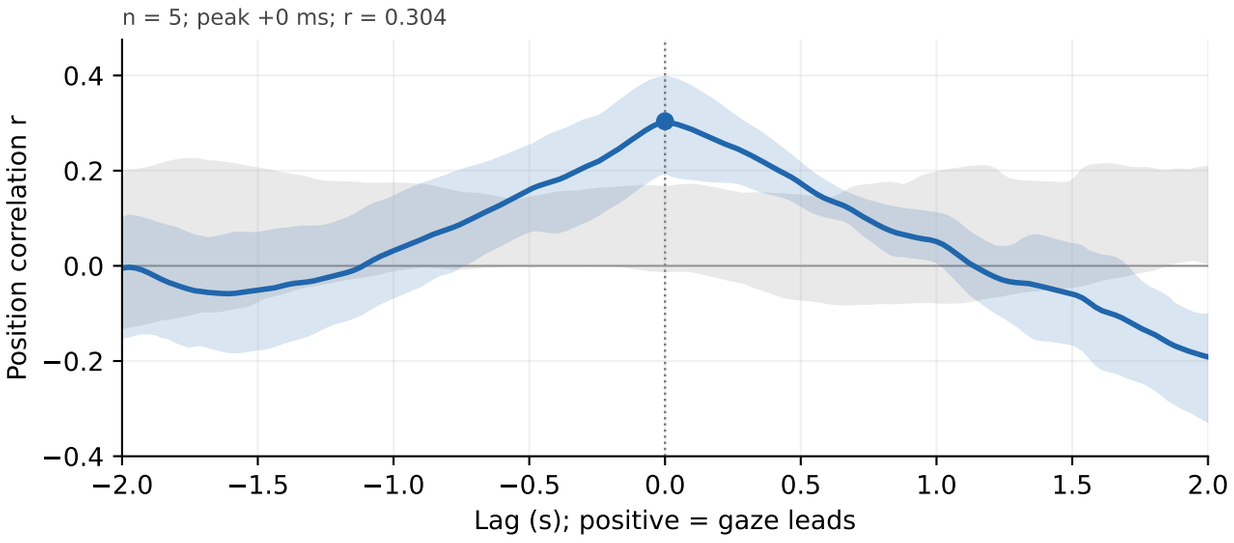
place ring / psm_stage

n = 5; peak -24 ms; r = 0.280



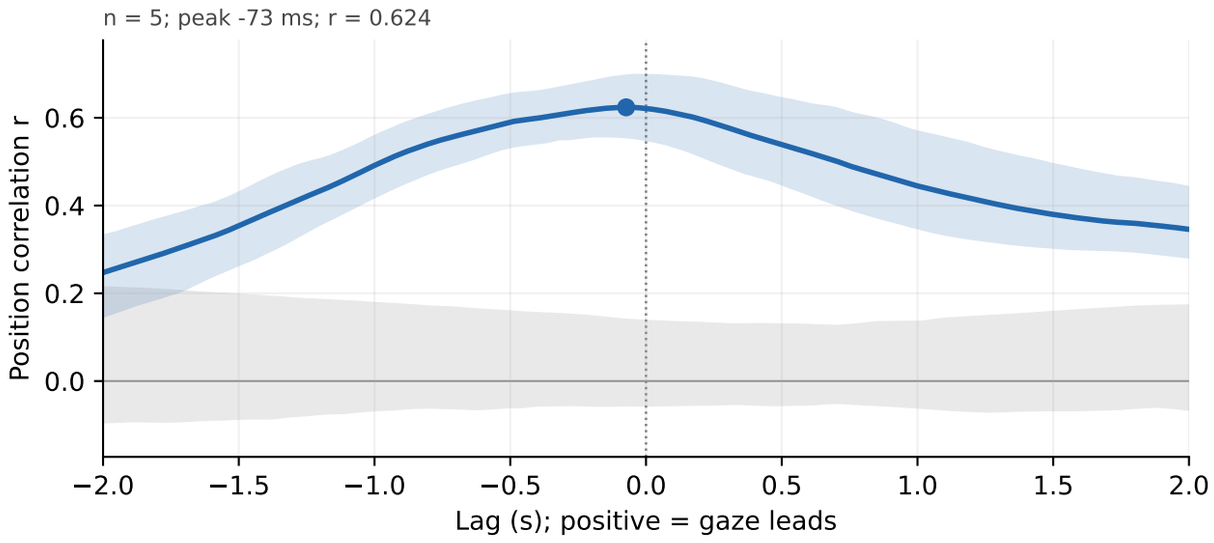
place ring / within_stage

n = 5; peak +0 ms; r = 0.304



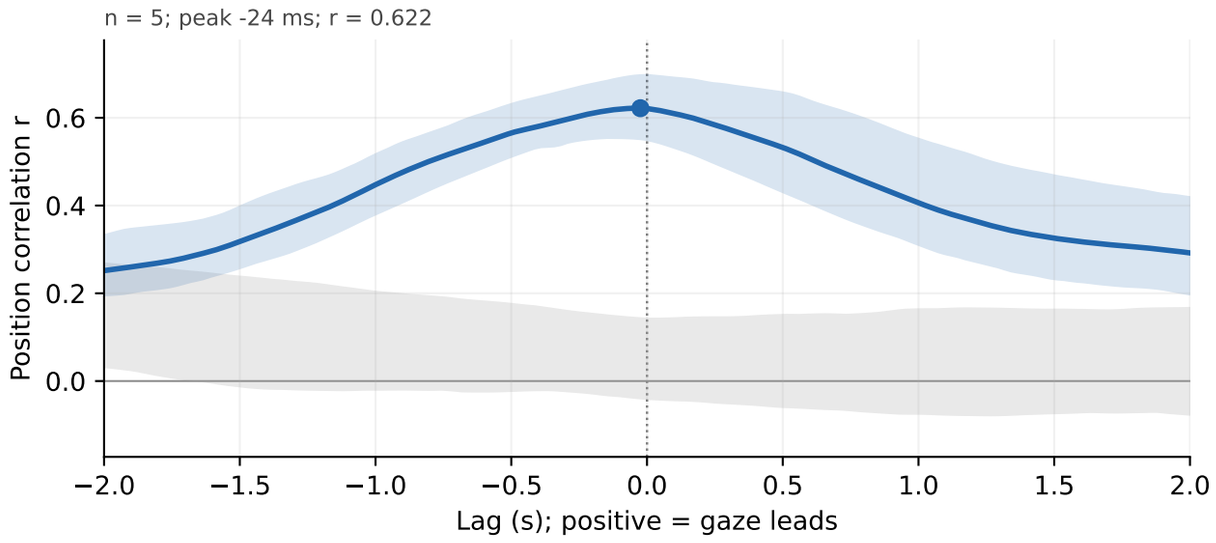
reach ring / psm_stage

n = 5; peak -73 ms; r = 0.624



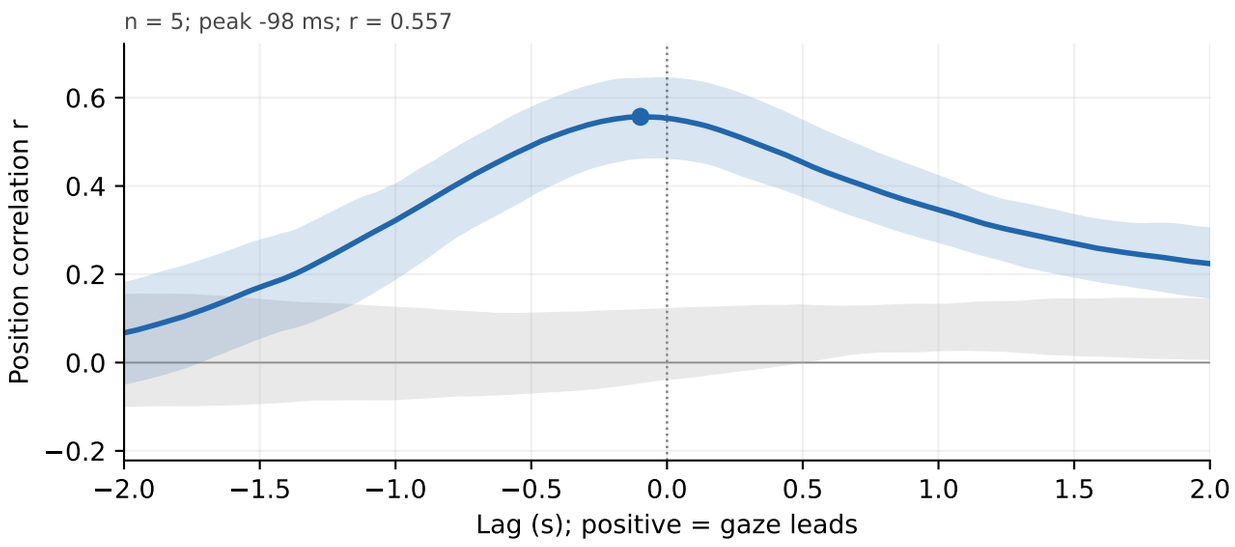
reach ring / within_stage

n = 5; peak -24 ms; r = 0.622



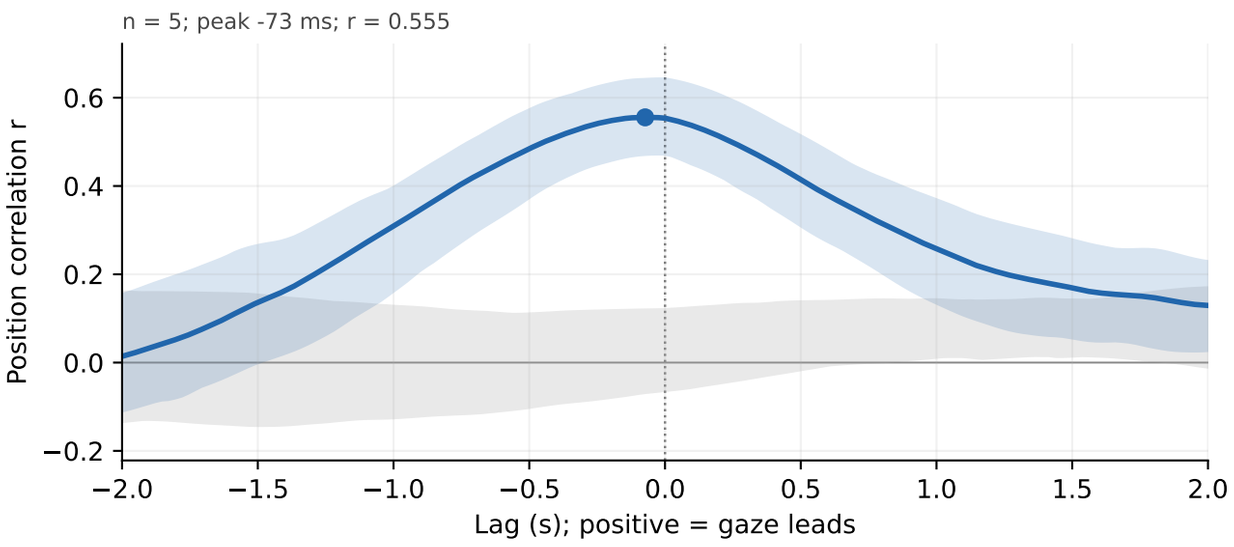
transfer ring / psm_stage

n = 5; peak -98 ms; r = 0.557



transfer ring / within_stage

n = 5; peak -73 ms; r = 0.555

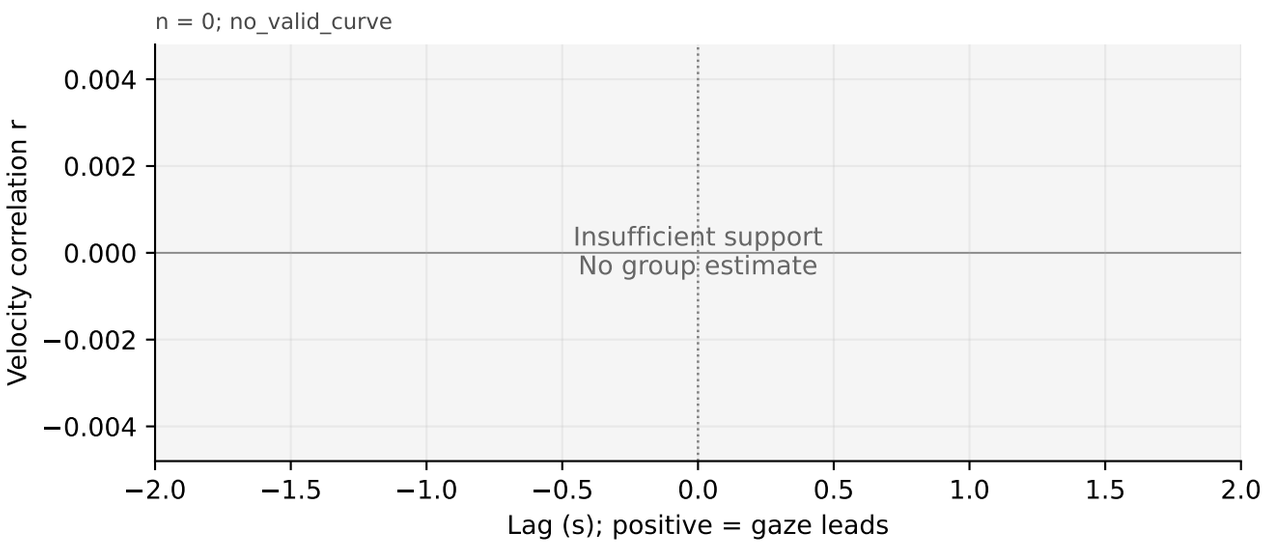


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

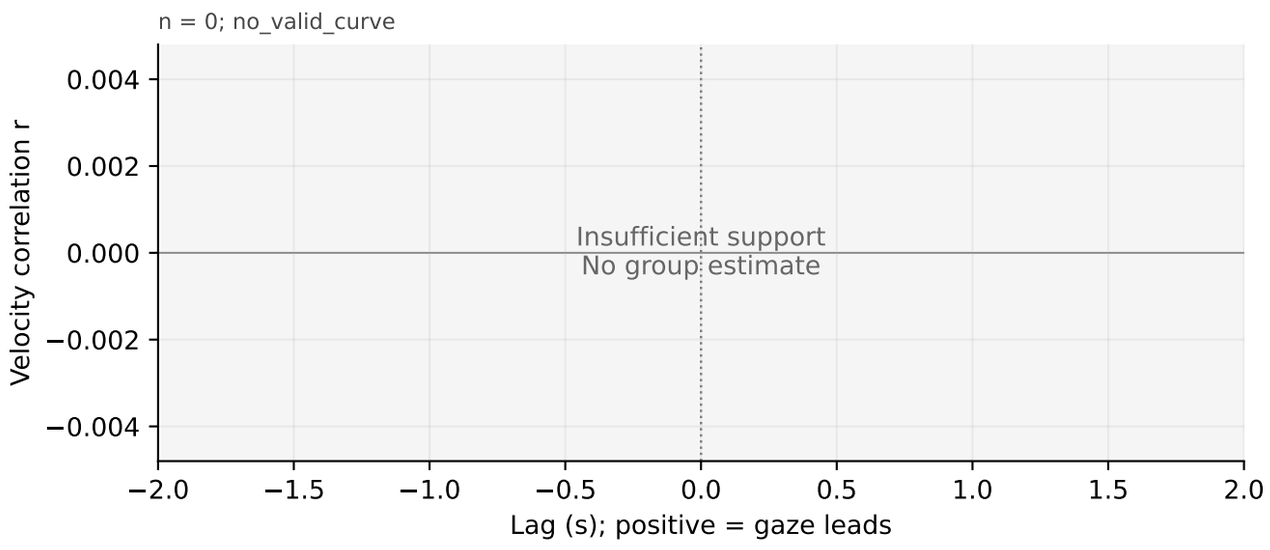
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

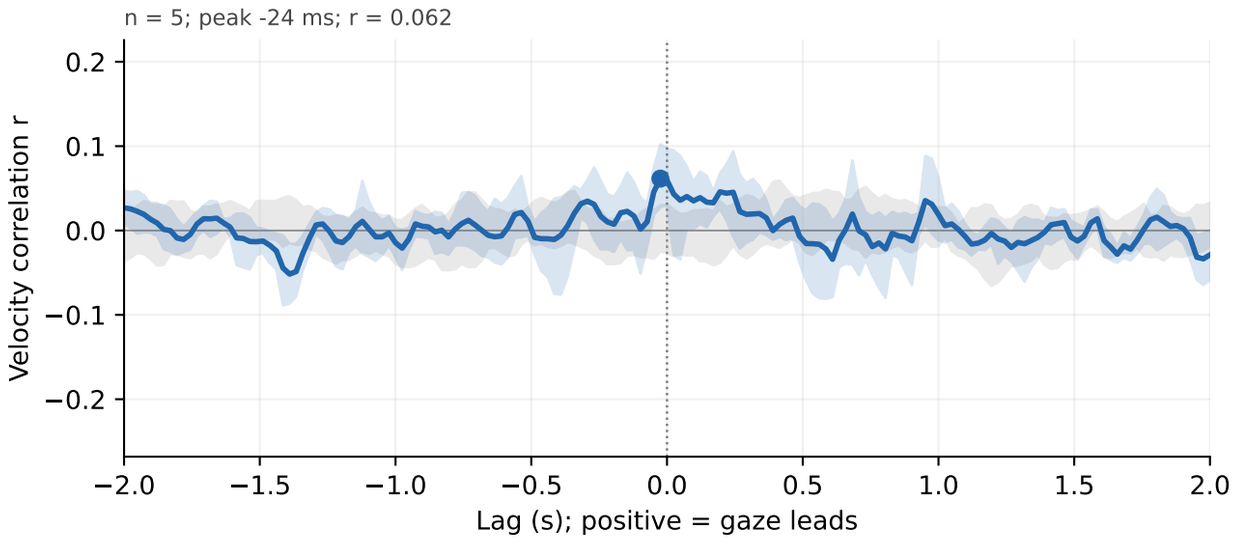
drop ring / psm_stage



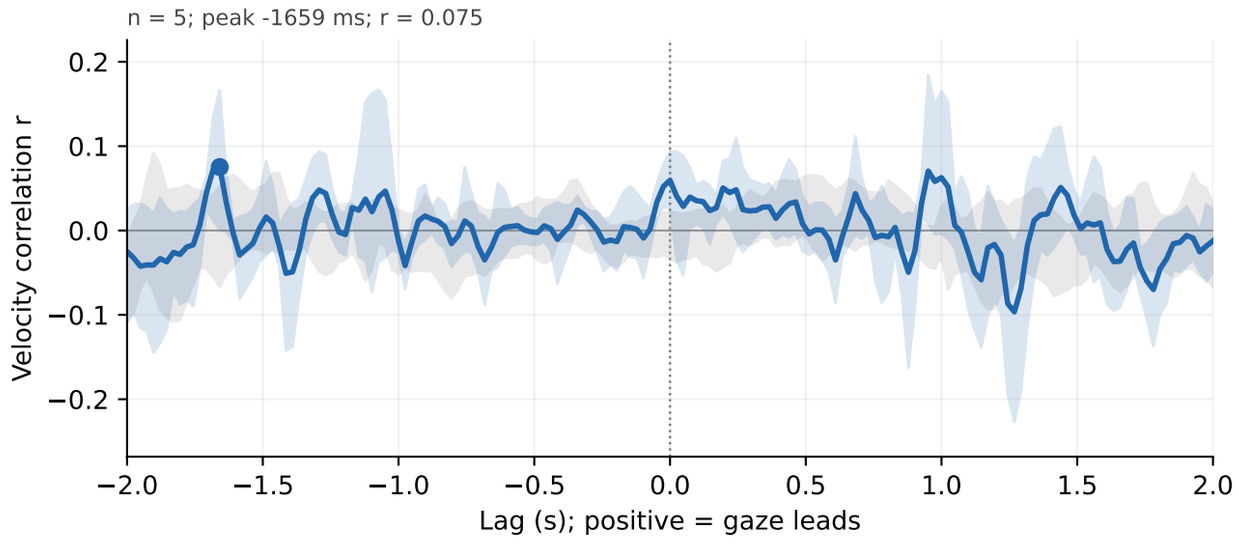
drop ring / within_stage



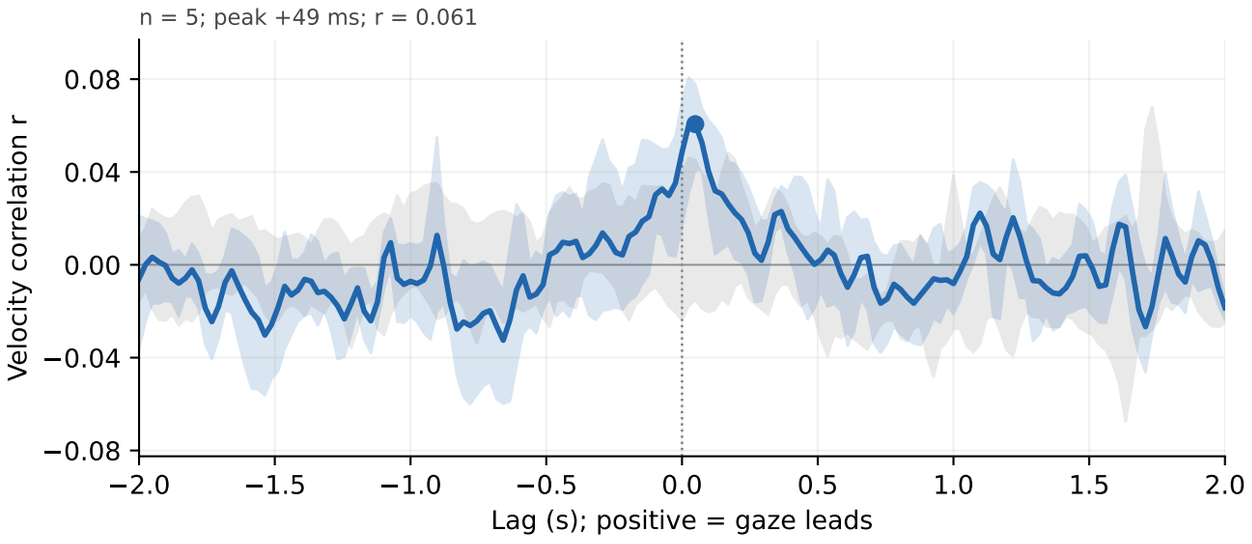
grab ring / psm_stage



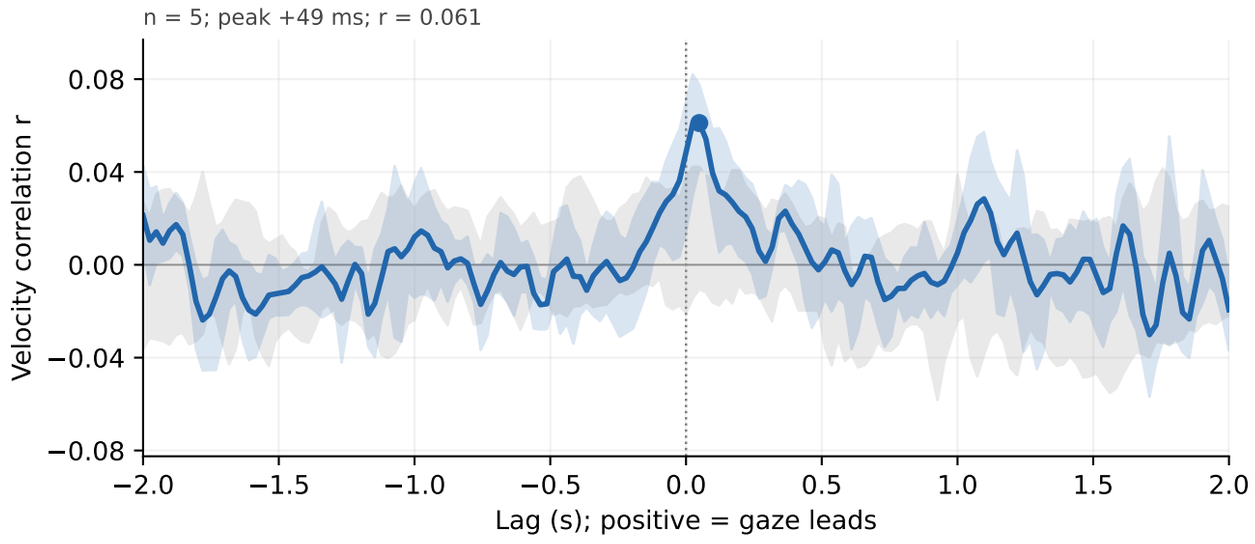
grab ring / within_stage



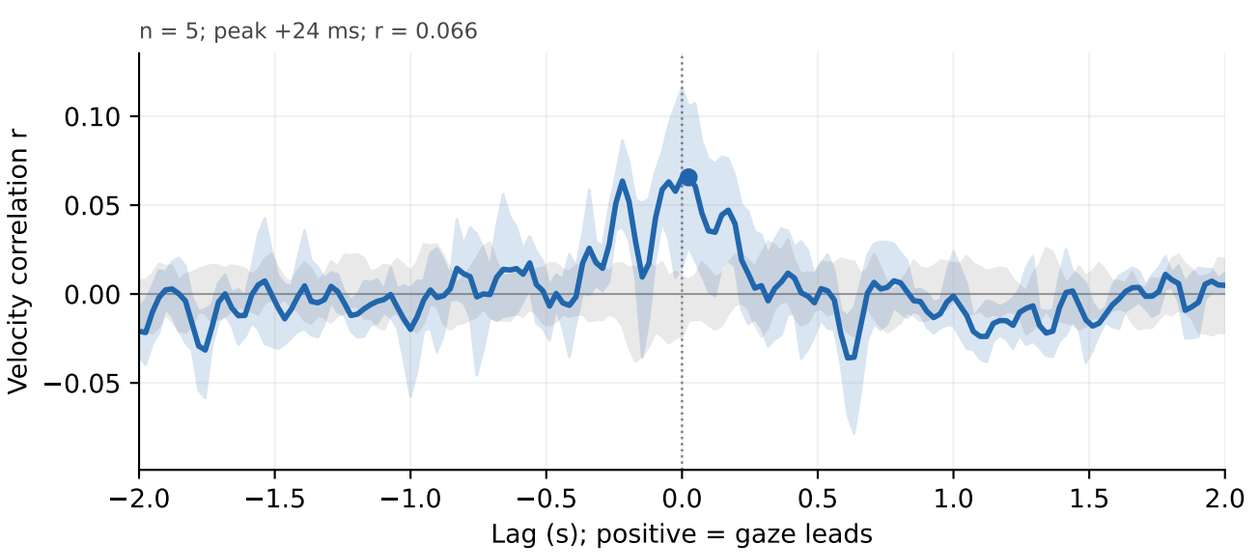
place ring / psm_stage



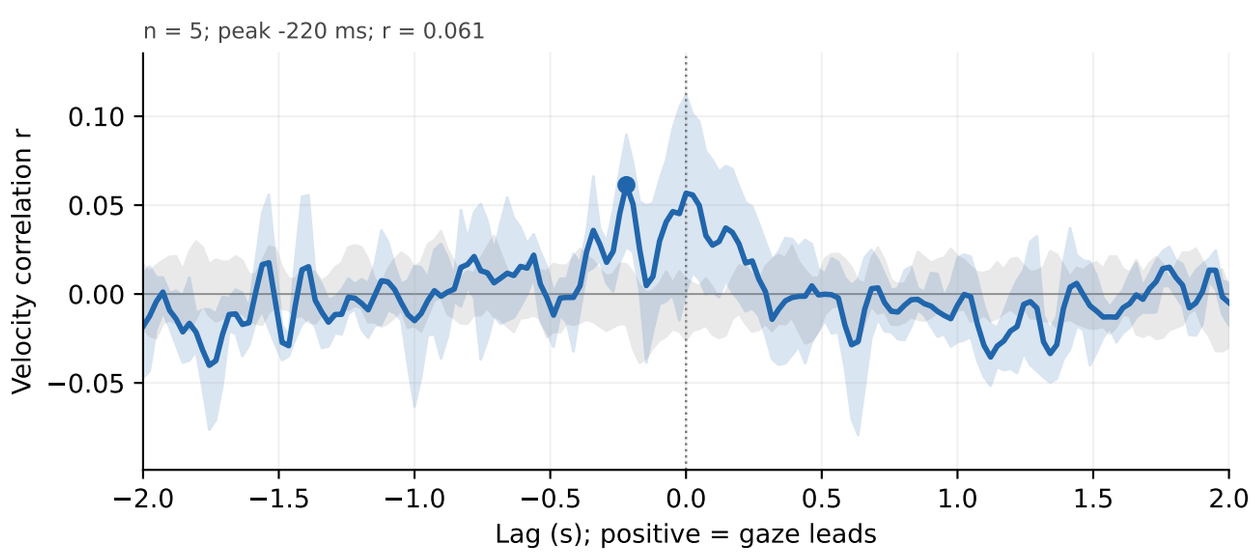
place ring / within_stage



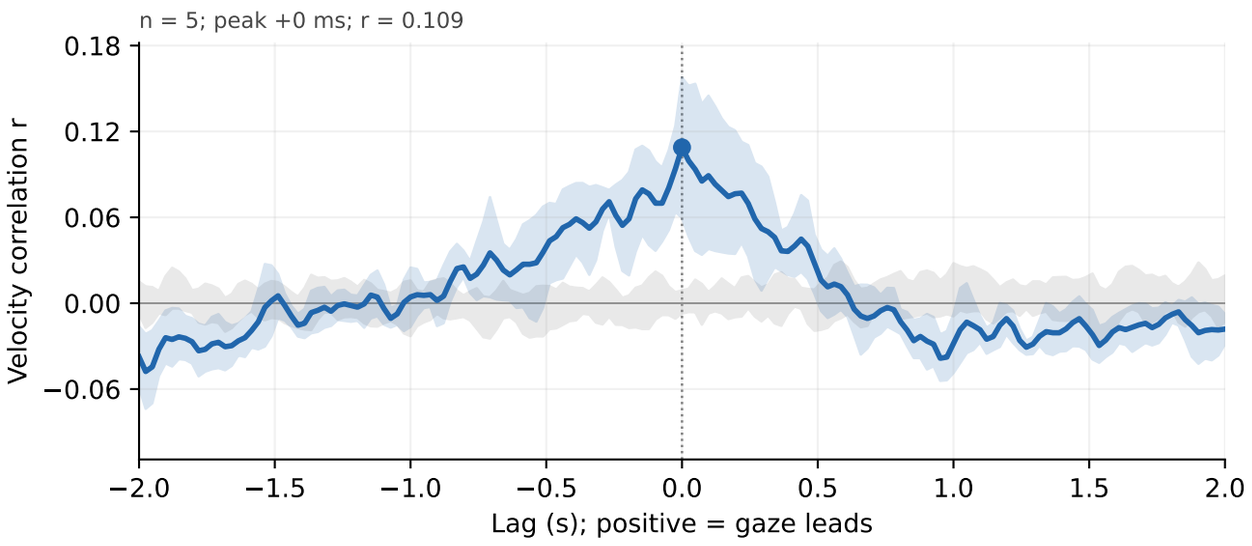
reach ring / psm_stage



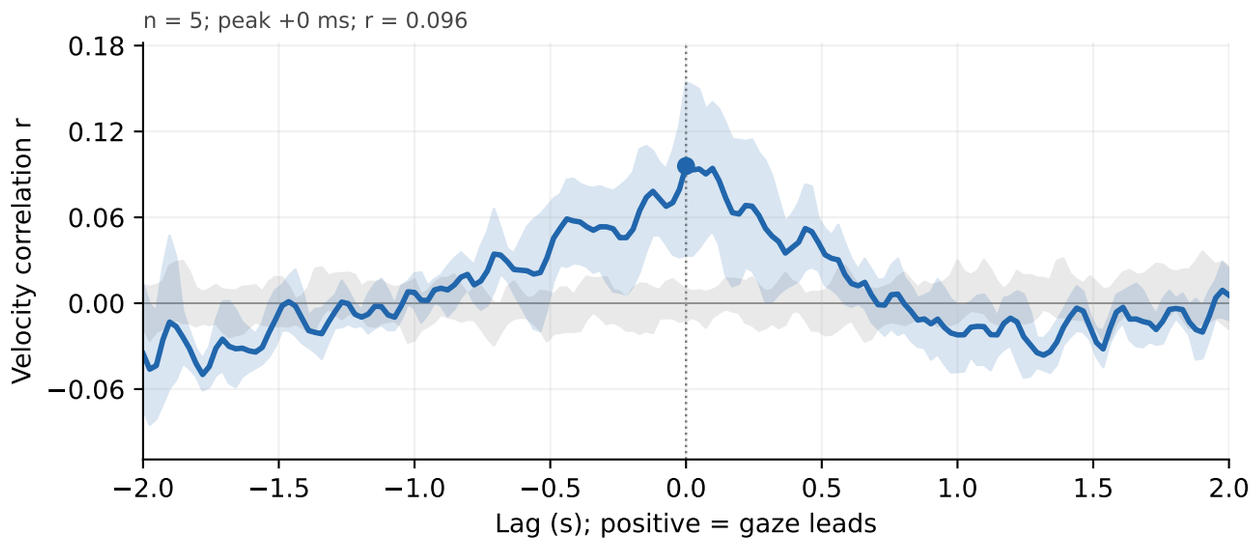
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

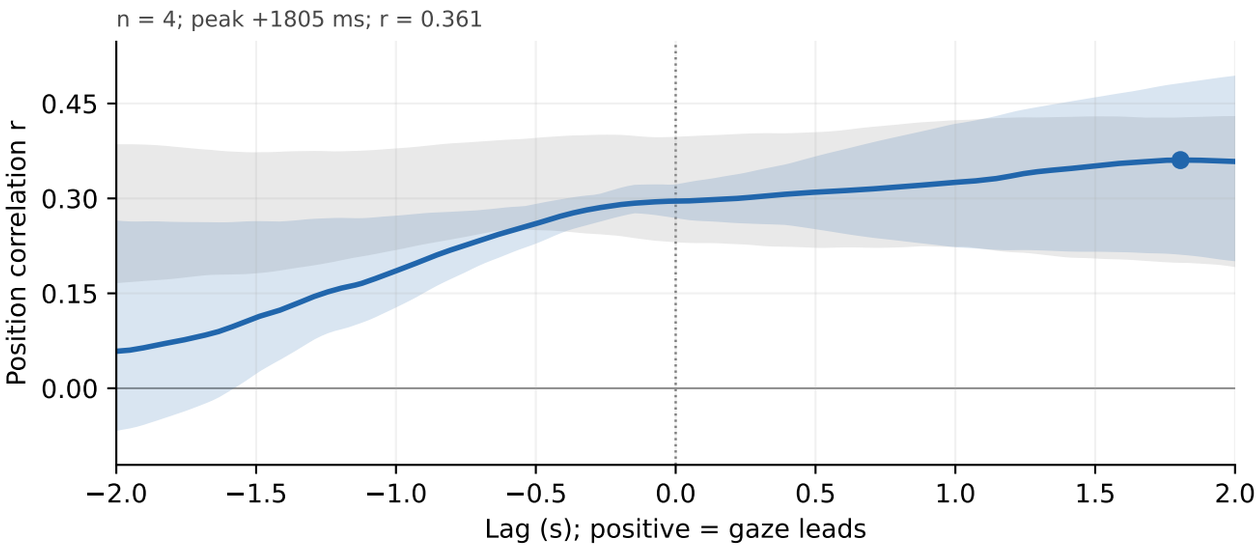


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

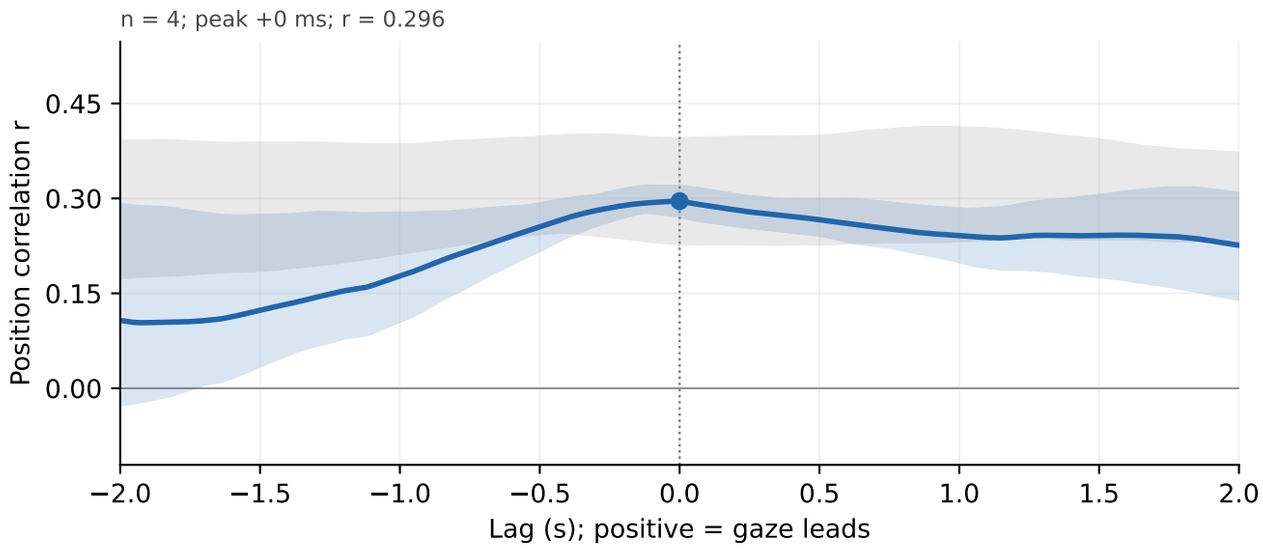
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

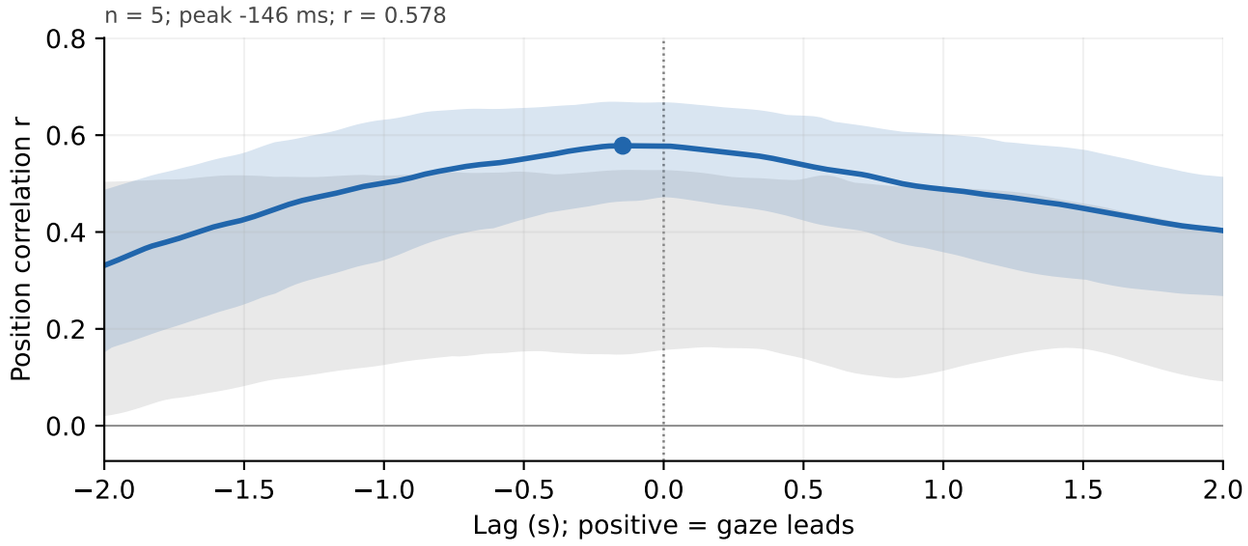
pull through and cinch / psm_stage



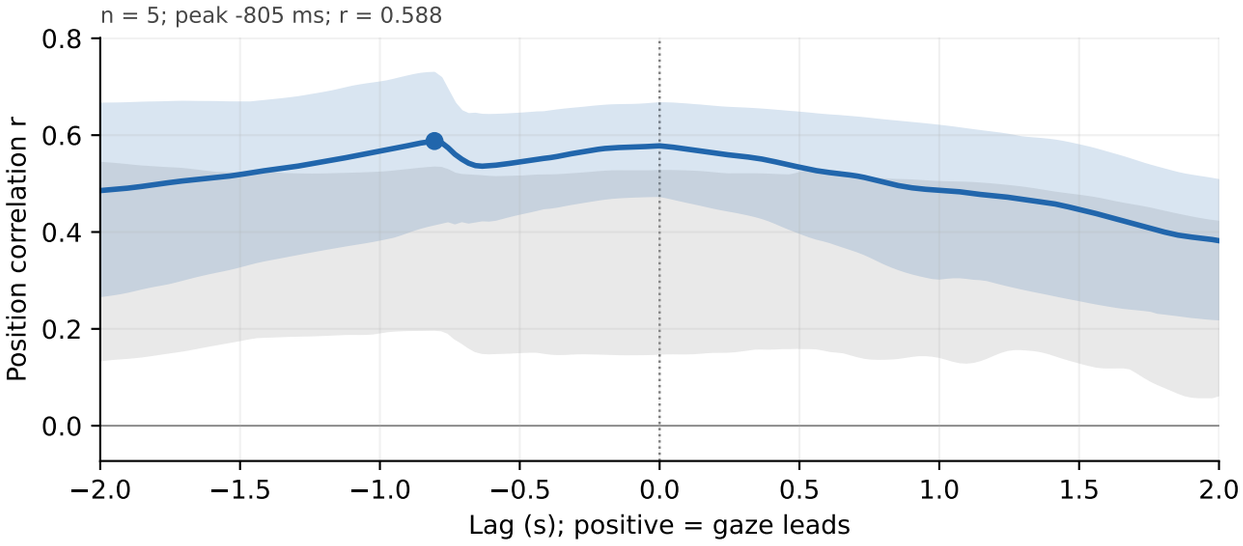
pull through and cinch / within_stage



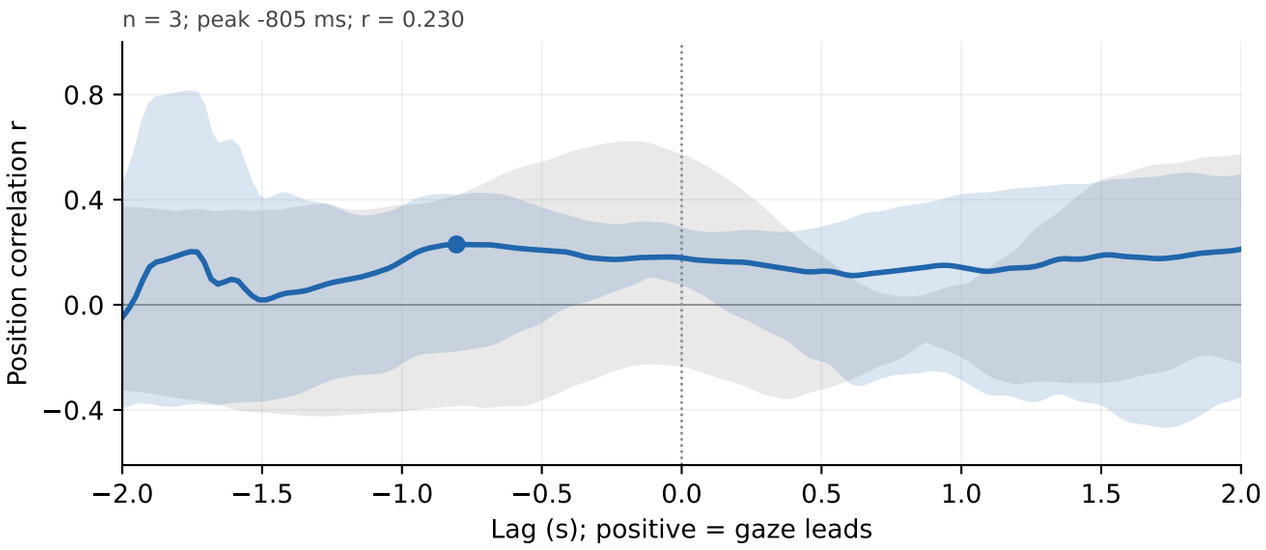
reach string / psm_stage



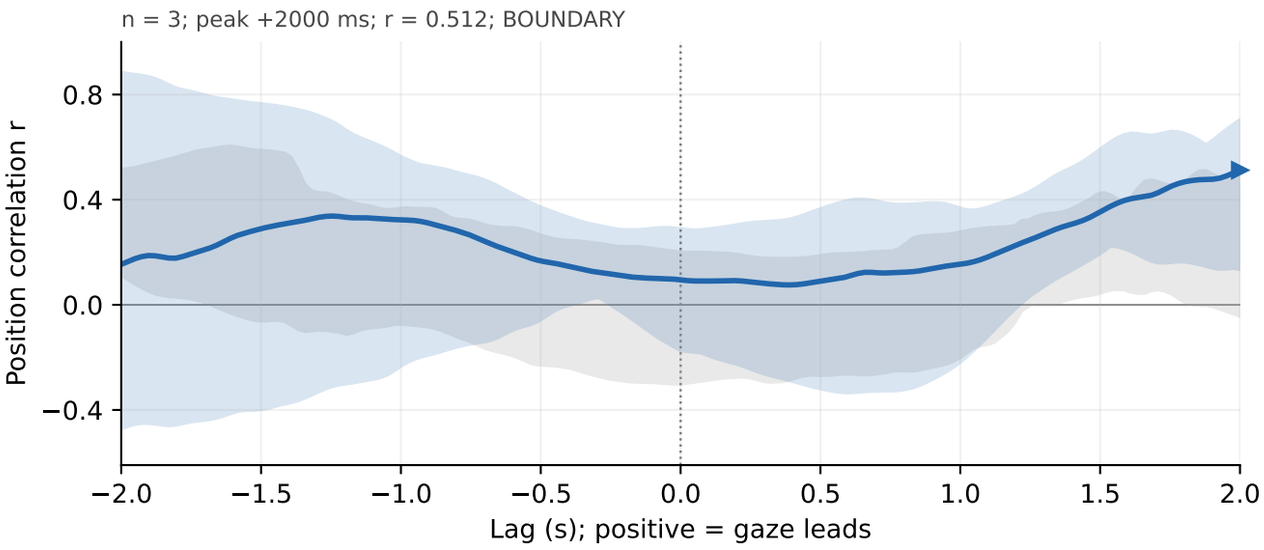
reach string / within_stage



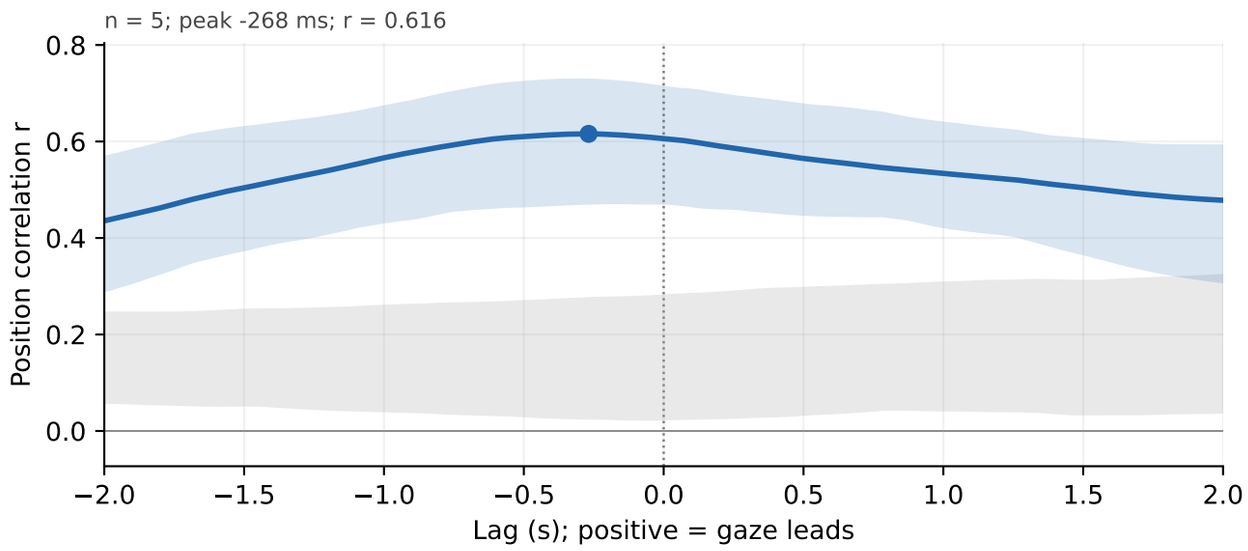
threading string through tower / psm_stage



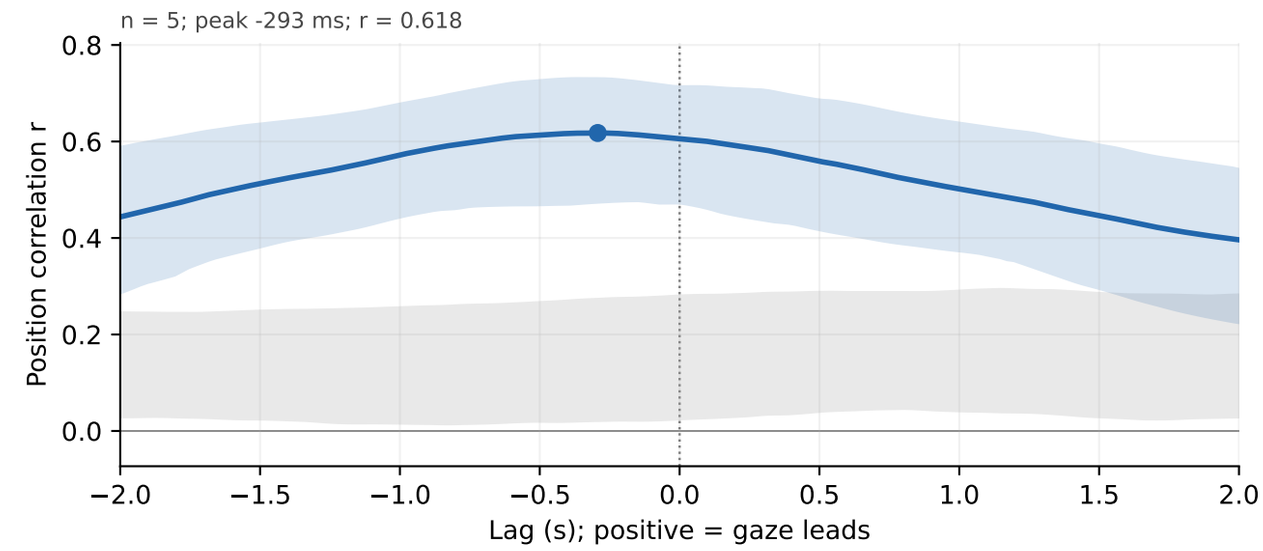
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

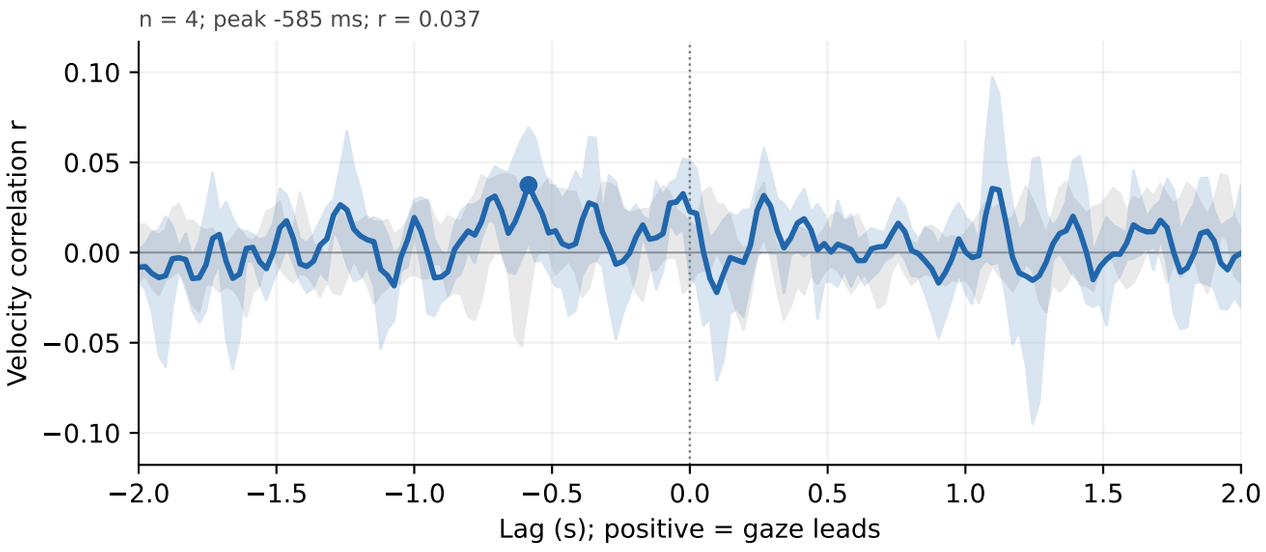


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

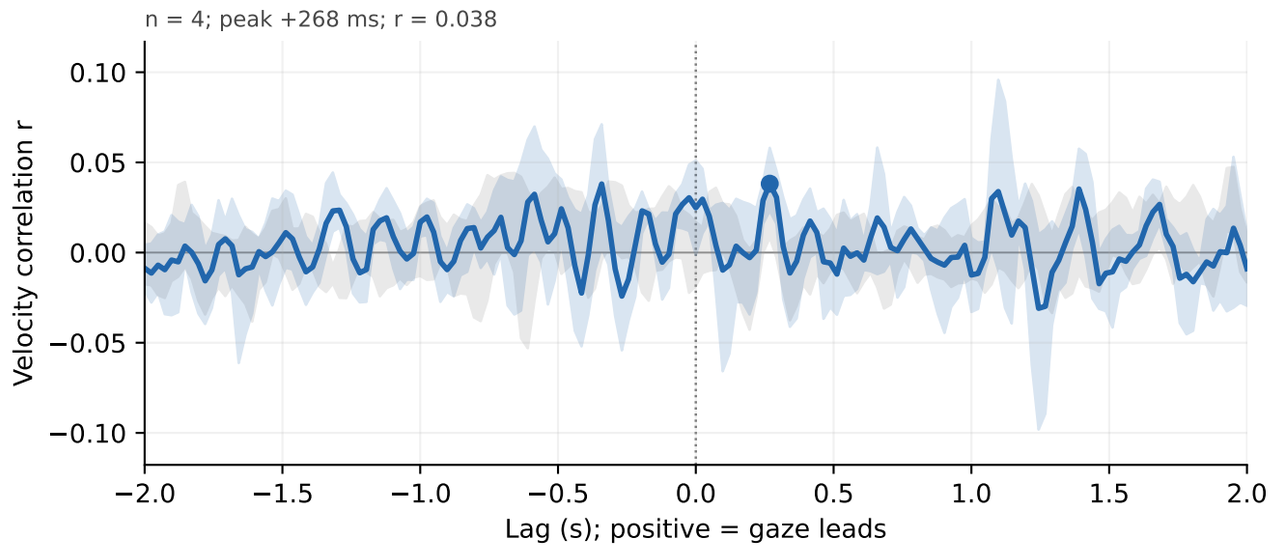
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

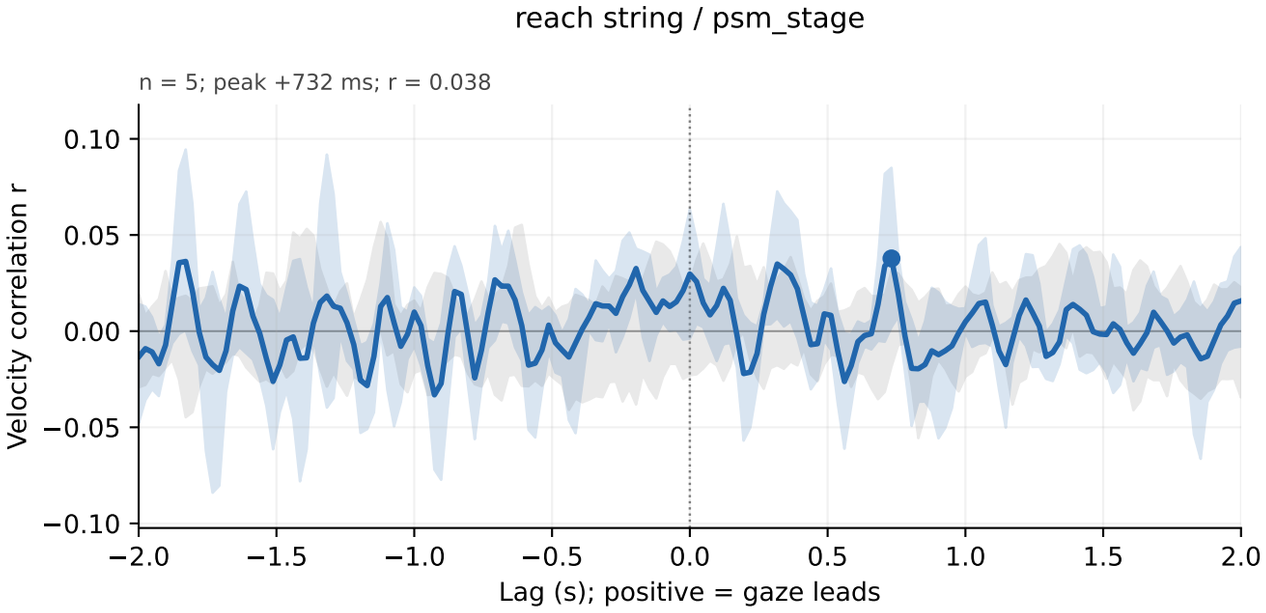
pull through and cinch / psm_stage



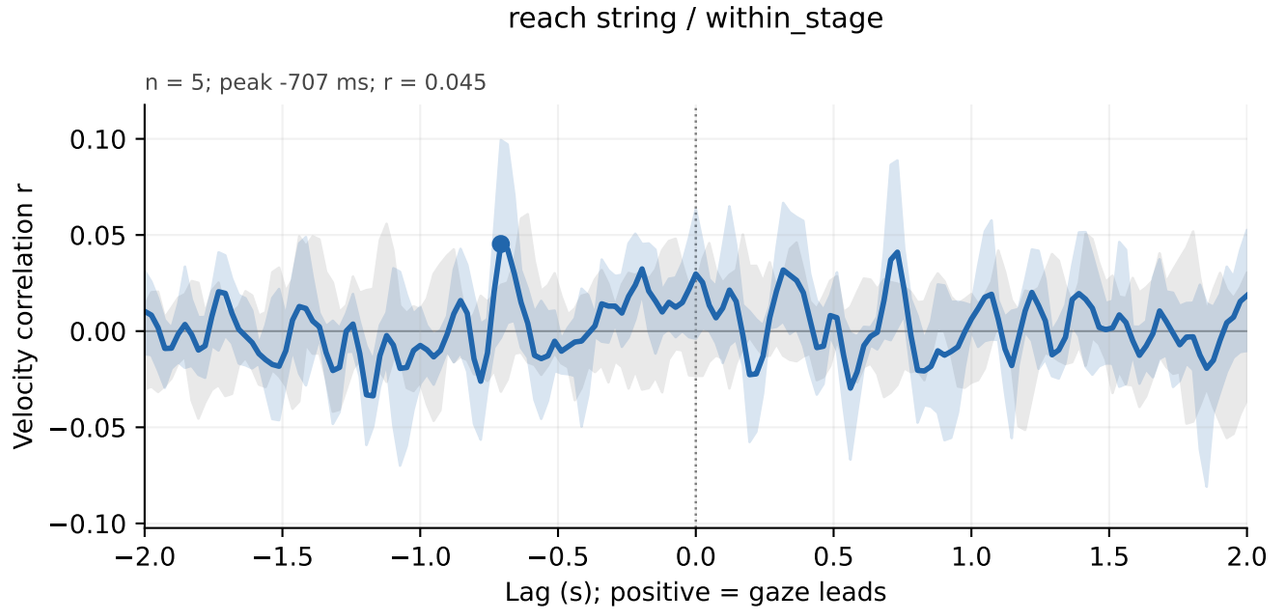
pull through and cinch / within_stage



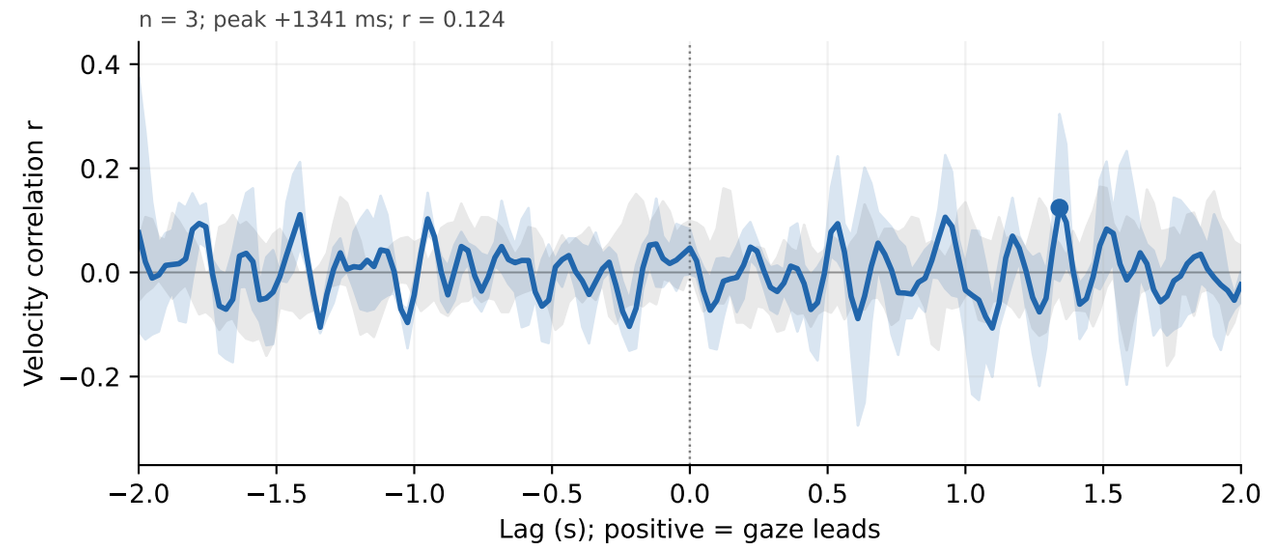
reach string / psm_stage



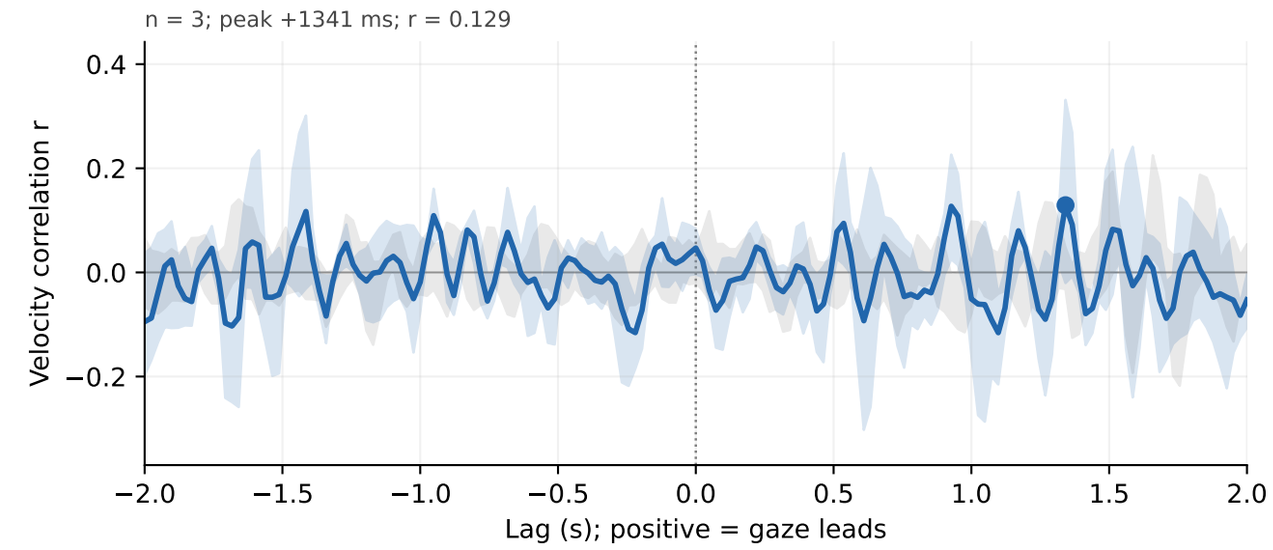
reach string / within_stage



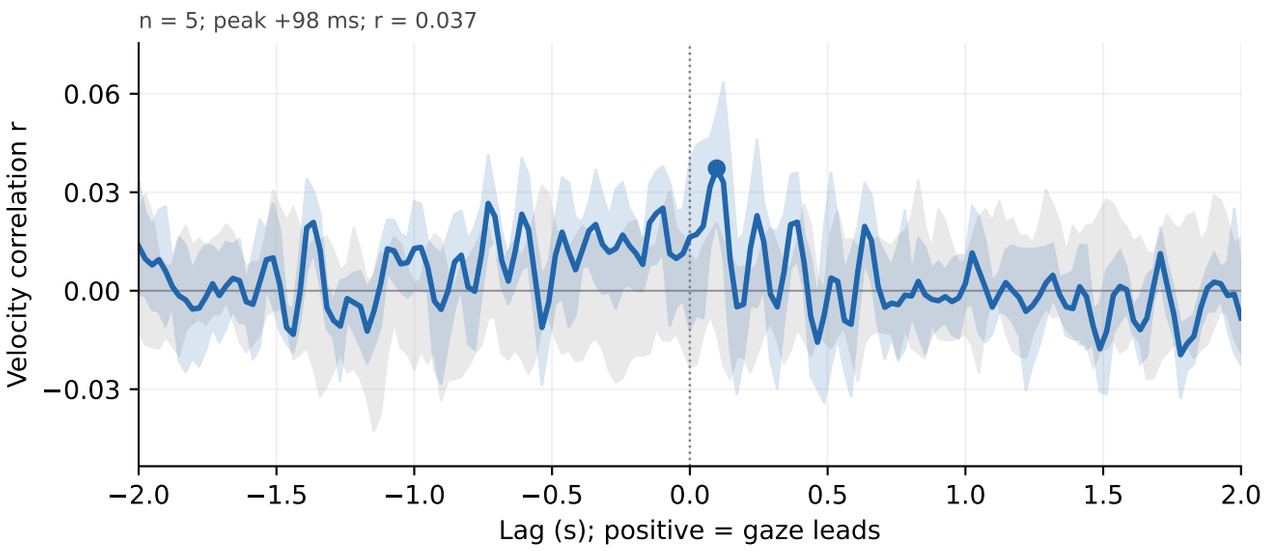
threading string through tower / psm_stage



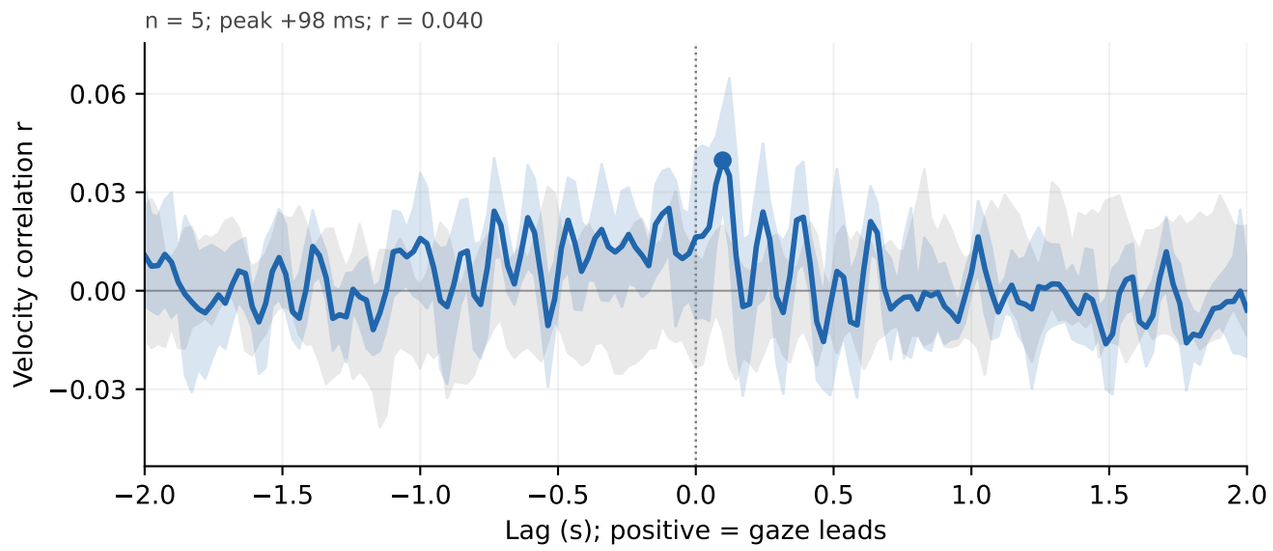
threading string through tower / within_stage



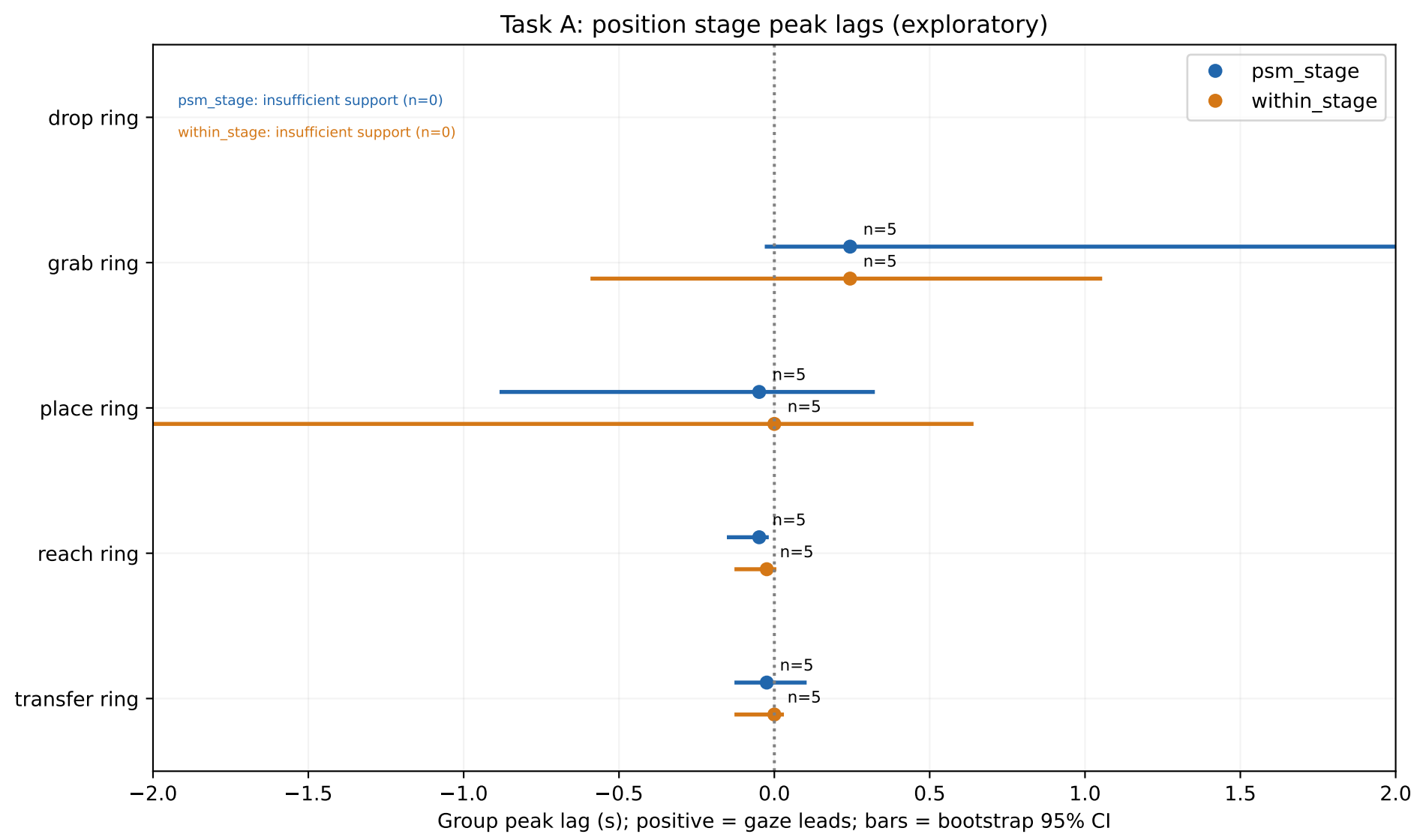
wrap around psm / psm_stage



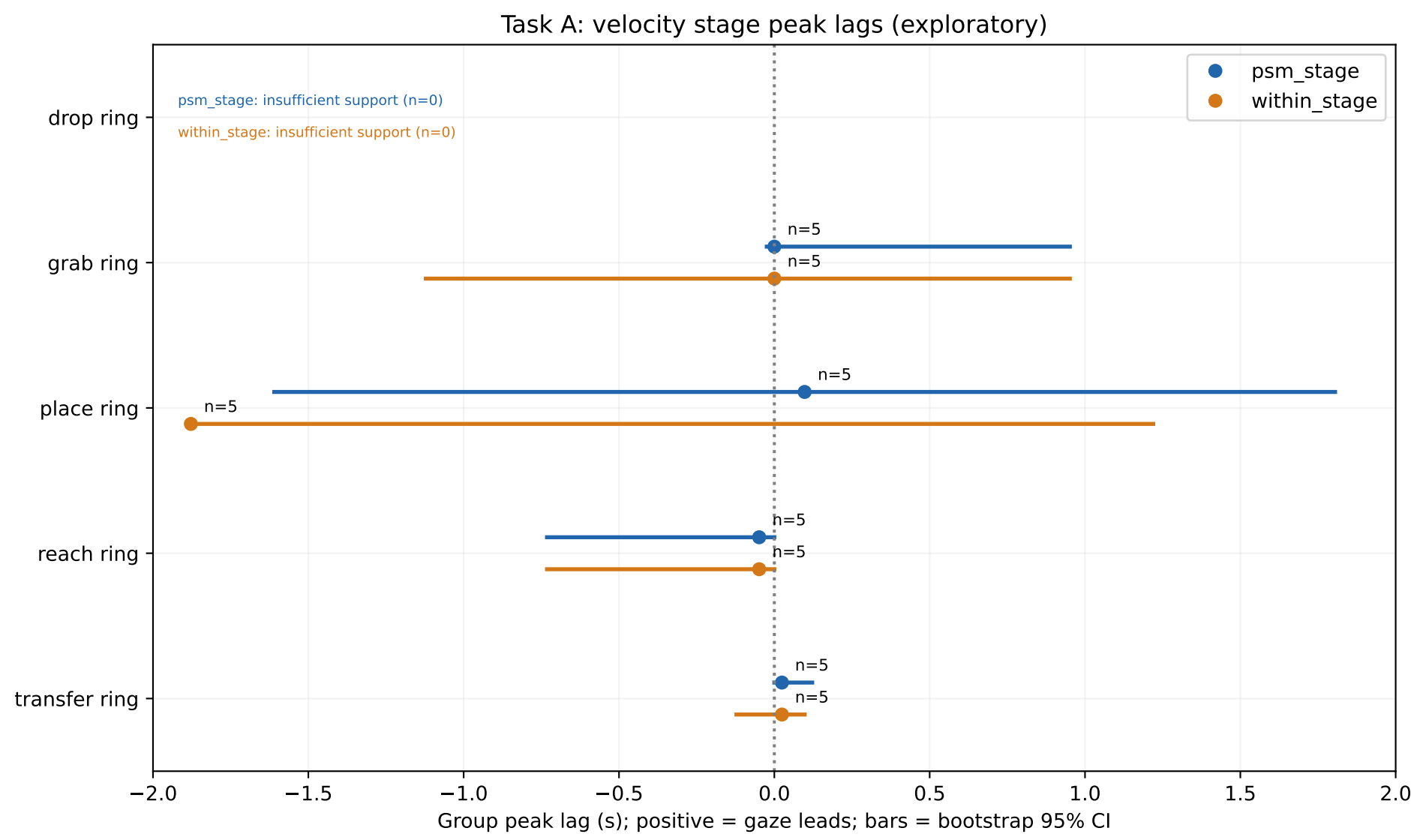
wrap around psm / within_stage



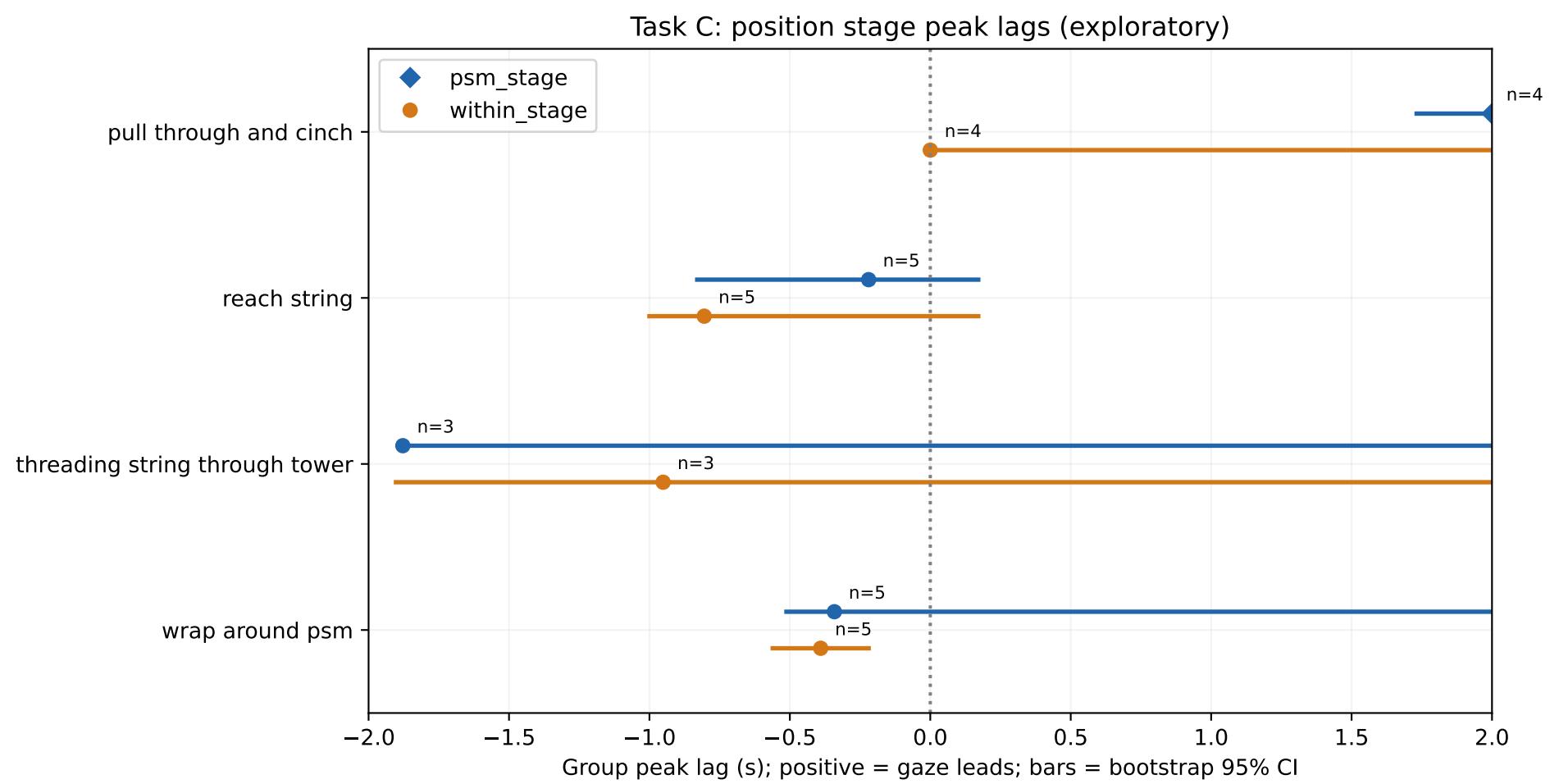
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



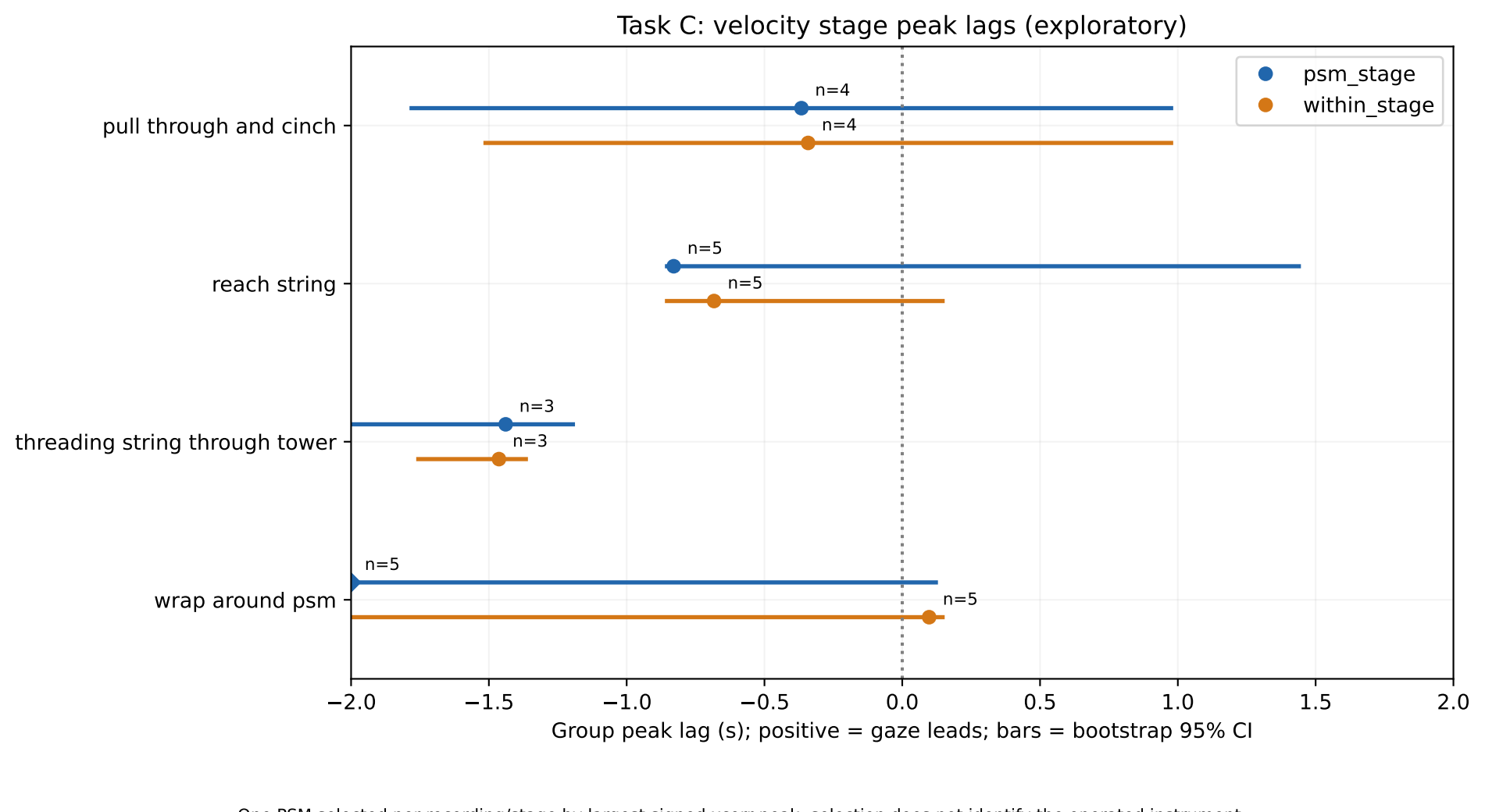
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



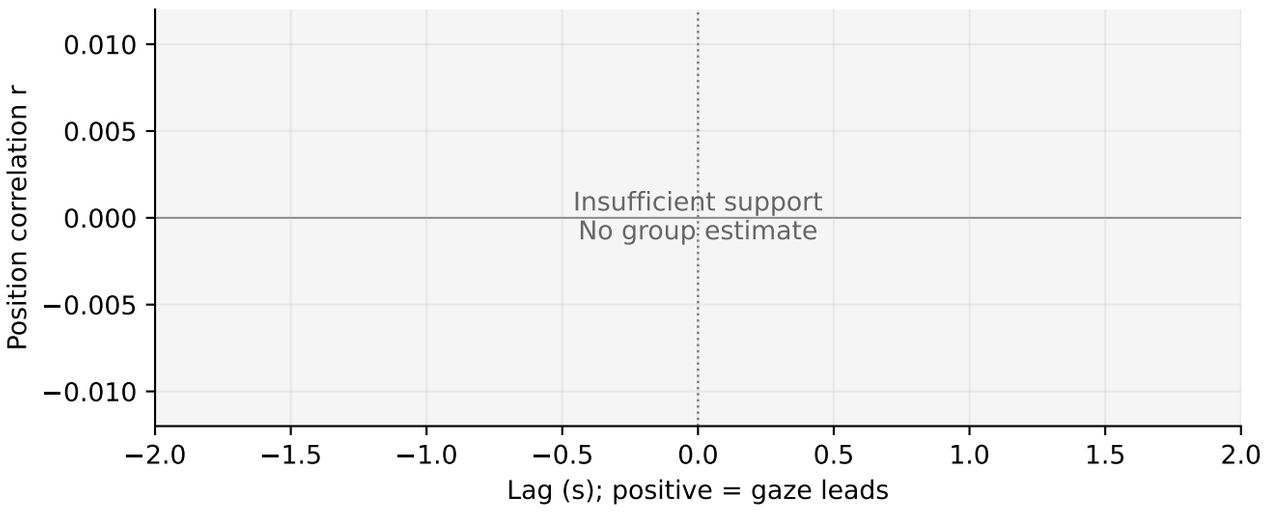
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

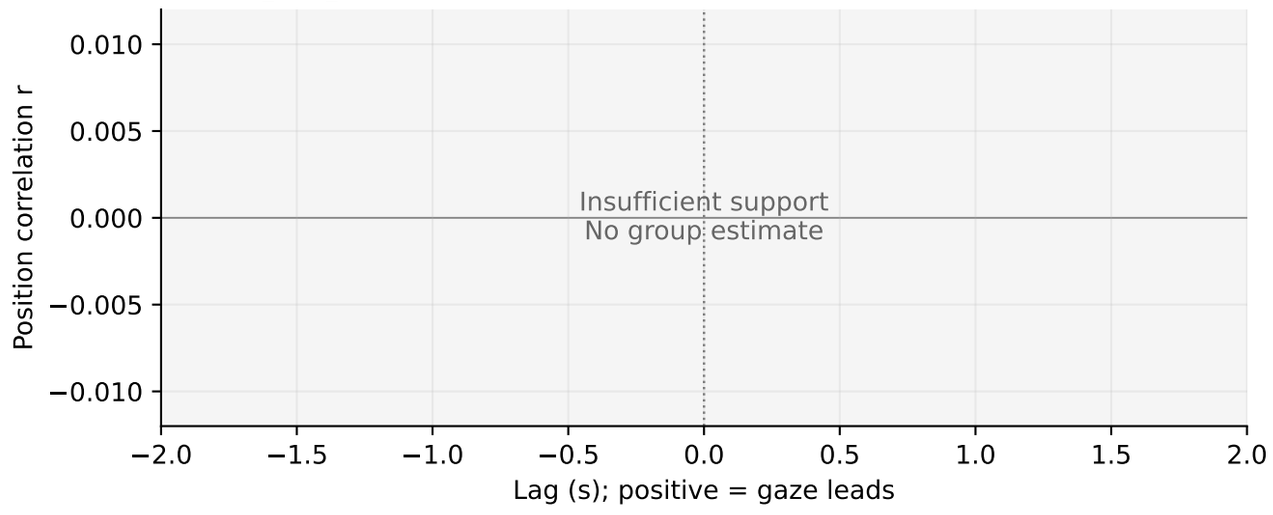
drop ring / psm_stage

n = 0; no_valid_curve



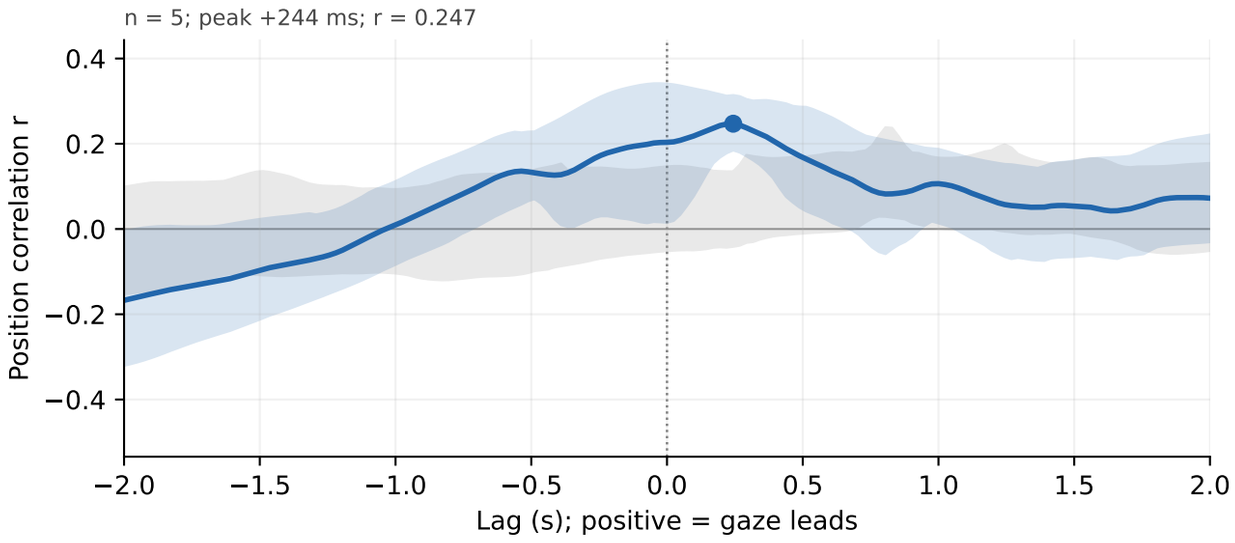
drop ring / within_stage

n = 0; no_valid_curve



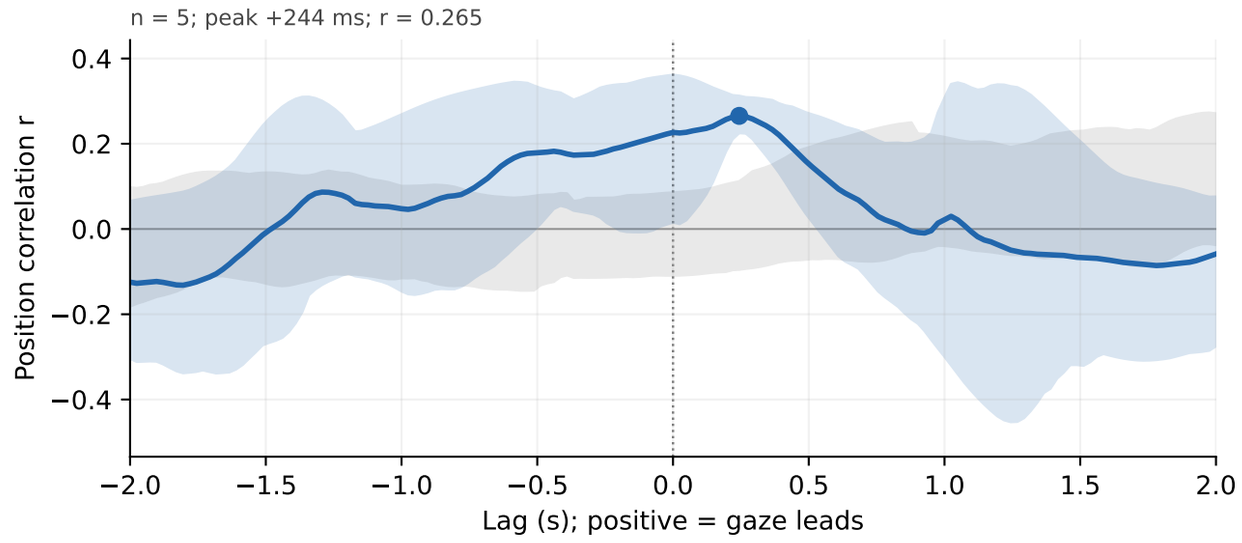
grab ring / psm_stage

n = 5; peak +244 ms; r = 0.247



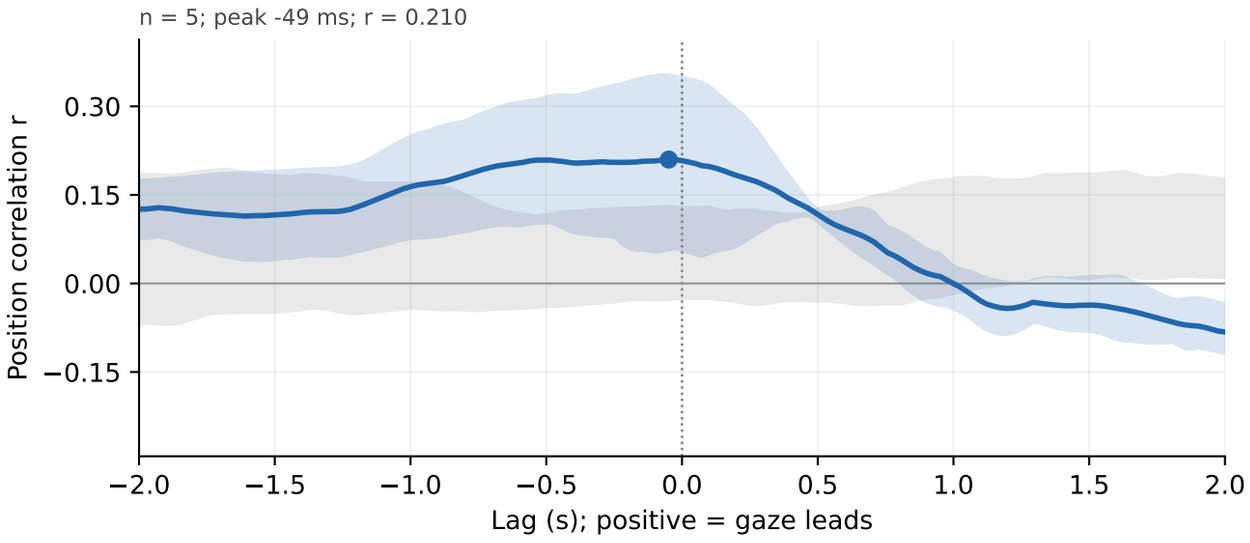
grab ring / within_stage

n = 5; peak +244 ms; r = 0.265



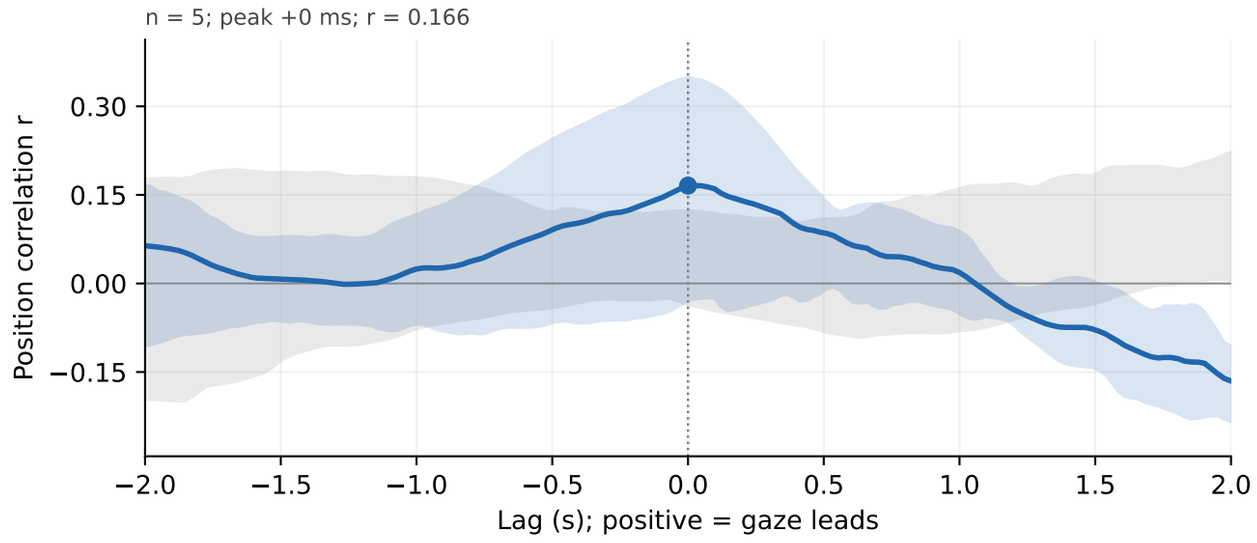
place ring / psm_stage

n = 5; peak -49 ms; r = 0.210



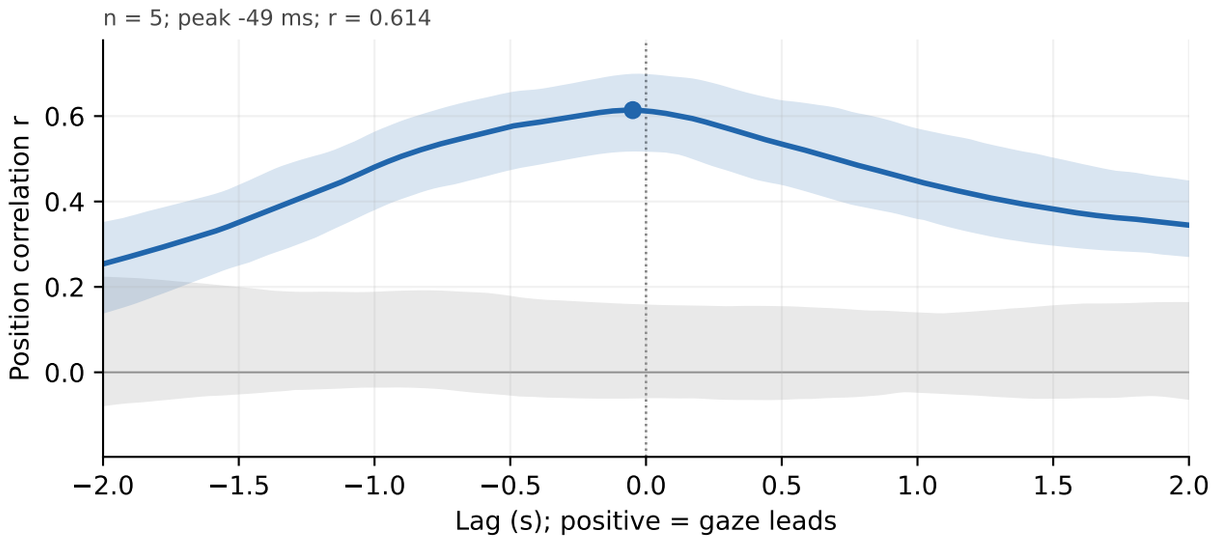
place ring / within_stage

n = 5; peak +0 ms; r = 0.166



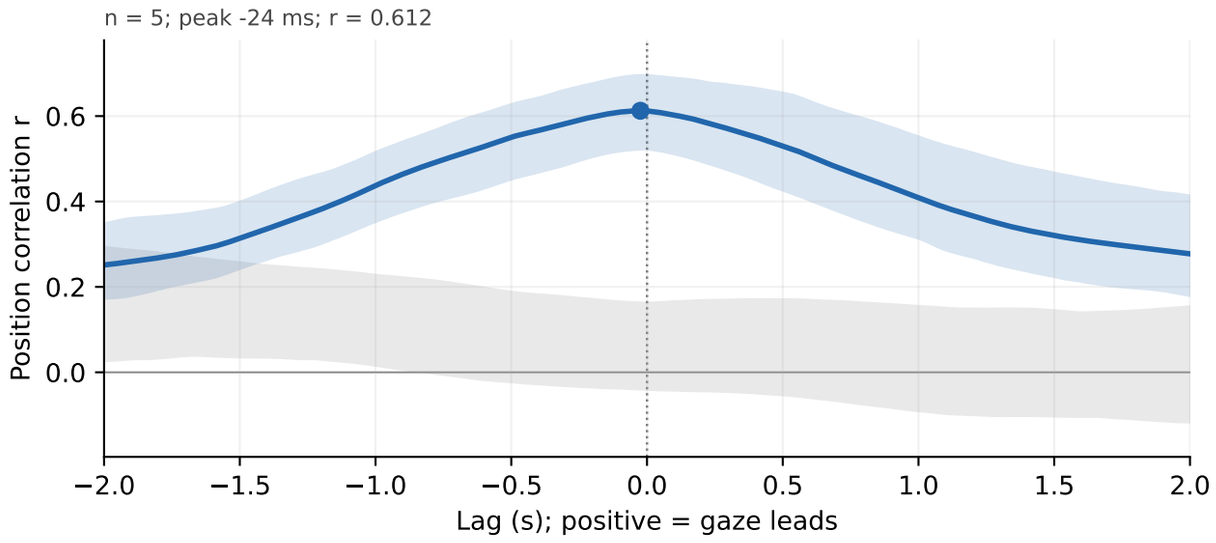
reach ring / psm_stage

n = 5; peak -49 ms; r = 0.614



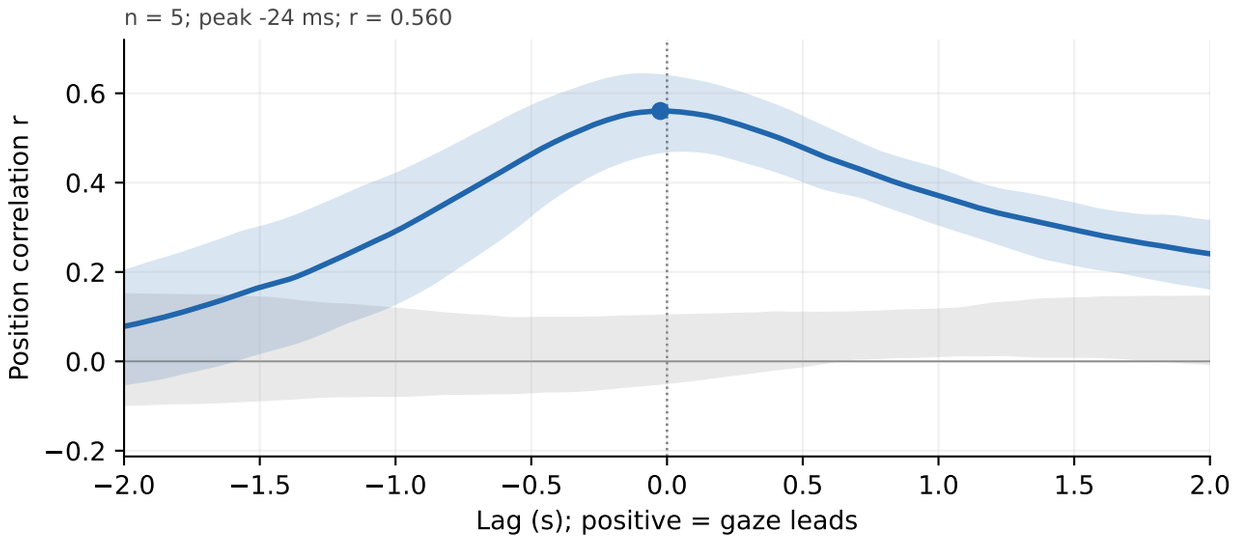
reach ring / within_stage

n = 5; peak -24 ms; r = 0.612



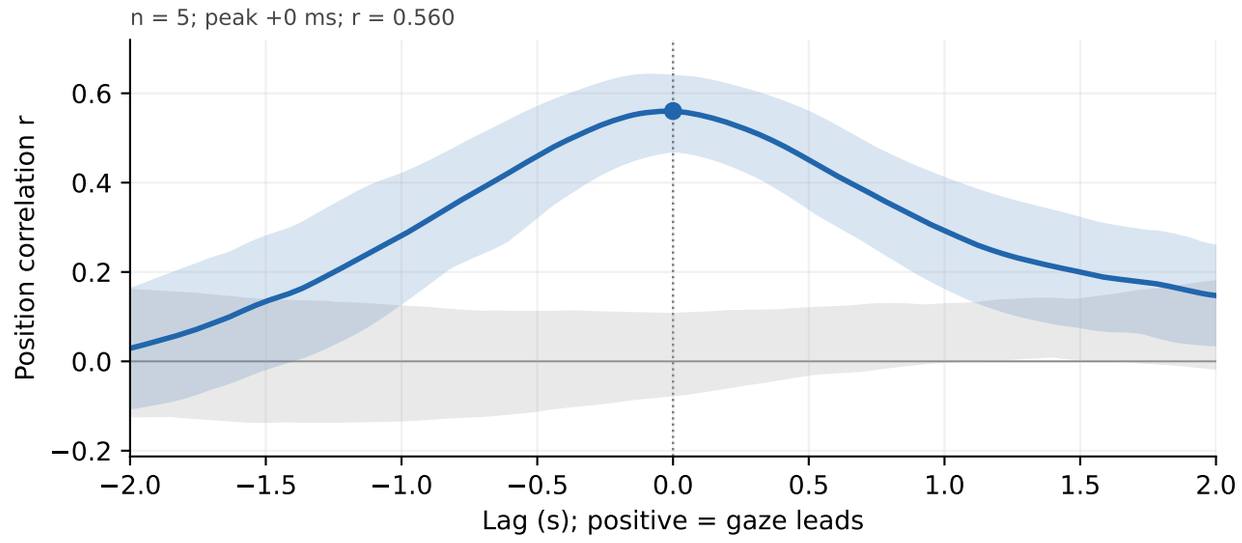
transfer ring / psm_stage

n = 5; peak -24 ms; r = 0.560



transfer ring / within_stage

n = 5; peak +0 ms; r = 0.560

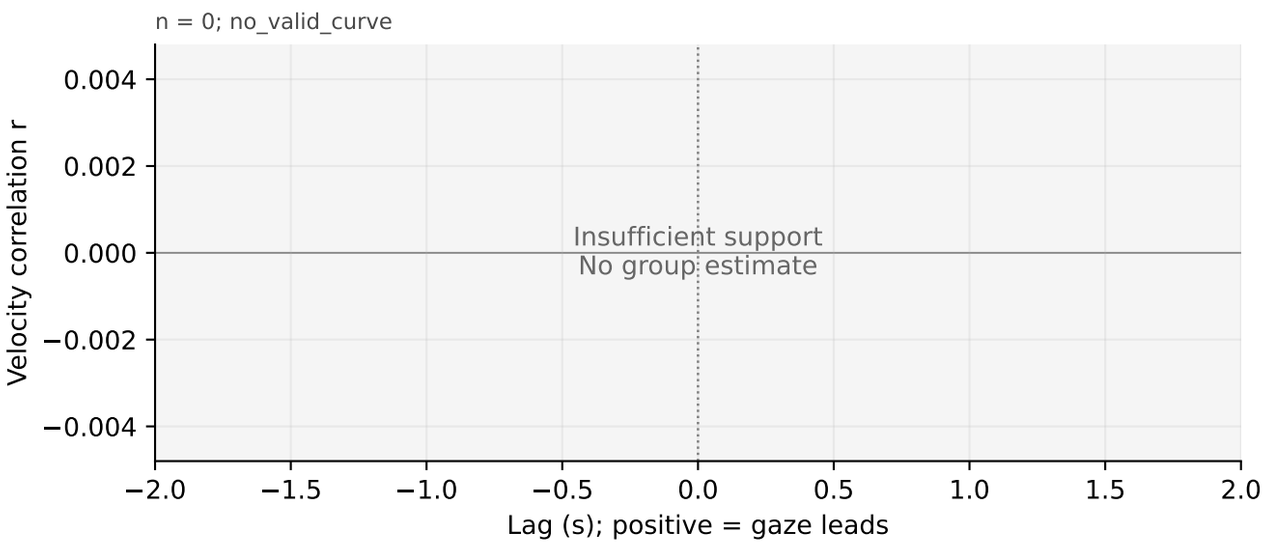


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

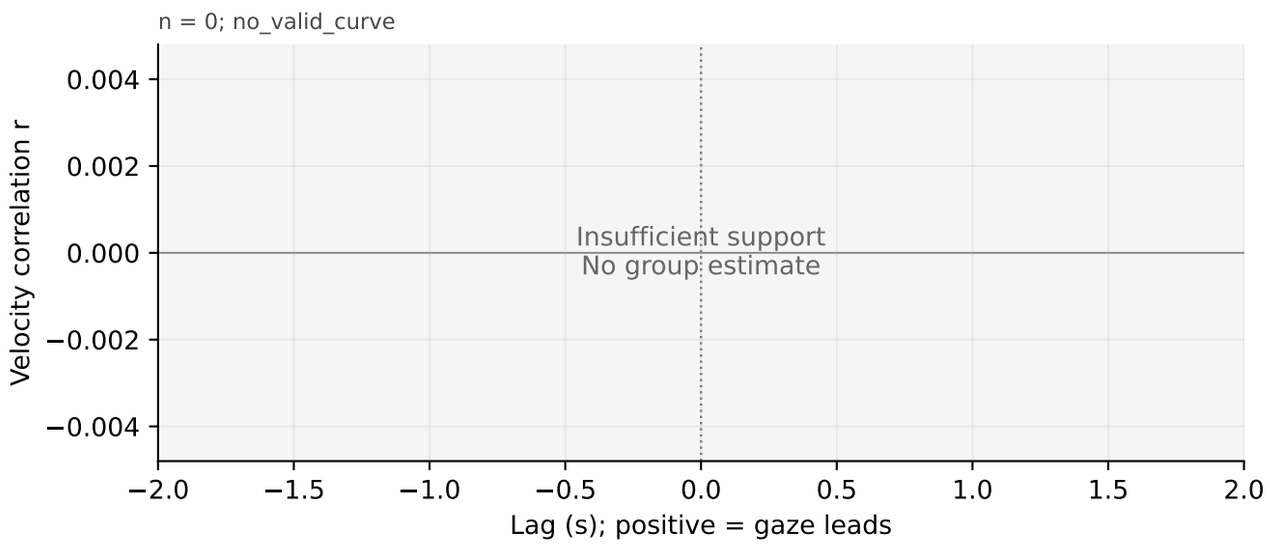
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

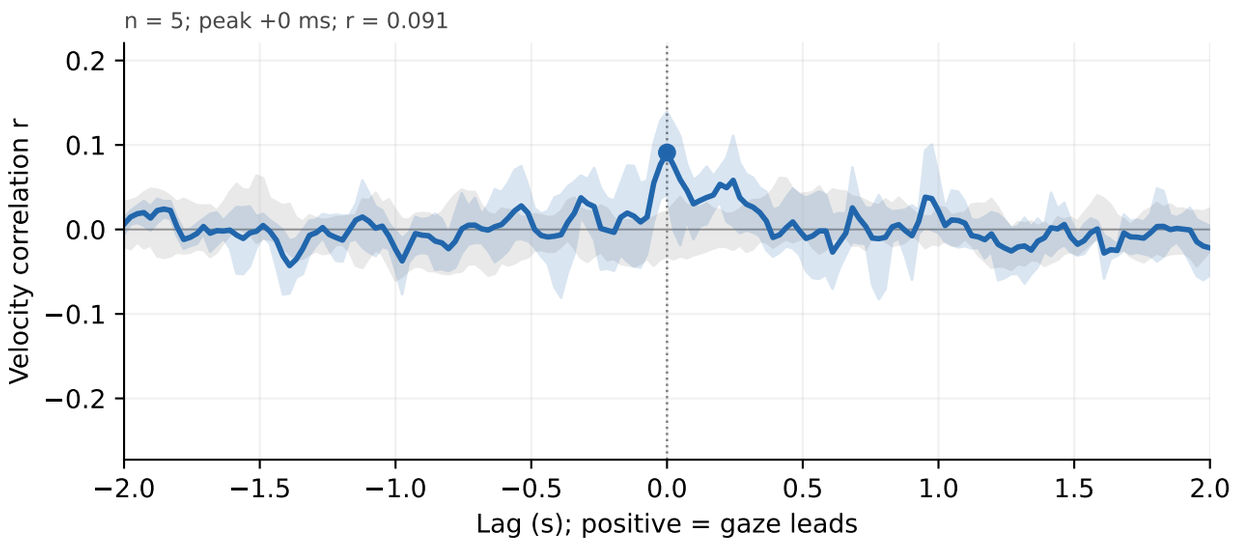
drop ring / psm_stage



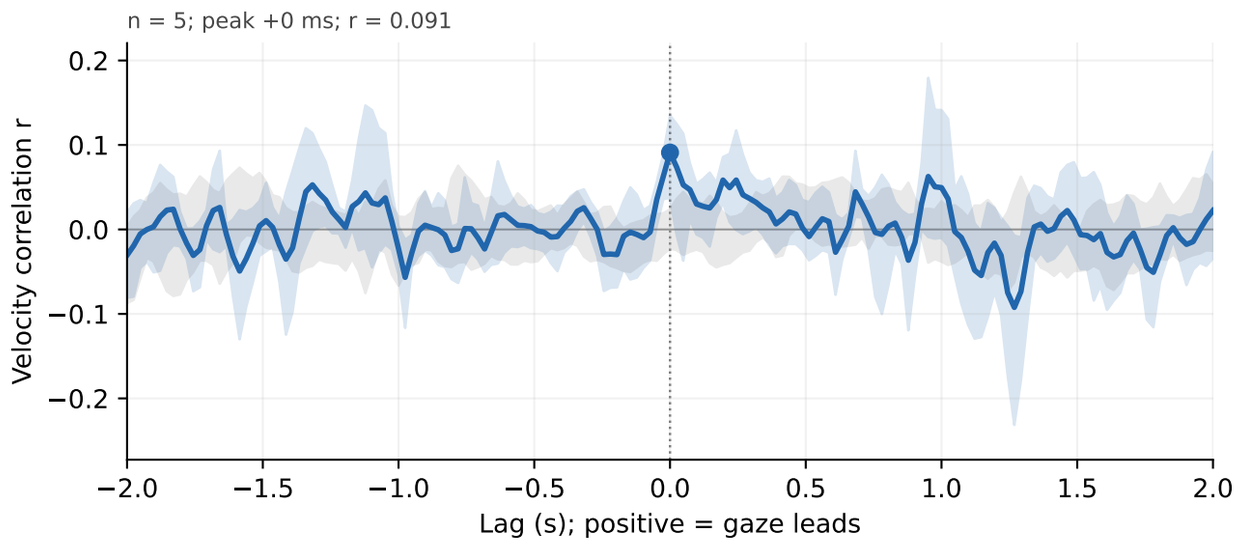
drop ring / within_stage



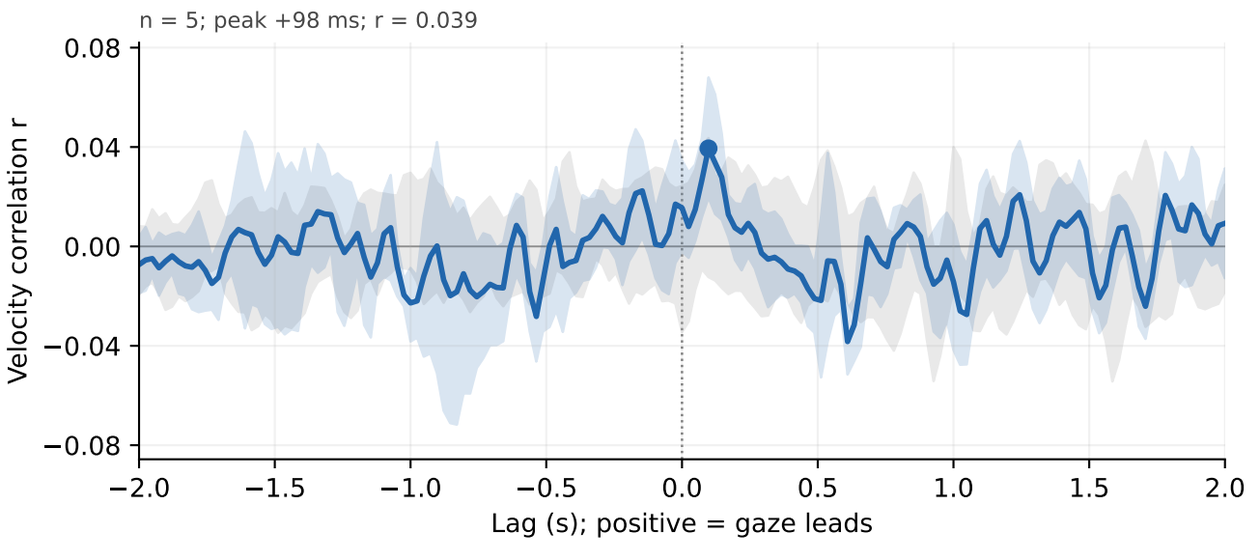
grab ring / psm_stage



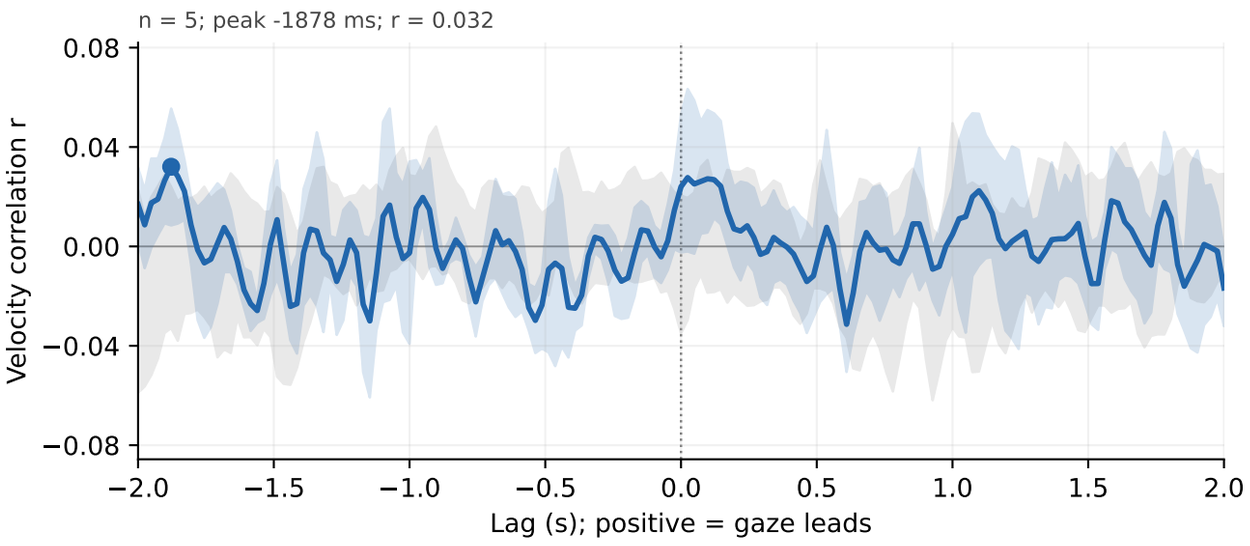
grab ring / within_stage



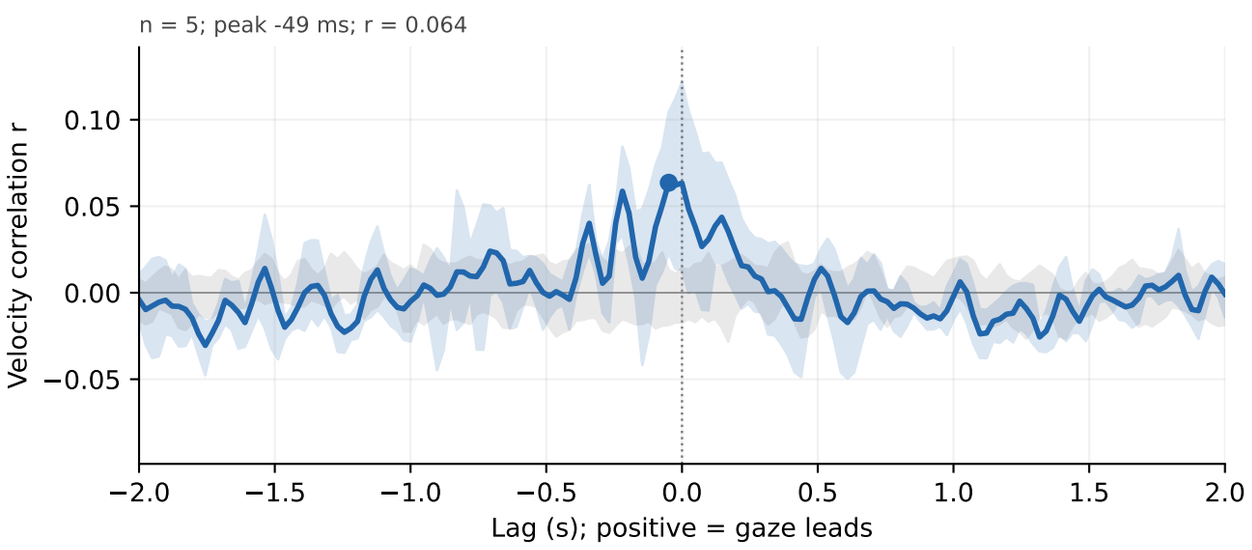
place ring / psm_stage



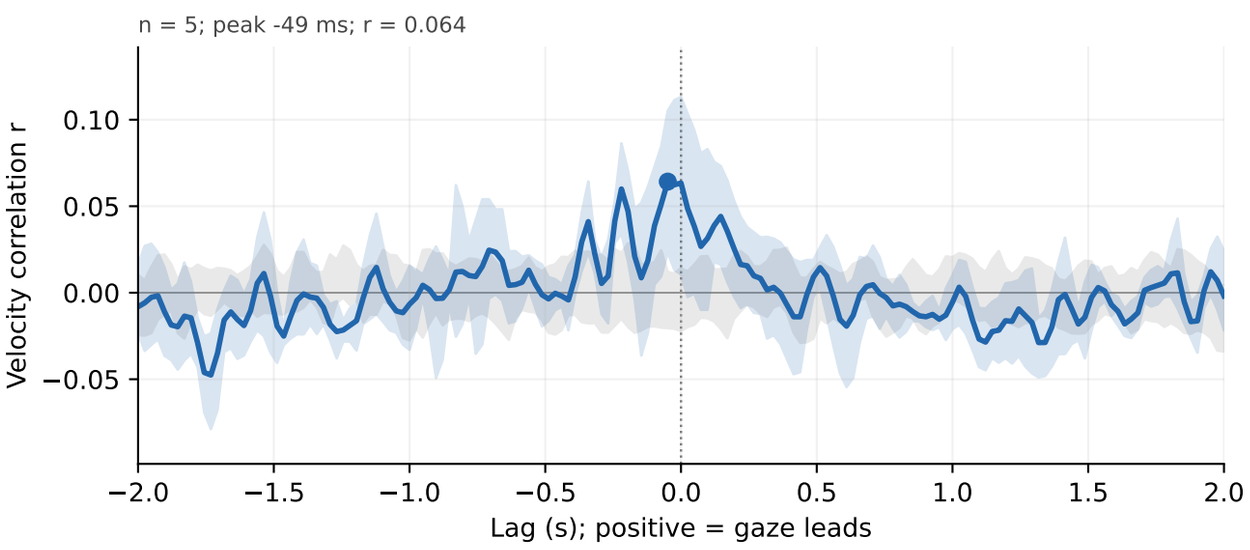
place ring / within_stage



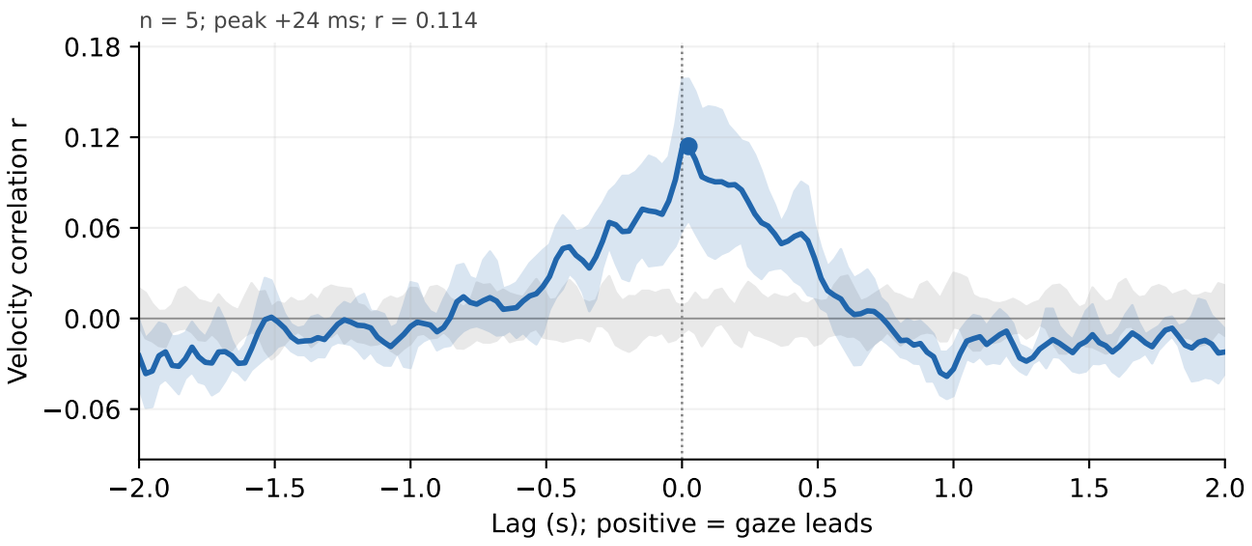
reach ring / psm_stage



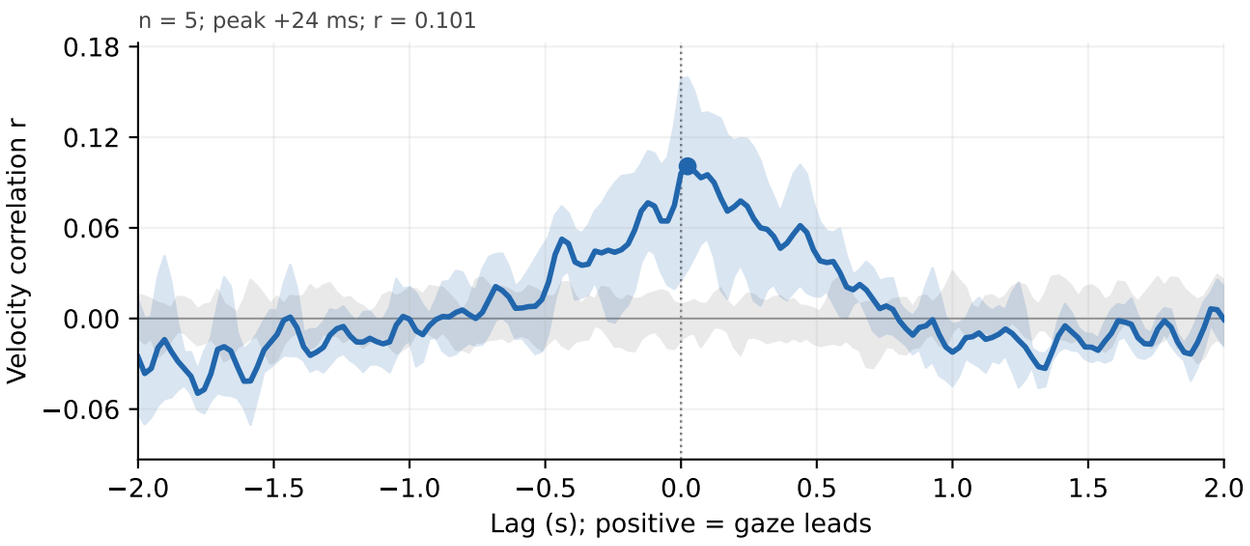
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

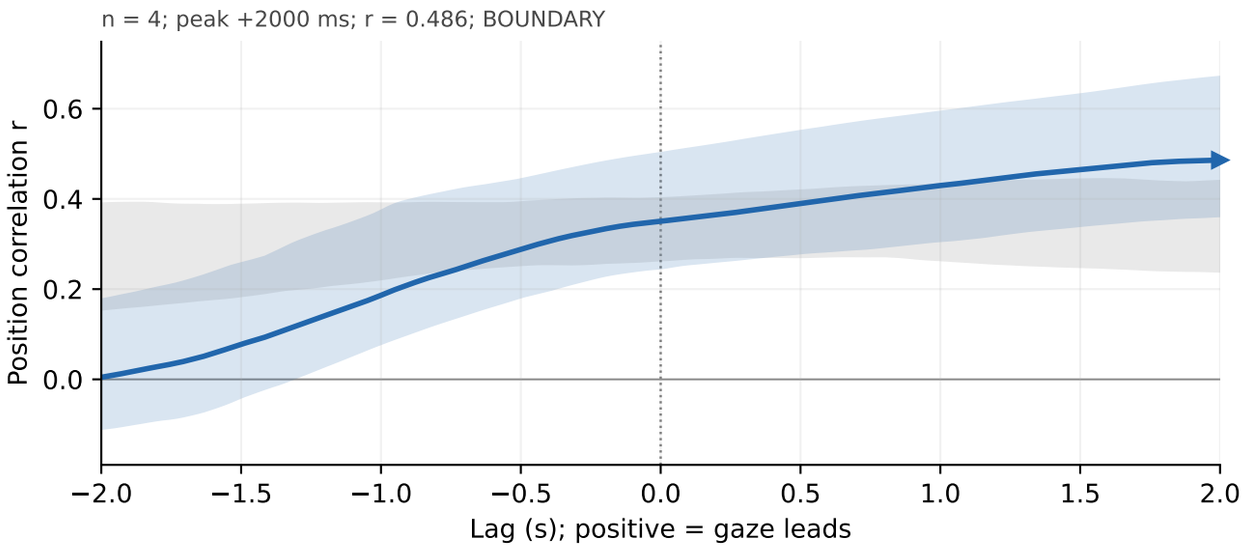


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

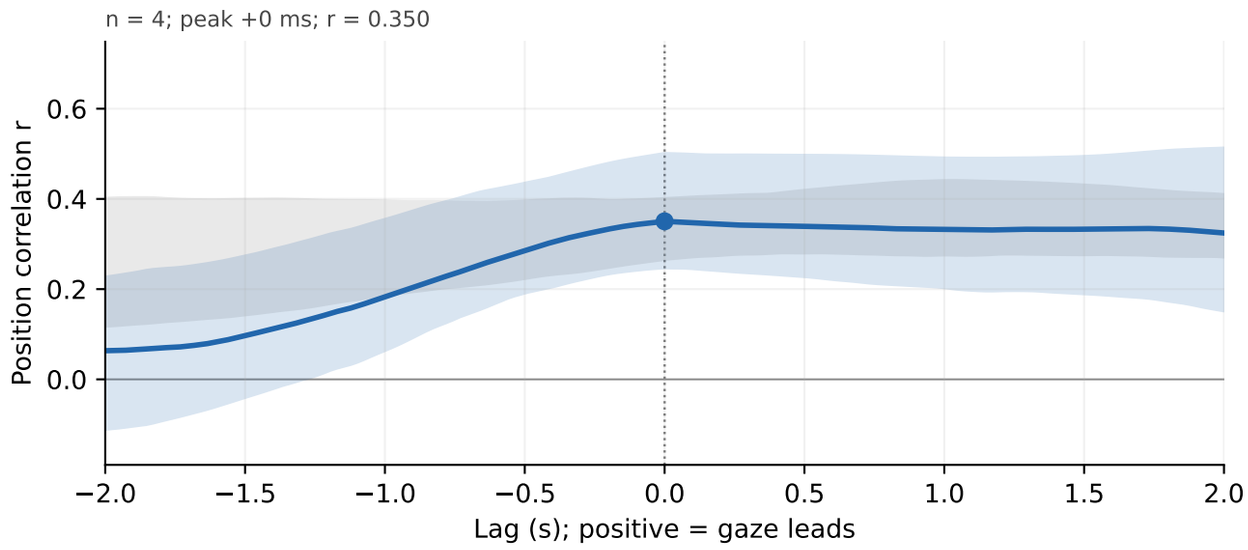
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

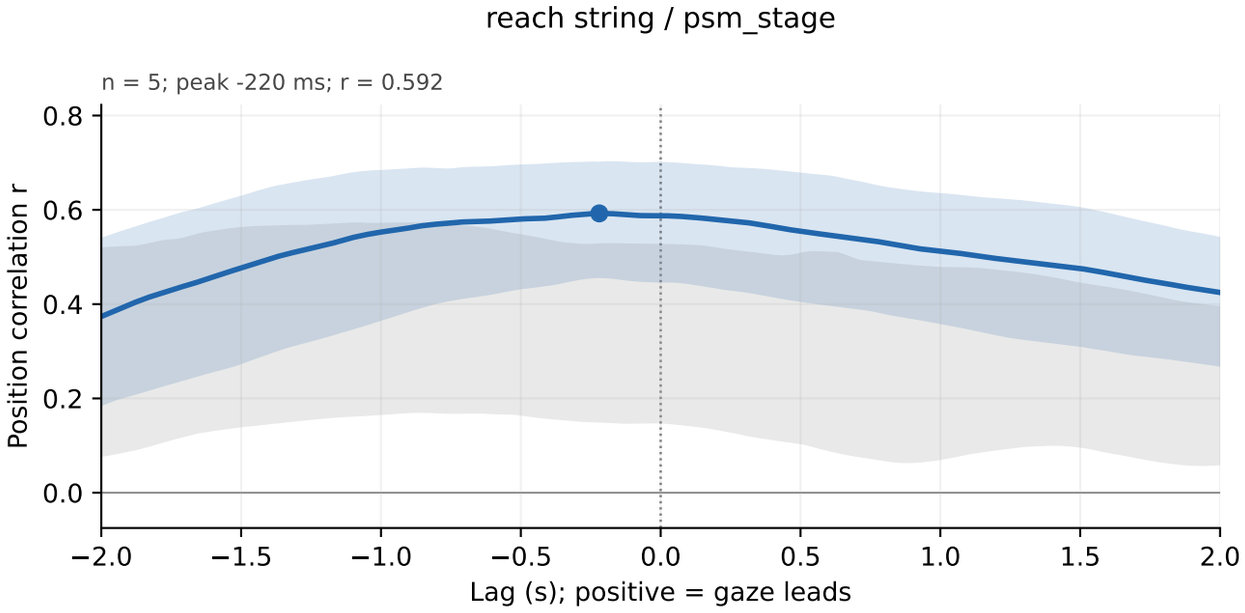
pull through and cinch / psm_stage



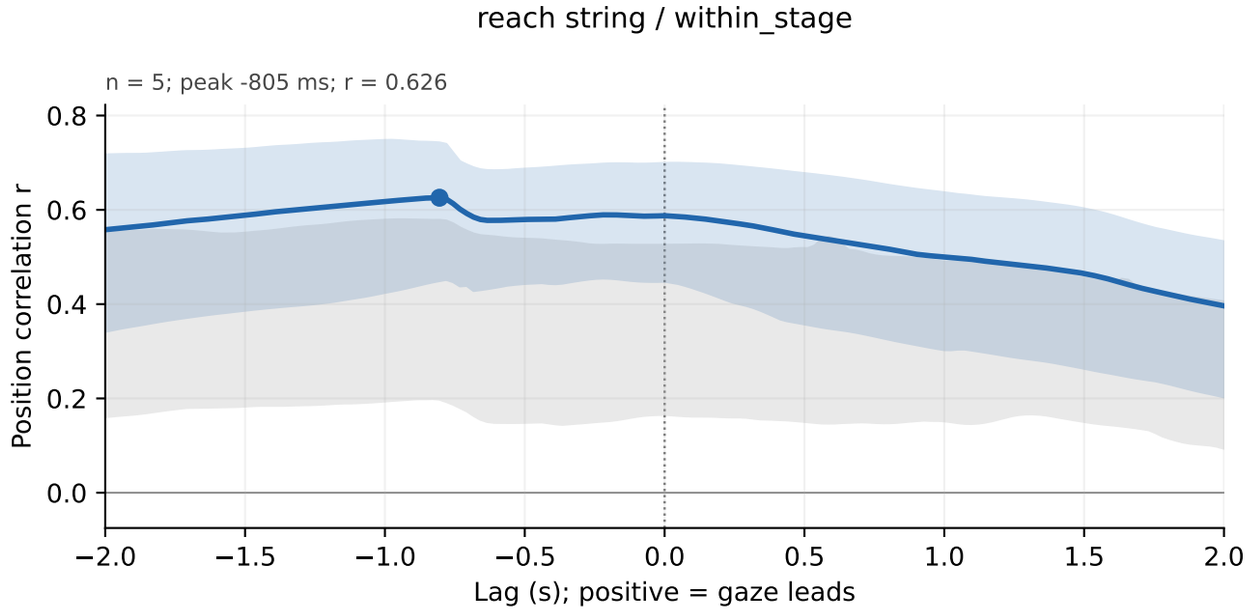
pull through and cinch / within_stage



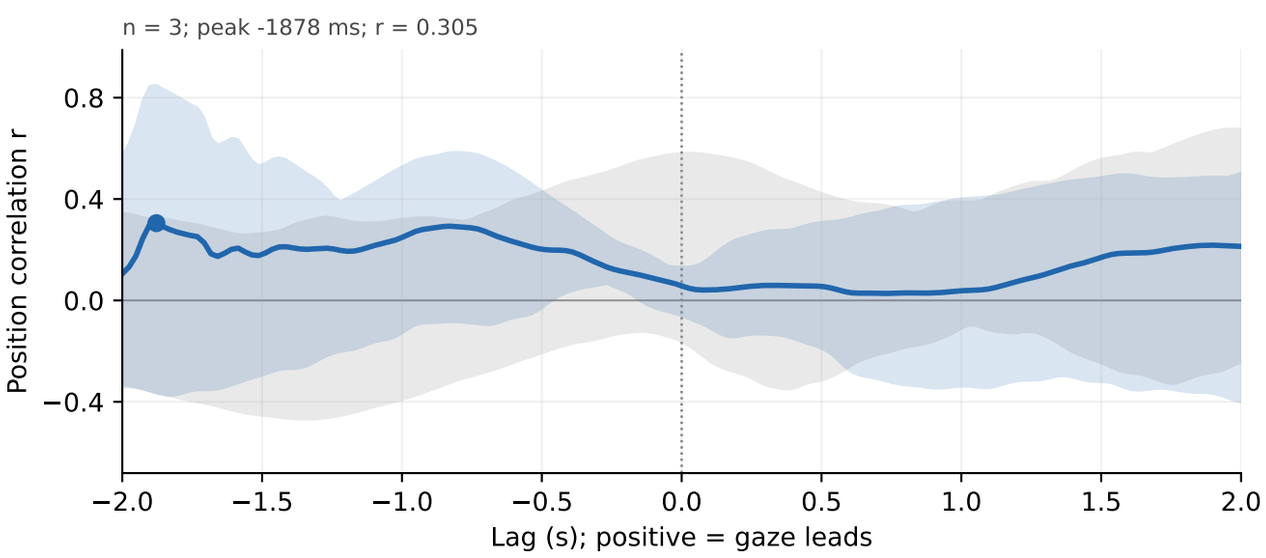
reach string / psm_stage



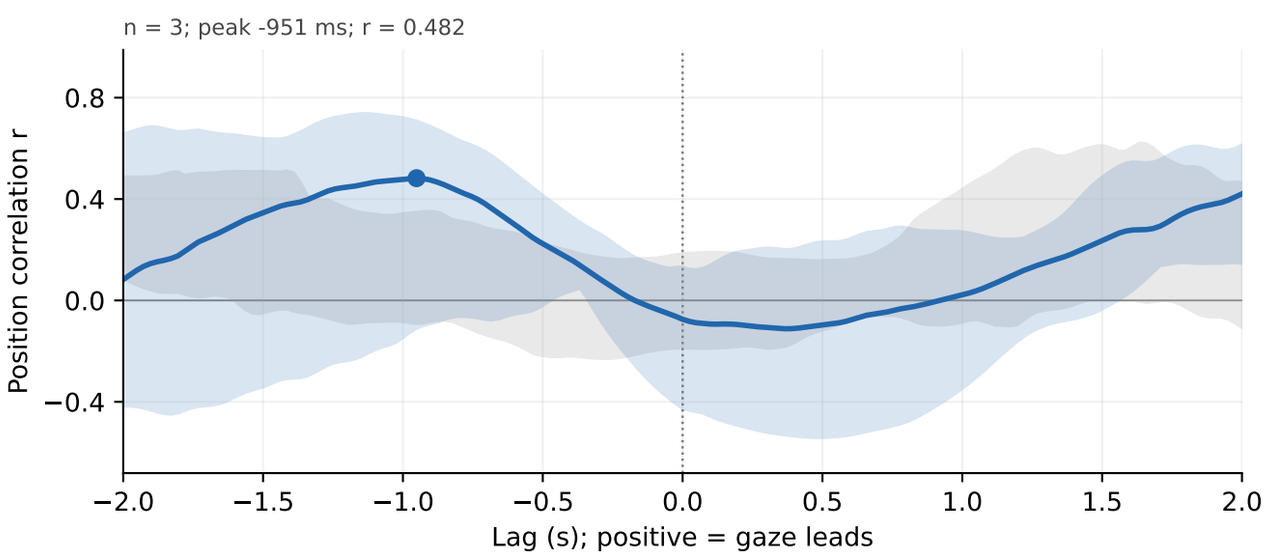
reach string / within_stage



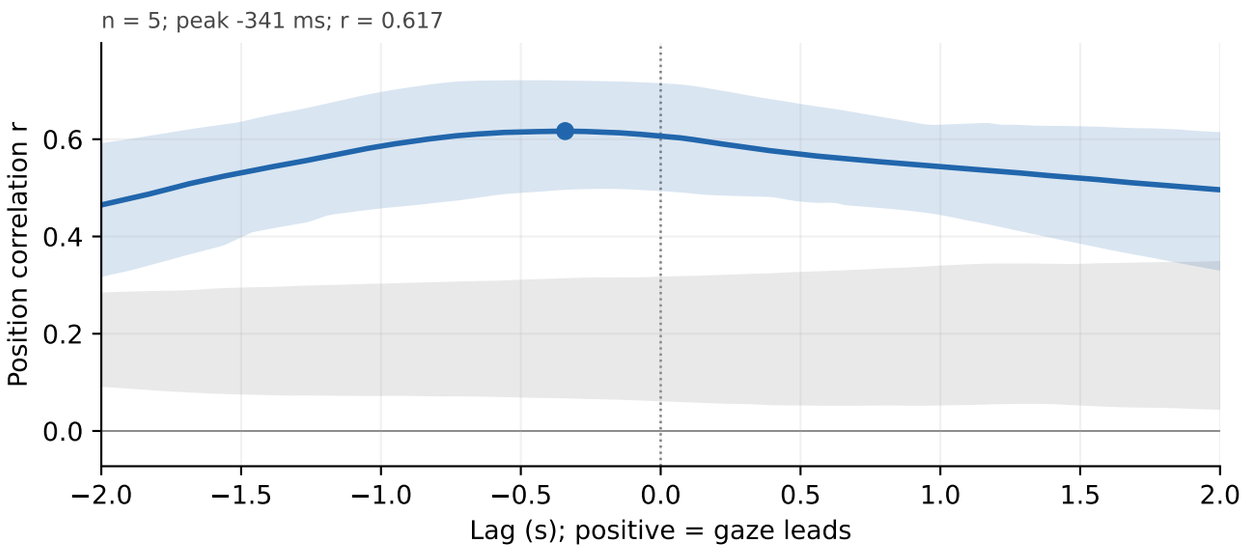
threading string through tower / psm_stage



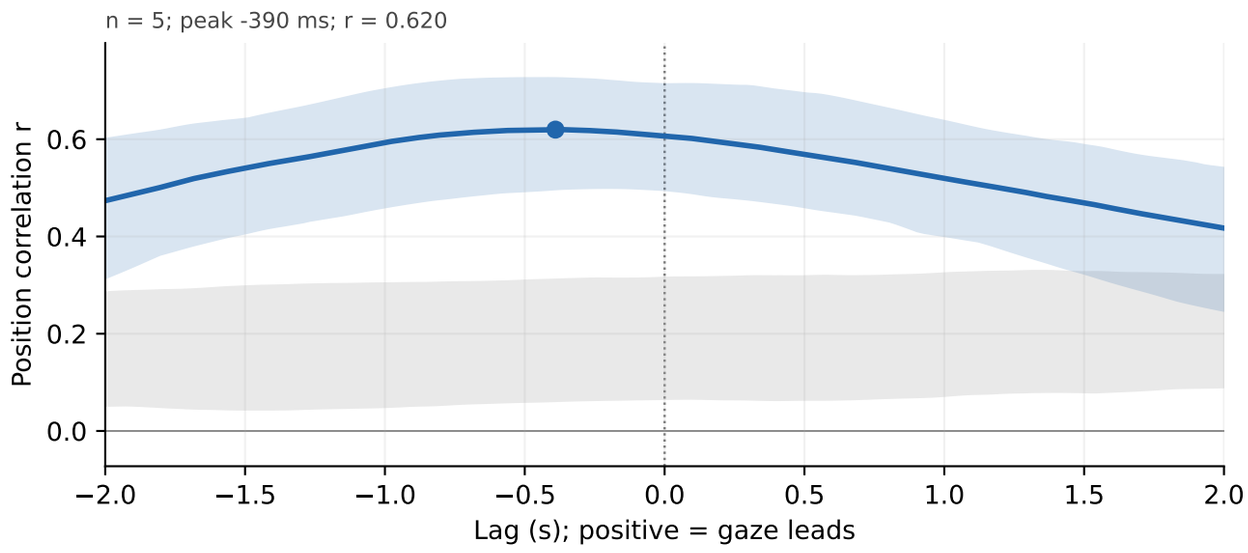
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

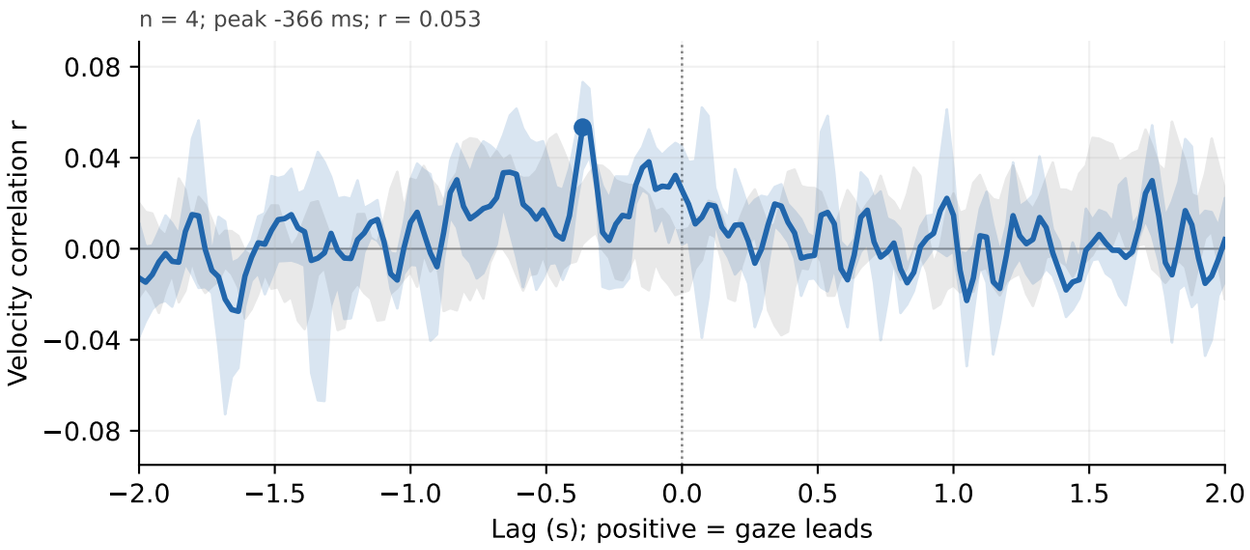


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

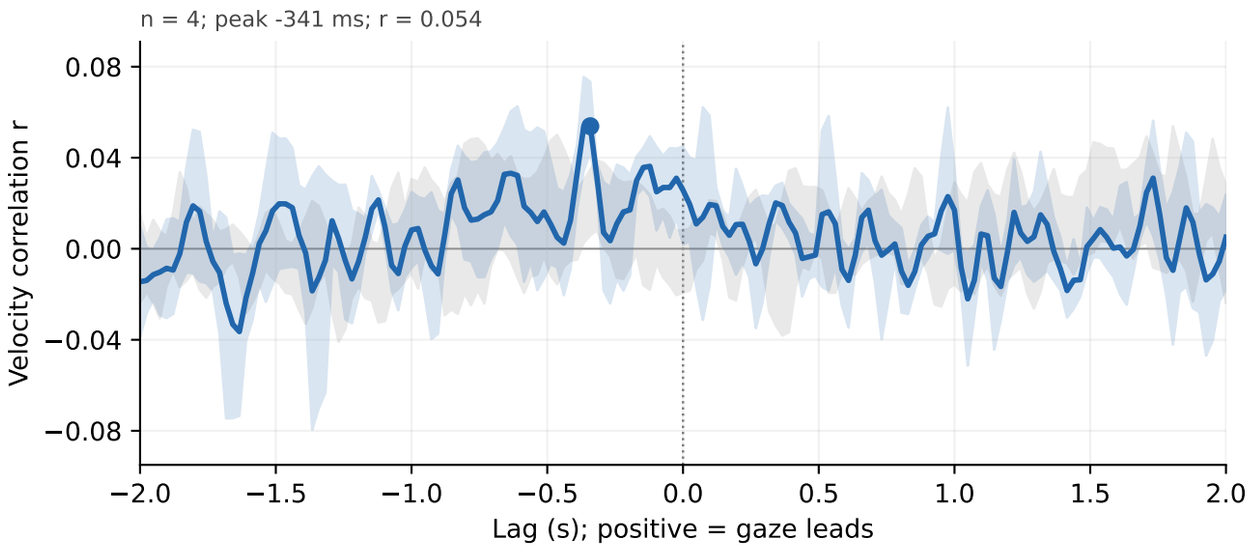
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

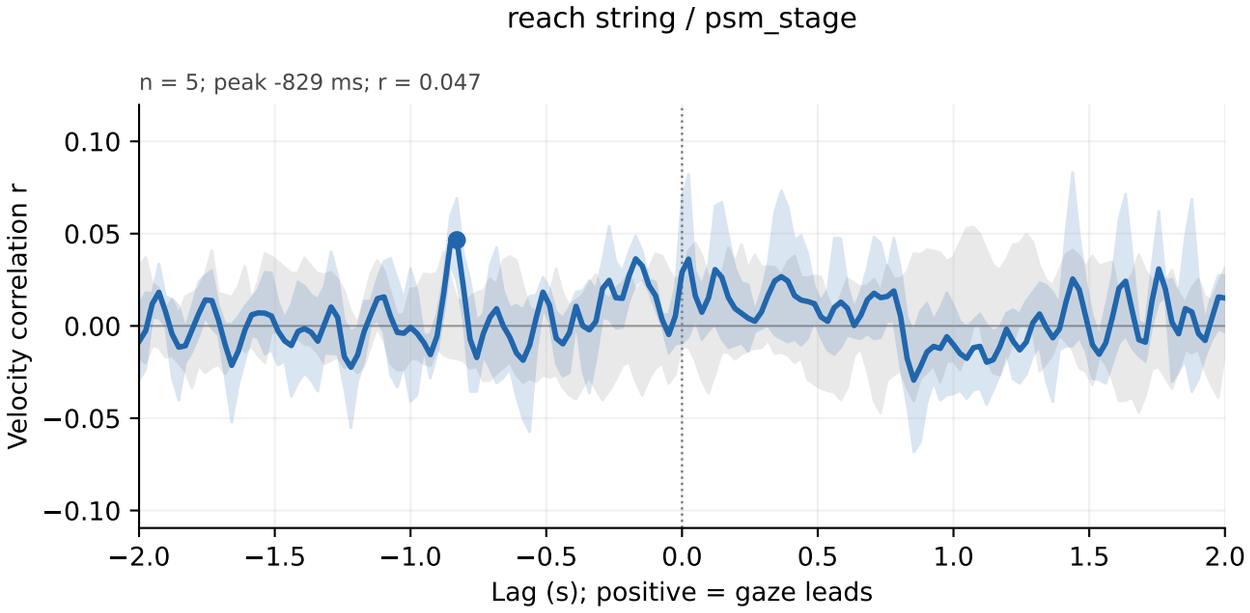
pull through and cinch / psm_stage



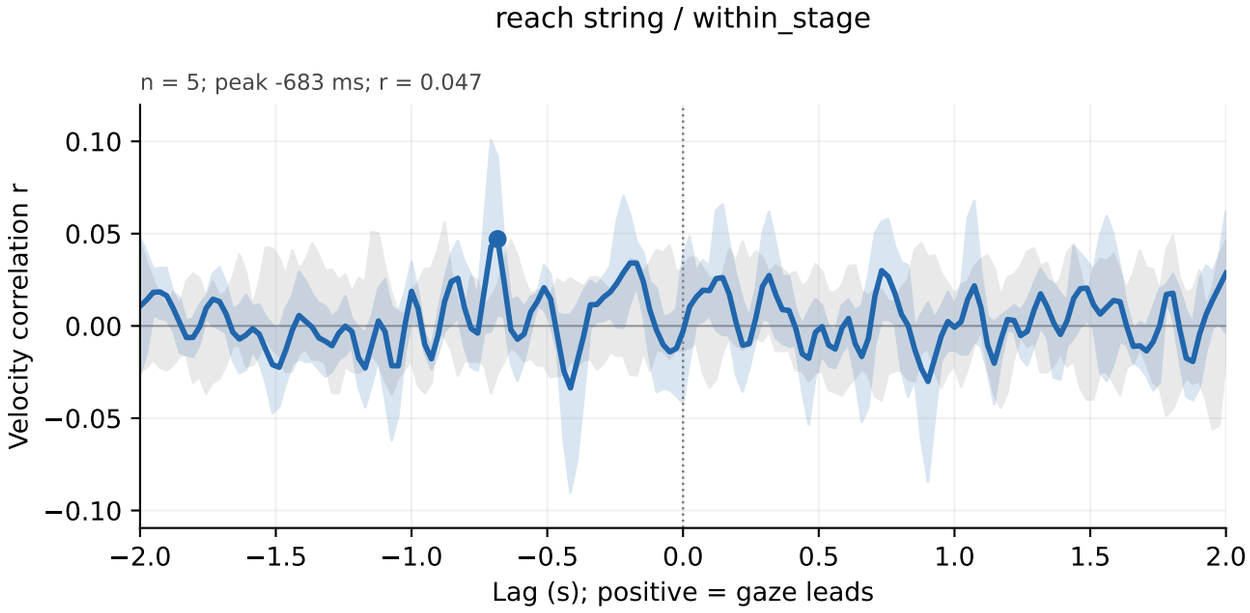
pull through and cinch / within_stage



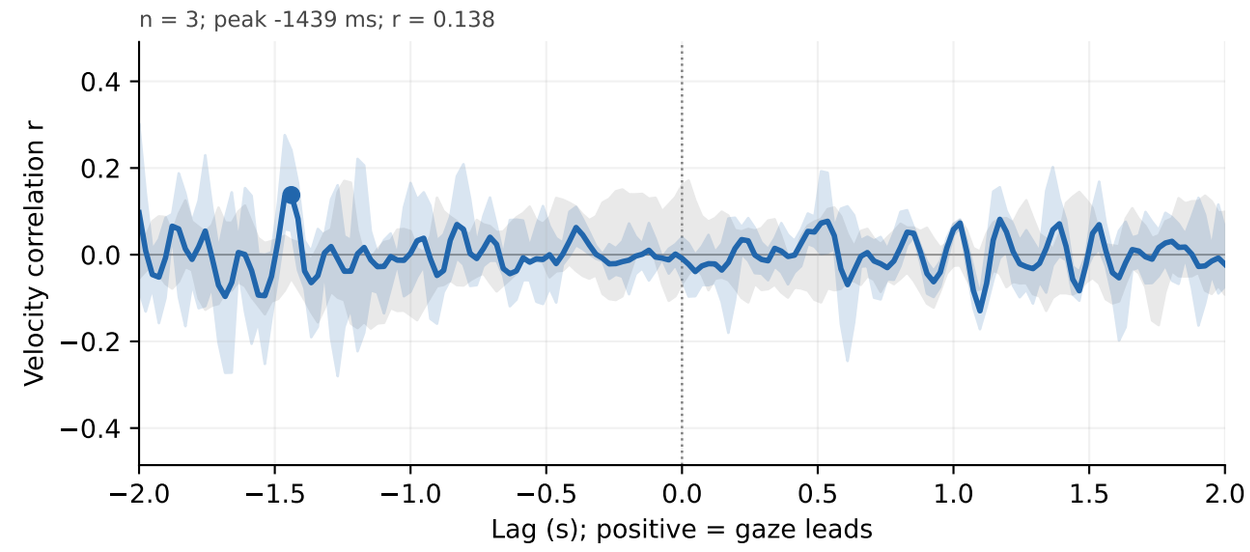
reach string / psm_stage



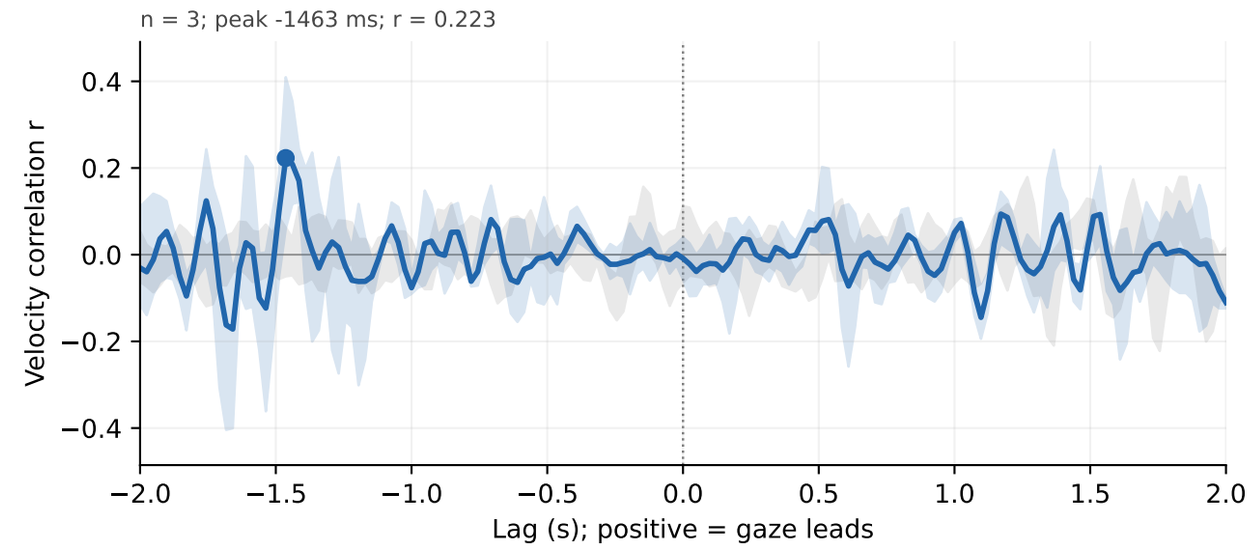
reach string / within_stage



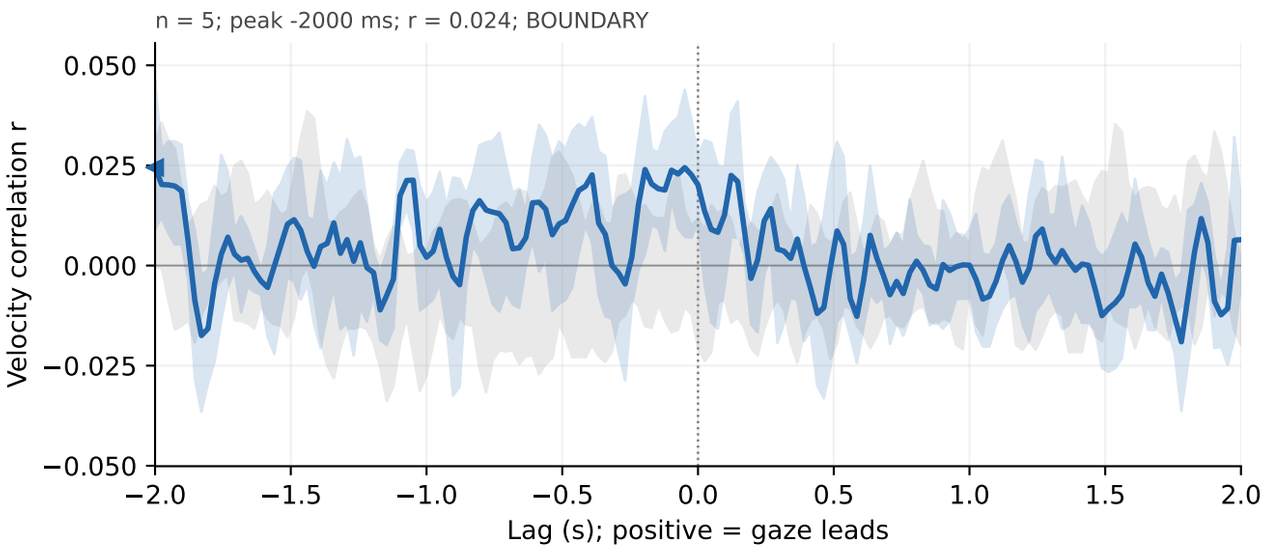
threading string through tower / psm_stage



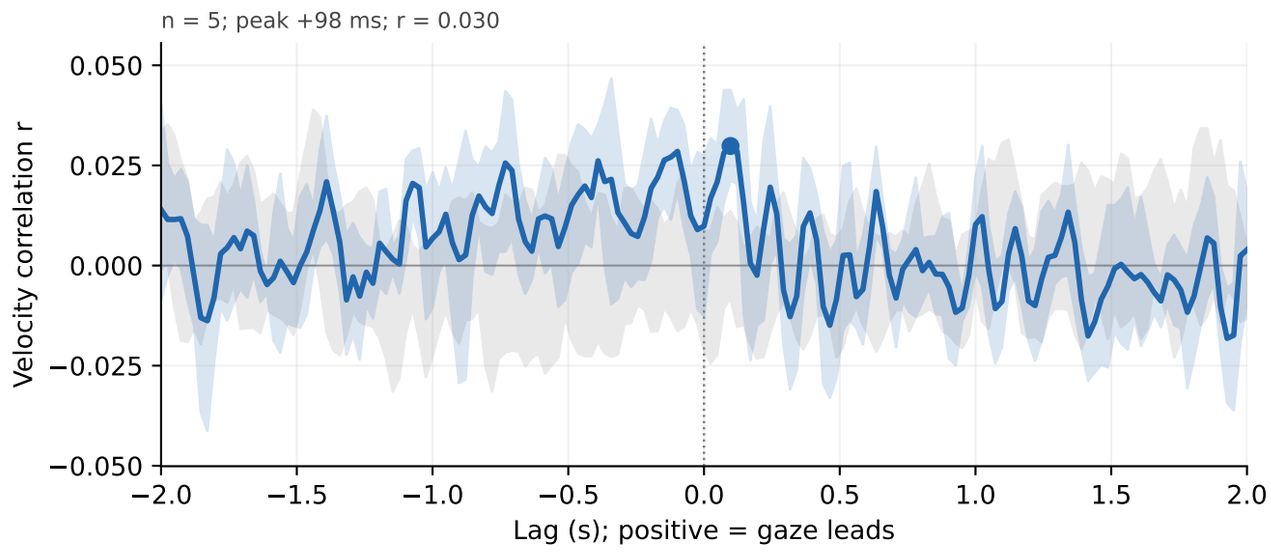
threading string through tower / within_stage



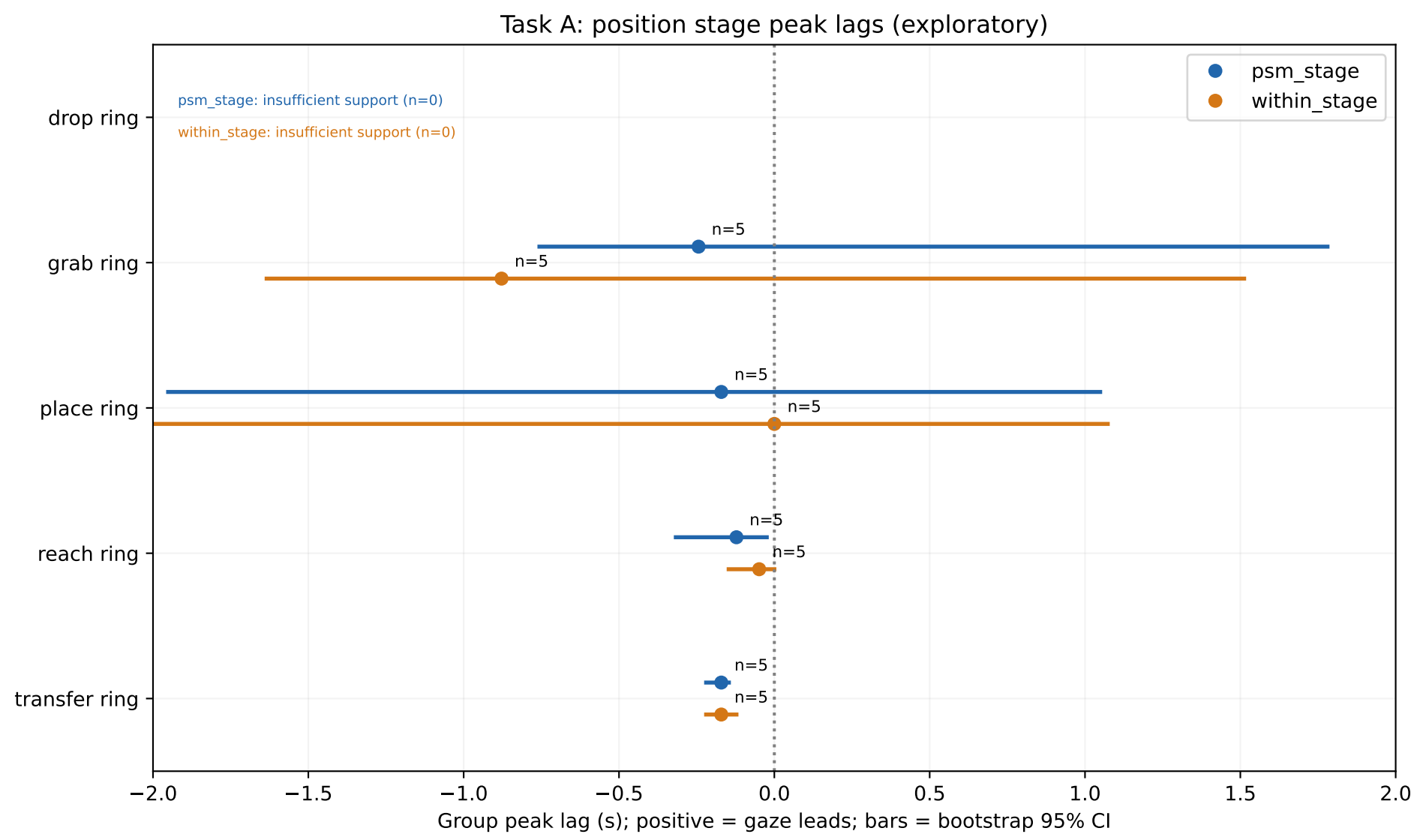
wrap around psm / psm_stage



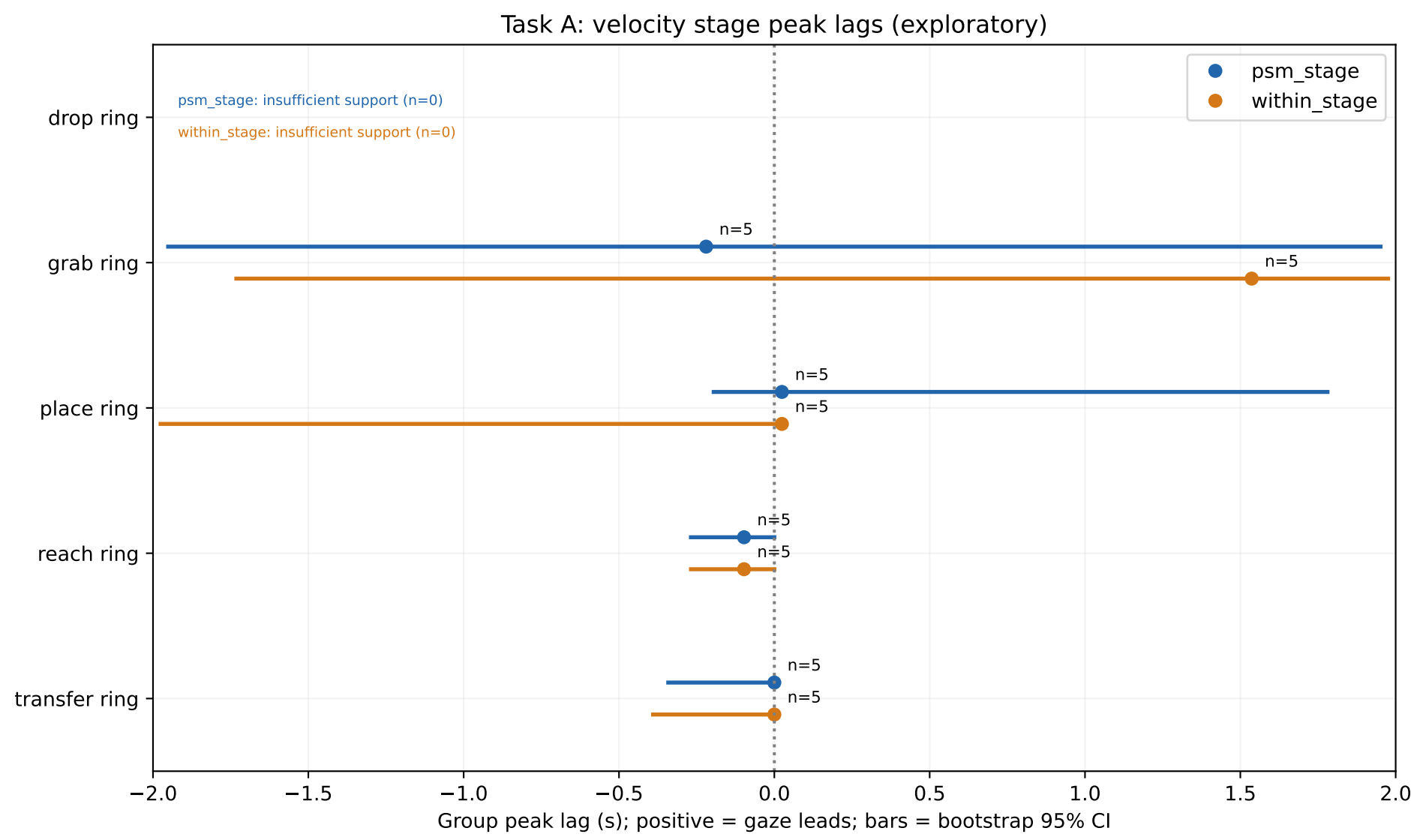
wrap around psm / within_stage



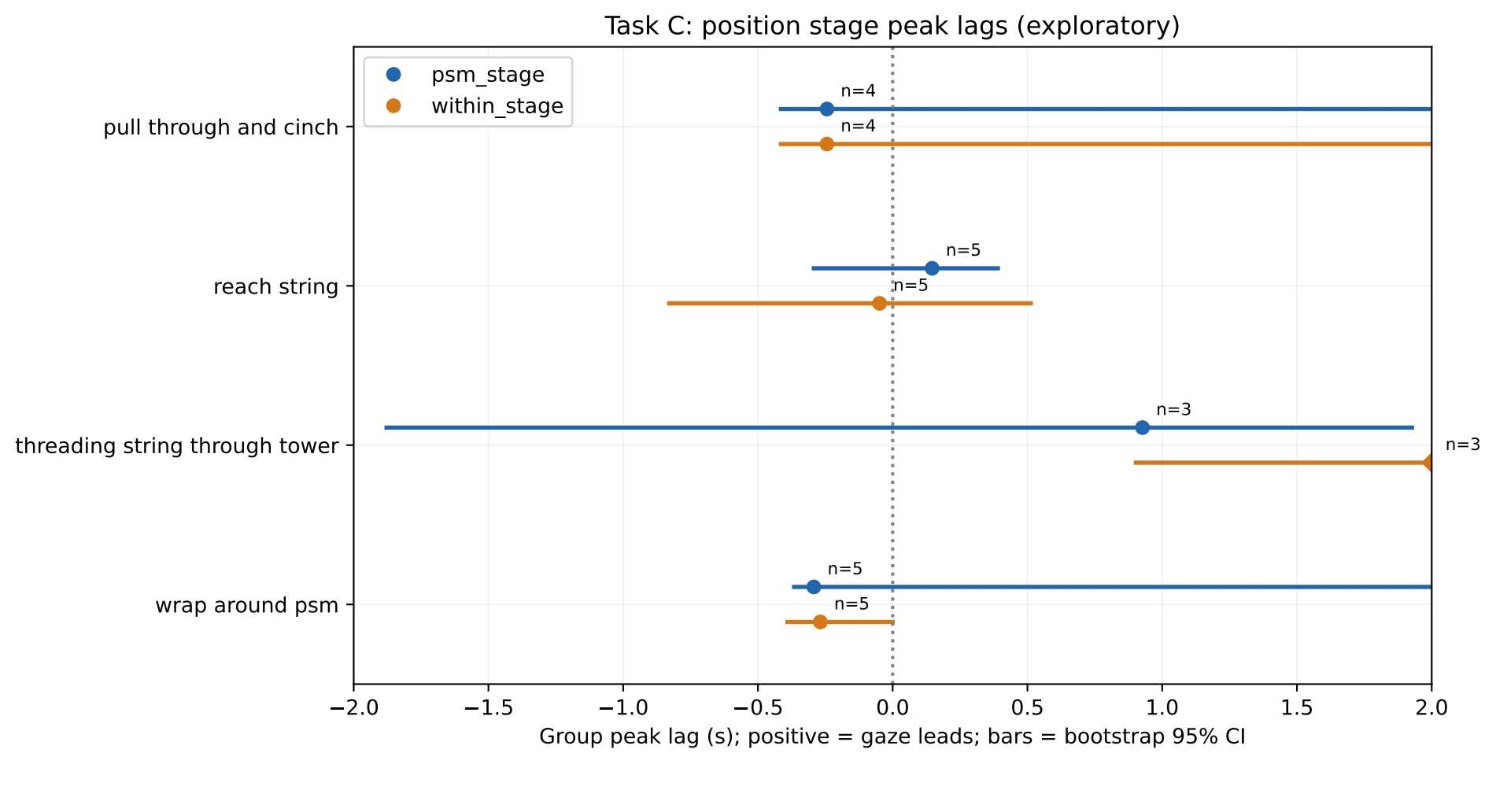
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



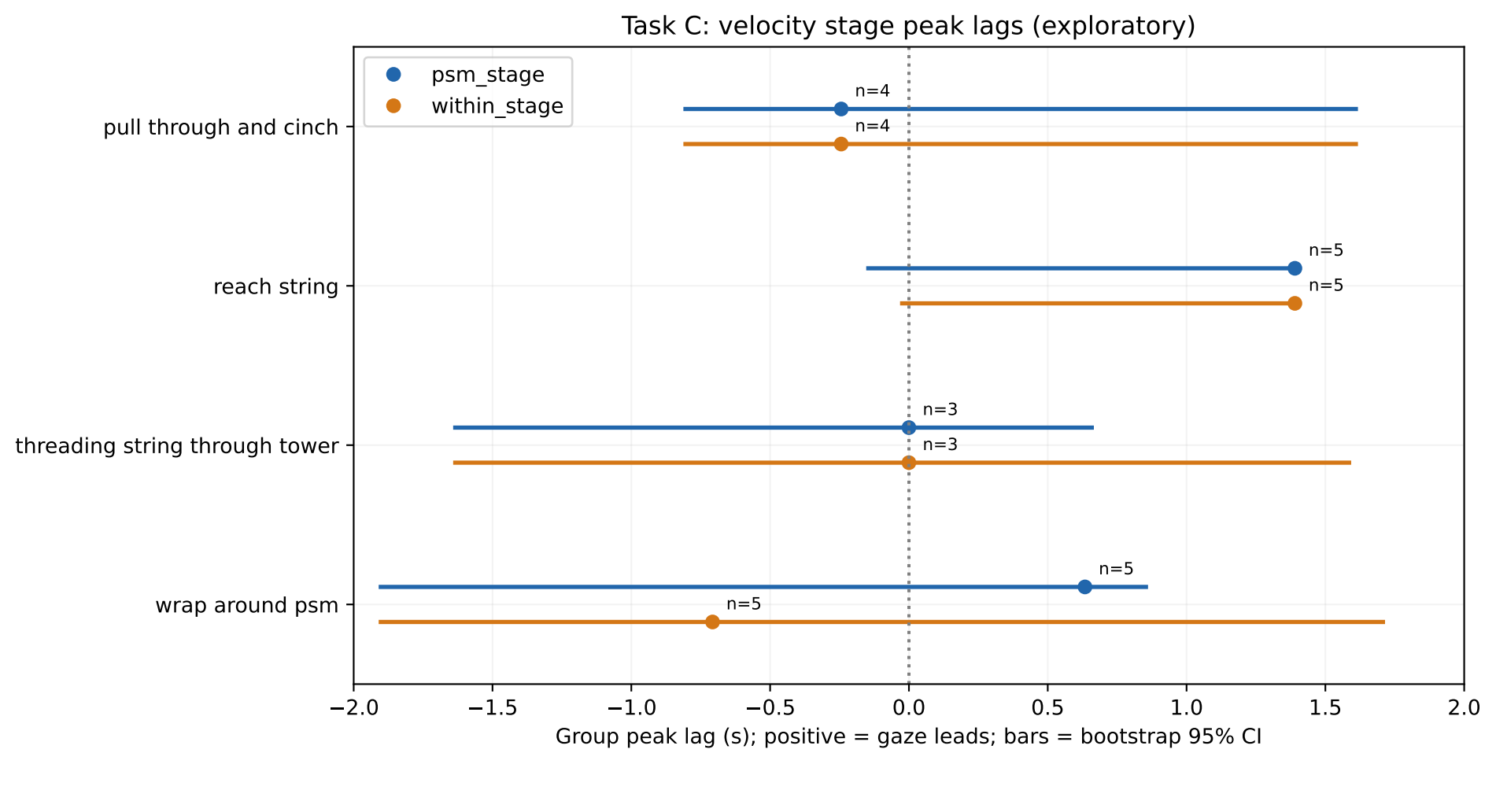
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



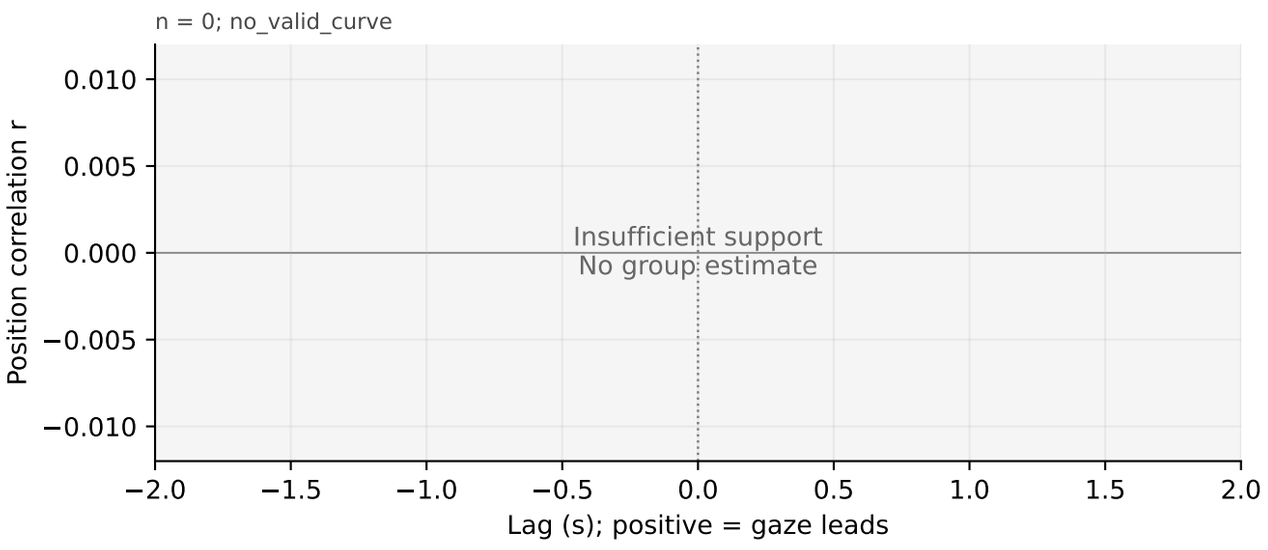
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) · Participant bootstrap (95%) · Group mean

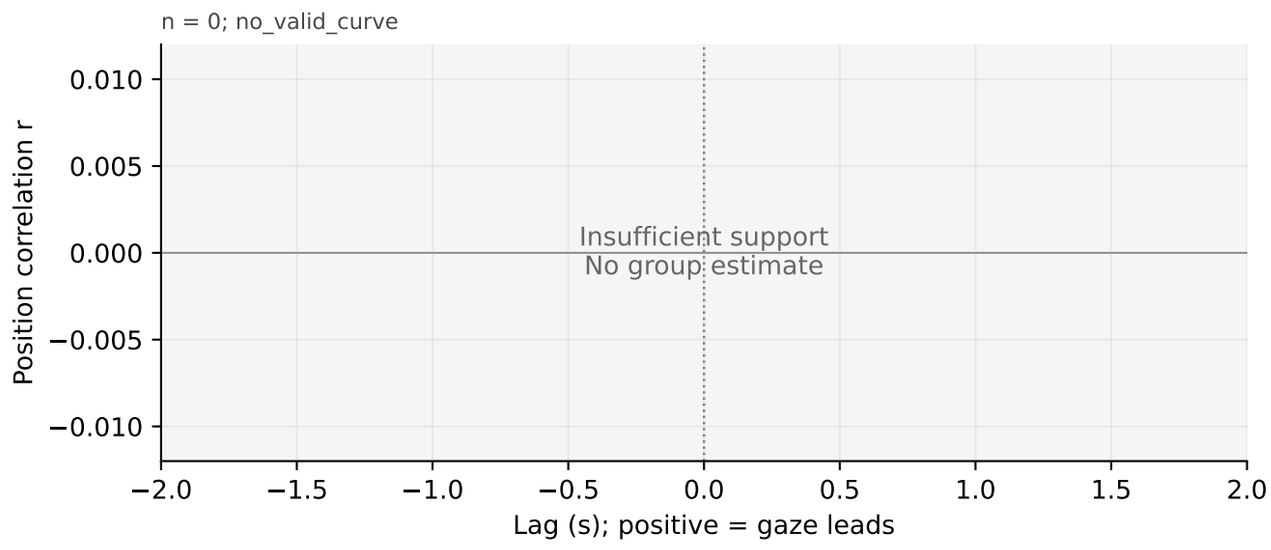
drop ring / psm_stage

n = 0; no_valid_curve



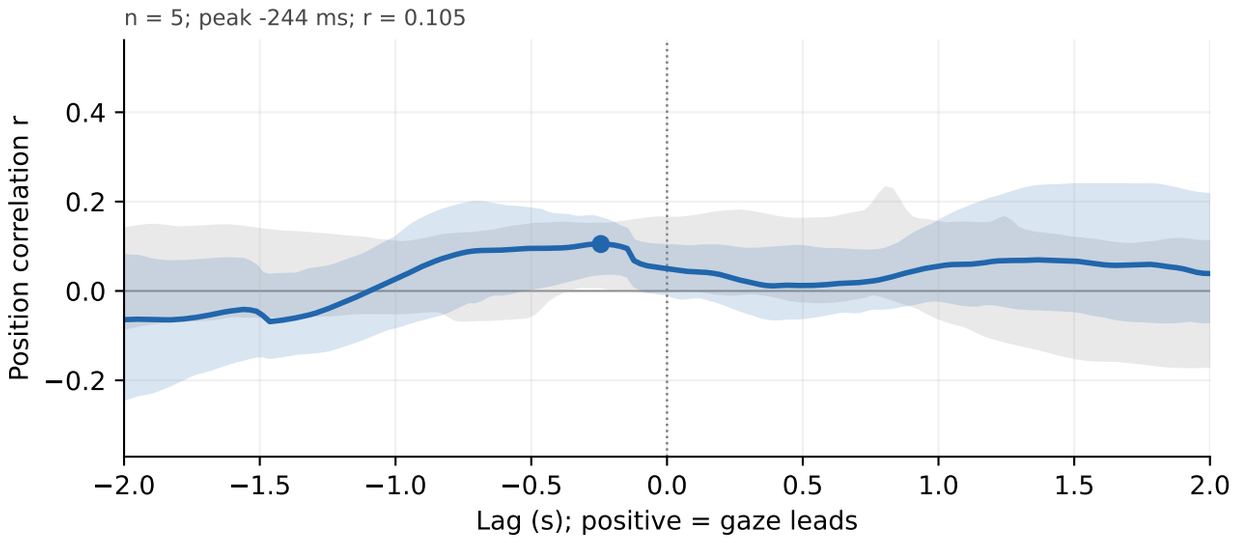
drop ring / within_stage

n = 0; no_valid_curve



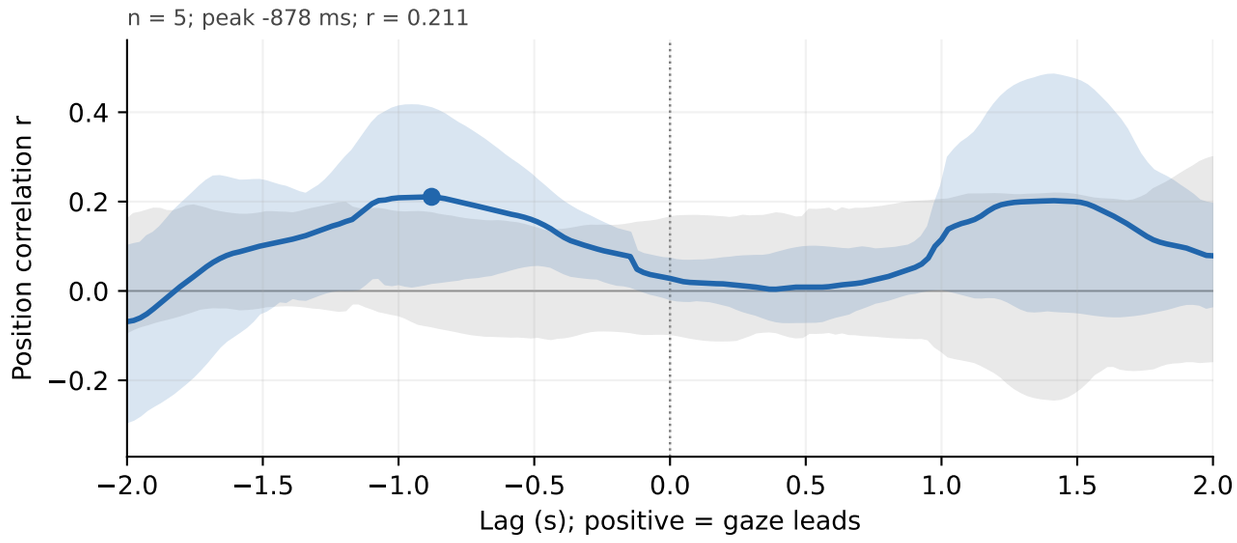
grab ring / psm_stage

n = 5; peak -244 ms; r = 0.105



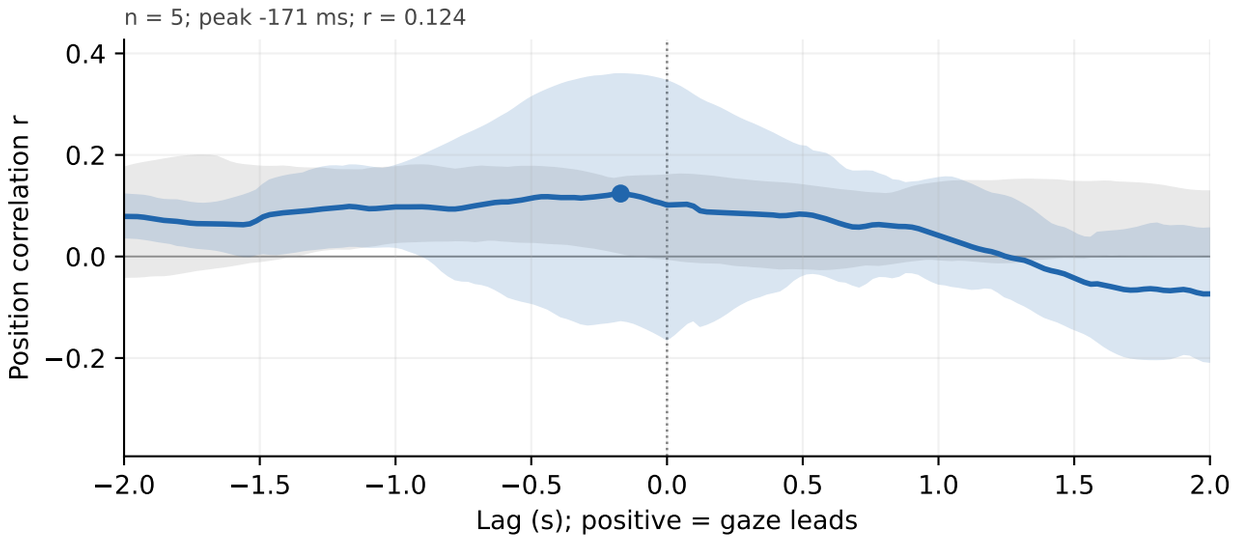
grab ring / within_stage

n = 5; peak -878 ms; r = 0.211



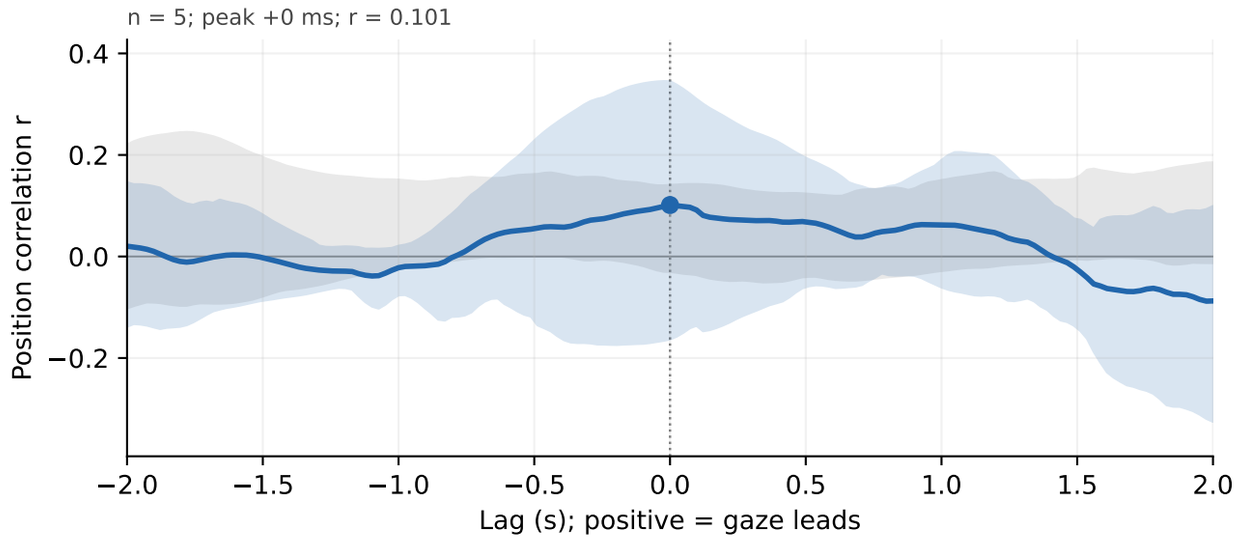
place ring / psm_stage

n = 5; peak -171 ms; r = 0.124



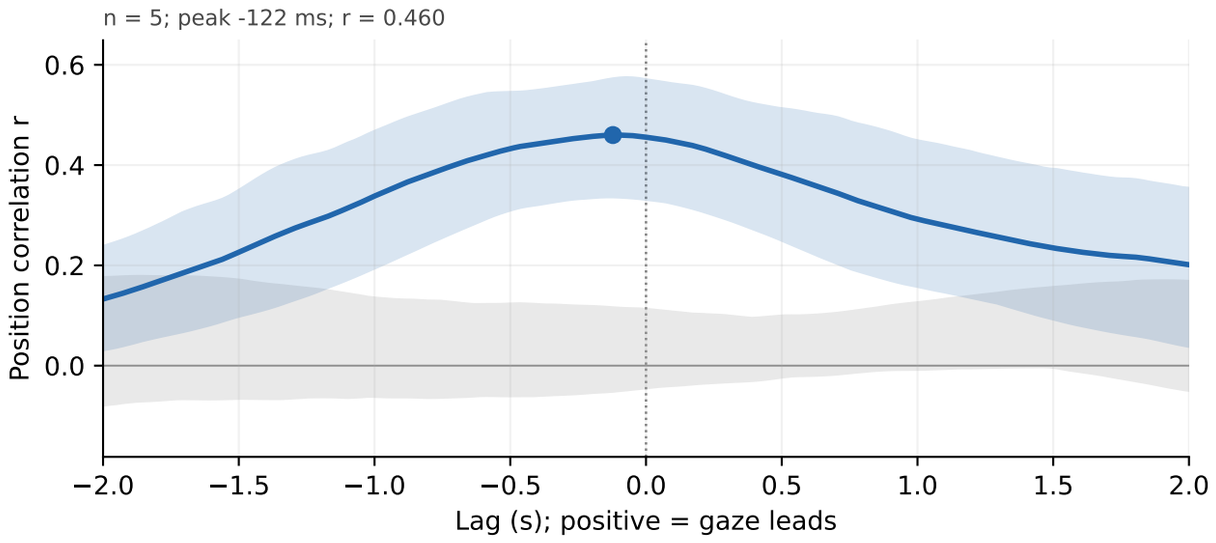
place ring / within_stage

n = 5; peak +0 ms; r = 0.101



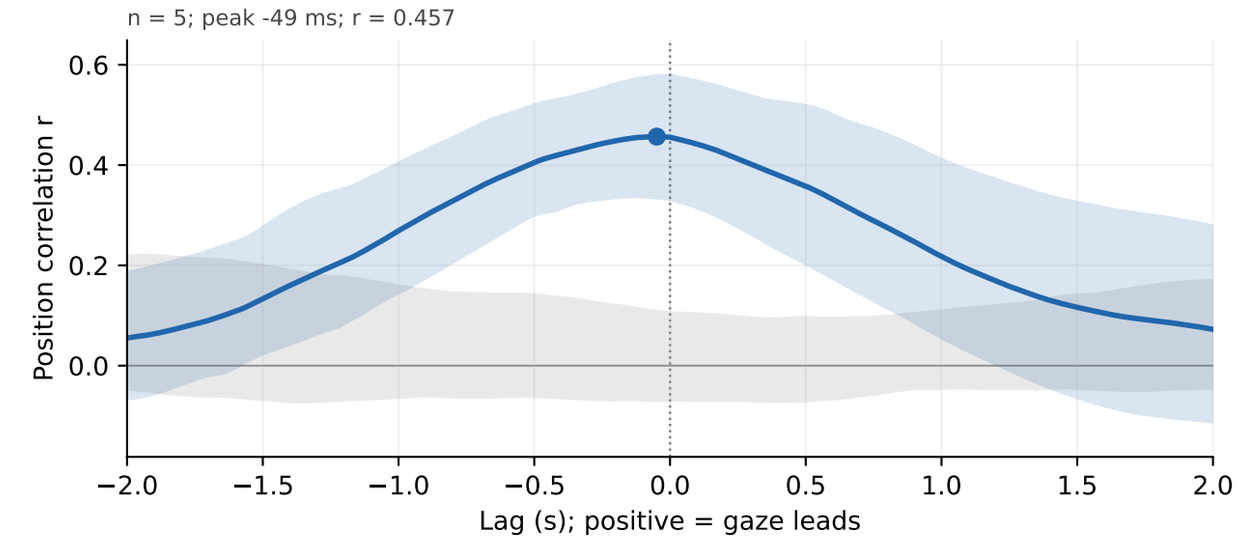
reach ring / psm_stage

n = 5; peak -122 ms; r = 0.460



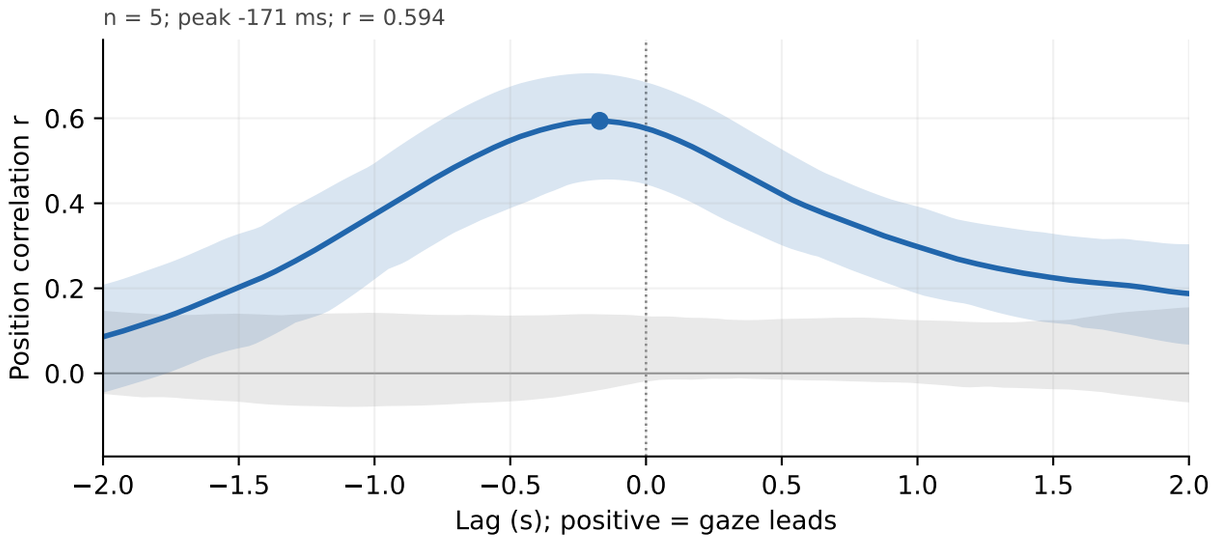
reach ring / within_stage

n = 5; peak -49 ms; r = 0.457



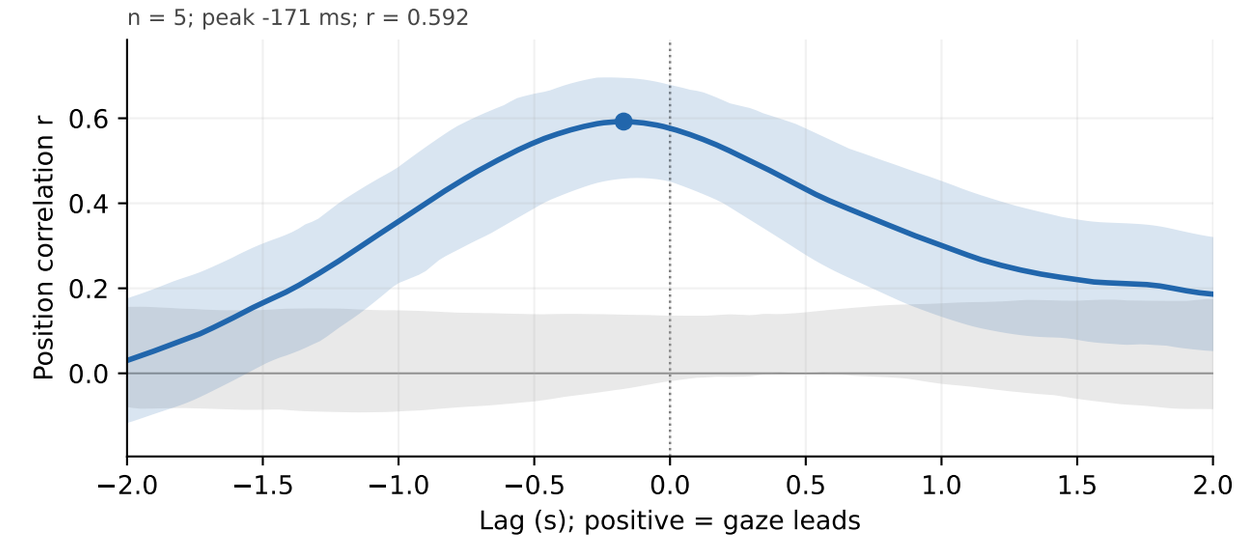
transfer ring / psm_stage

n = 5; peak -171 ms; r = 0.594



transfer ring / within_stage

n = 5; peak -171 ms; r = 0.592



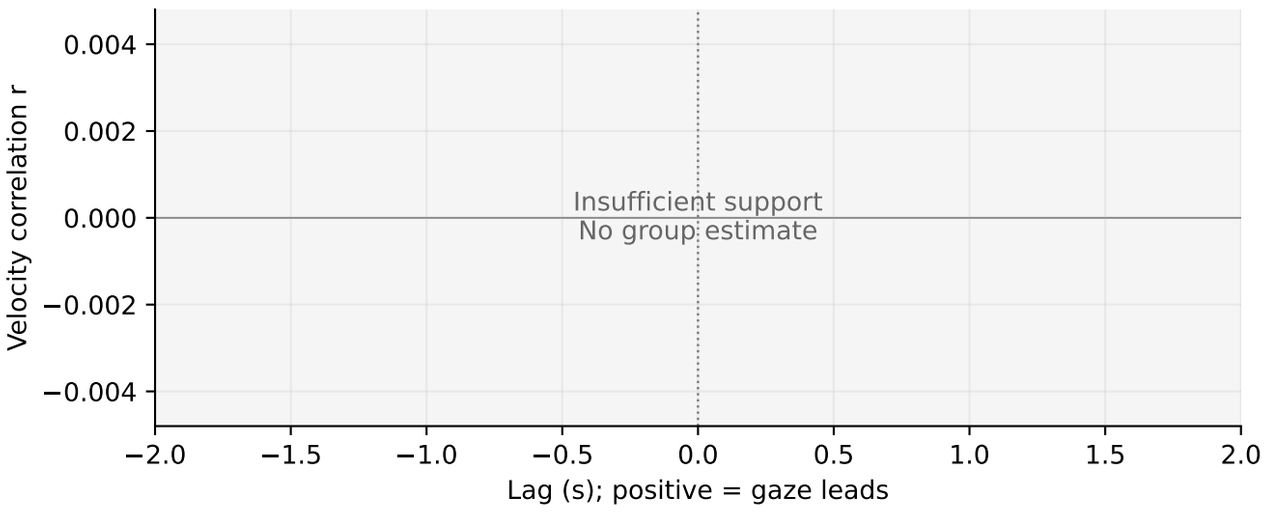
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

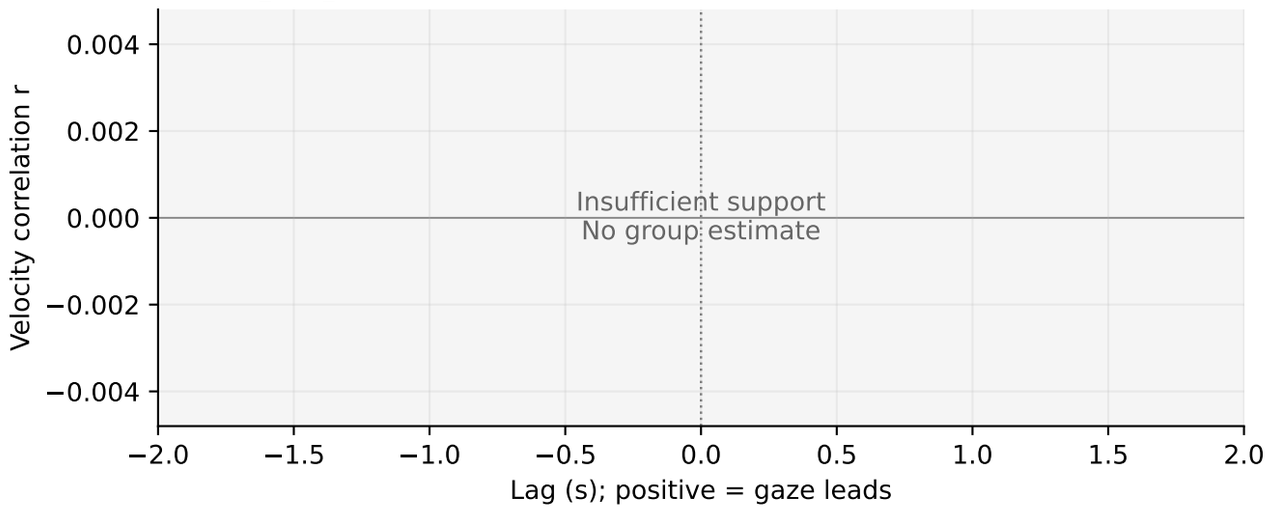
drop ring / psm_stage

n = 0; no_valid_curve



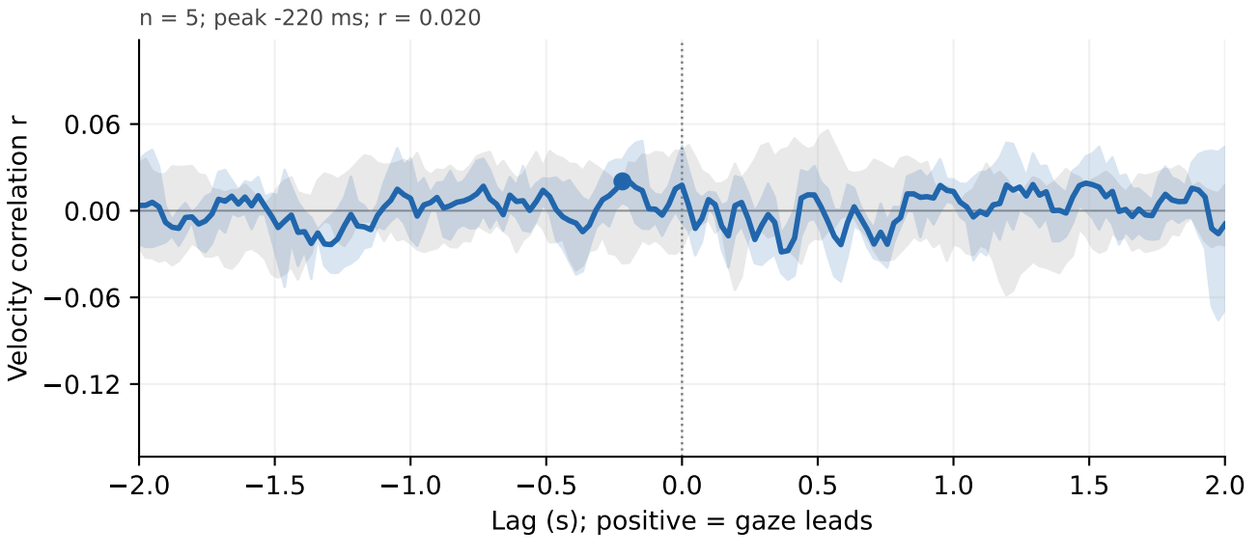
drop ring / within_stage

n = 0; no_valid_curve



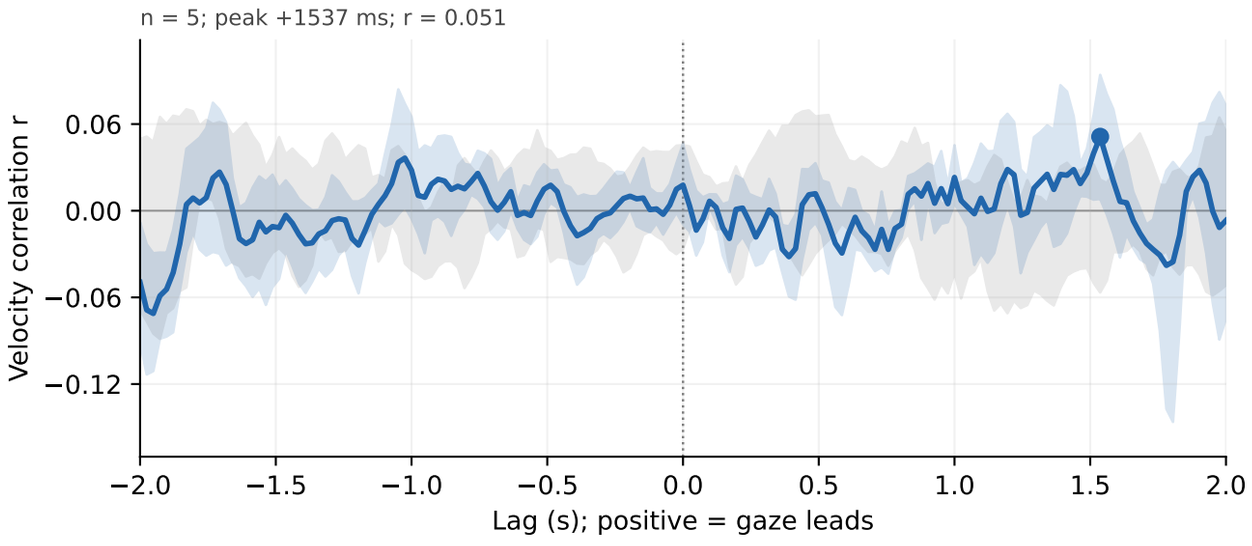
grab ring / psm_stage

n = 5; peak -220 ms; r = 0.020



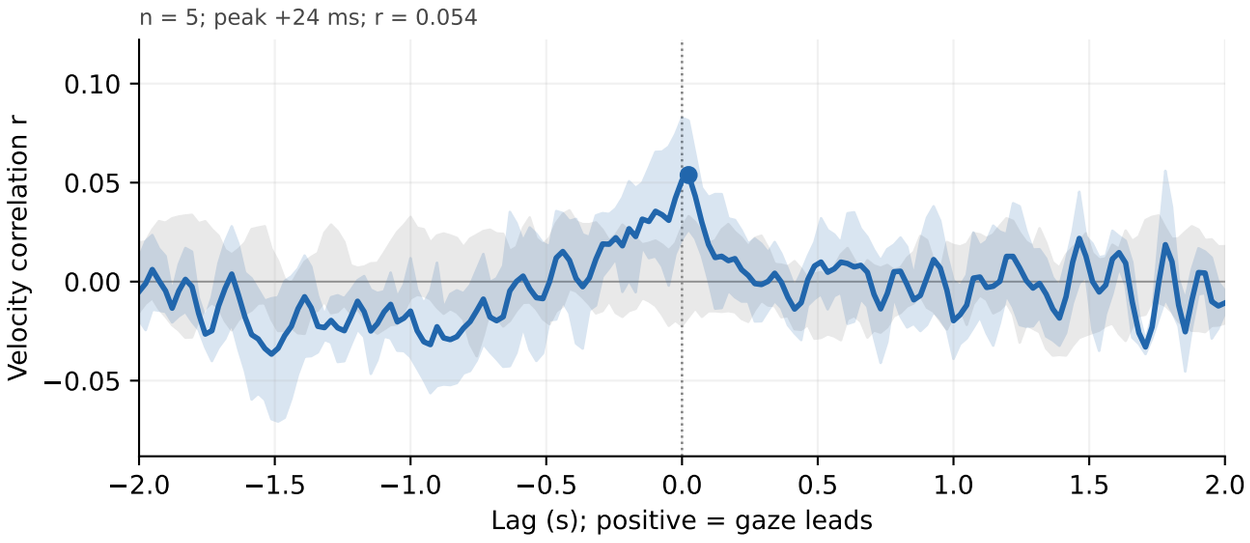
grab ring / within_stage

n = 5; peak +1537 ms; r = 0.051



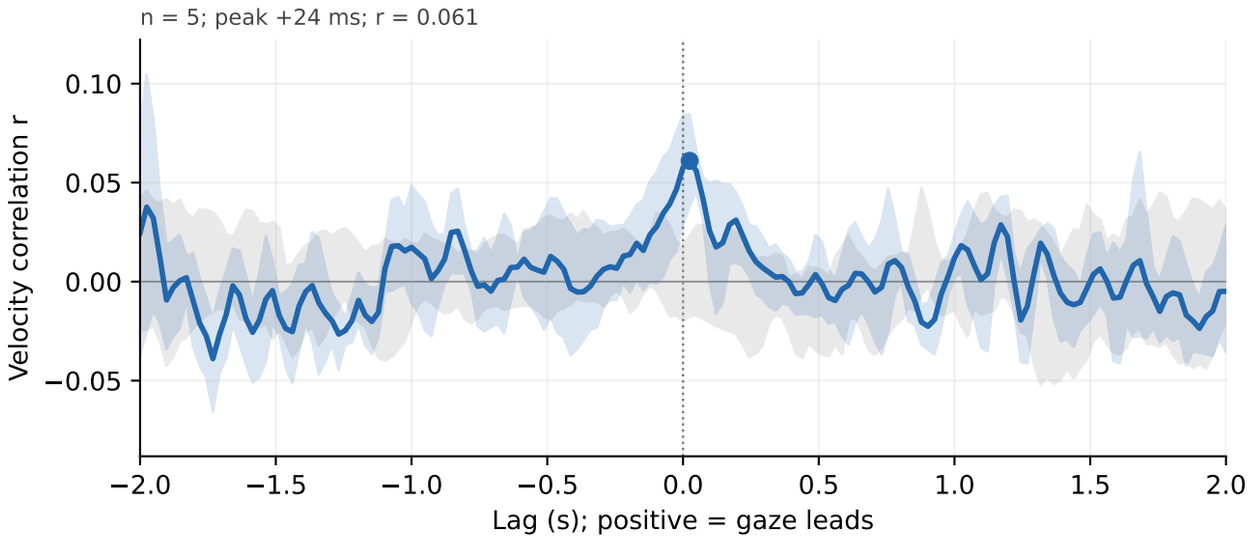
place ring / psm_stage

n = 5; peak +24 ms; r = 0.054



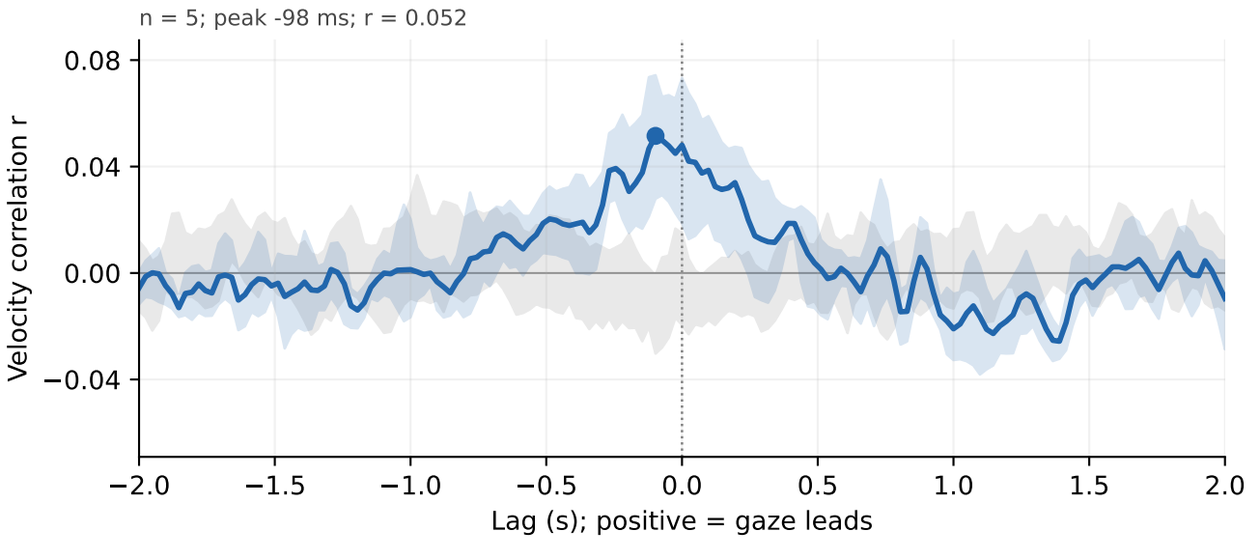
place ring / within_stage

n = 5; peak +24 ms; r = 0.061



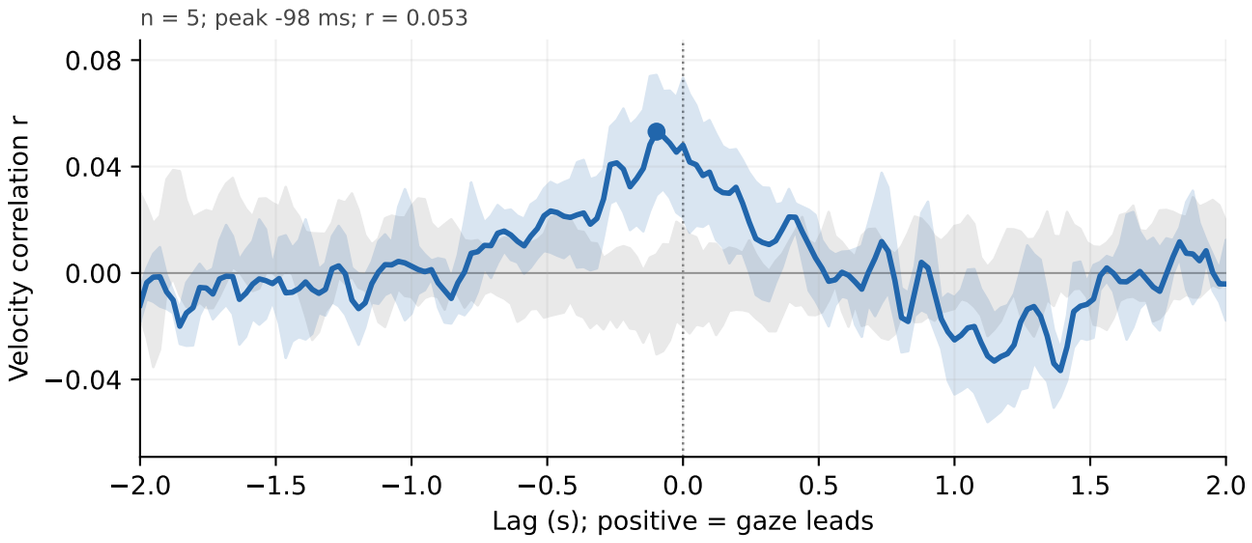
reach ring / psm_stage

n = 5; peak -98 ms; r = 0.052



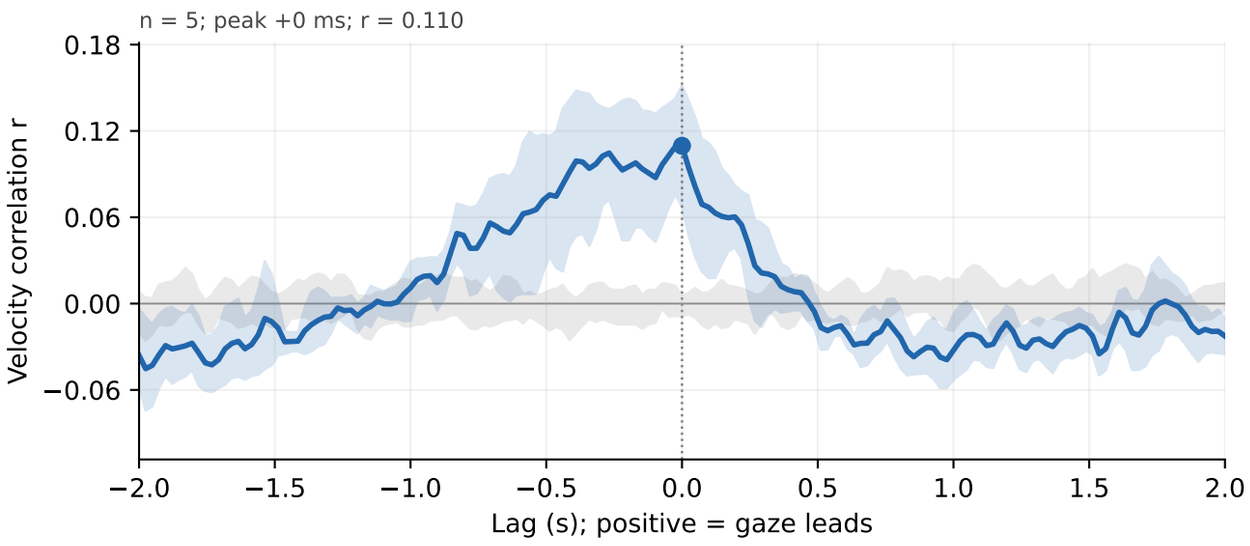
reach ring / within_stage

n = 5; peak -98 ms; r = 0.053



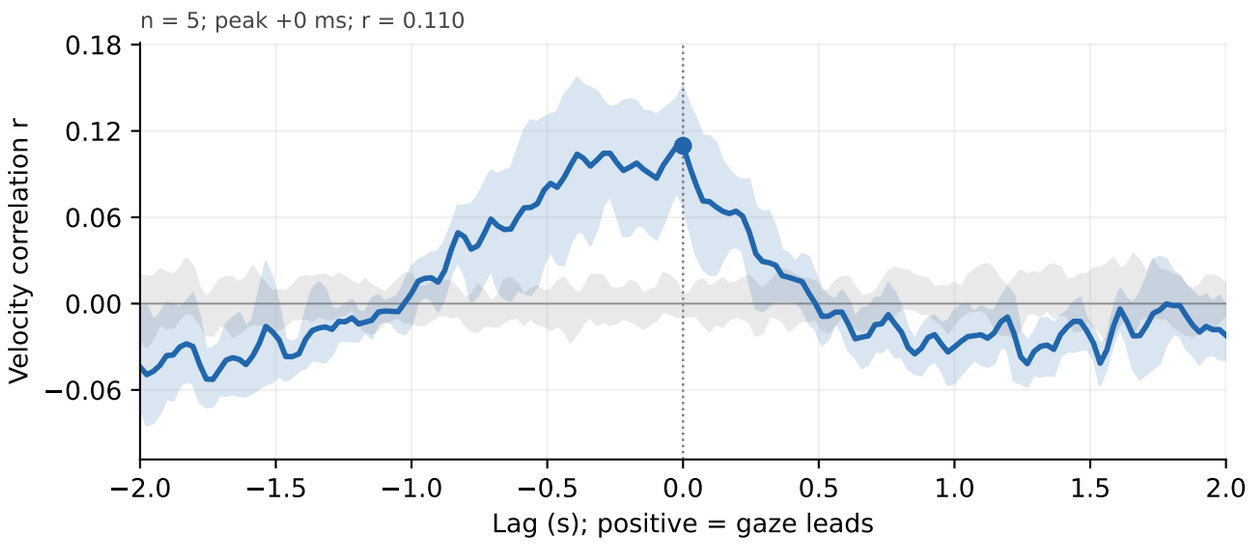
transfer ring / psm_stage

n = 5; peak +0 ms; r = 0.110



transfer ring / within_stage

n = 5; peak +0 ms; r = 0.110

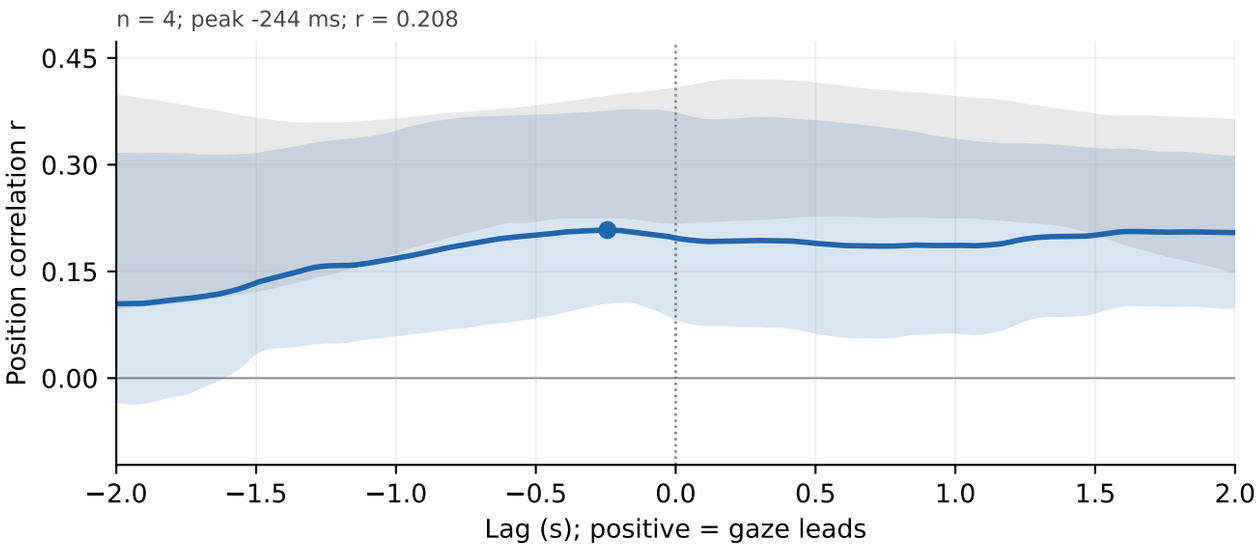


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

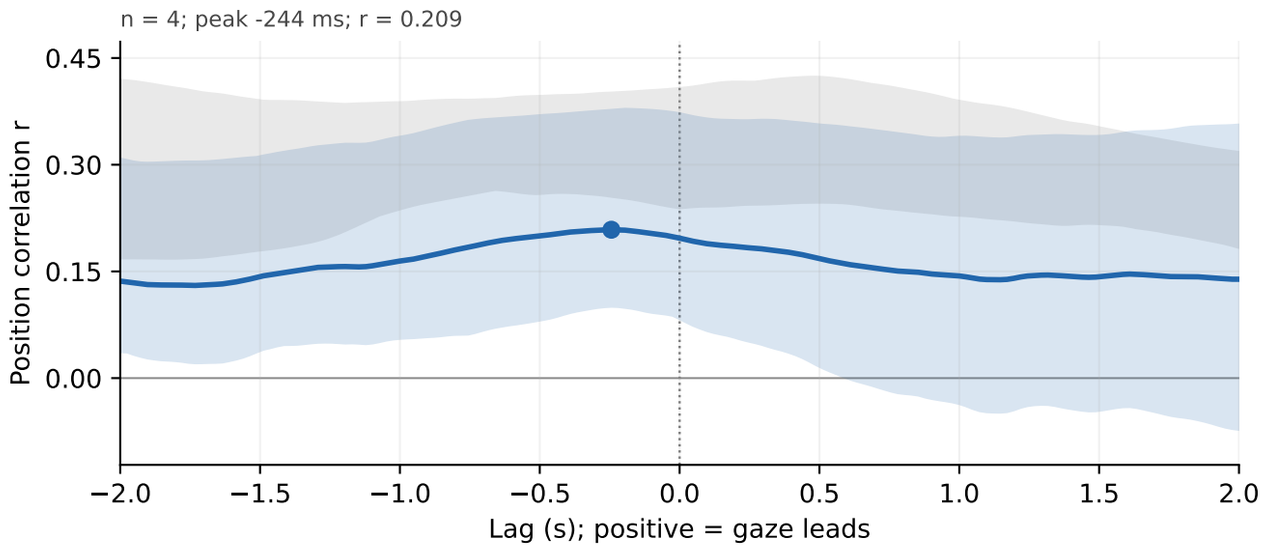
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

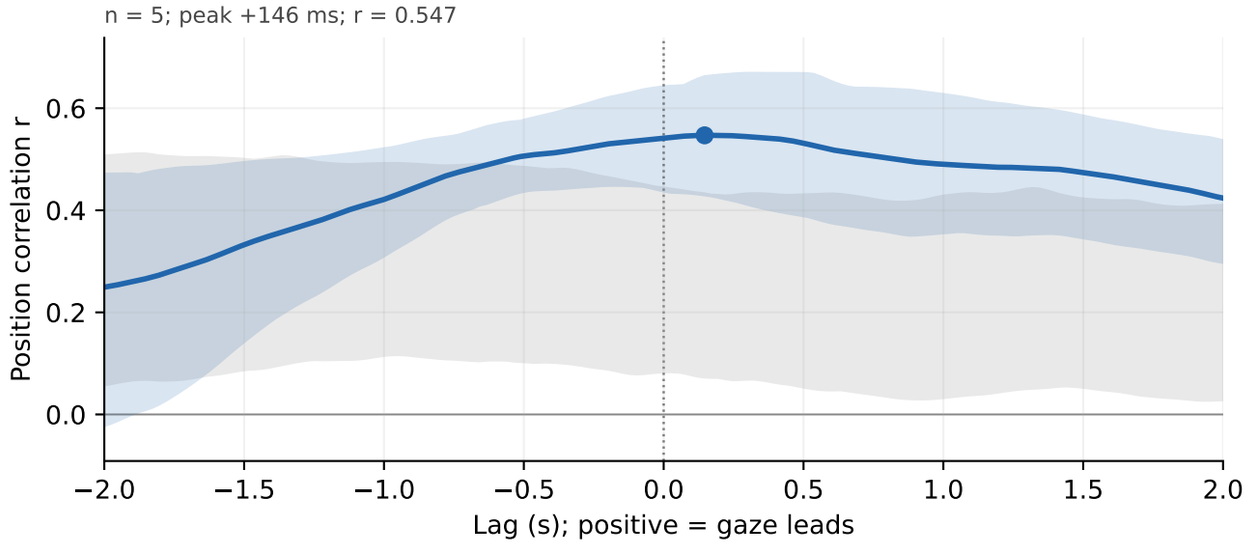
pull through and cinch / psm_stage



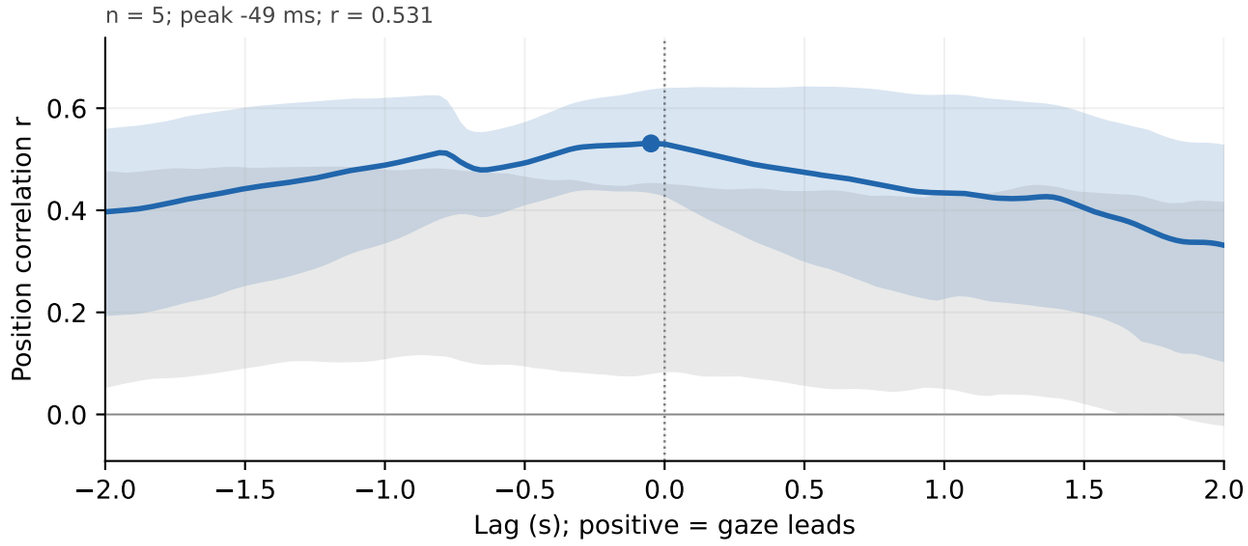
pull through and cinch / within_stage



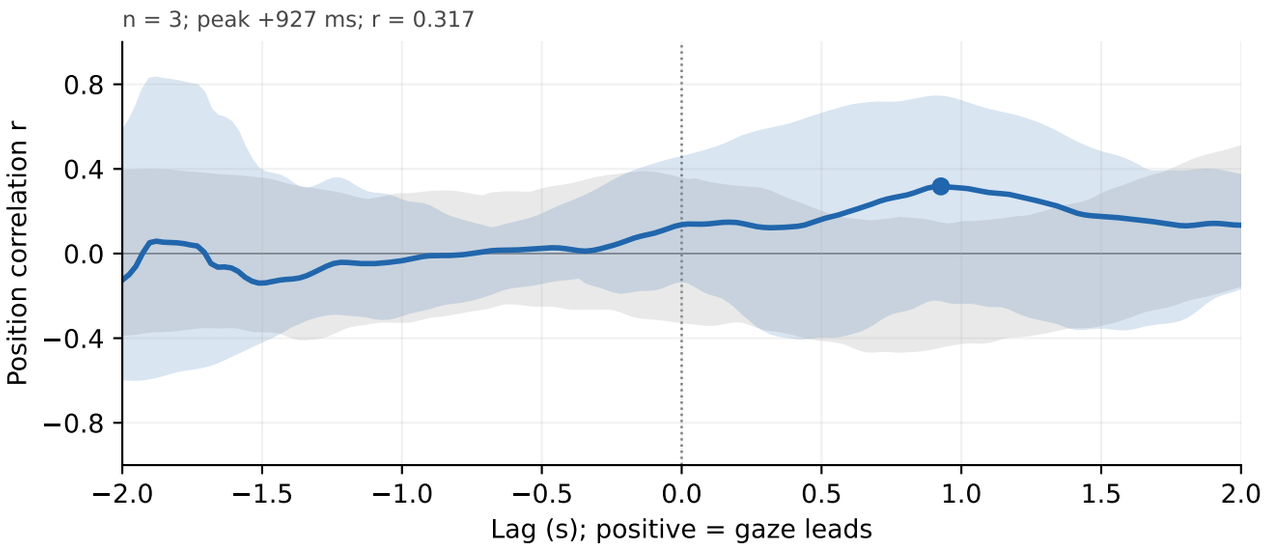
reach string / psm_stage



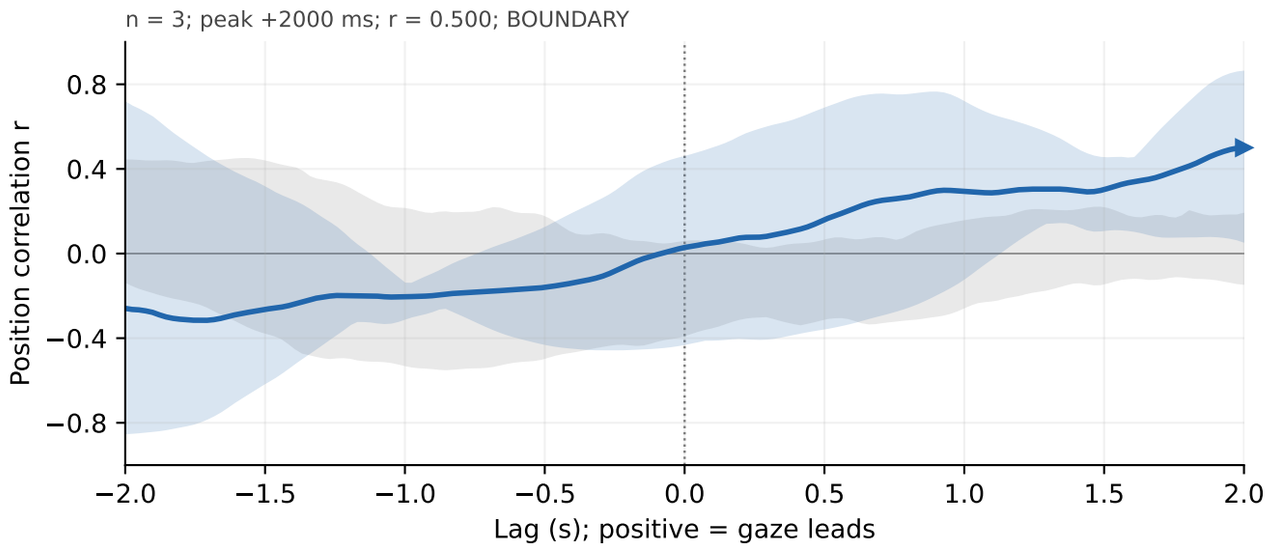
reach string / within_stage



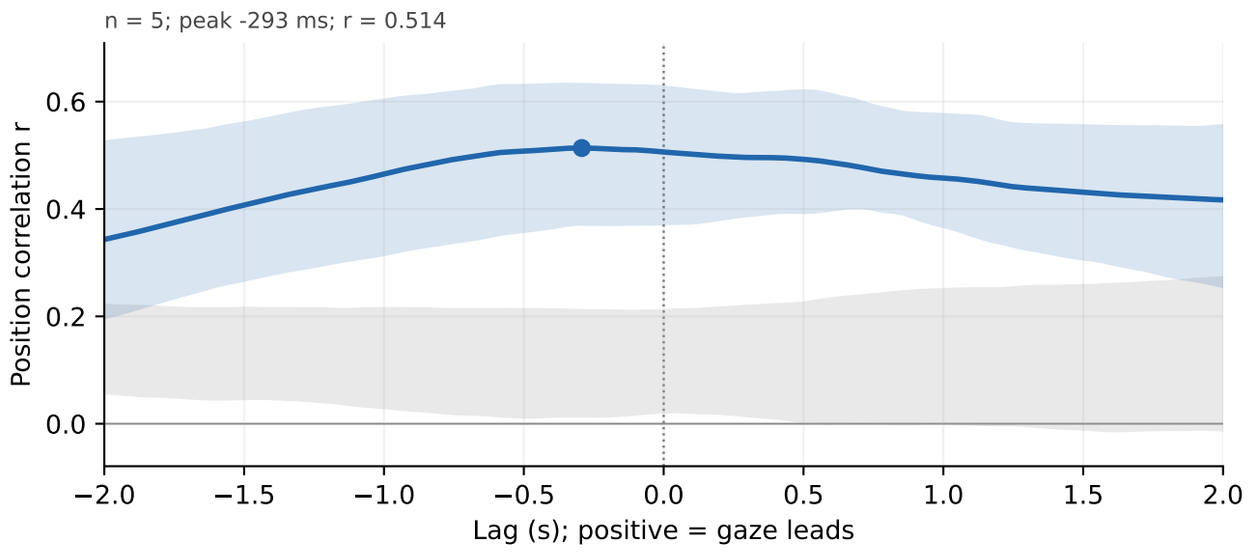
threading string through tower / psm_stage



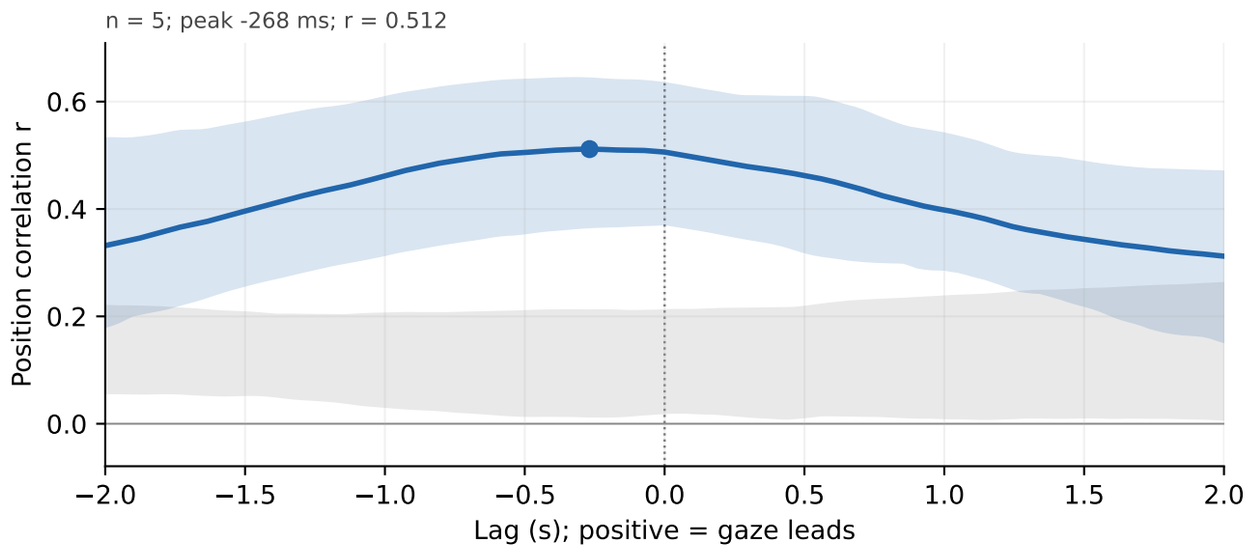
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

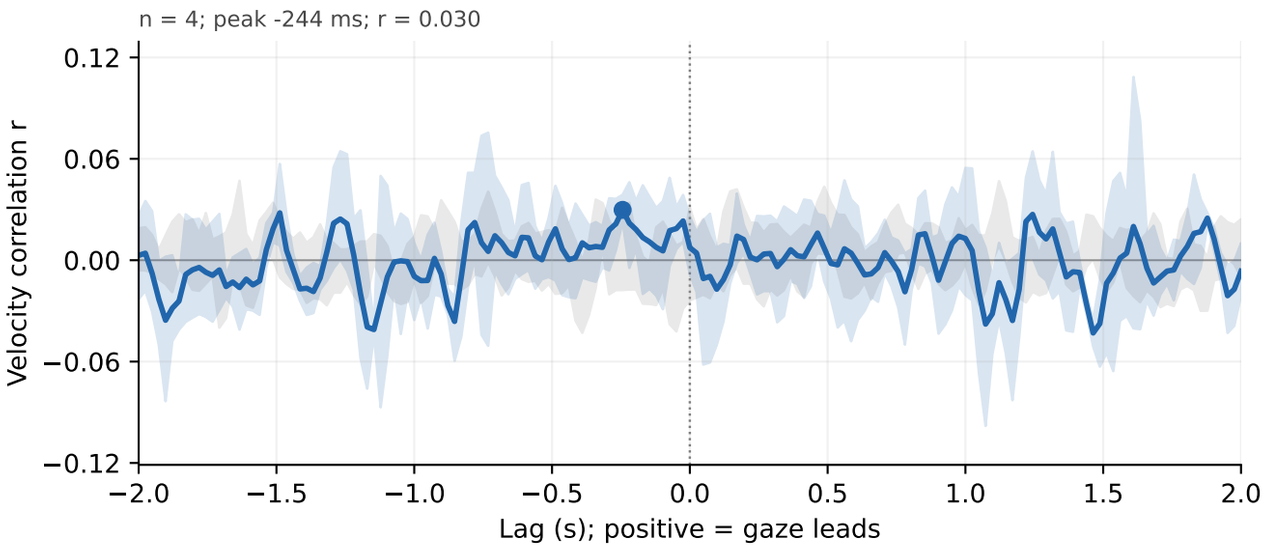


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

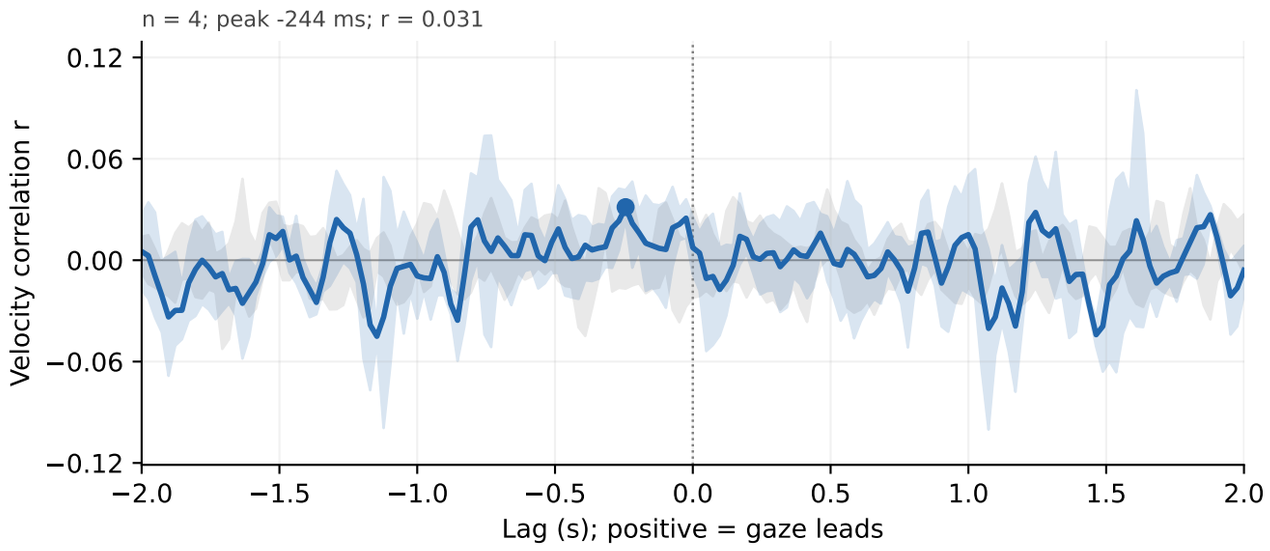
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

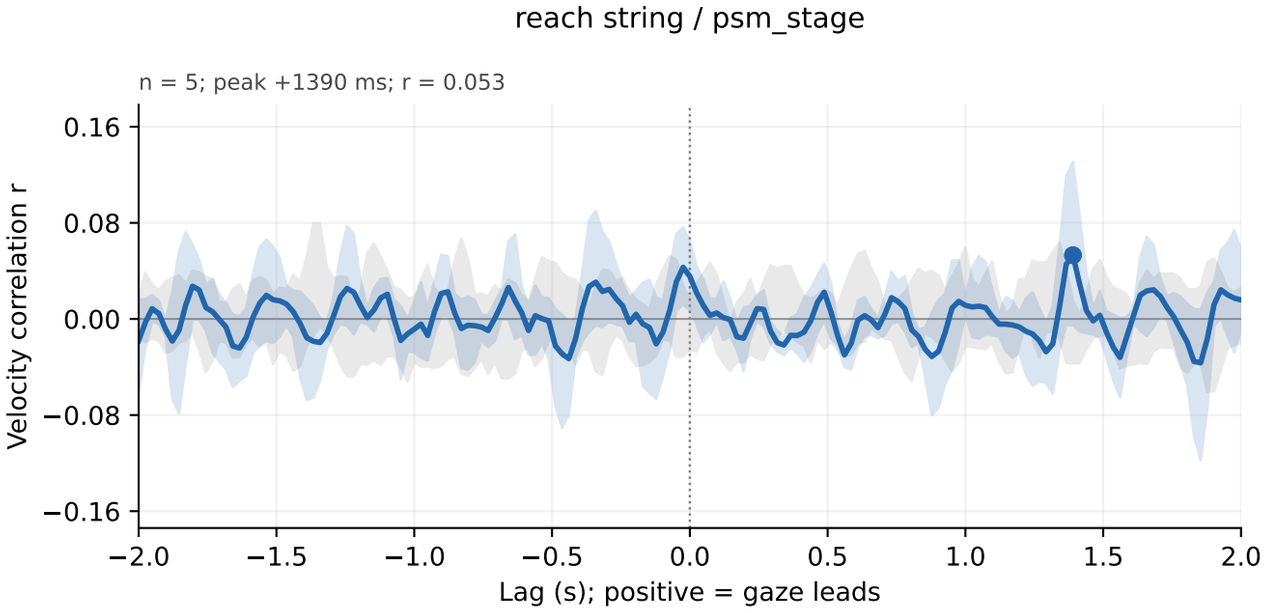
pull through and cinch / psm_stage



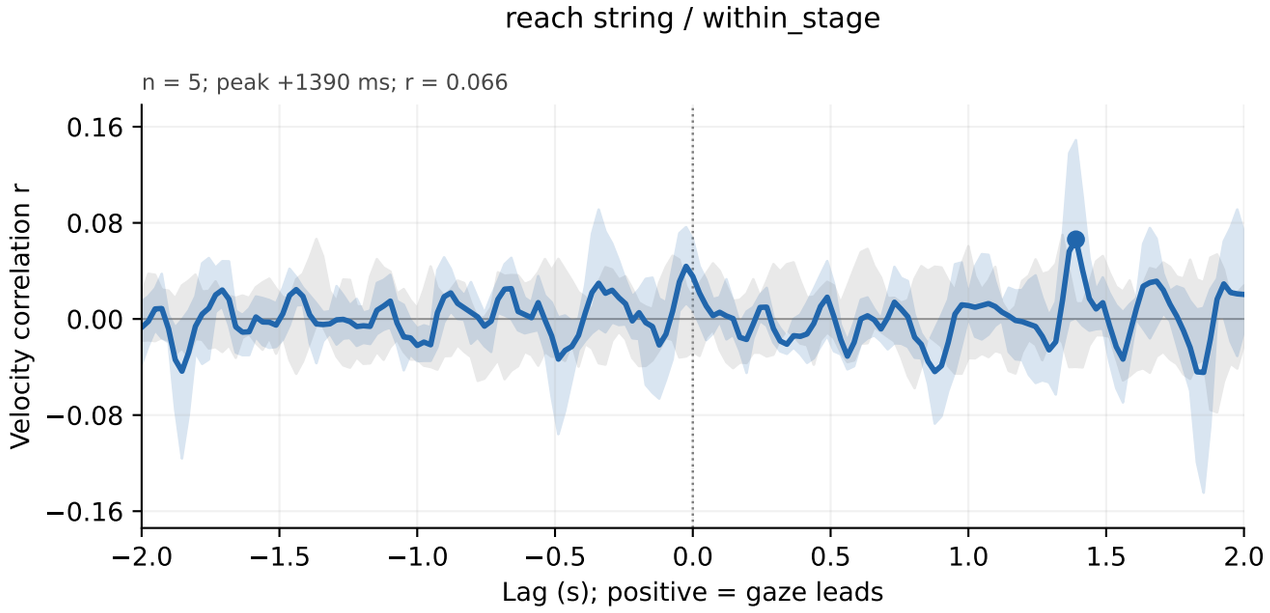
pull through and cinch / within_stage



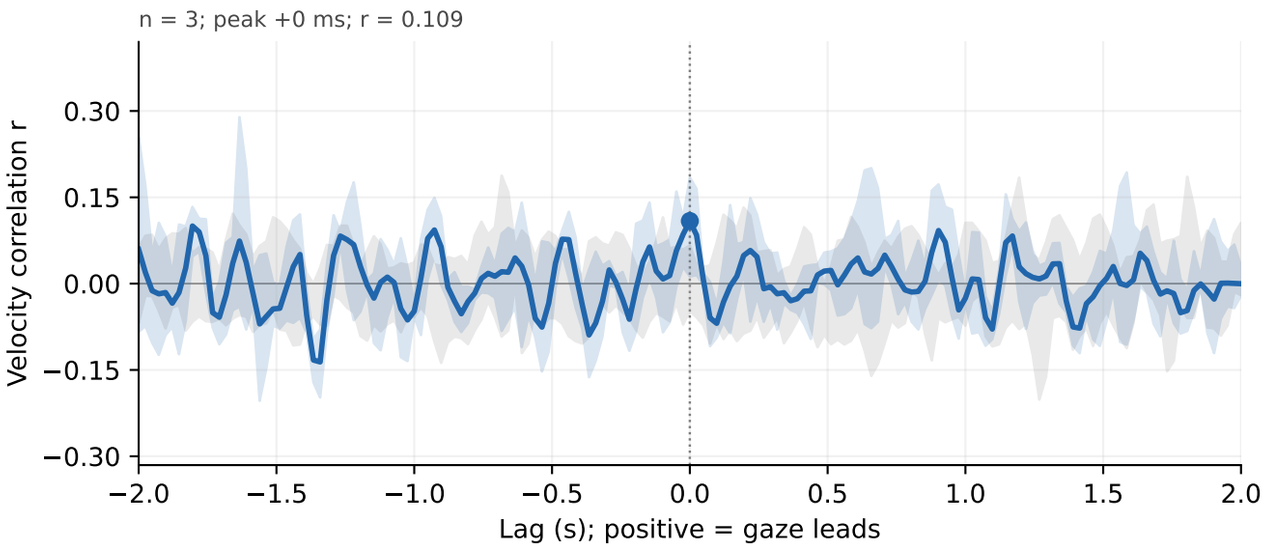
reach string / psm_stage



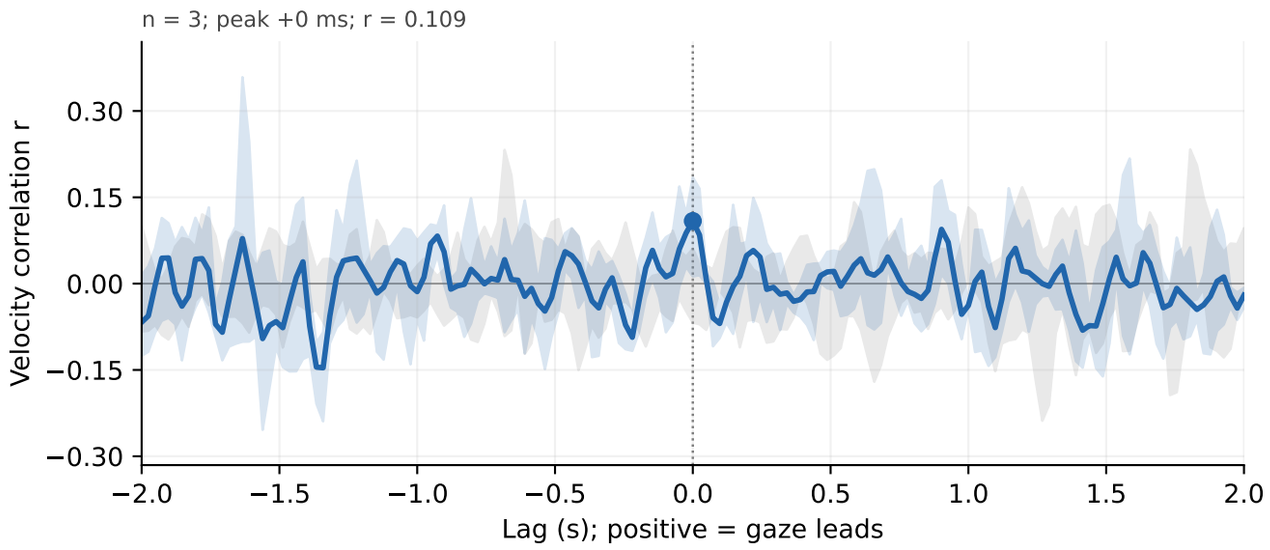
reach string / within_stage



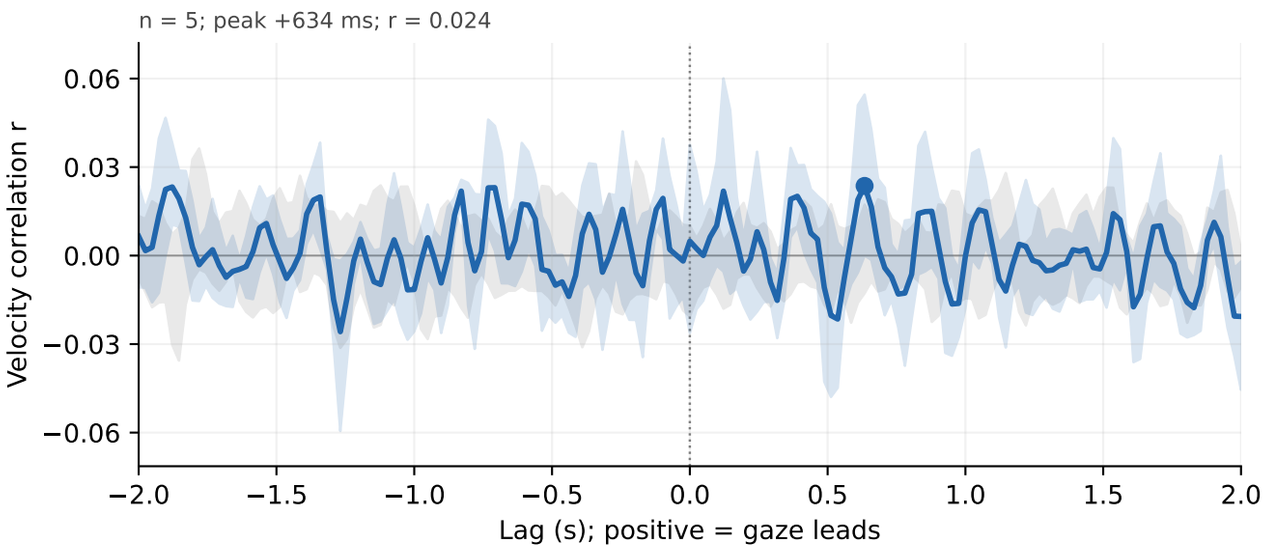
threading string through tower / psm_stage



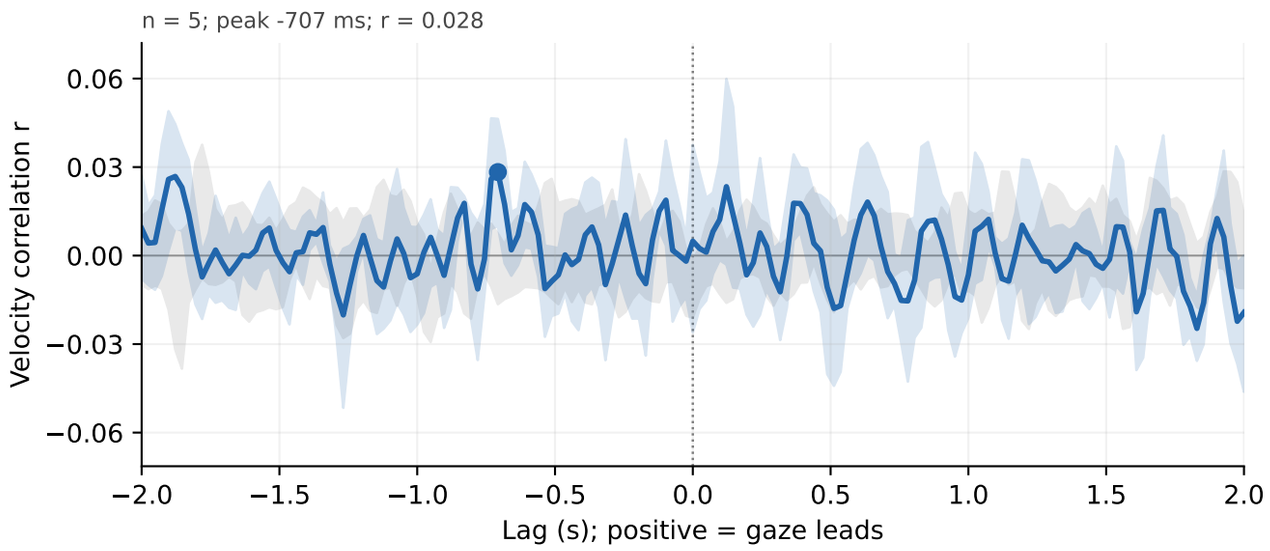
threading string through tower / within_stage



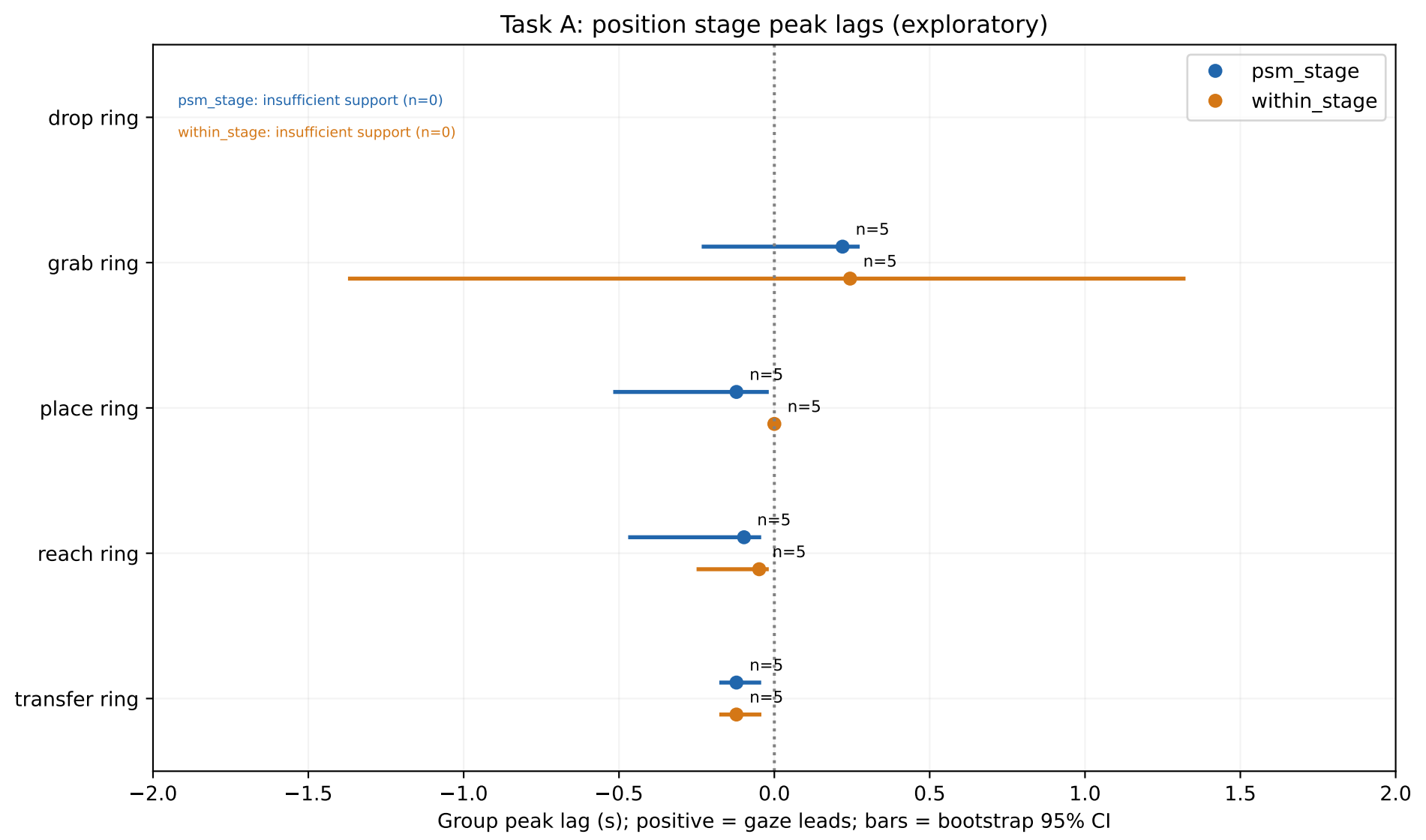
wrap around psm / psm_stage



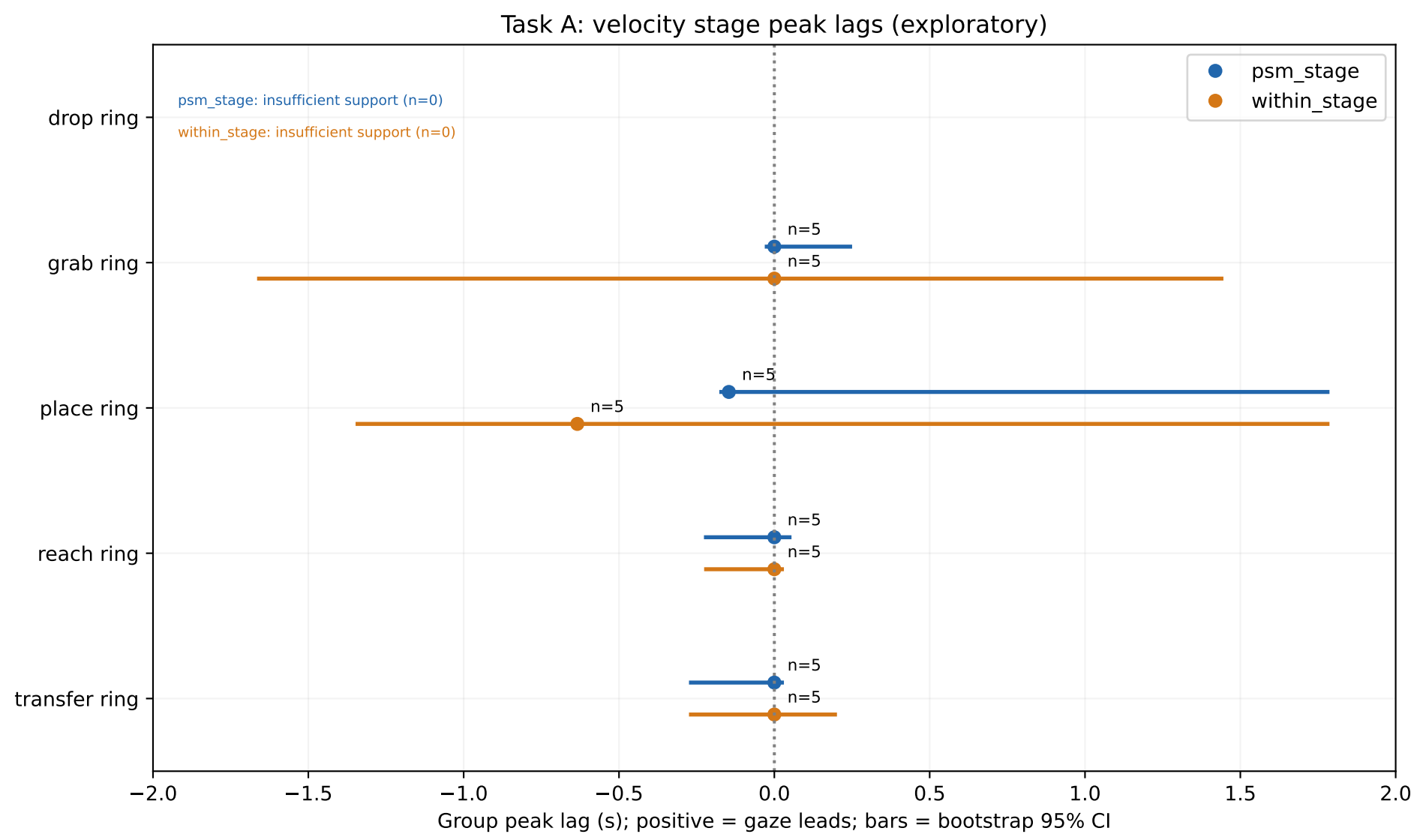
wrap around psm / within_stage



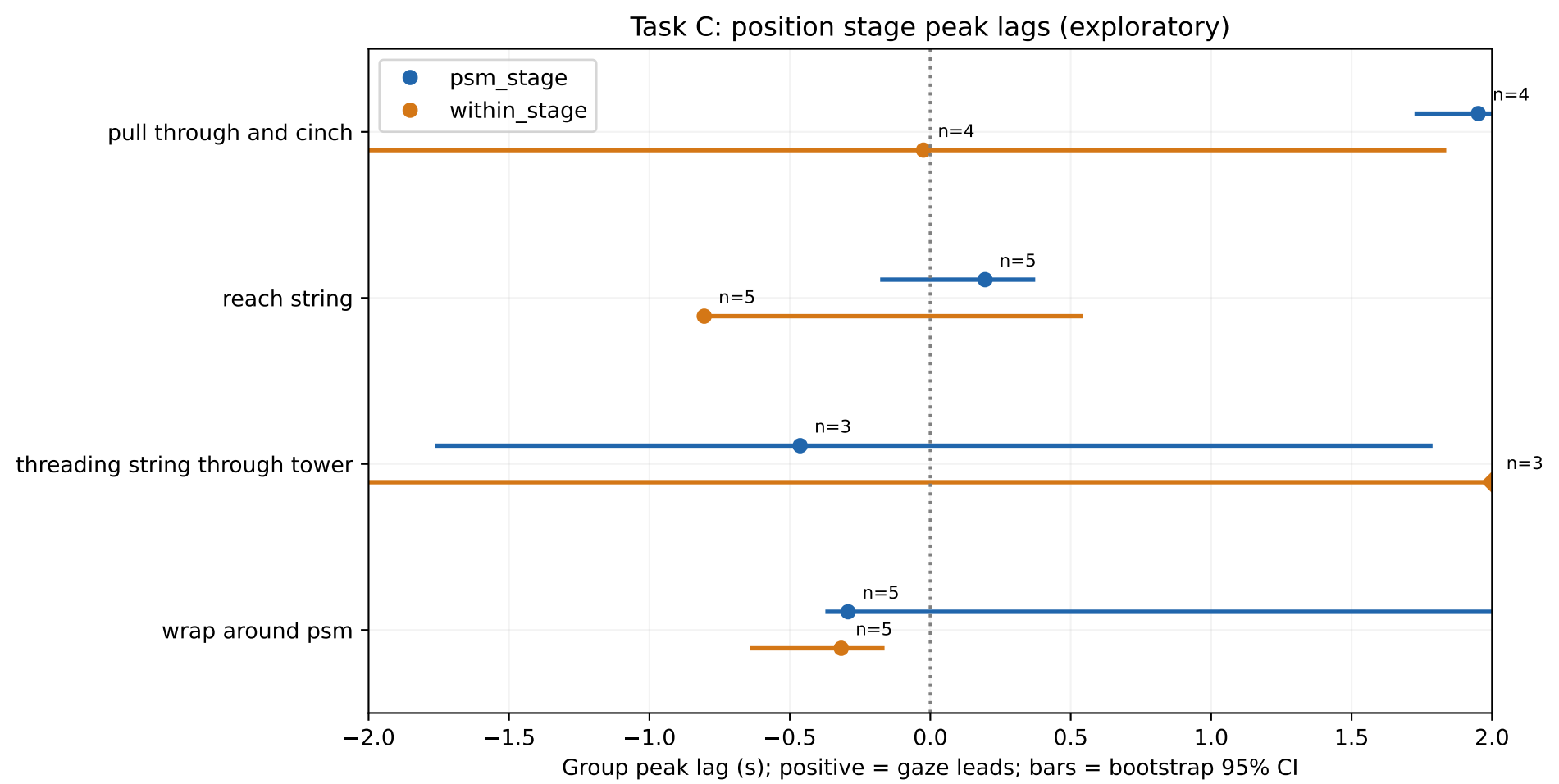
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



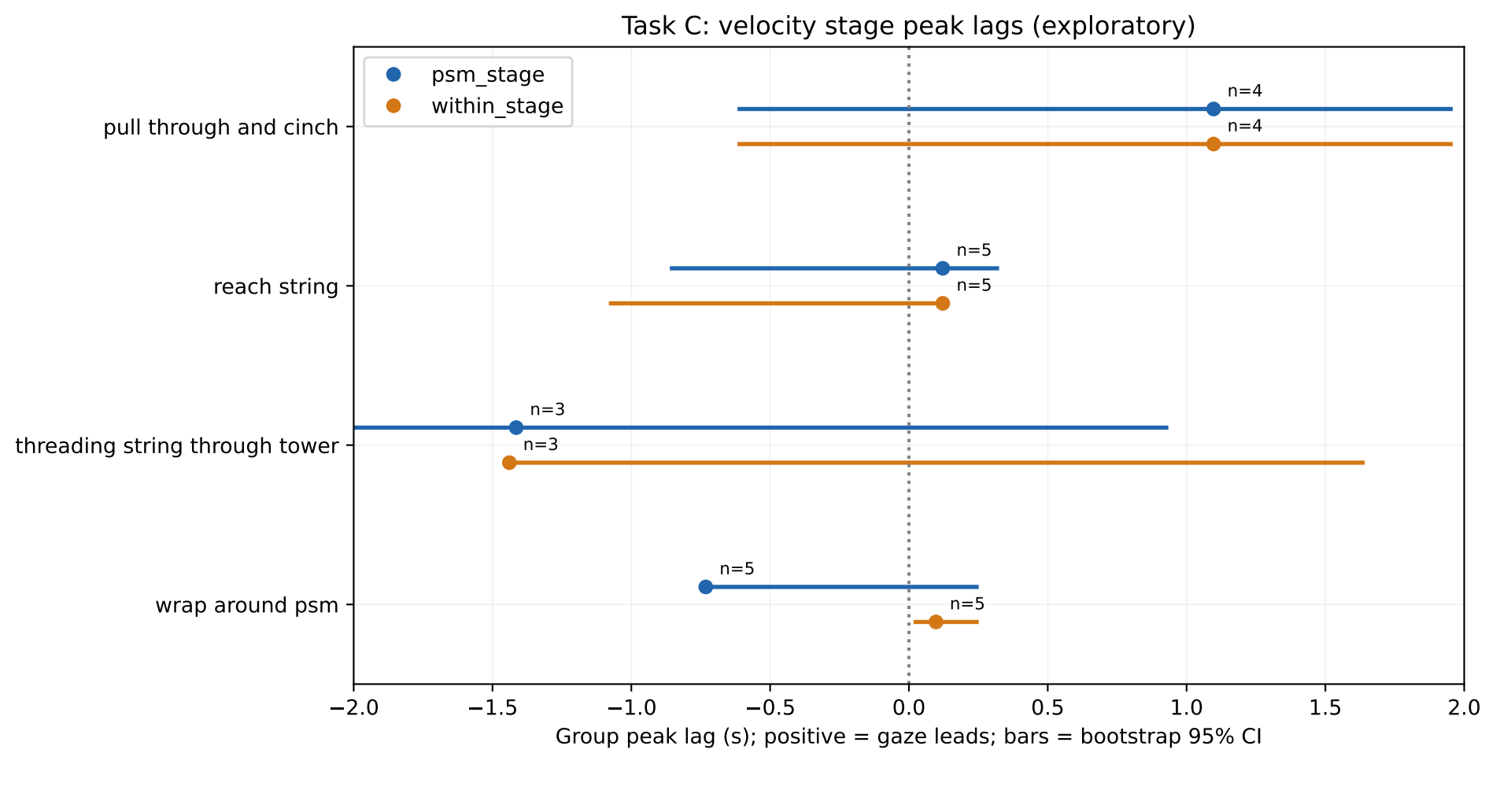
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



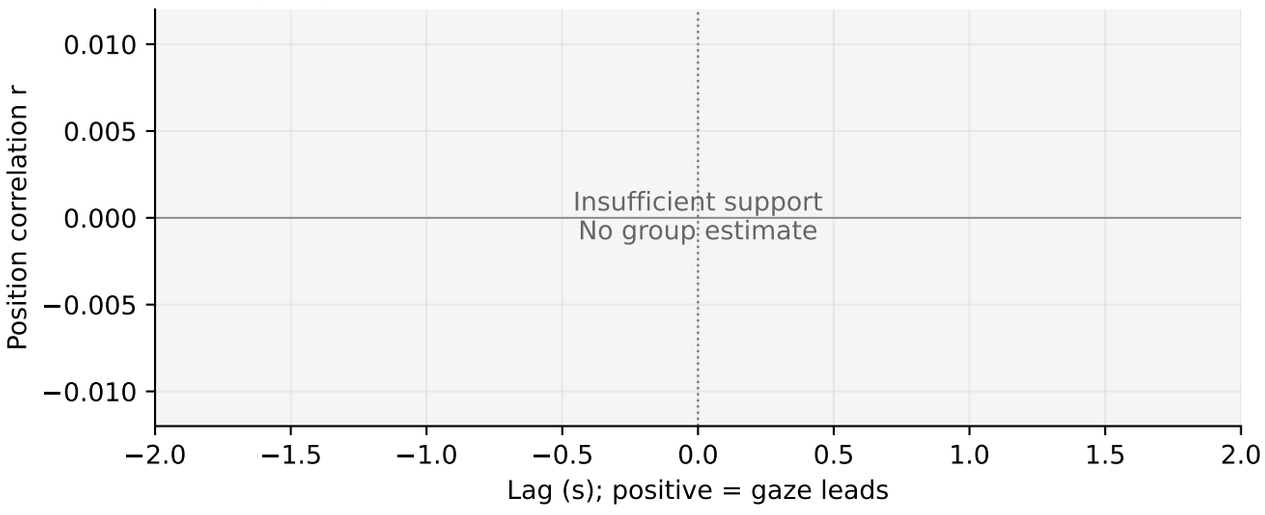
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

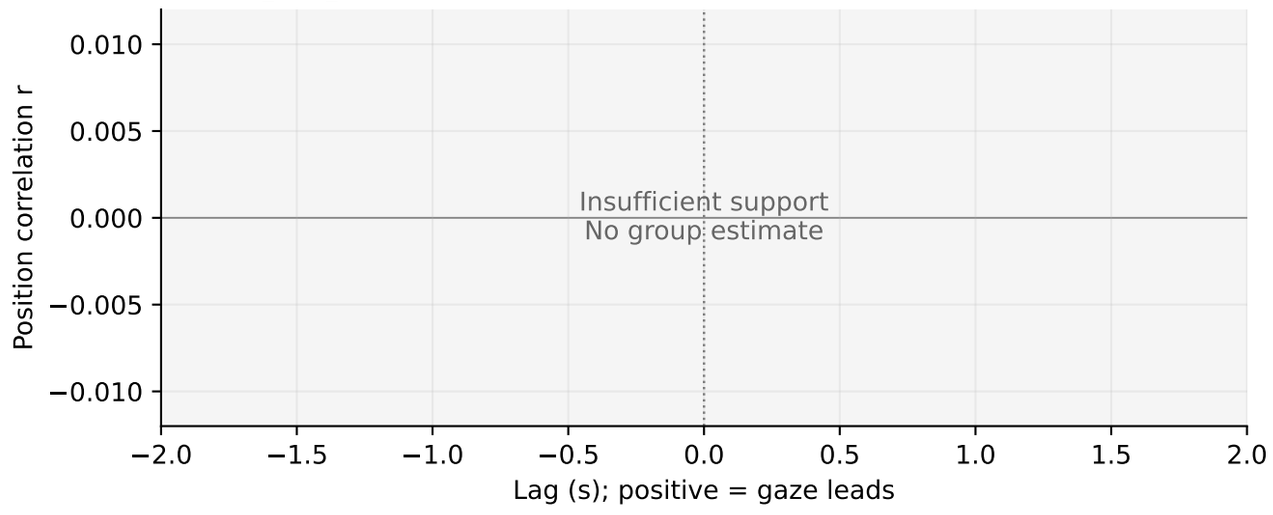
drop ring / psm_stage

n = 0; no_valid_curve



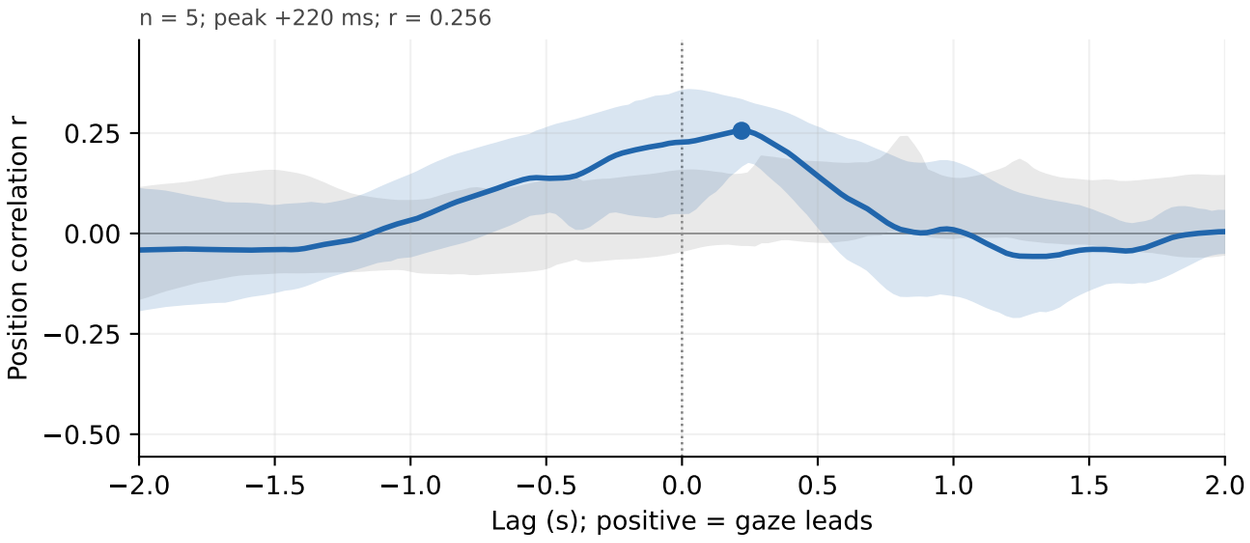
drop ring / within_stage

n = 0; no_valid_curve



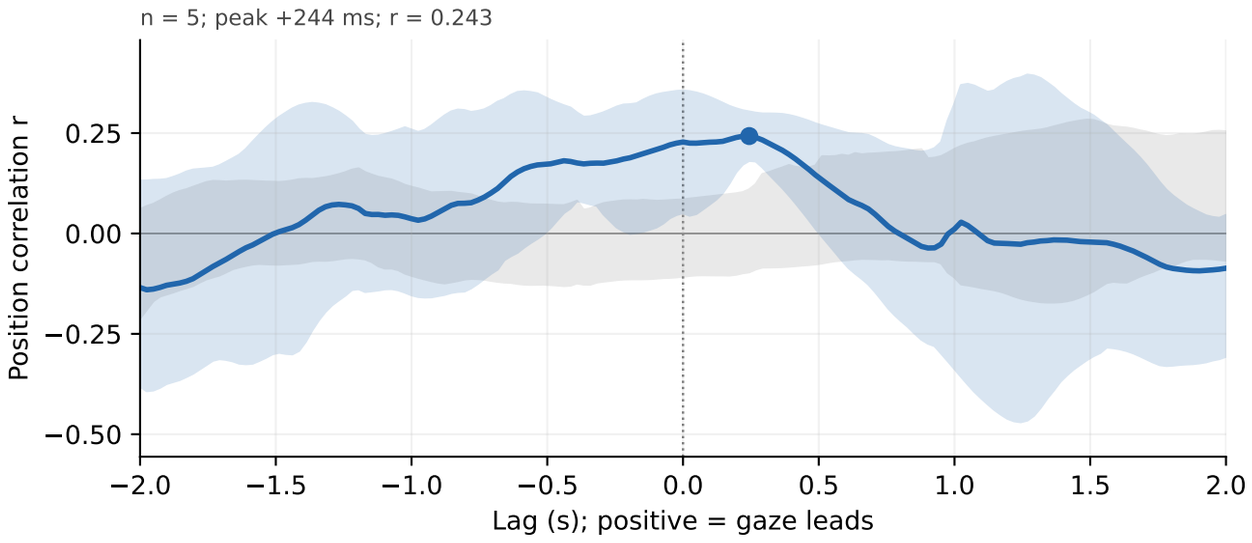
grab ring / psm_stage

n = 5; peak +220 ms; r = 0.256



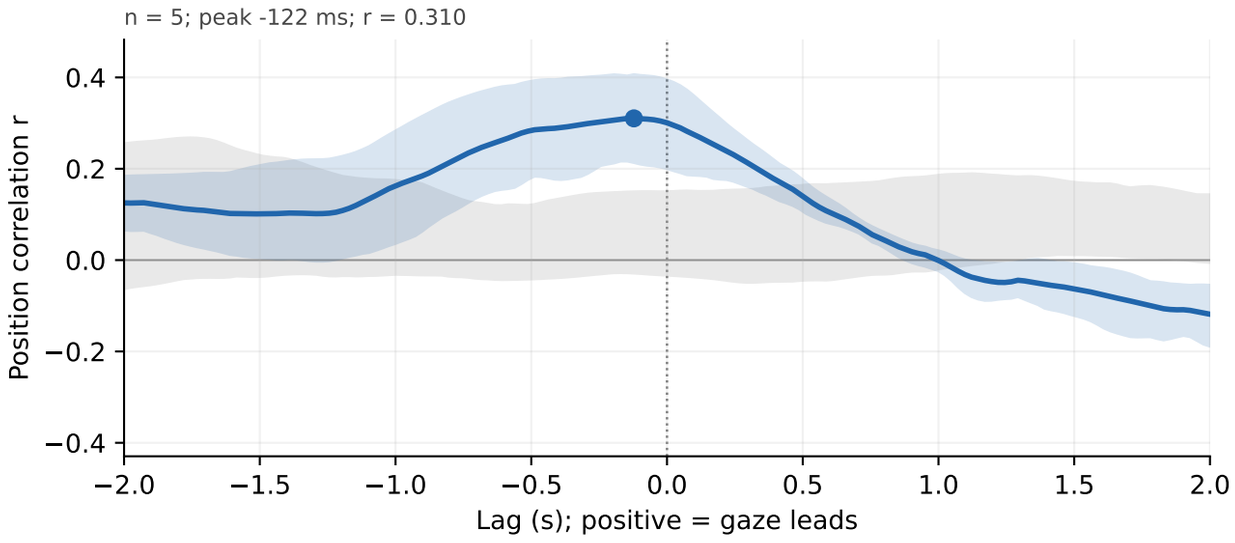
grab ring / within_stage

n = 5; peak +244 ms; r = 0.243



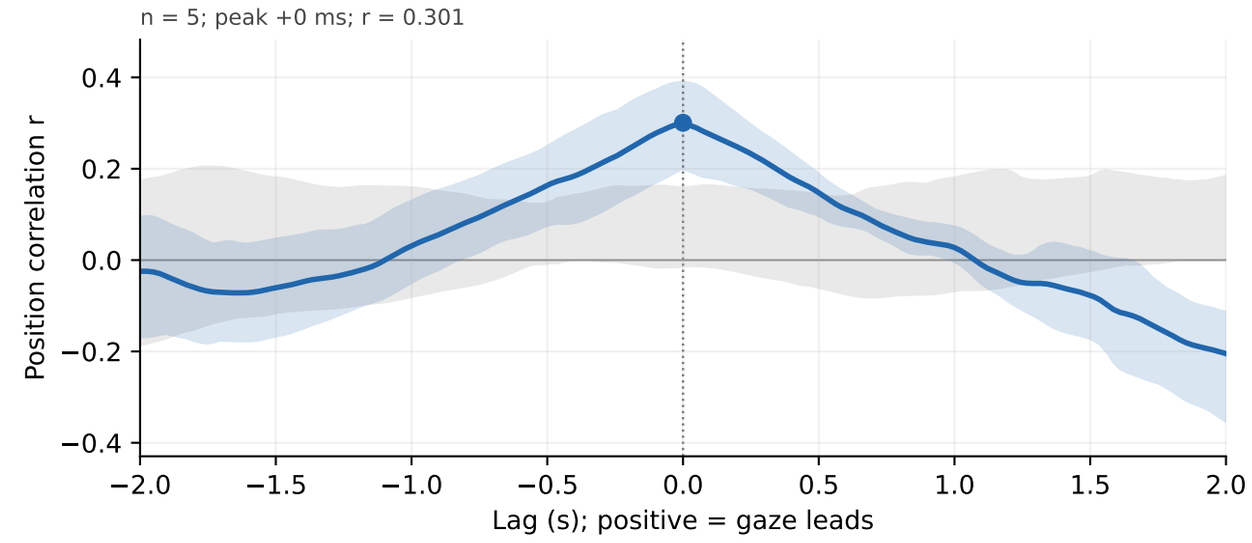
place ring / psm_stage

n = 5; peak -122 ms; r = 0.310



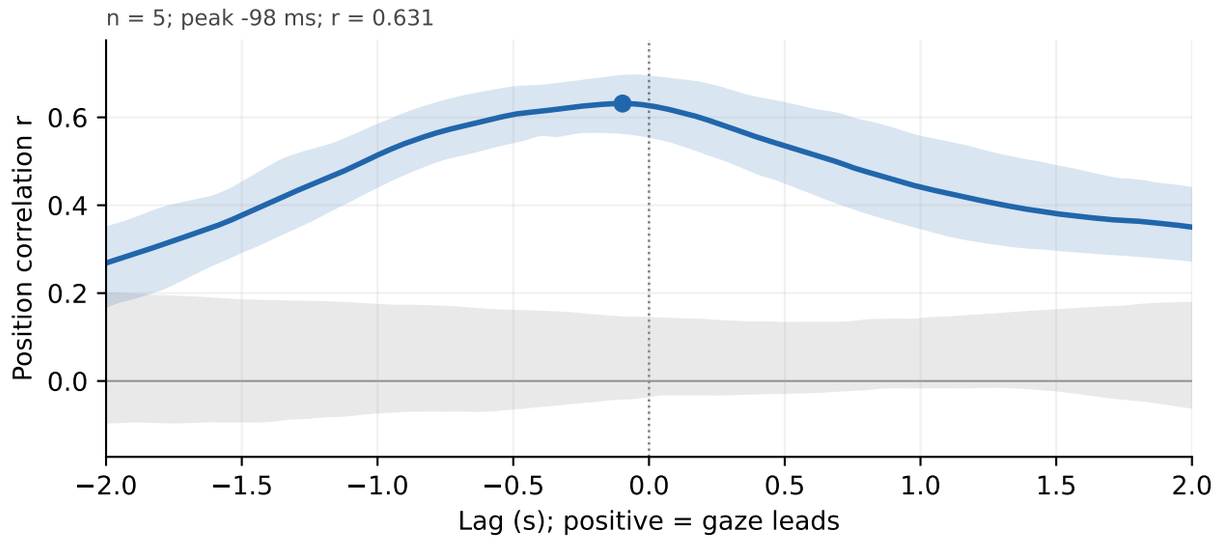
place ring / within_stage

n = 5; peak +0 ms; r = 0.301



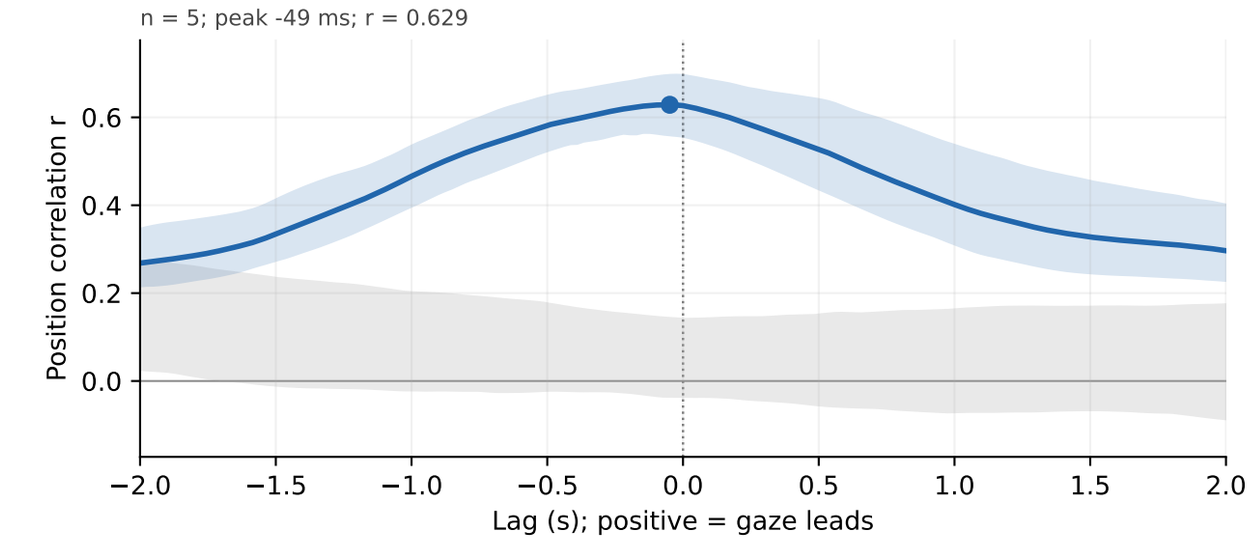
reach ring / psm_stage

n = 5; peak -98 ms; r = 0.631



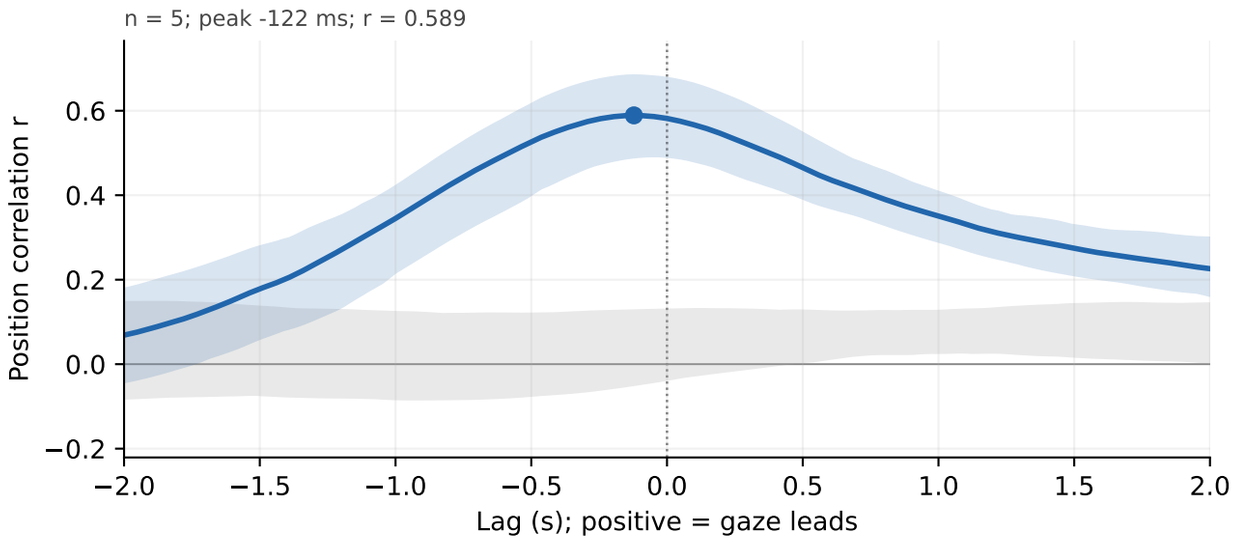
reach ring / within_stage

n = 5; peak -49 ms; r = 0.629



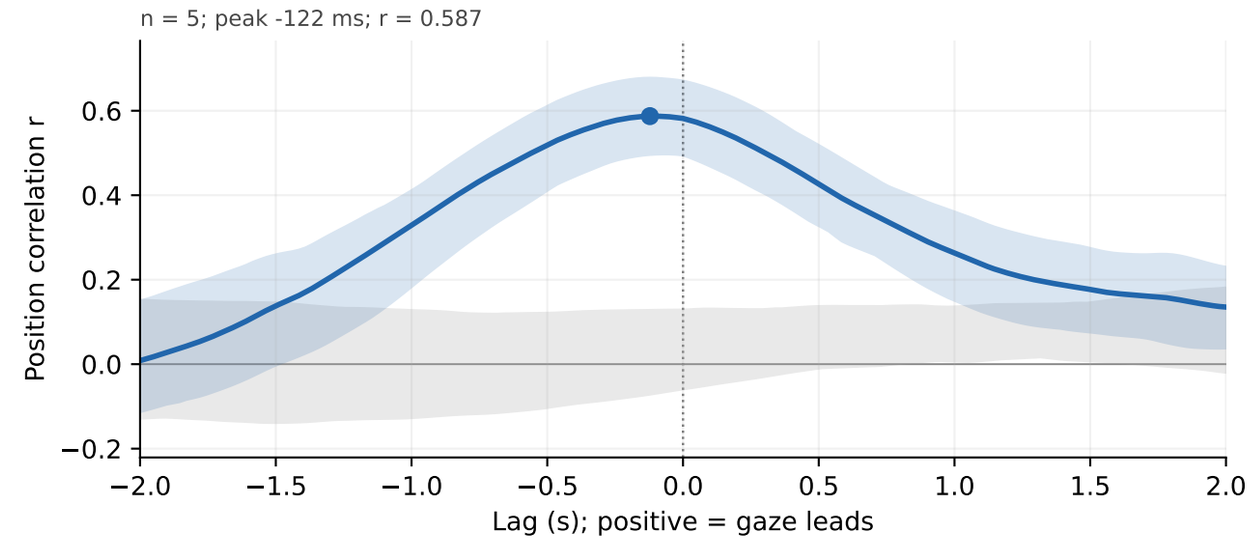
transfer ring / psm_stage

n = 5; peak -122 ms; r = 0.589



transfer ring / within_stage

n = 5; peak -122 ms; r = 0.587

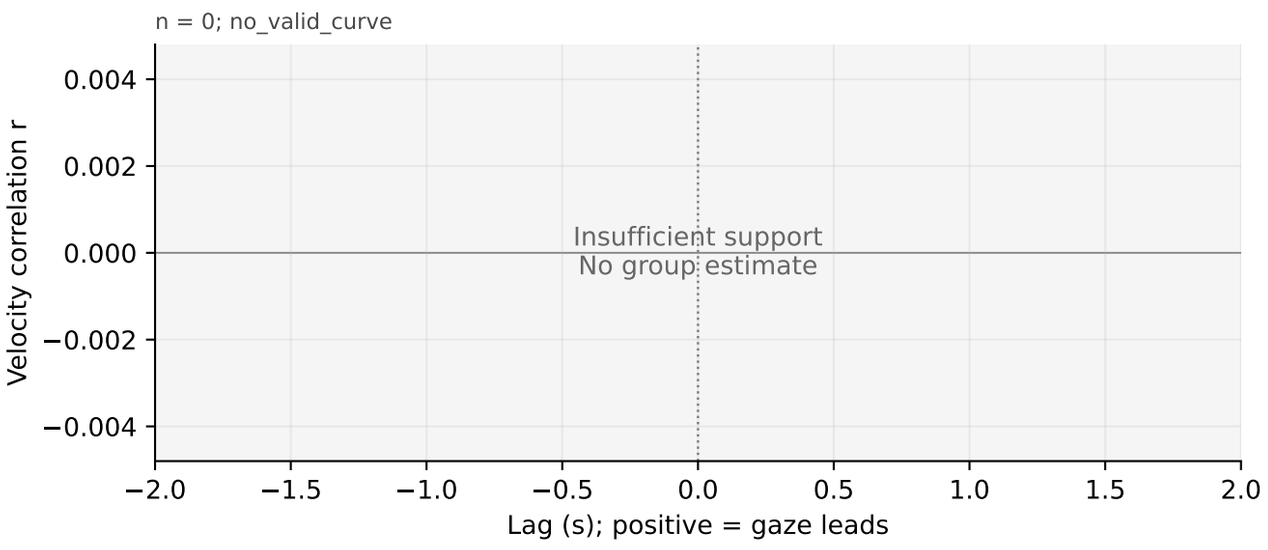


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

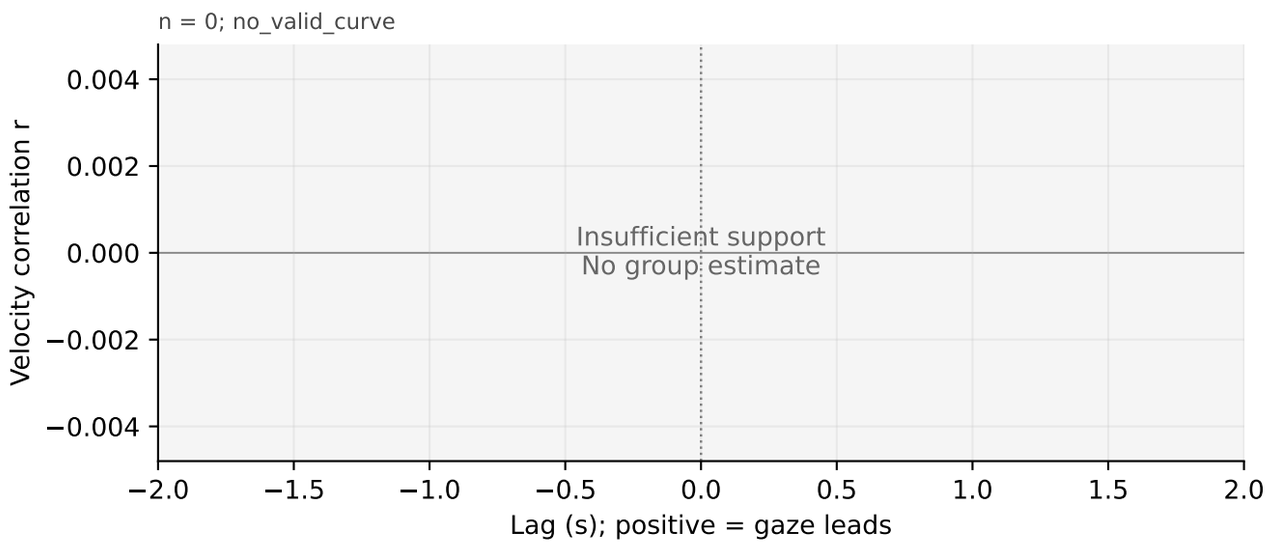
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

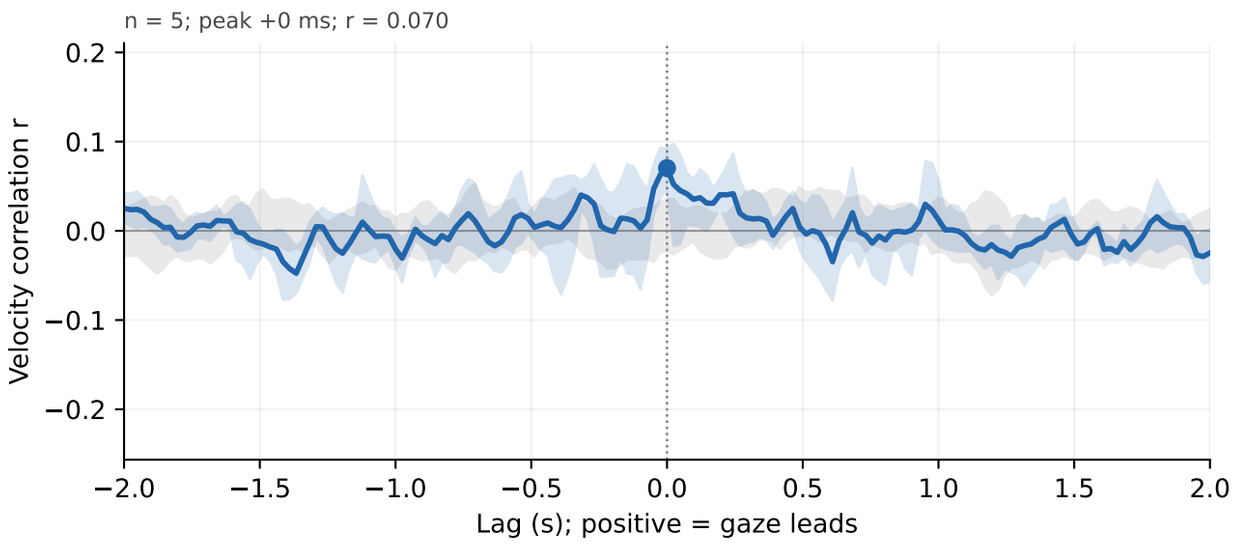
drop ring / psm_stage



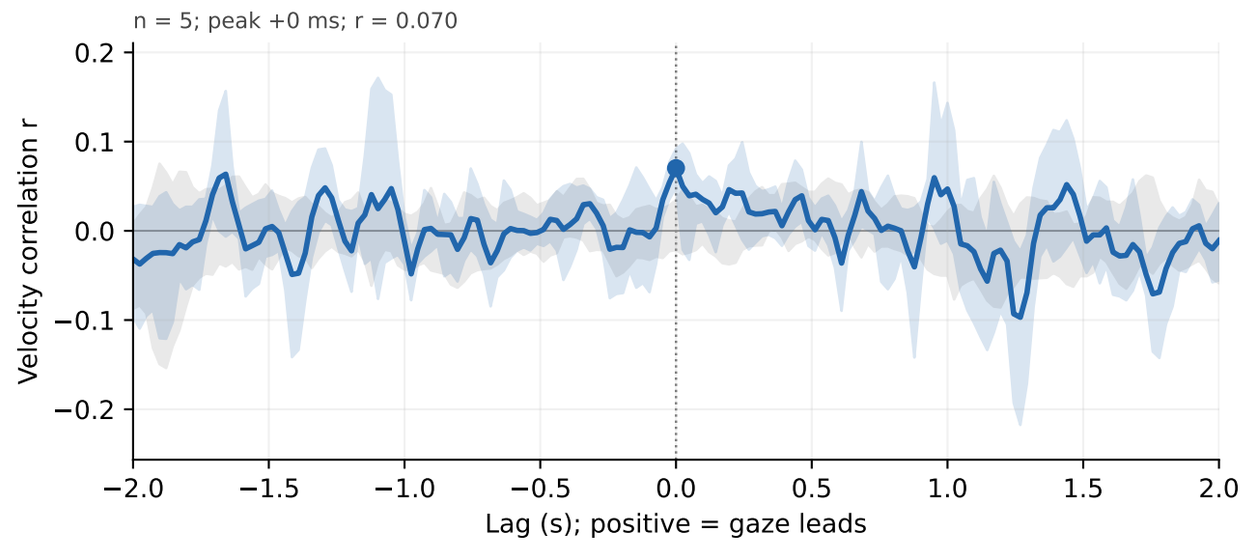
drop ring / within_stage



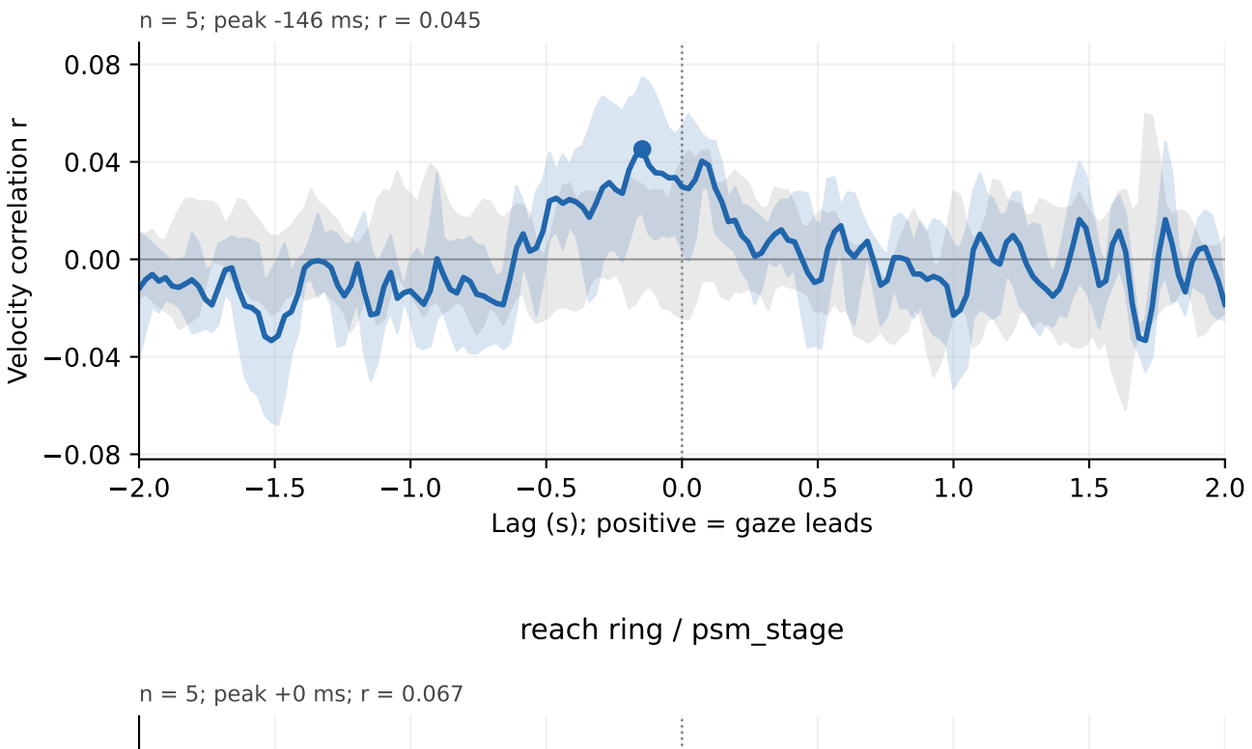
grab ring / psm_stage



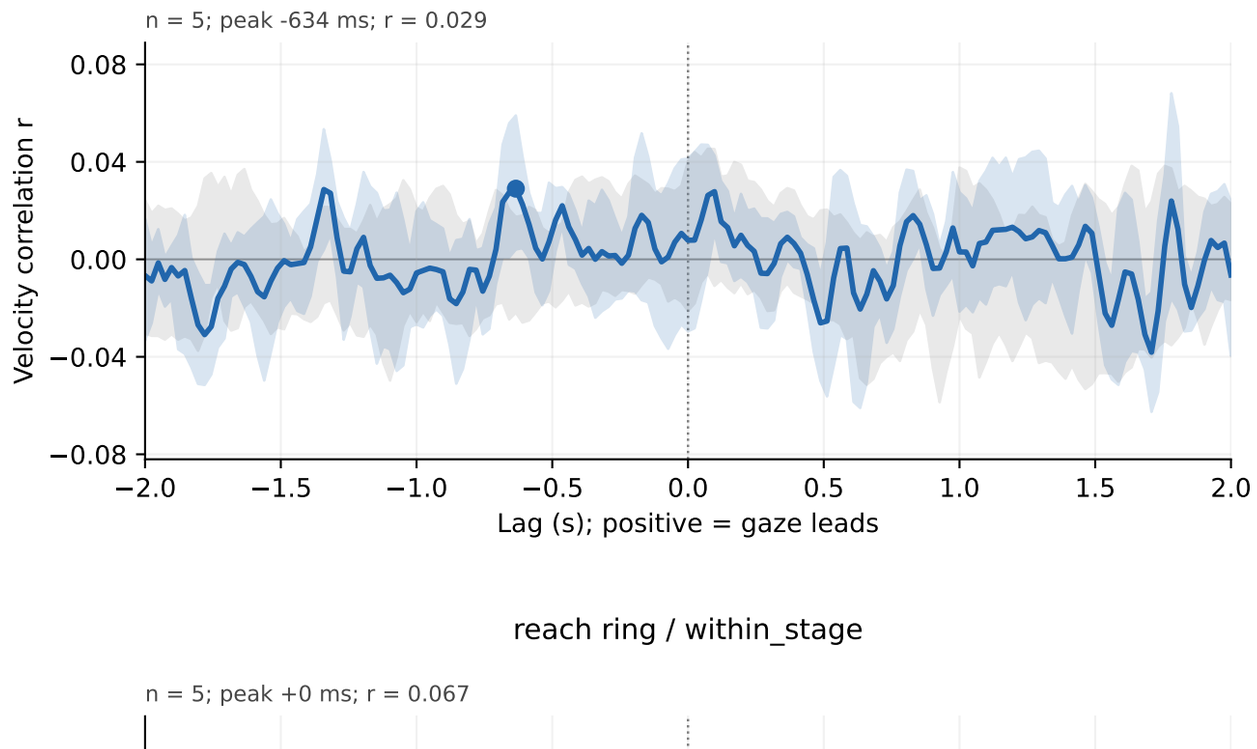
grab ring / within_stage



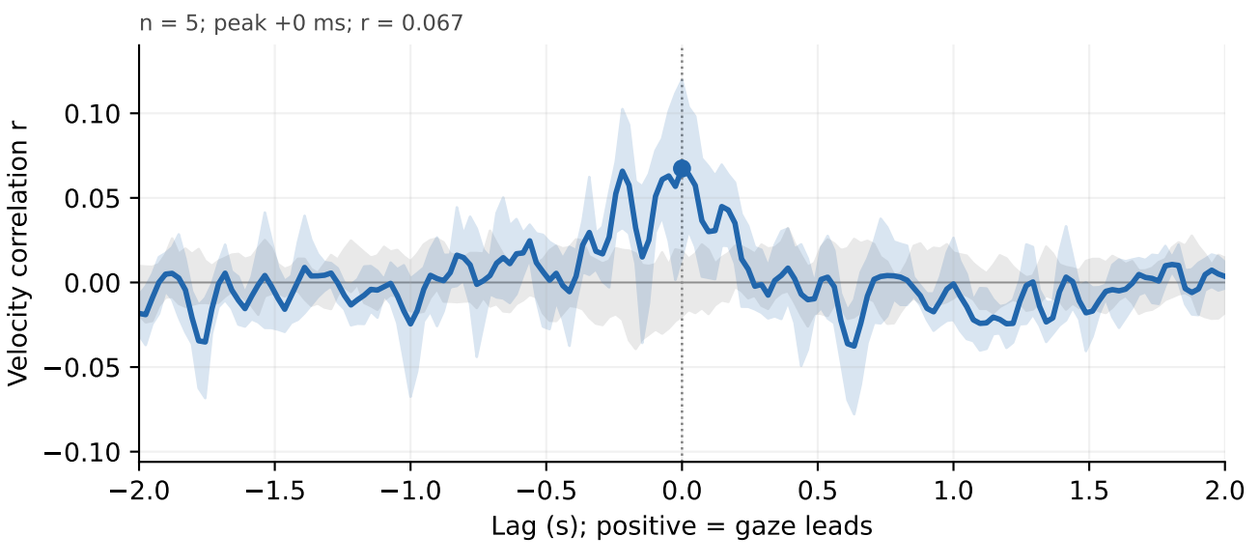
place ring / psm_stage



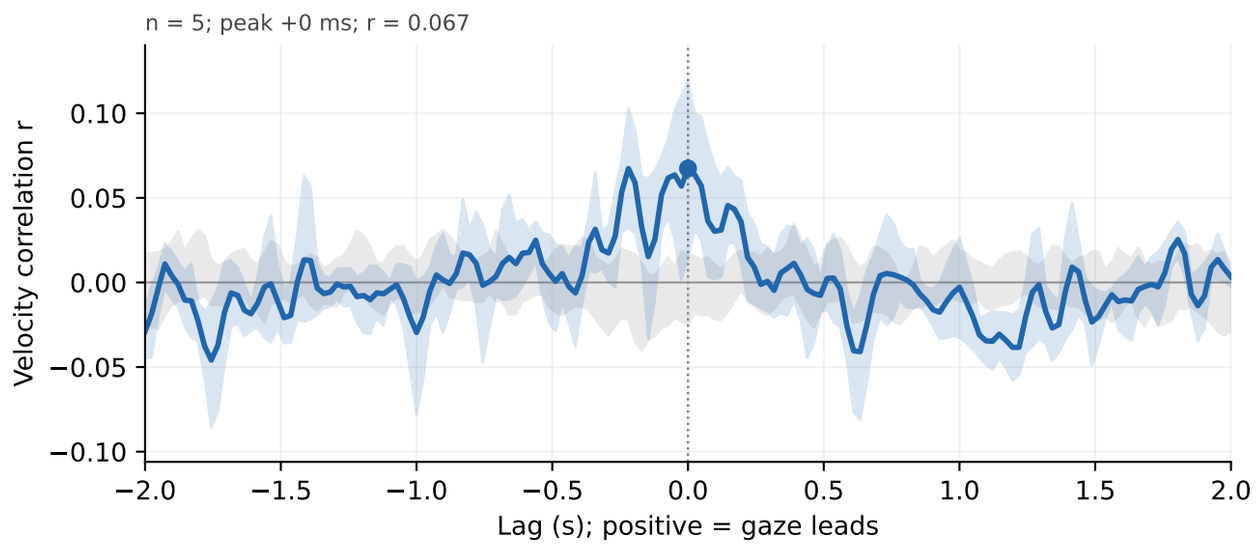
place ring / within_stage



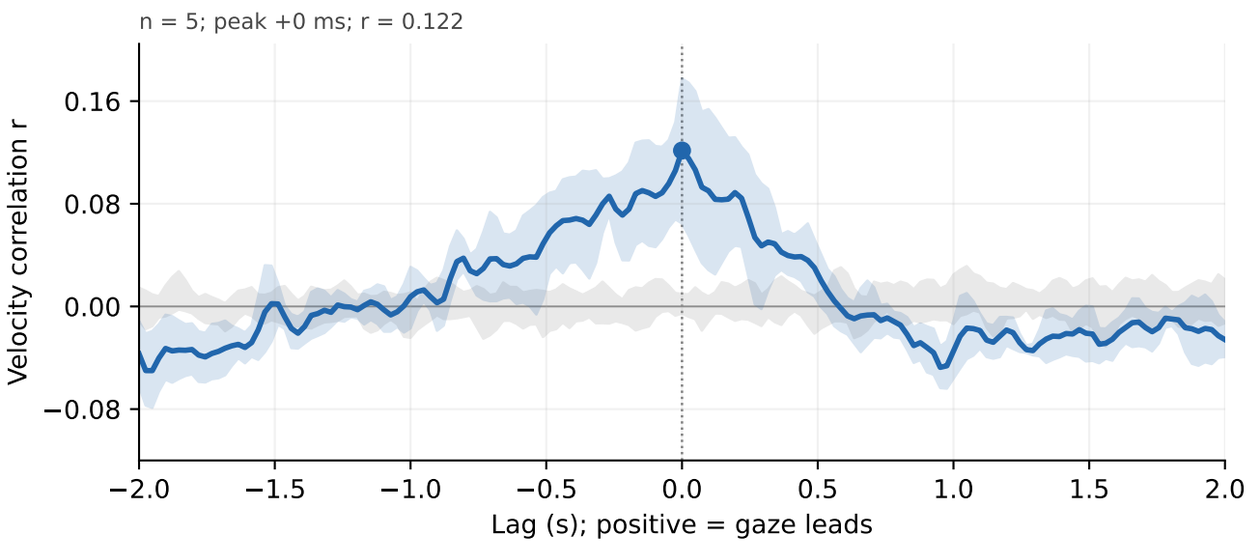
reach ring / psm_stage



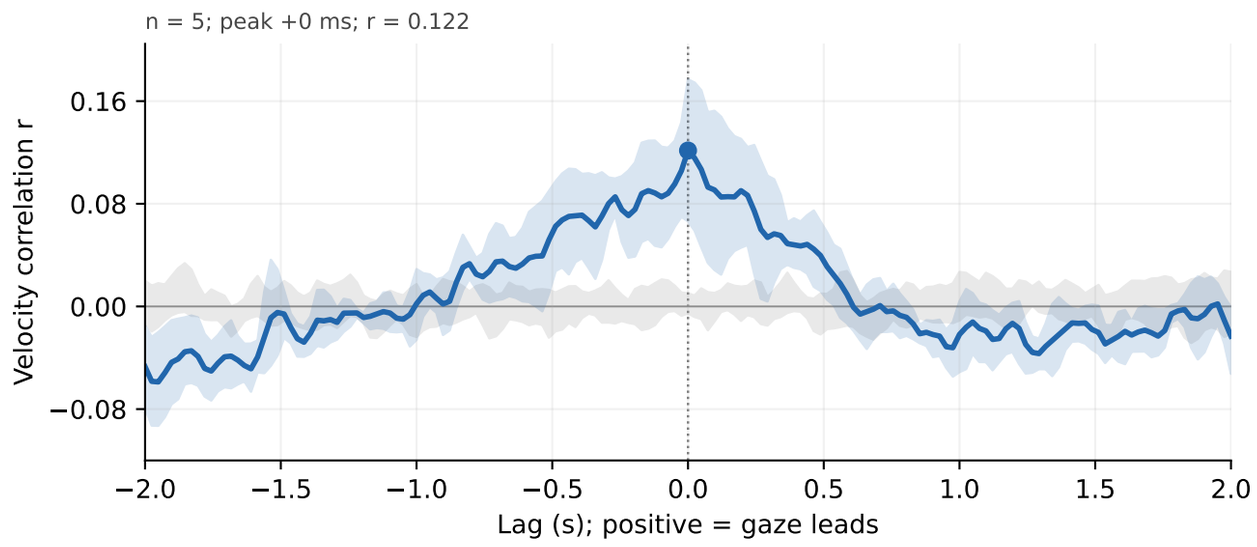
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

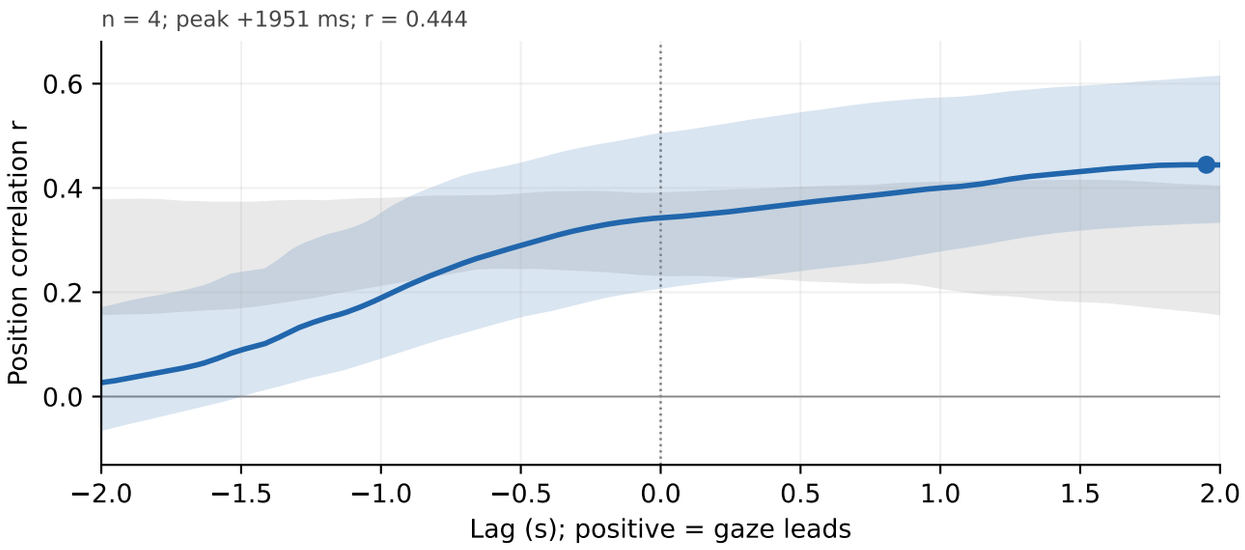


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

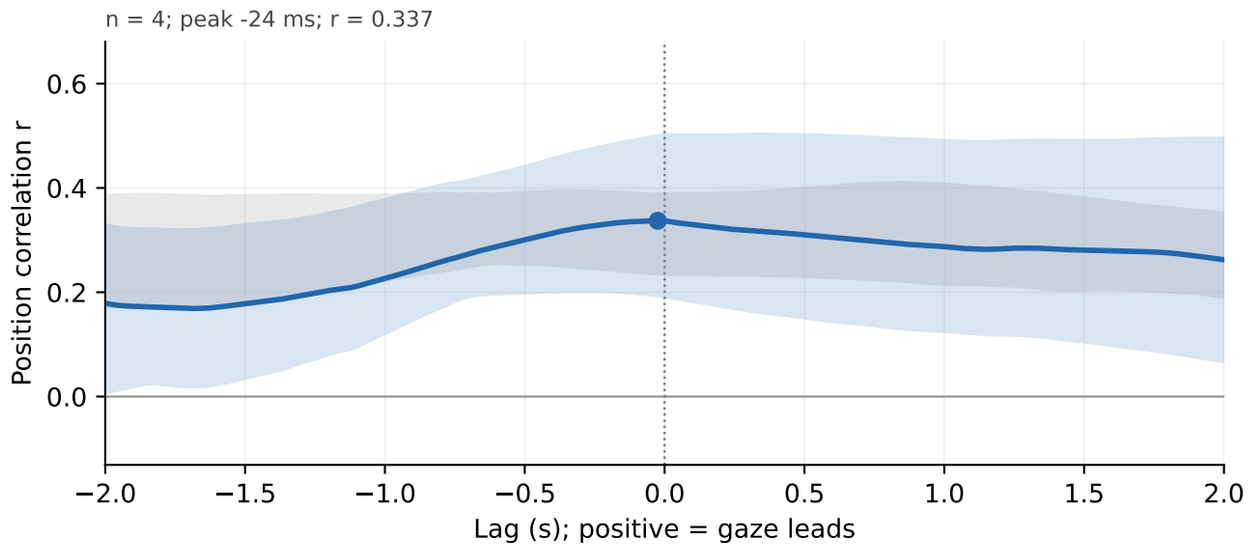
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

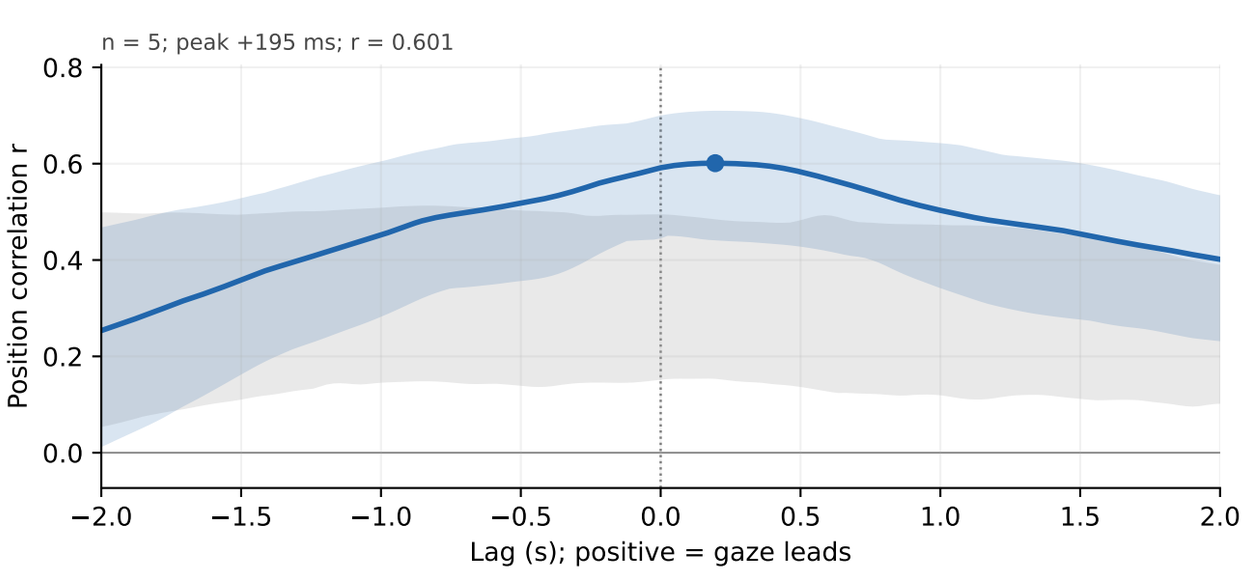
pull through and cinch / psm_stage



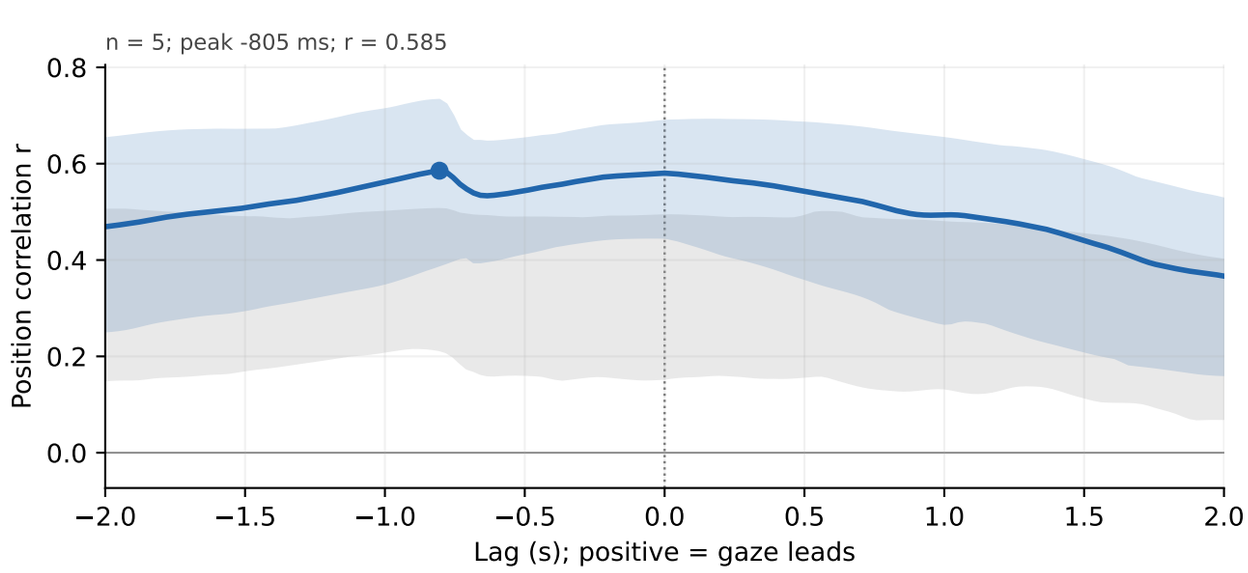
pull through and cinch / within_stage



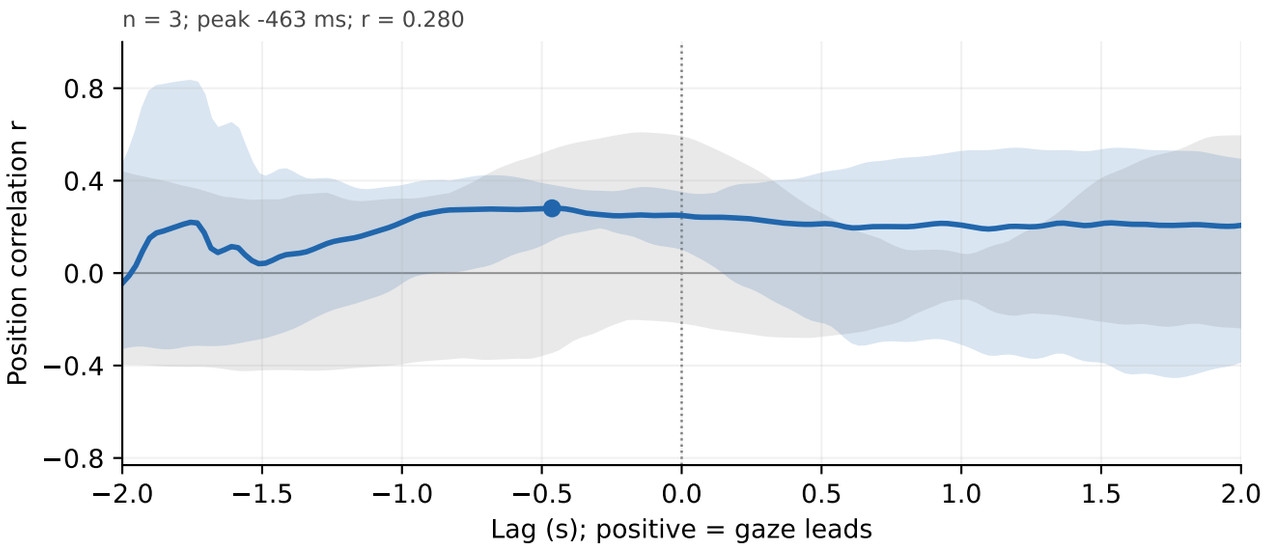
reach string / psm_stage



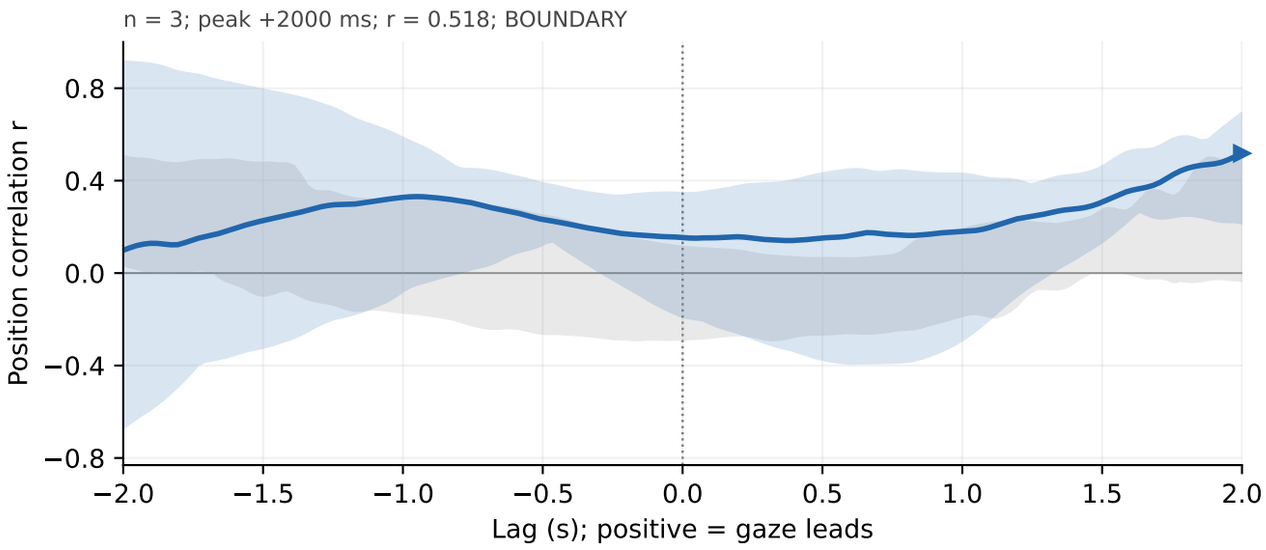
reach string / within_stage



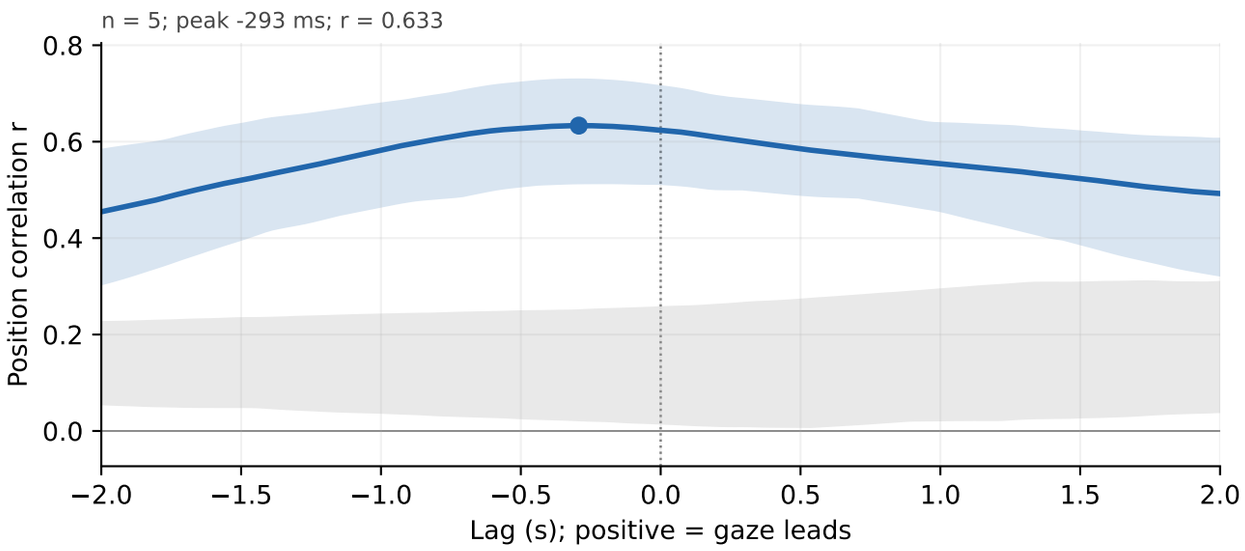
threading string through tower / psm_stage



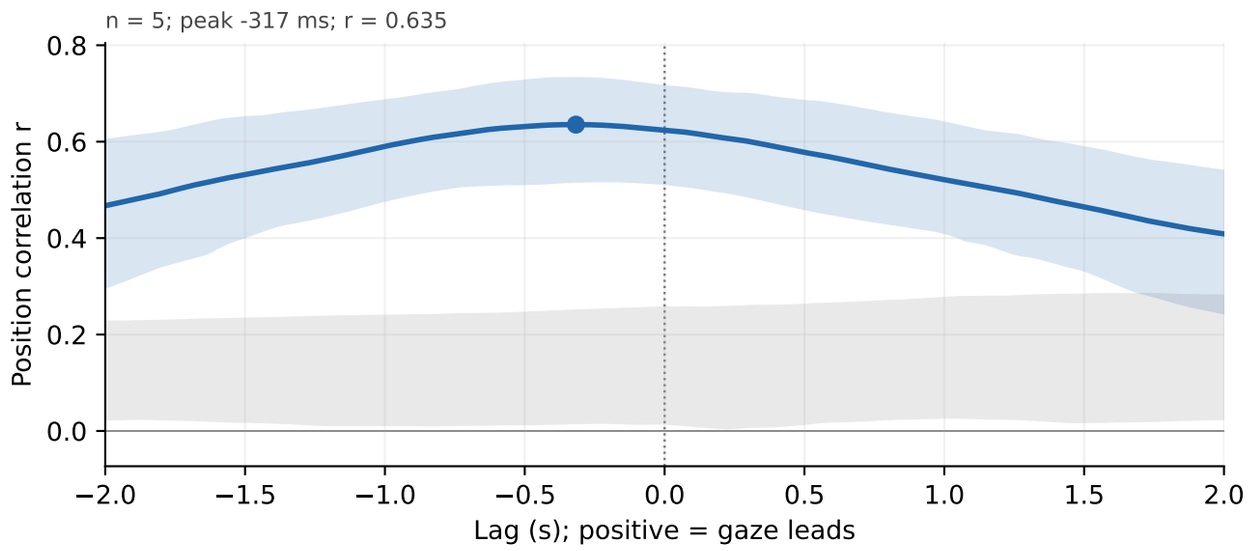
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

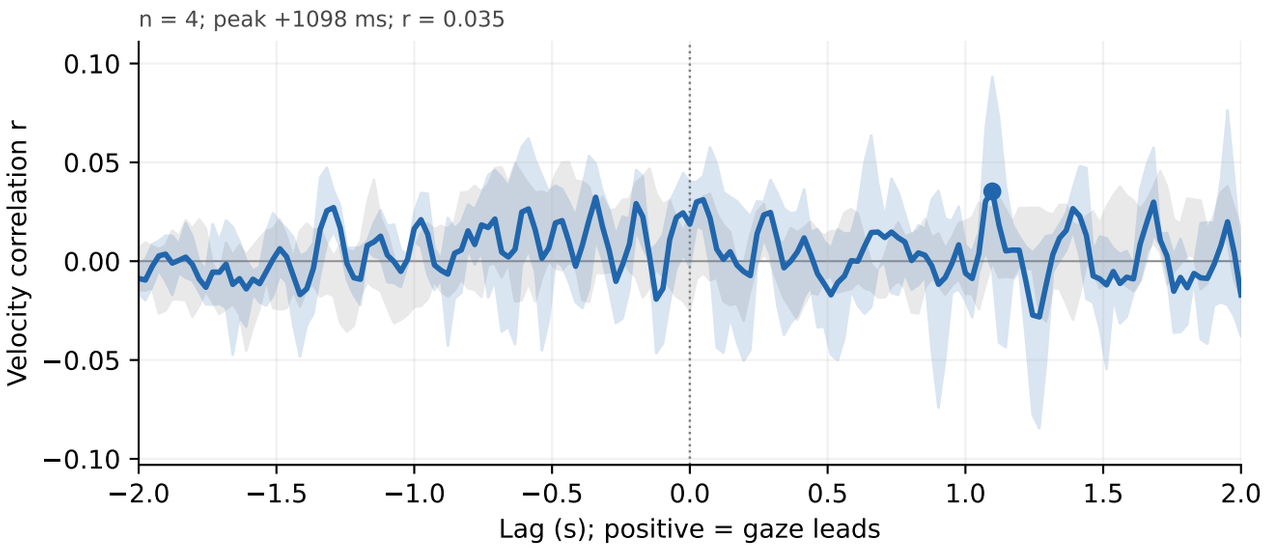


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

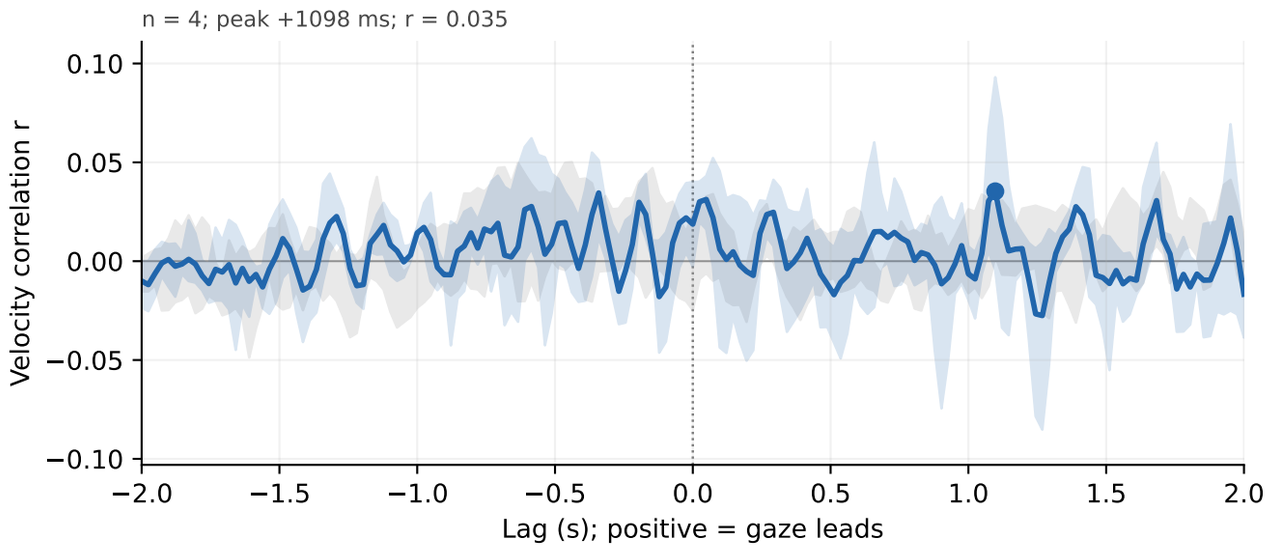
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

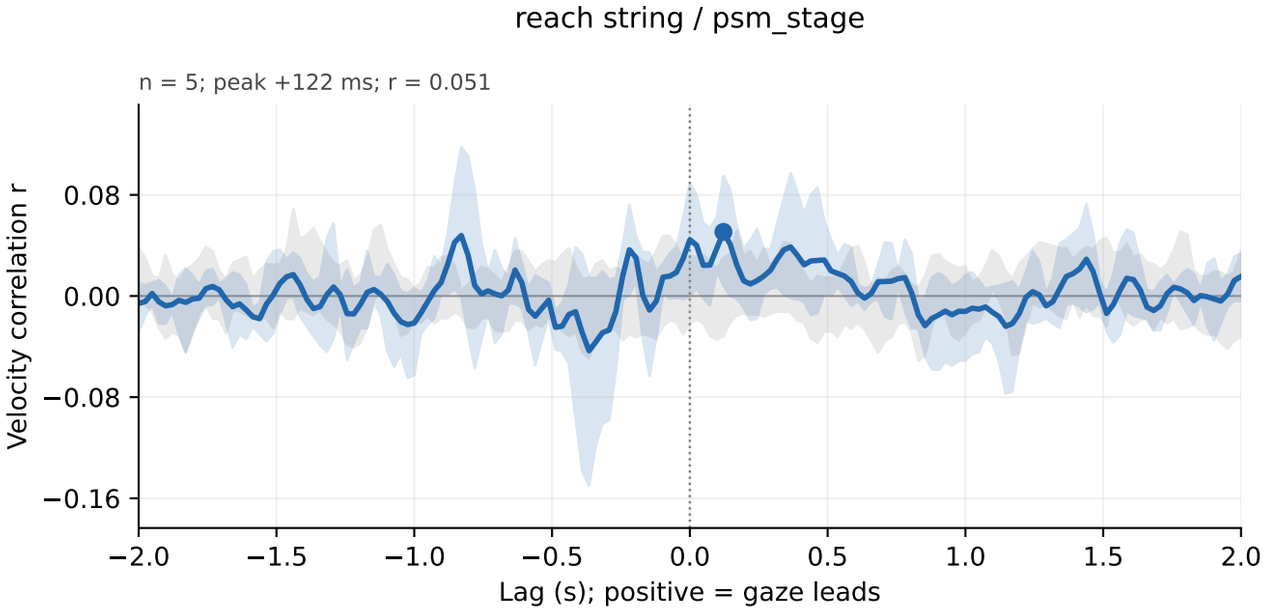
pull through and cinch / psm_stage



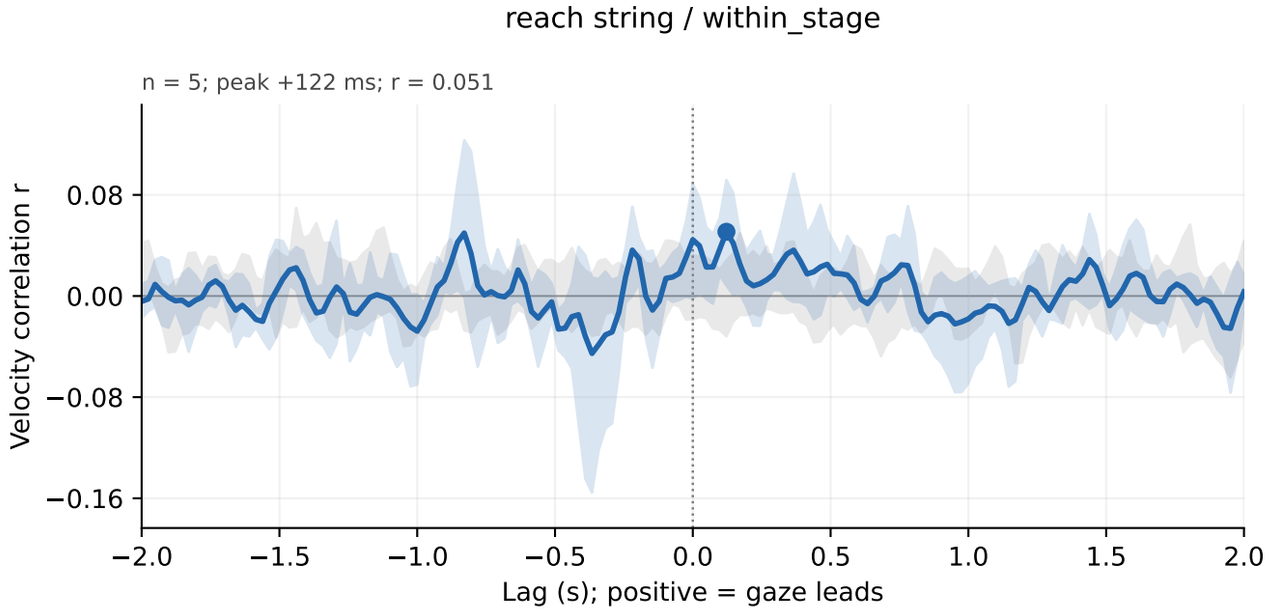
pull through and cinch / within_stage



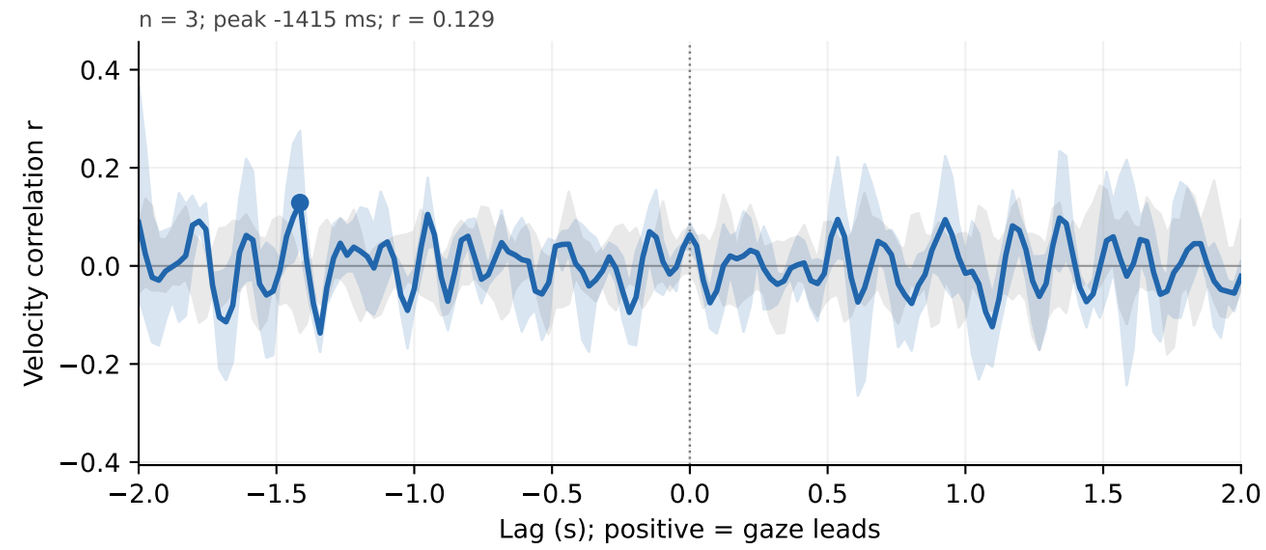
reach string / psm_stage



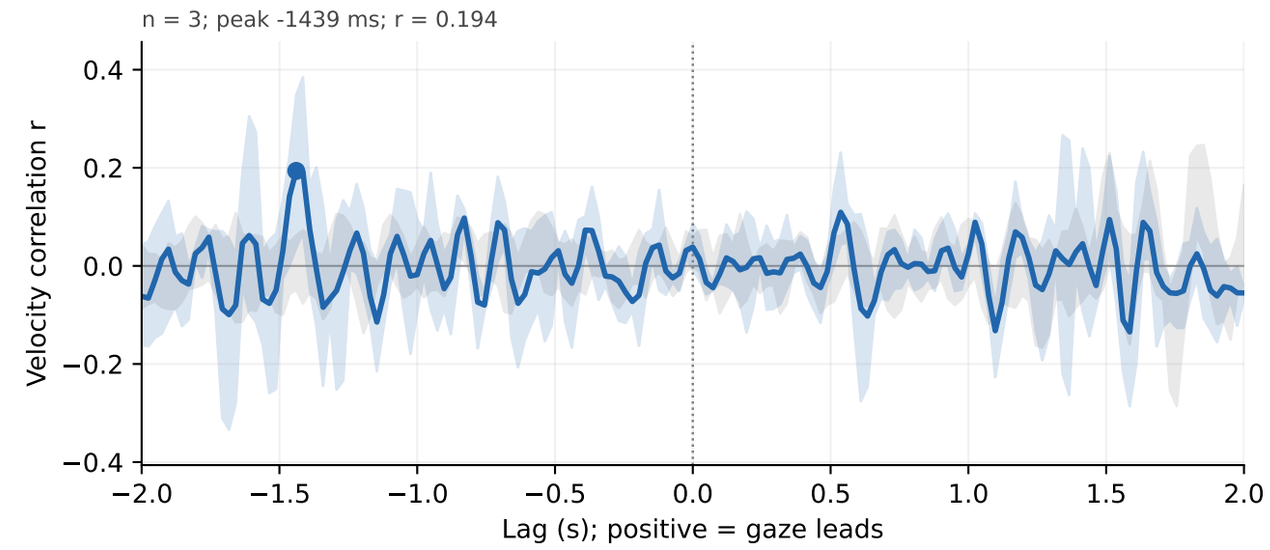
reach string / within_stage



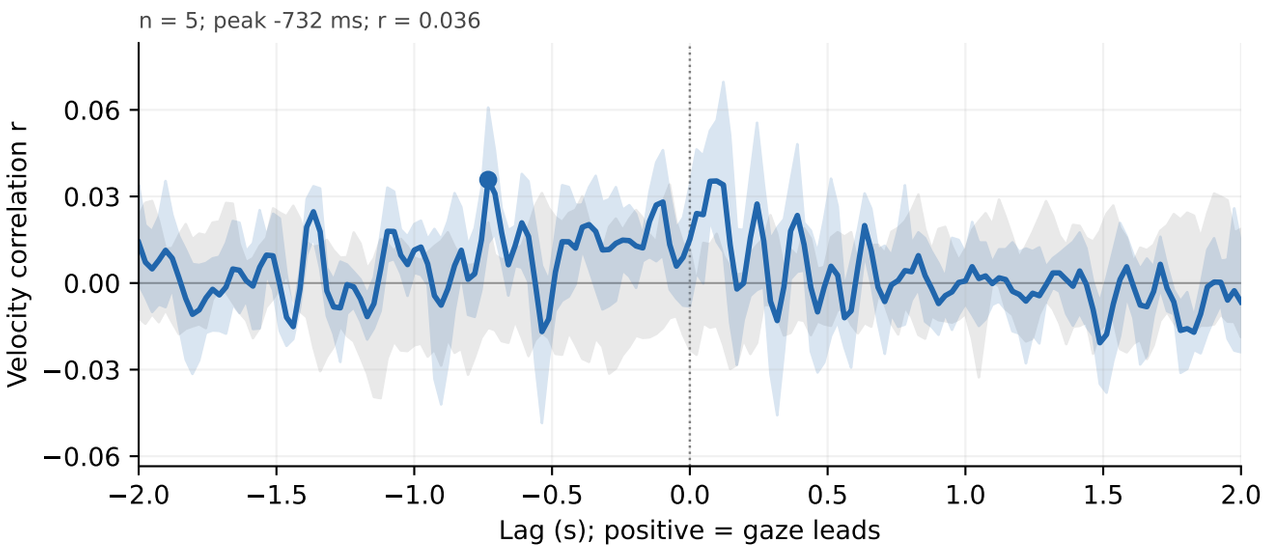
threading string through tower / psm_stage



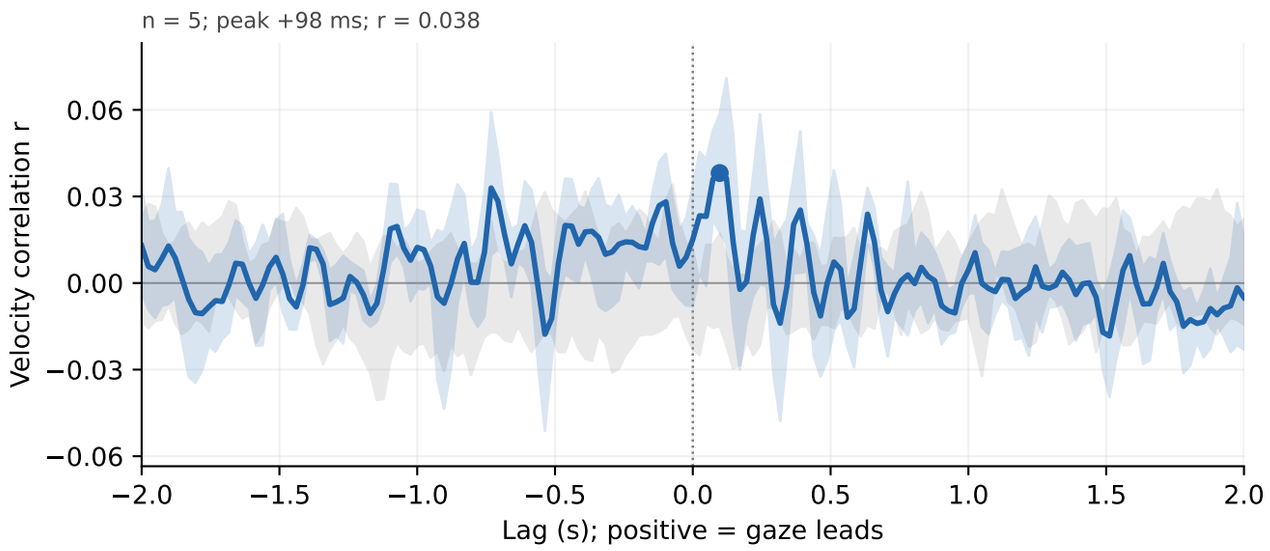
threading string through tower / within_stage



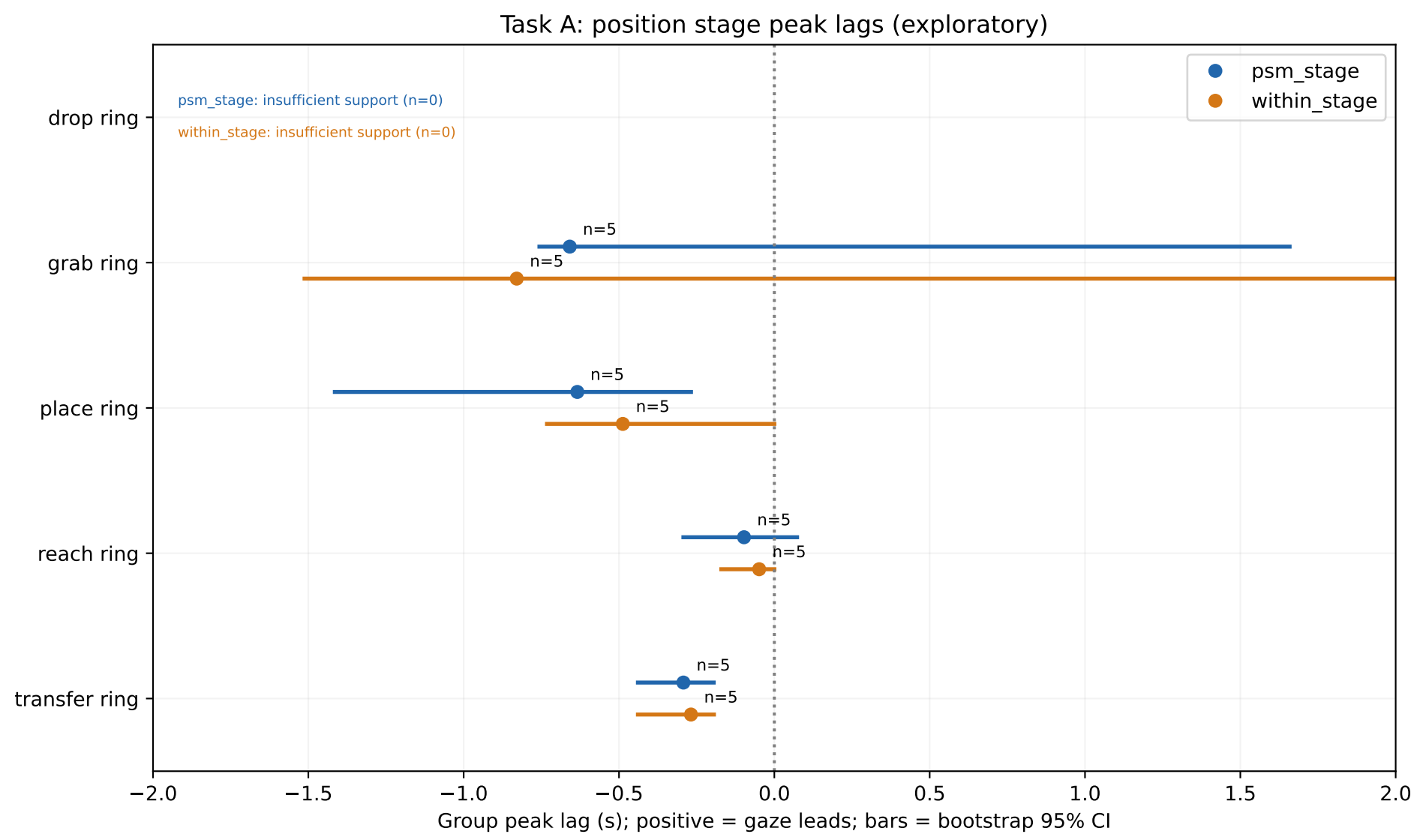
wrap around psm / psm_stage



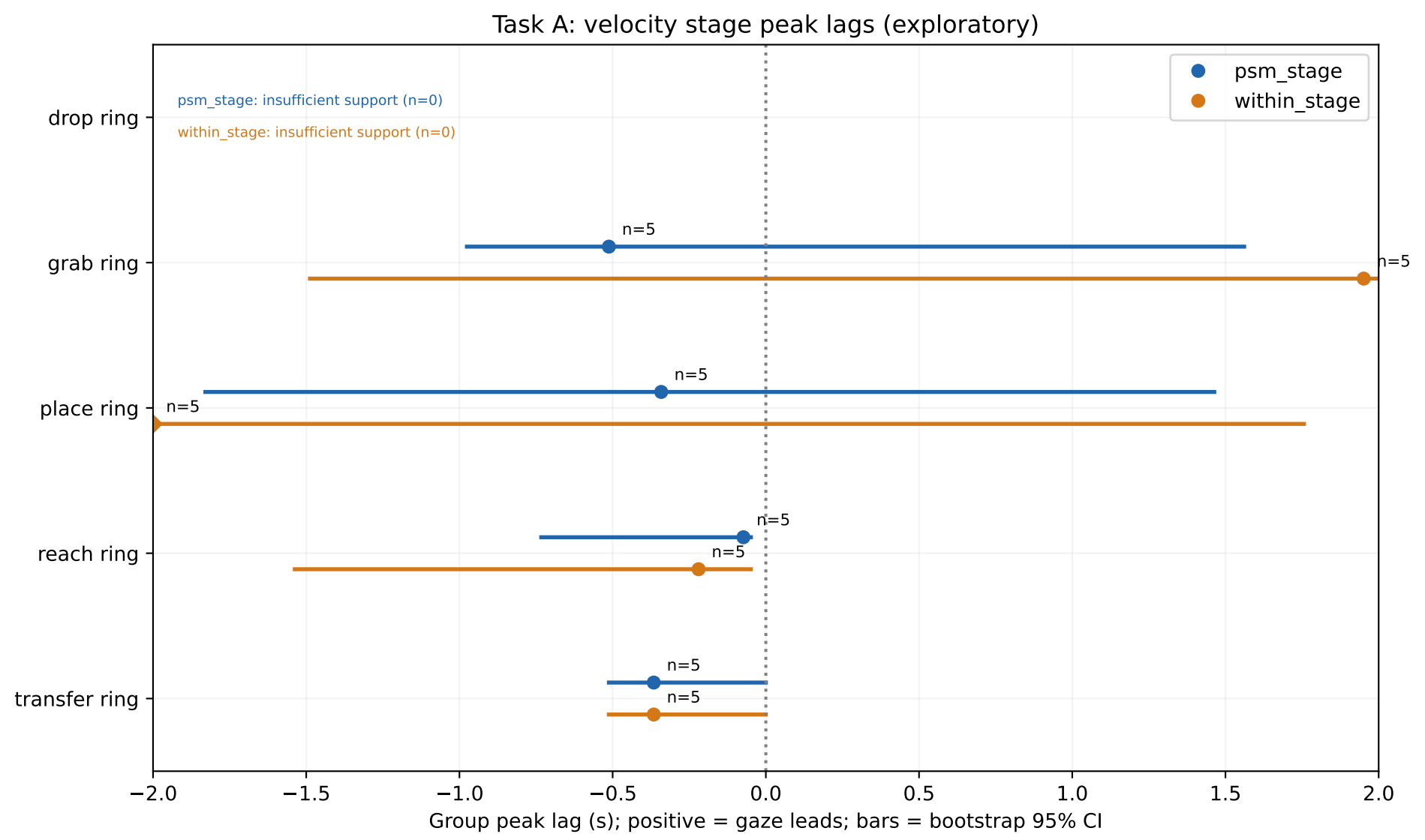
wrap around psm / within_stage



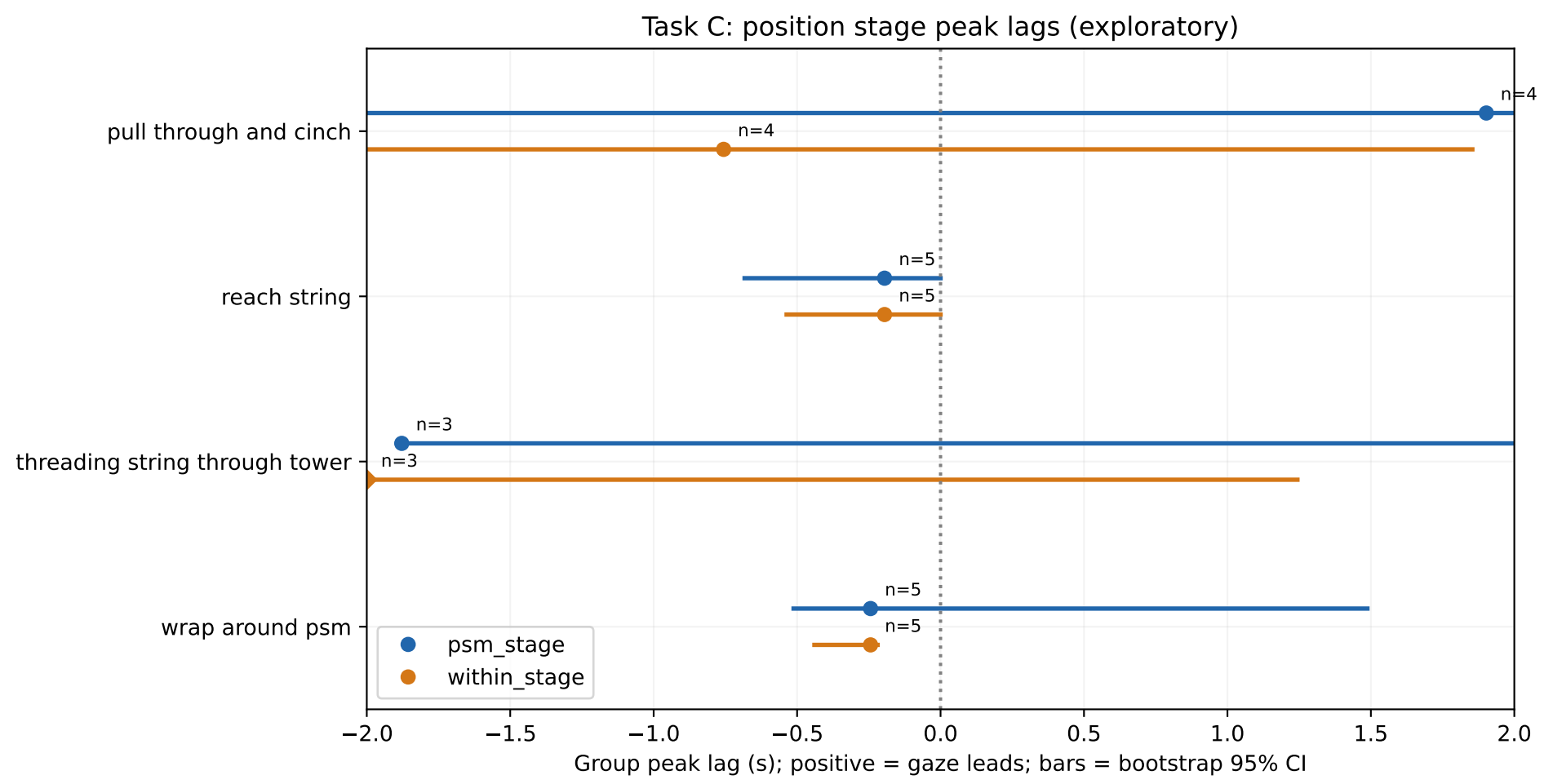
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



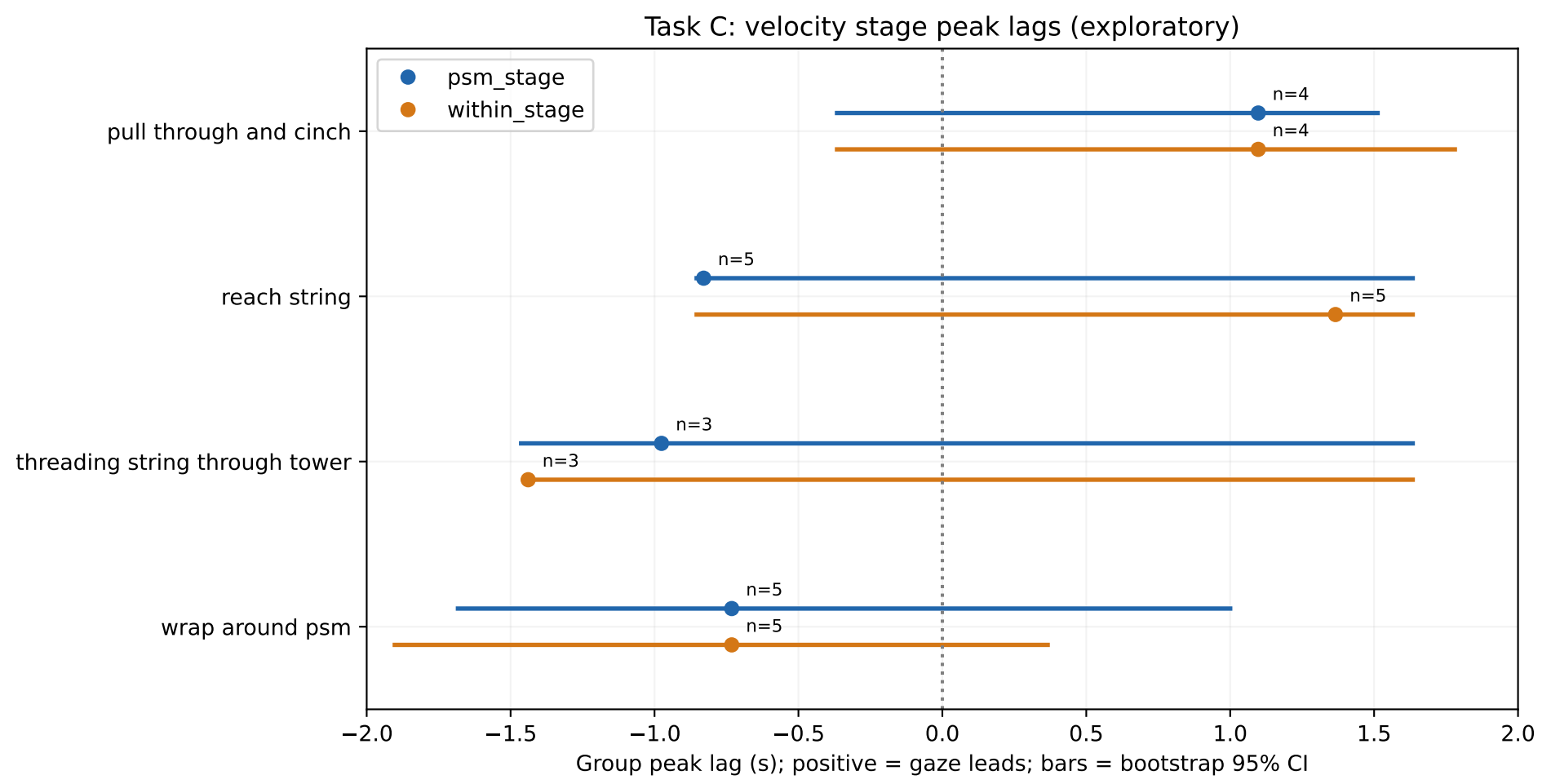
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



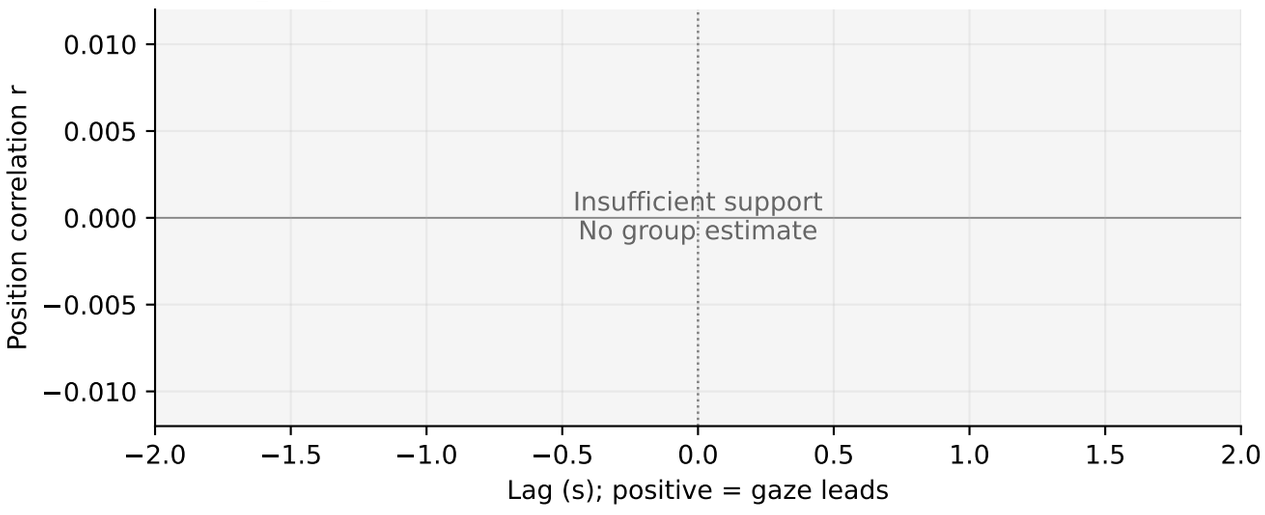
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

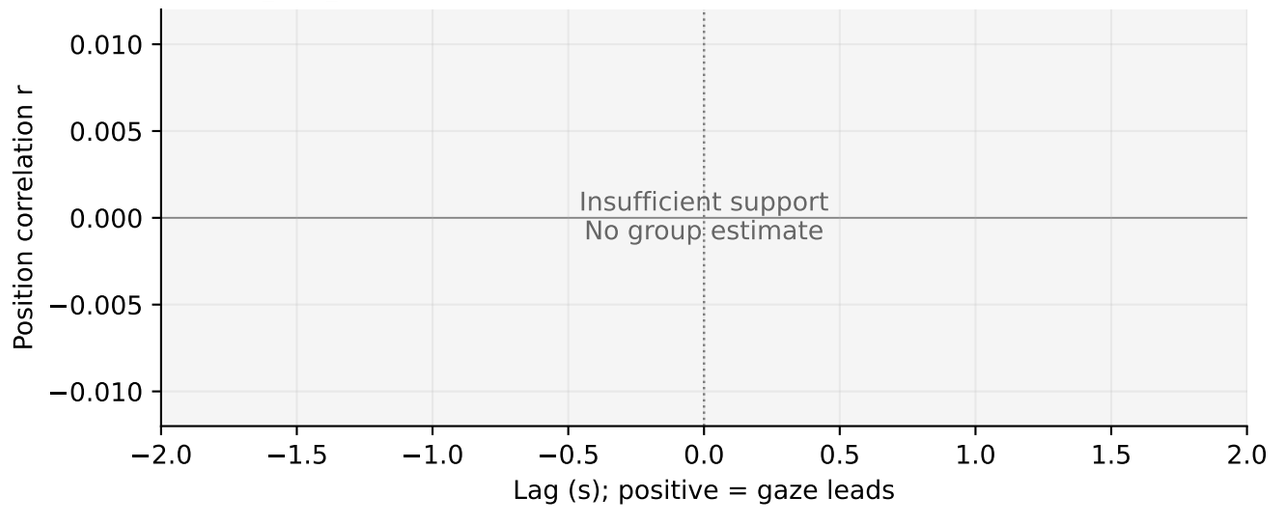
drop ring / psm_stage

n = 0; no_valid_curve



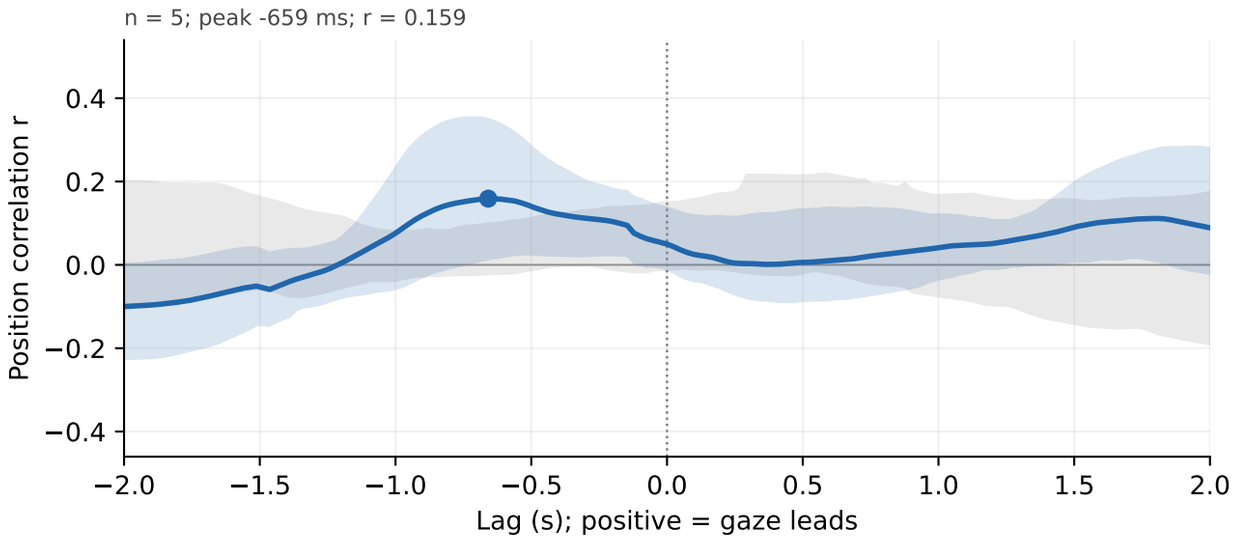
drop ring / within_stage

n = 0; no_valid_curve



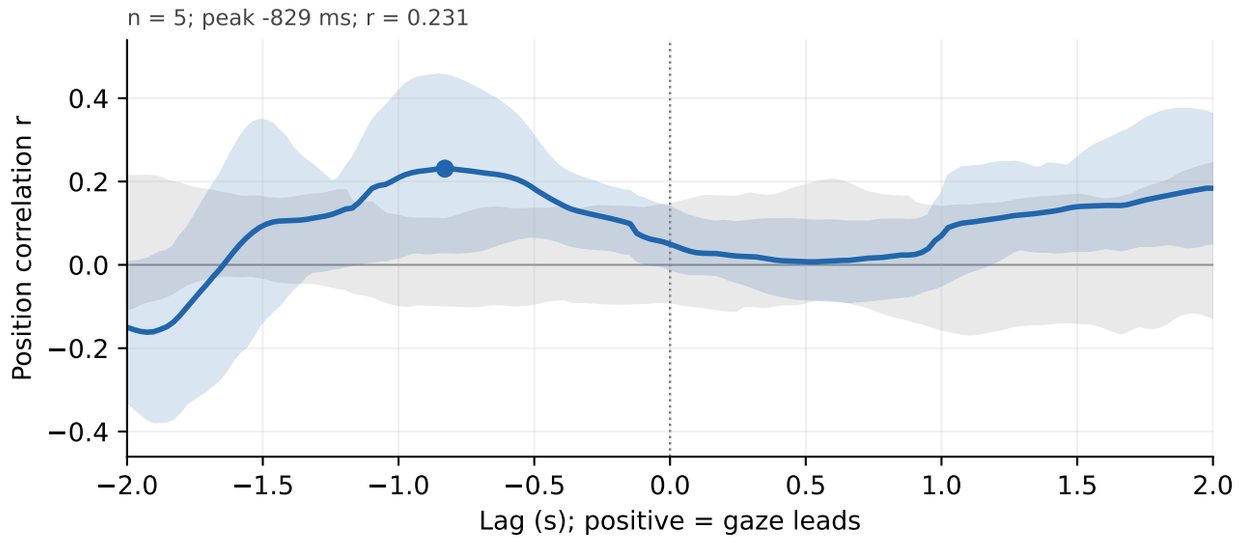
grab ring / psm_stage

n = 5; peak -659 ms; r = 0.159



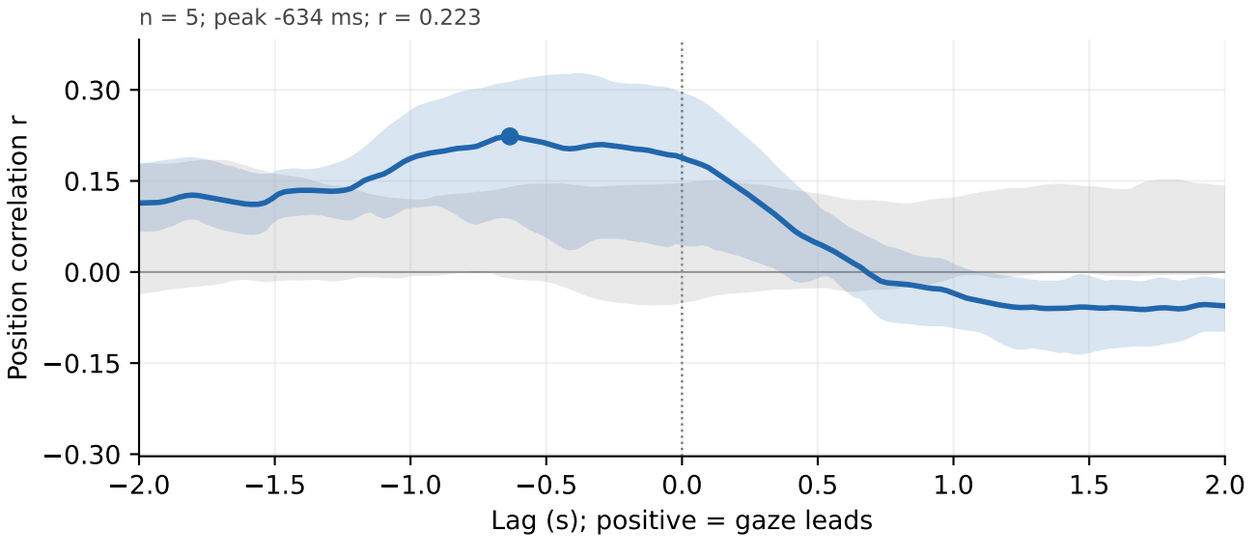
grab ring / within_stage

n = 5; peak -829 ms; r = 0.231



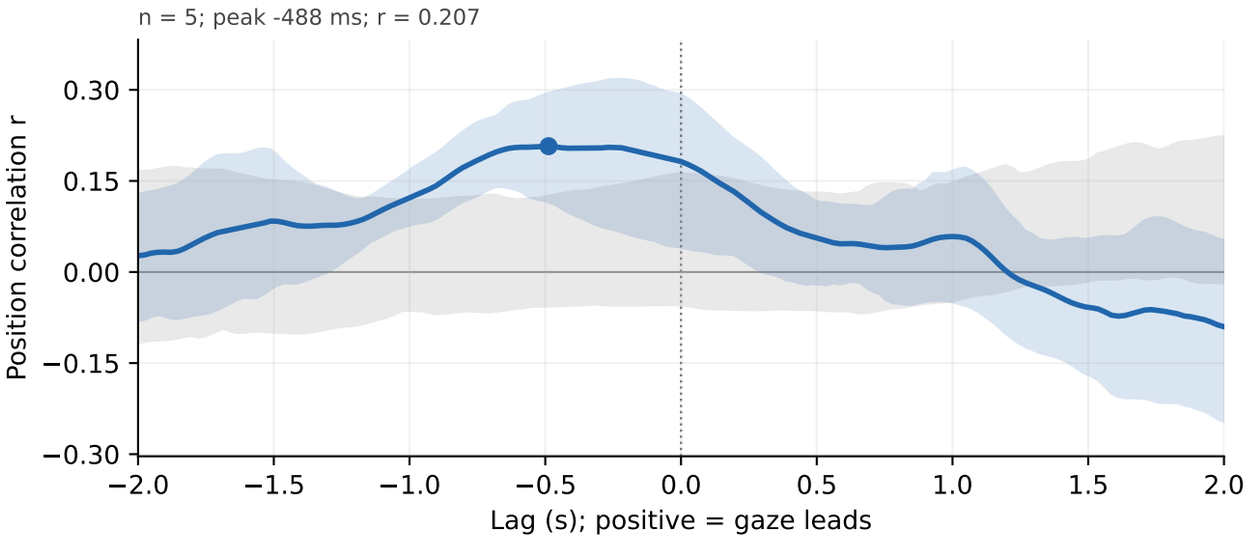
place ring / psm_stage

n = 5; peak -634 ms; r = 0.223



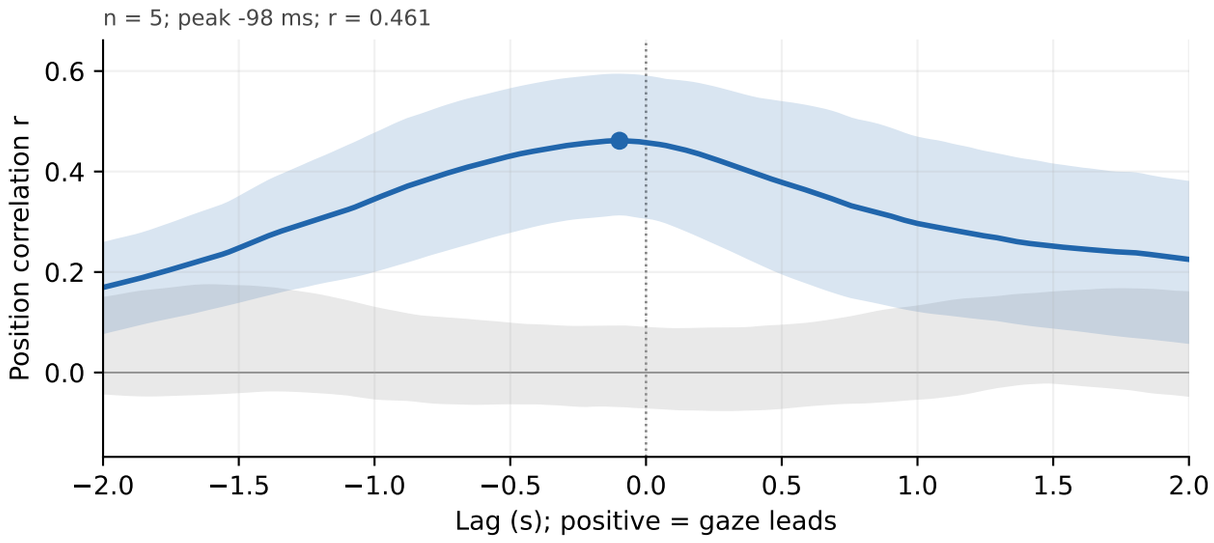
place ring / within_stage

n = 5; peak -488 ms; r = 0.207



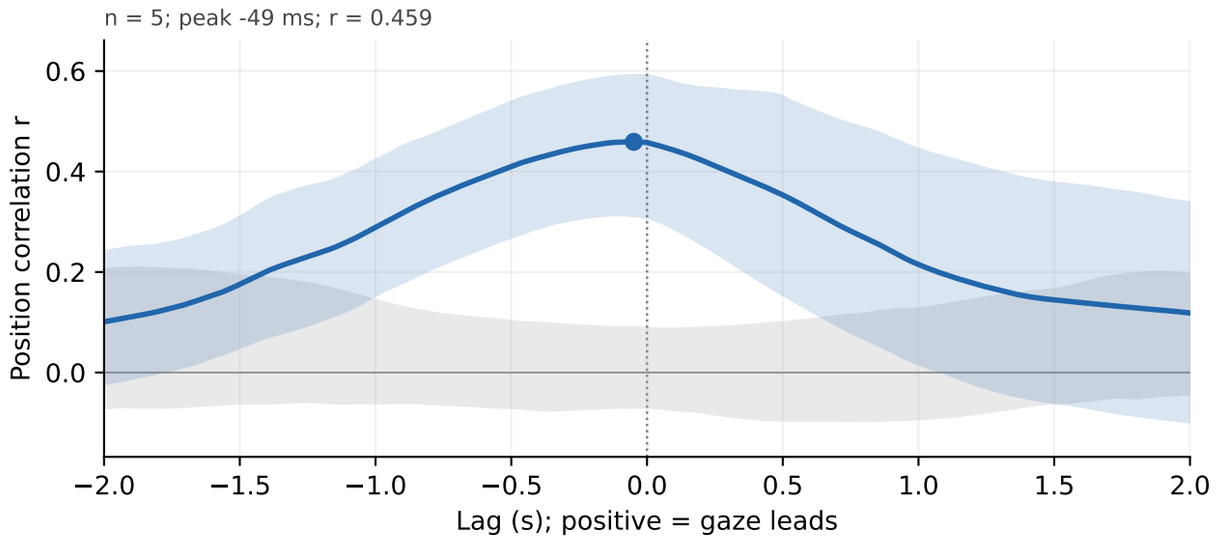
reach ring / psm_stage

n = 5; peak -98 ms; r = 0.461



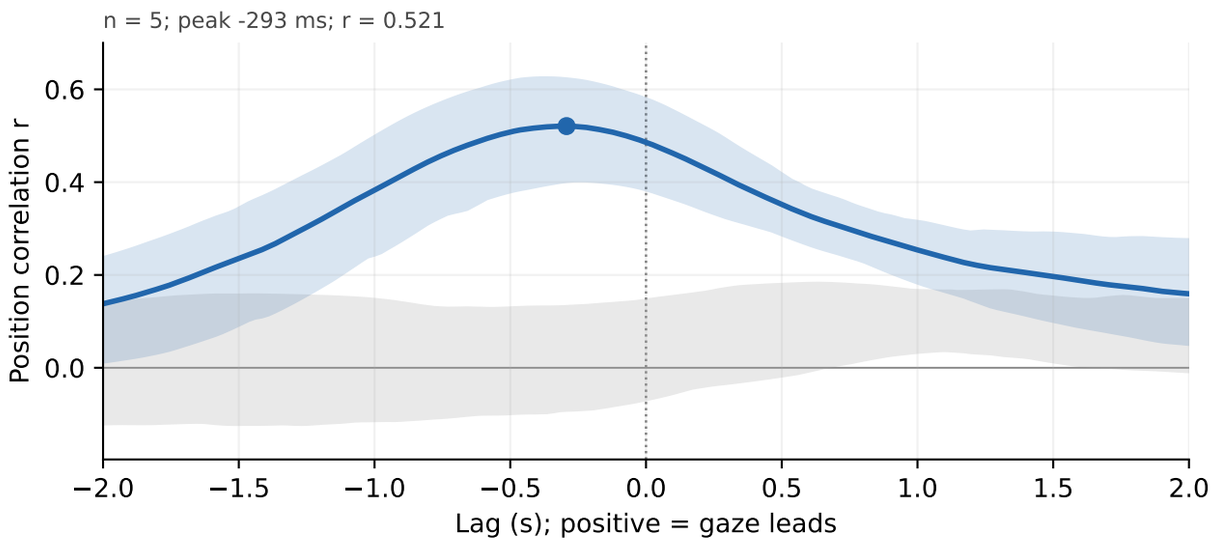
reach ring / within_stage

n = 5; peak -49 ms; r = 0.459



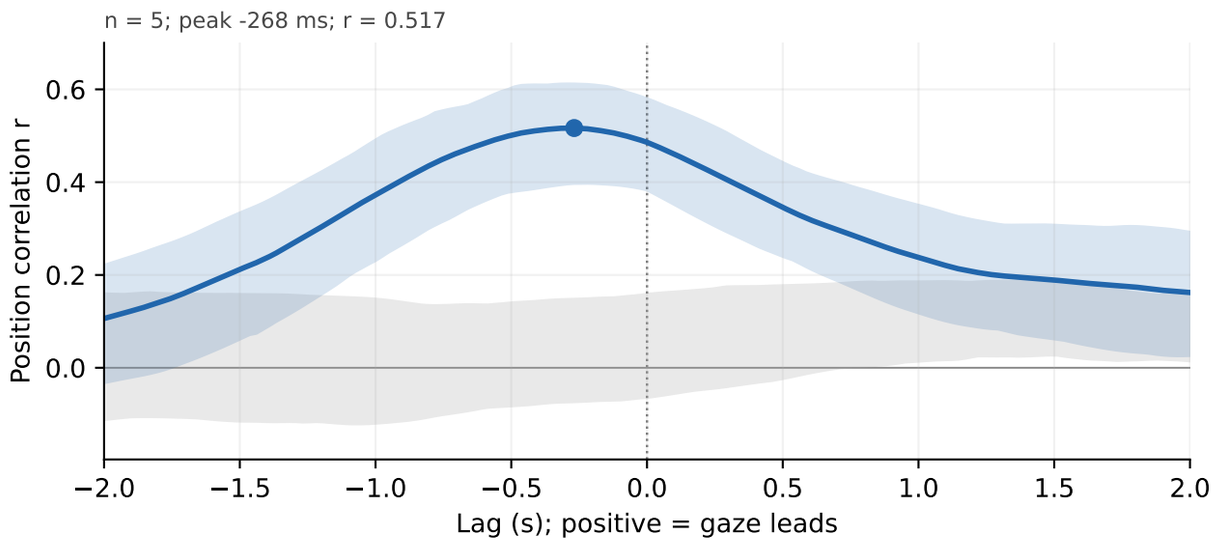
transfer ring / psm_stage

n = 5; peak -293 ms; r = 0.521



transfer ring / within_stage

n = 5; peak -268 ms; r = 0.517

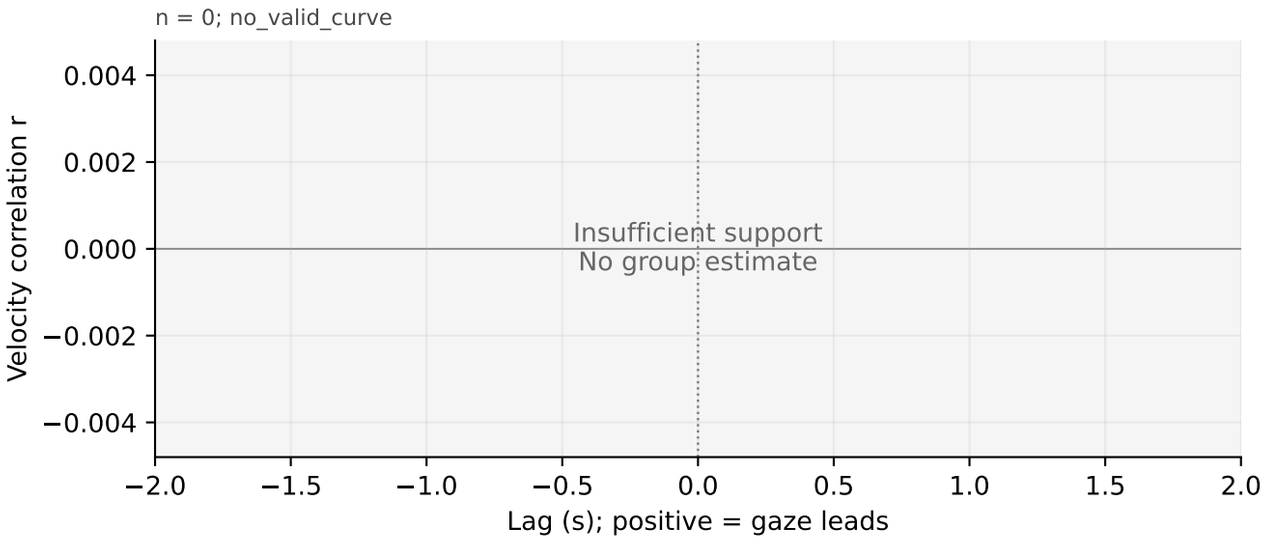


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

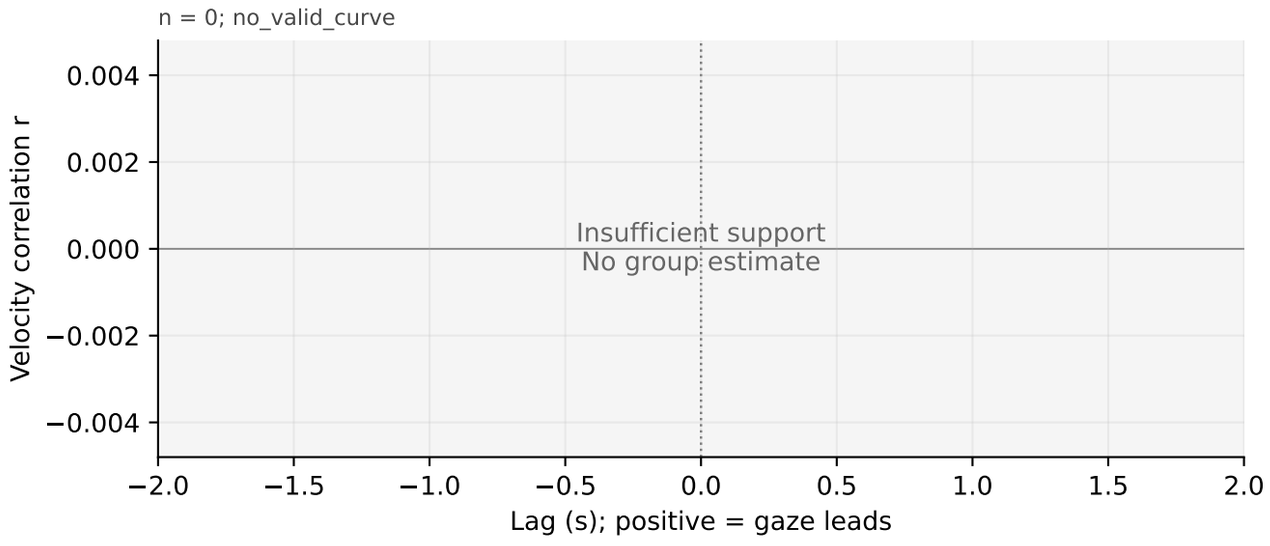
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) · Participant bootstrap (95%) · Group mean

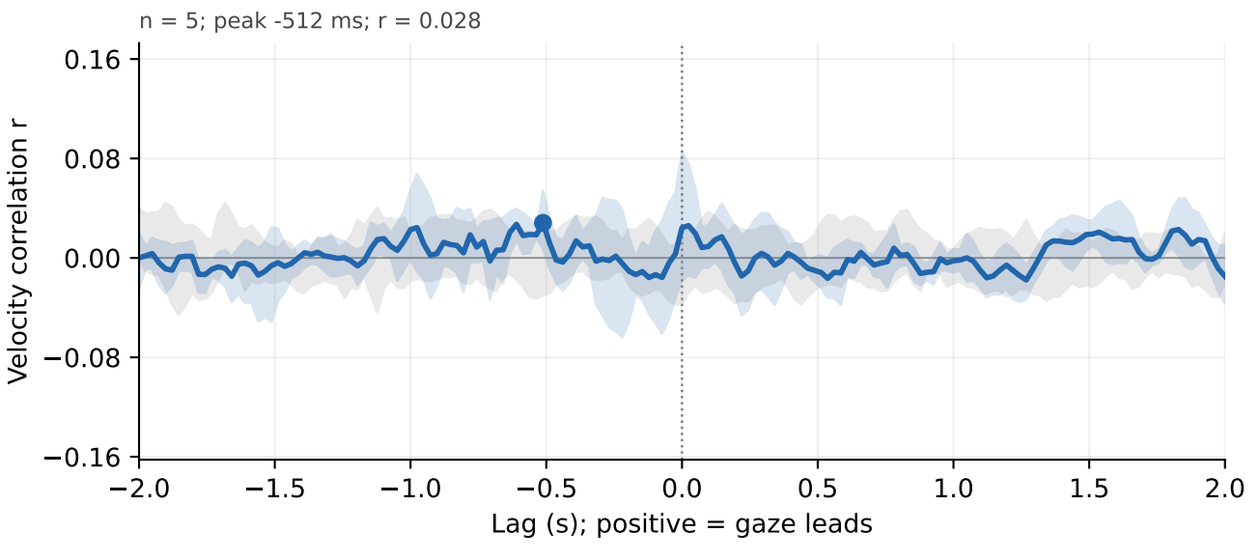
drop ring / psm_stage



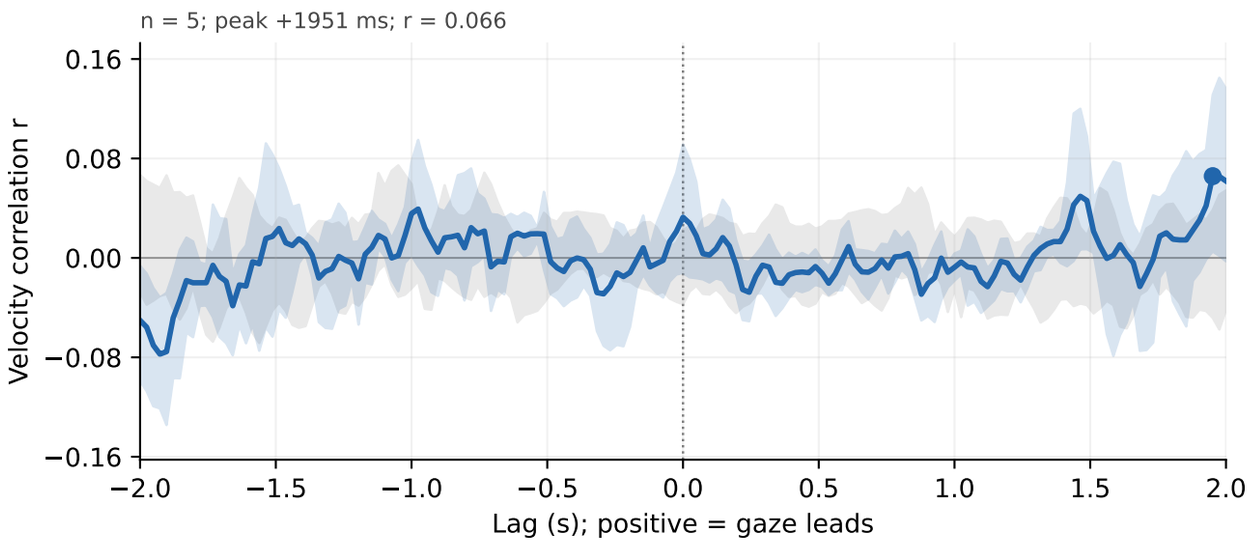
drop ring / within_stage



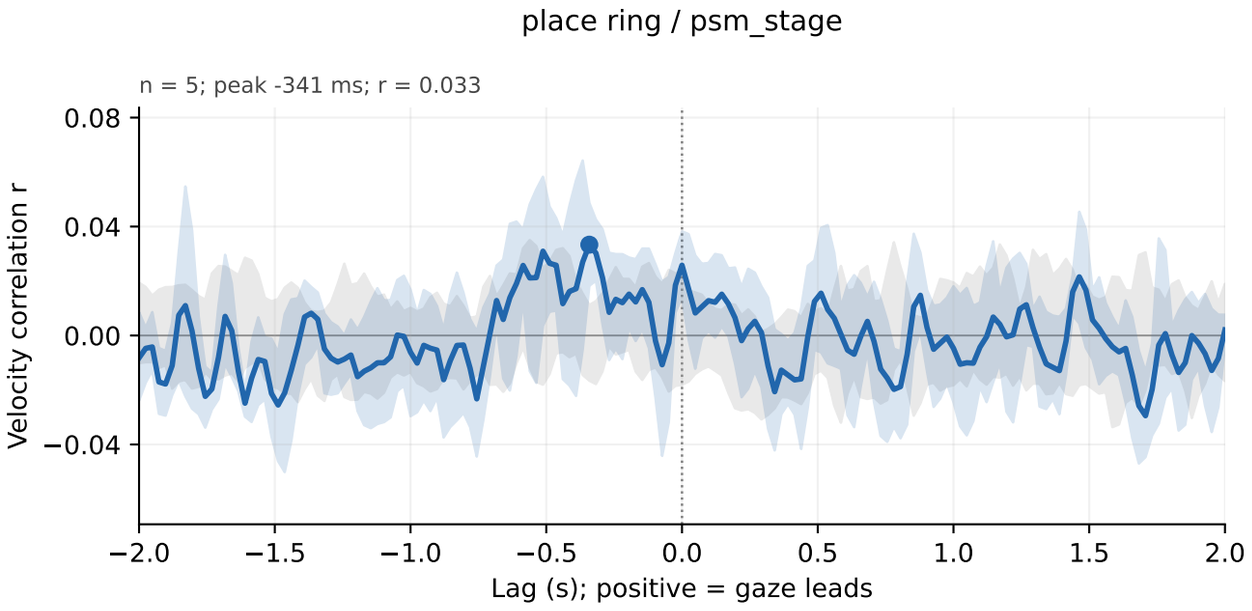
grab ring / psm_stage



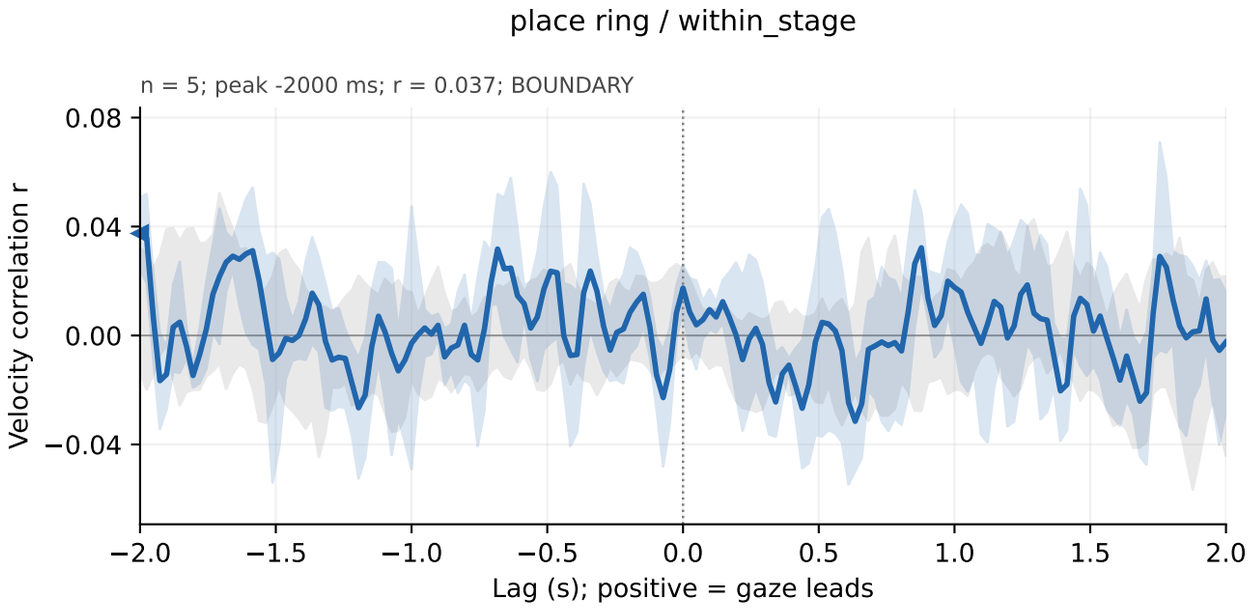
grab ring / within_stage



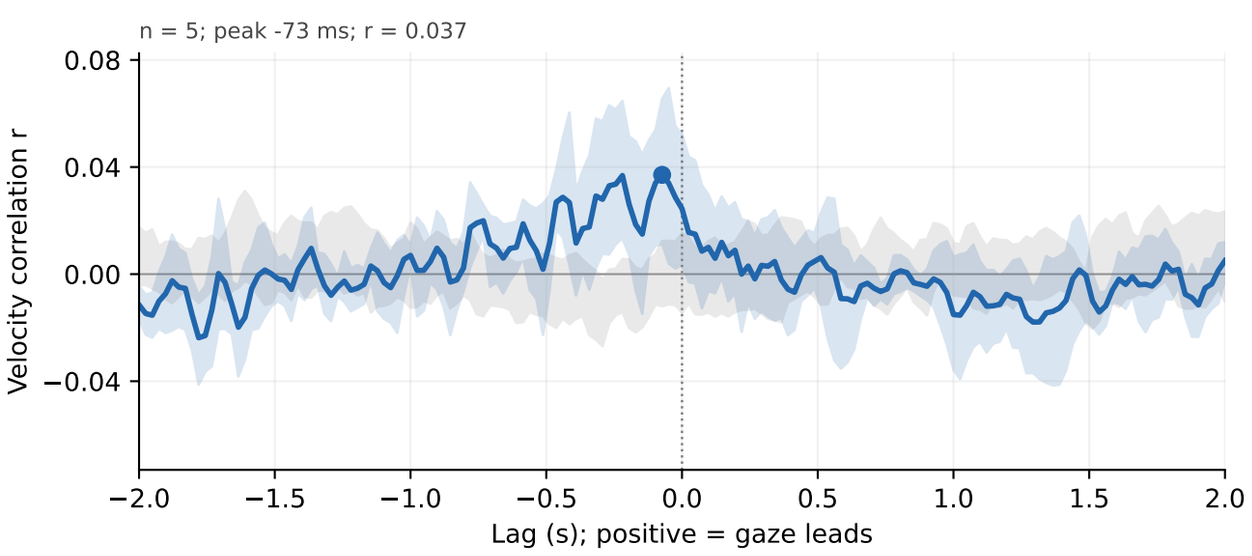
place ring / psm_stage



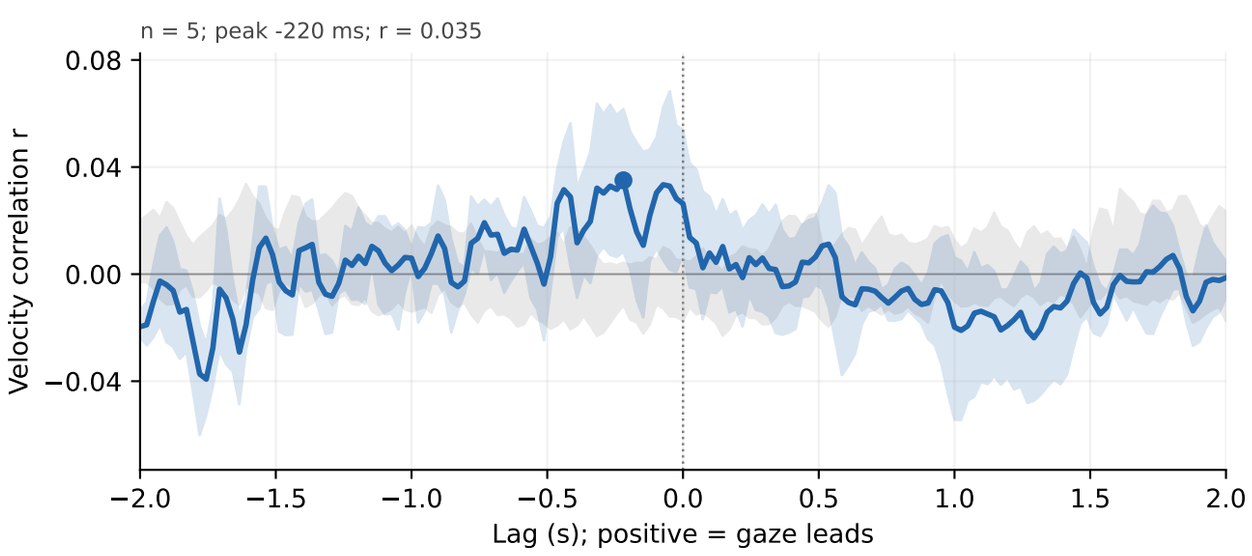
place ring / within_stage



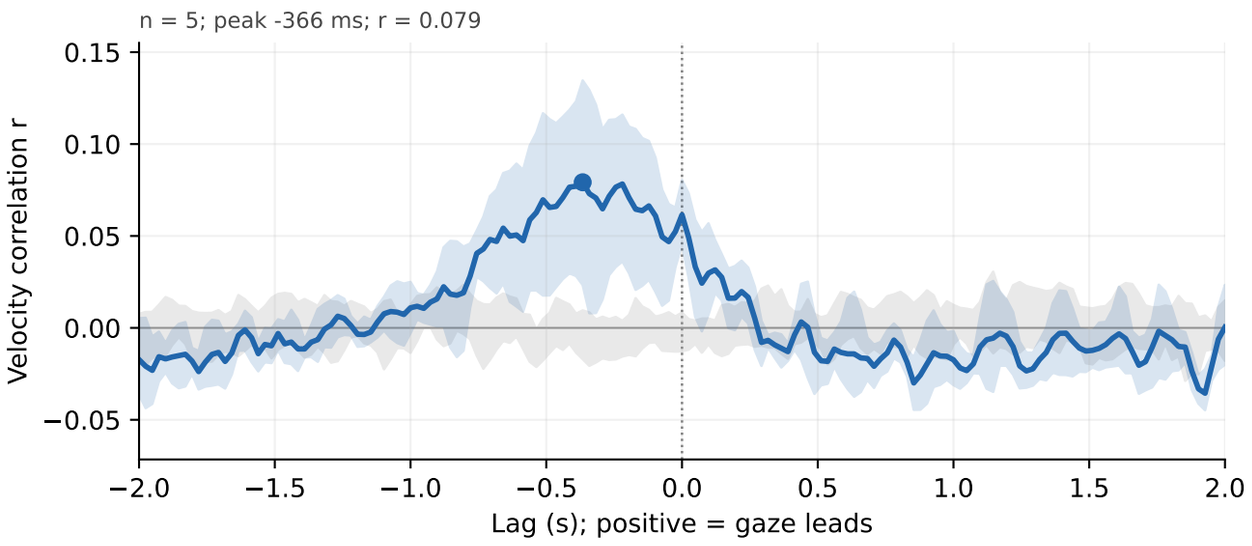
reach ring / psm_stage



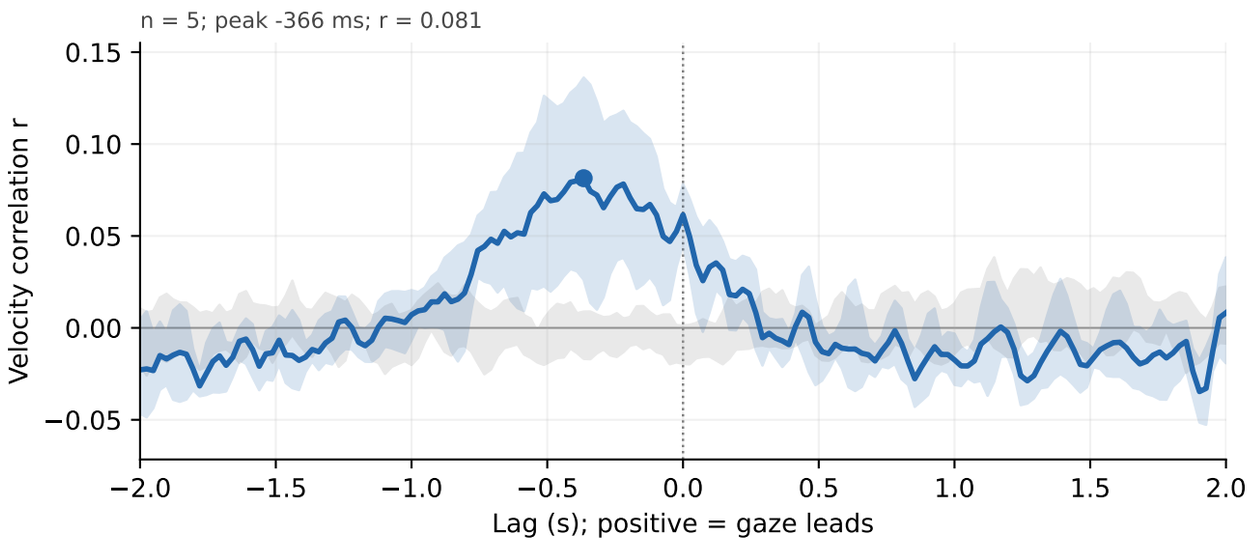
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

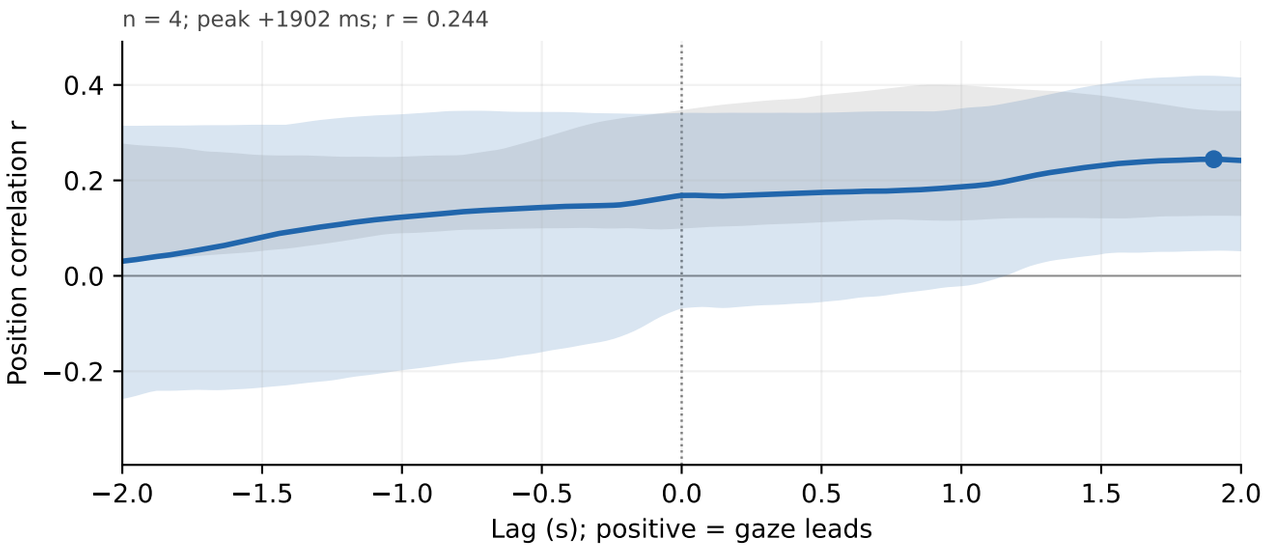


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

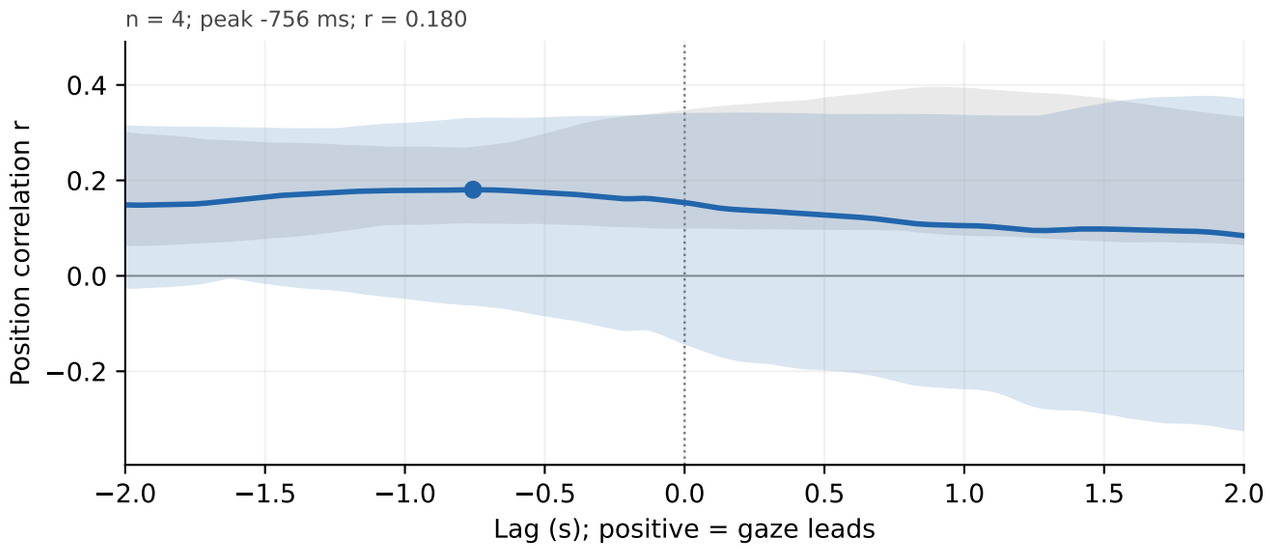
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

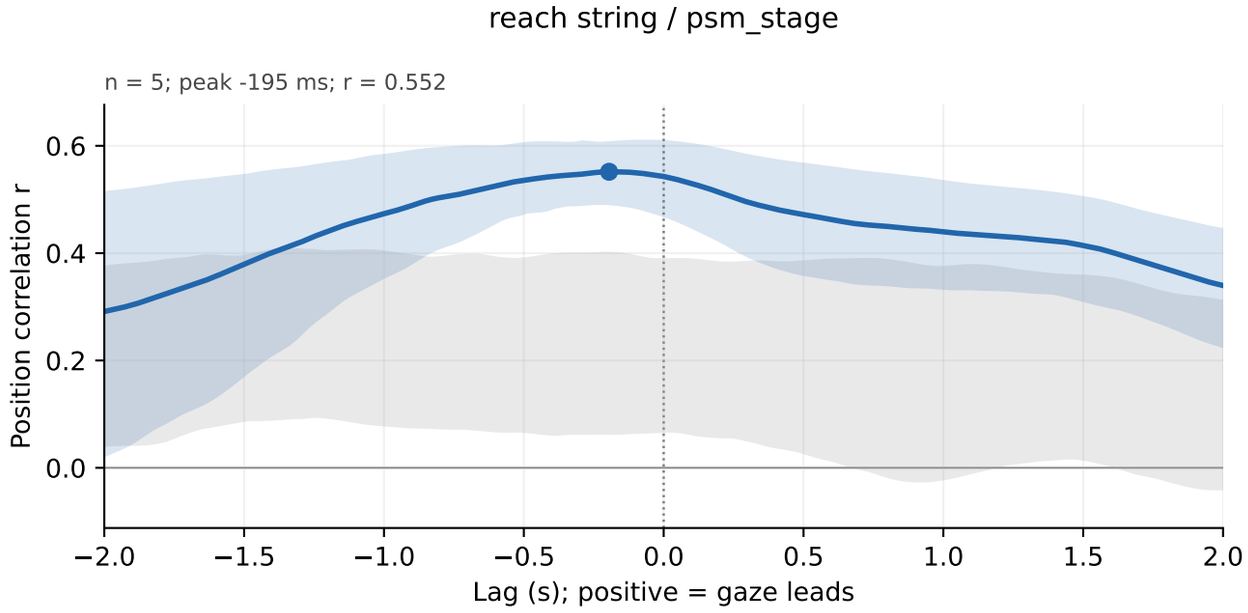
pull through and cinch / psm_stage



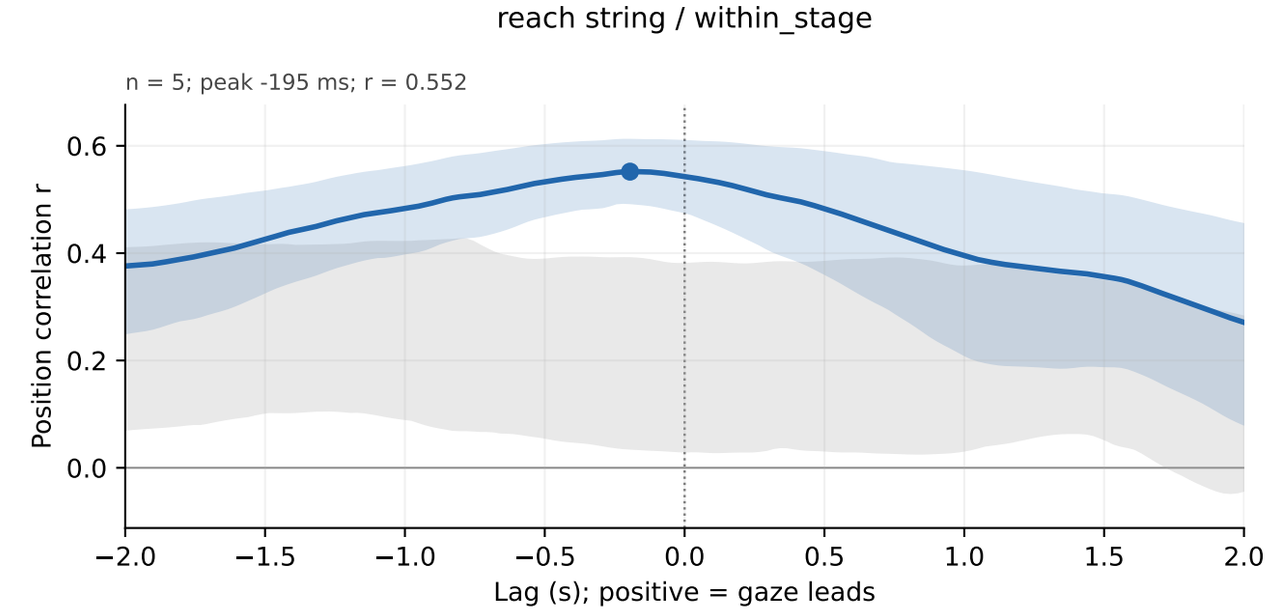
pull through and cinch / within_stage



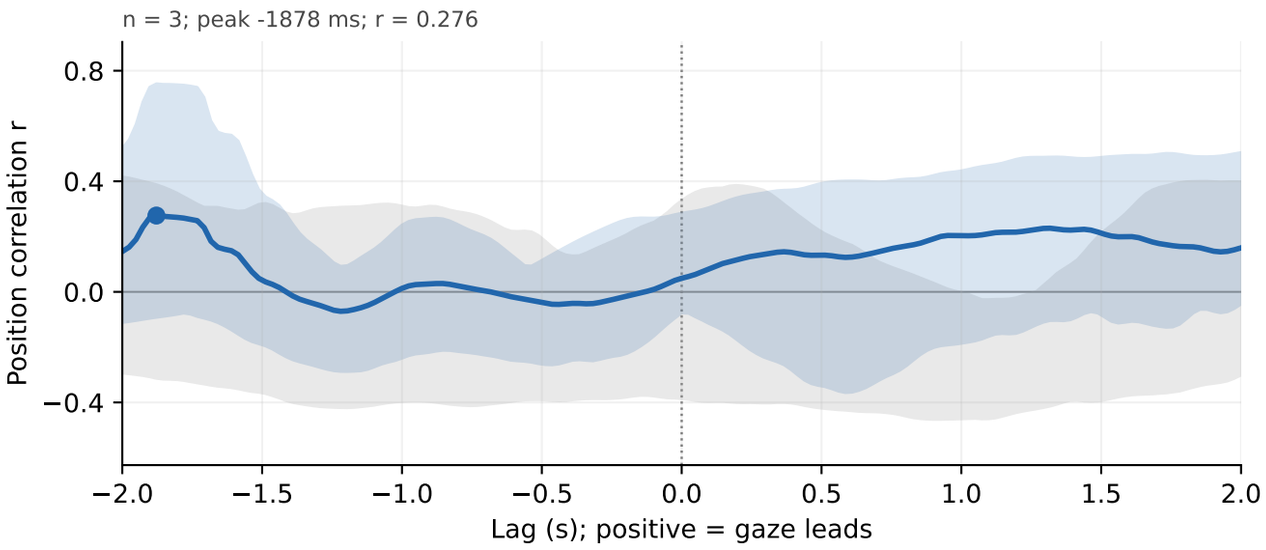
reach string / psm_stage



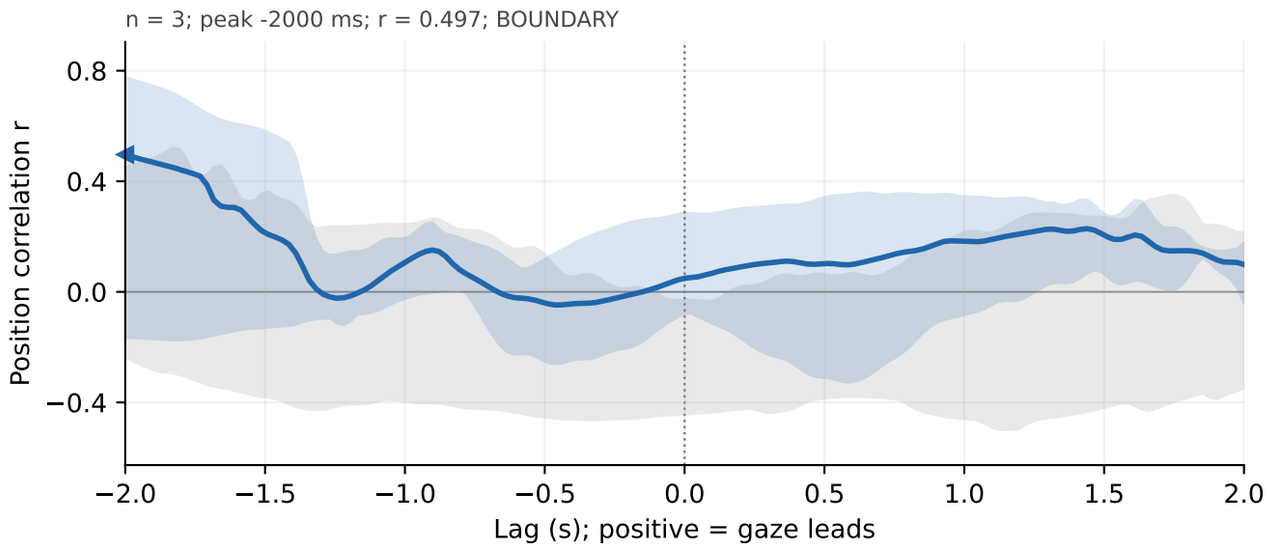
reach string / within_stage



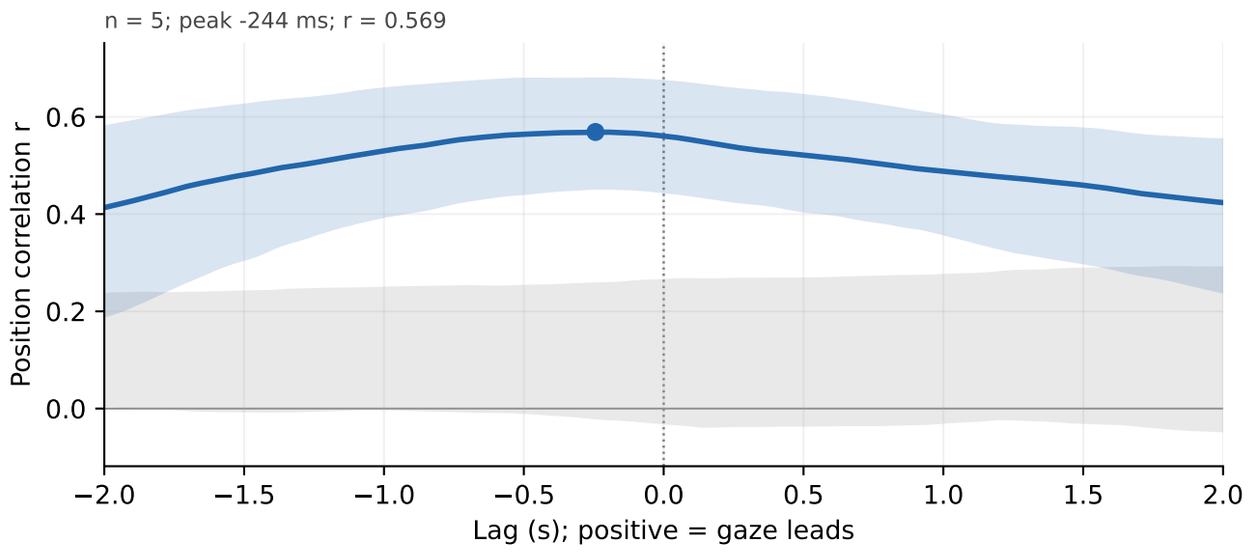
threading string through tower / psm_stage



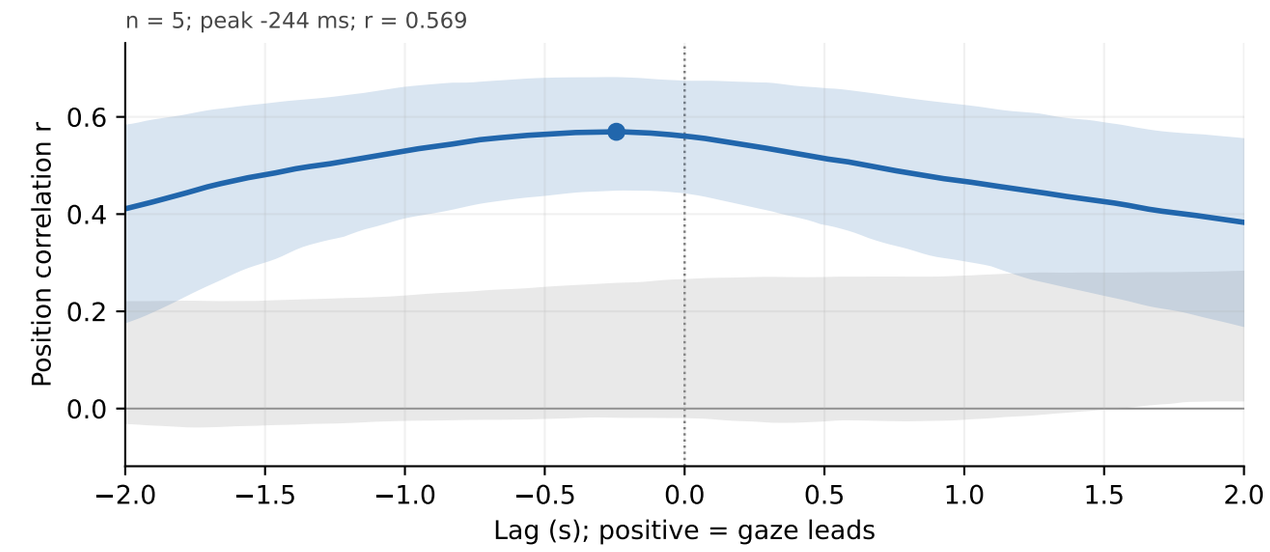
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

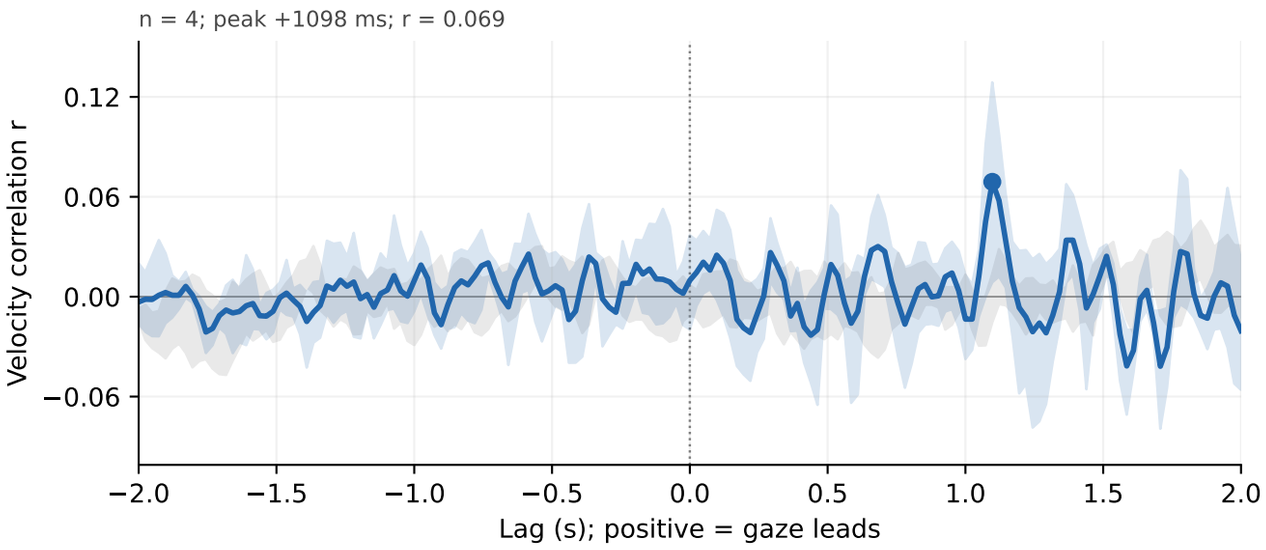


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

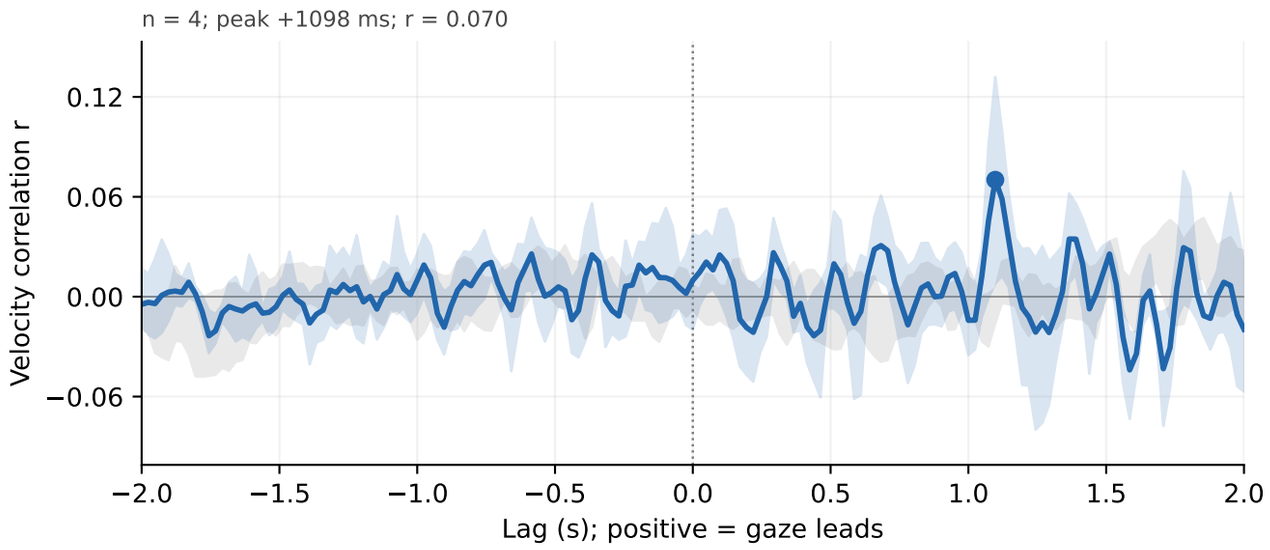
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

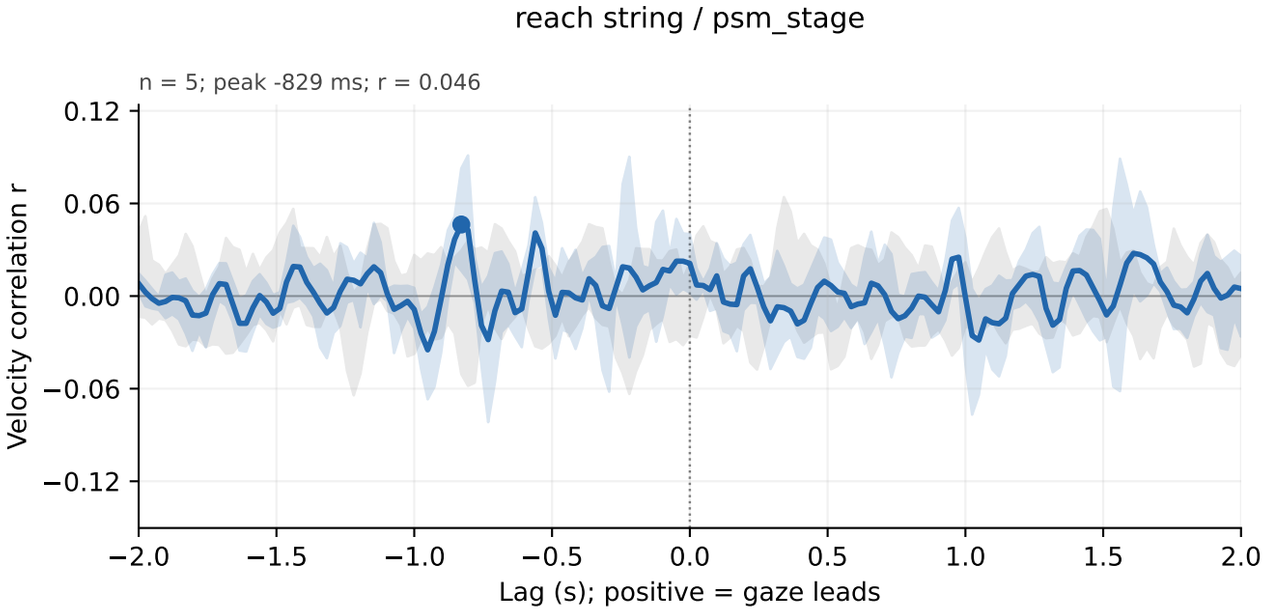
pull through and cinch / psm_stage



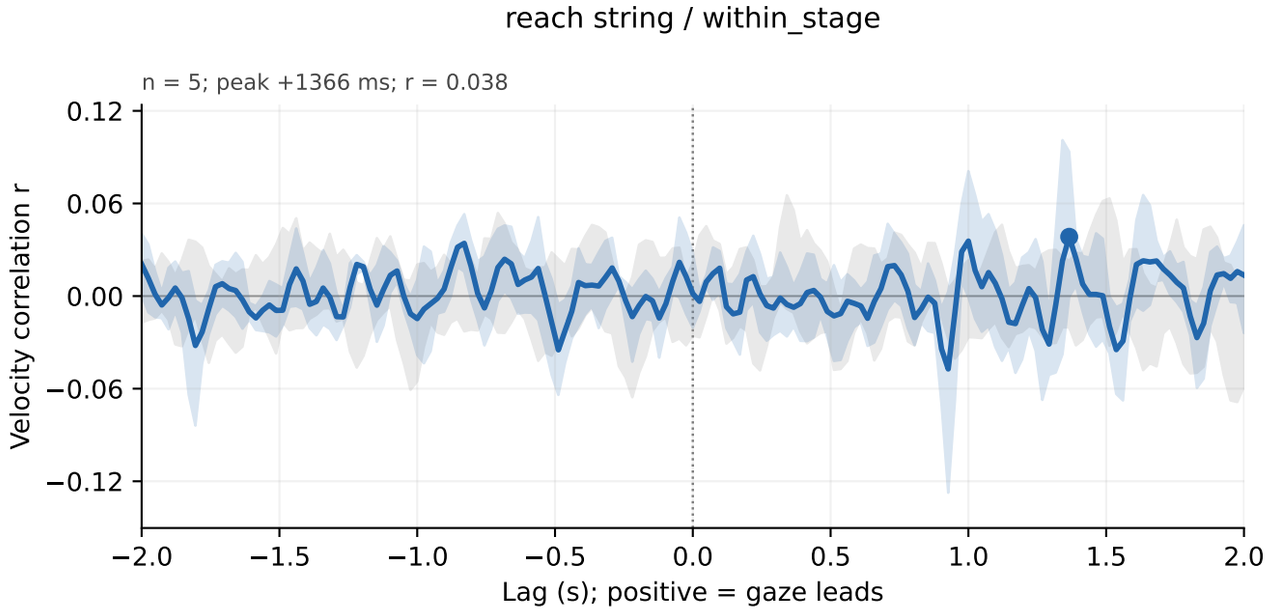
pull through and cinch / within_stage



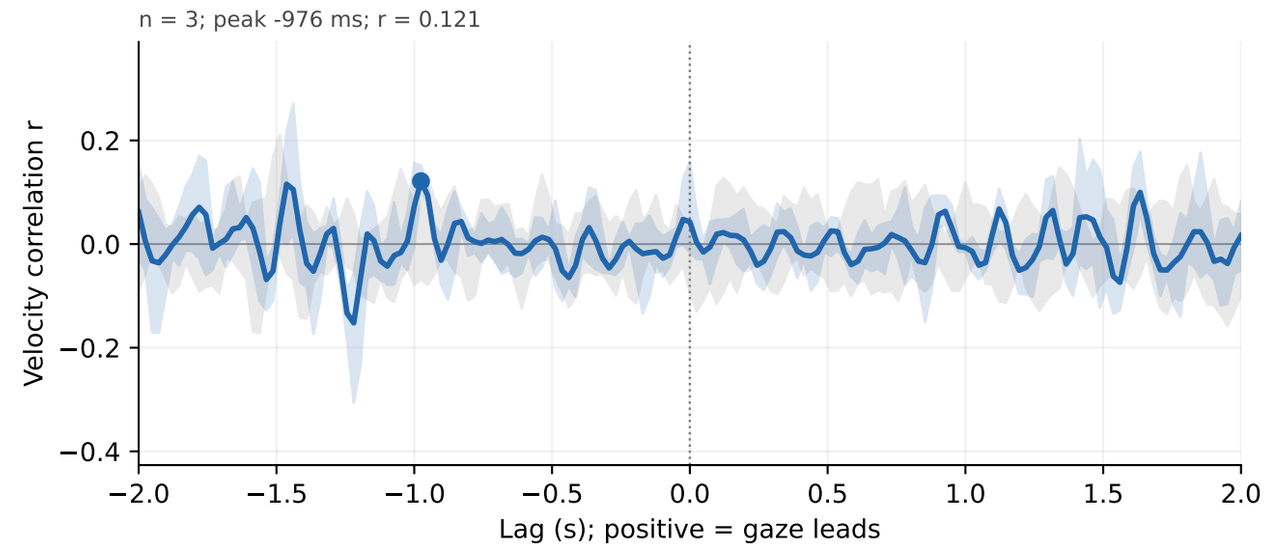
reach string / psm_stage



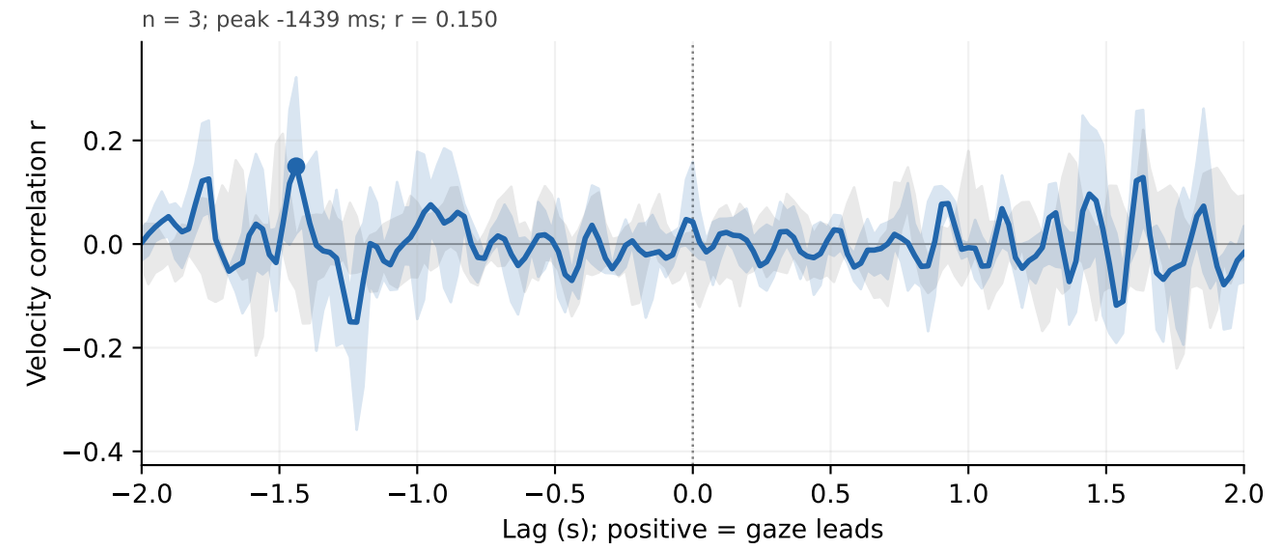
reach string / within_stage



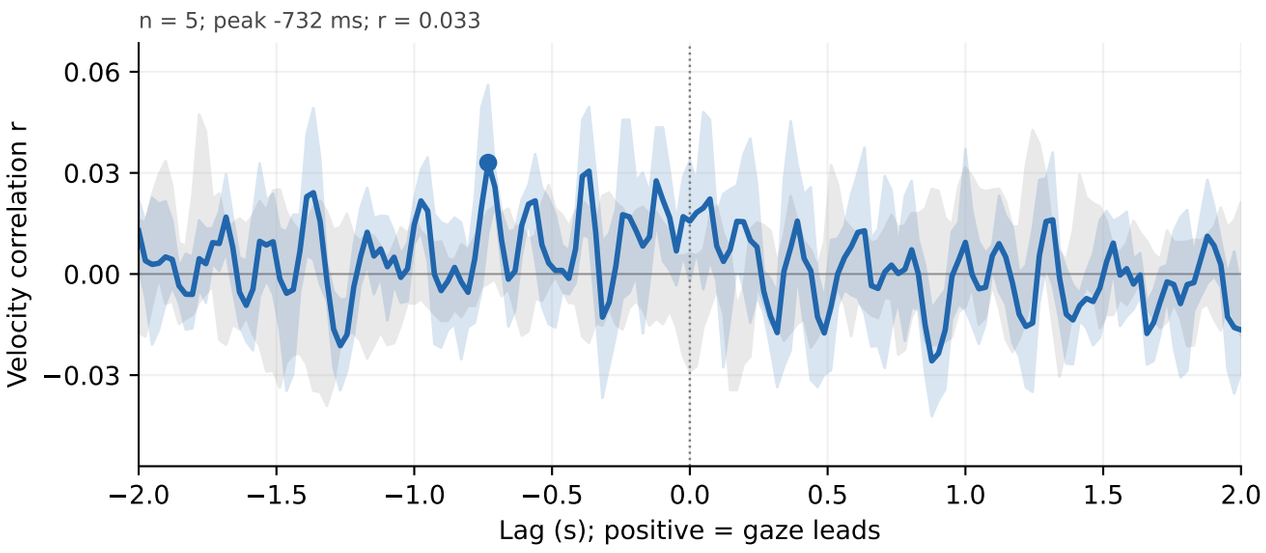
threading string through tower / psm_stage



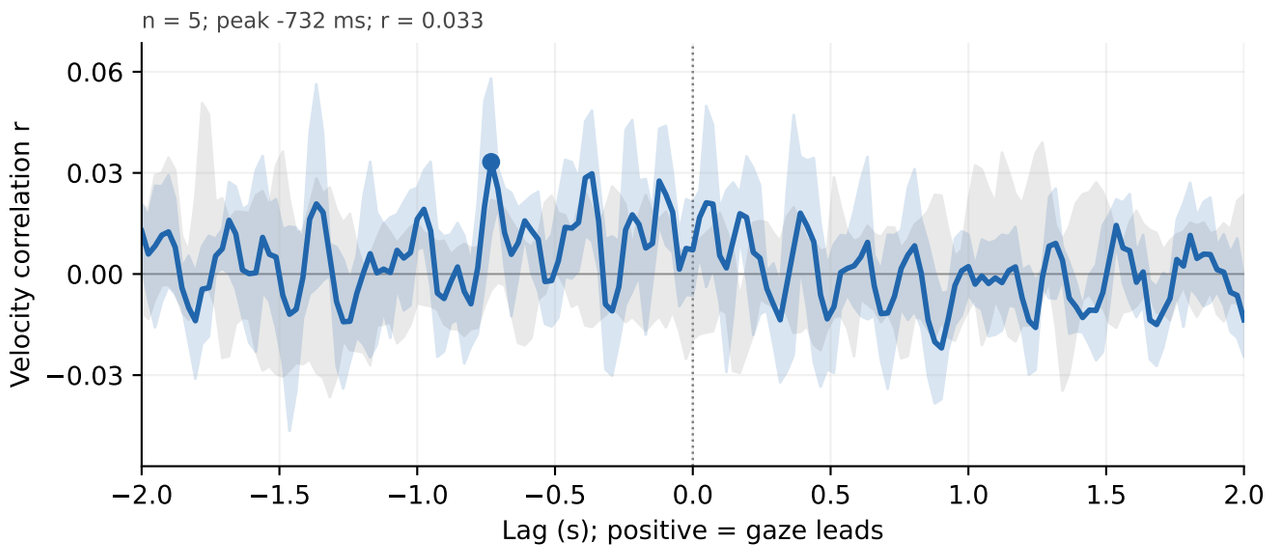
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.